MANAGERIAL ACCOUNTING

MBA, SEMESTER-I

According to the New Syllabus of **'Savitribai Phule Pune University, Pune'**

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MANAGERIAL ACCOUNTING

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"Dedicated to

*my all **Students**"*

- Dr. Mandakini R. Bhosale

"Dedicated to

*my **Wife** and **Parents***
who always stood by my side in all the ups and downs.
It's just because of their blessings and support, this research is been possible."

- Dr. Surinder Sethi

Preface

"Managerial Accounting" and used as a means of communicating pertinent facts to managers as well as to others which helps in making the right and fruitful decisions for the organisation.

This book, **Managerial Accounting** has been specially designed as per the New Syllabus of **MBA First Semester** of **Savitribai Phule Pune University, Pune** and gives exposure to basics of accounting with introducing different branches of accounting by giving details to cost accounting. It also explicates the understanding of financial statements.

Along with text serves as a foundation course which is coupled with exercise, model paper, solved question papers and index is also included in this book to enhance the understanding of the students.

With all the new features included and extensive revision done, we feverishly hope that the book would appeal to the students, the teachers and all the interested readers. All the suggestions and feedbacks are welcomed to further improve the quality of the content to achieve the objective of presenting this book.

Please e-mail us at, **thakurpublication@gmail.com**

Website, **www.tppl.org.in**

Acknowledgement

I would like to express my gratitude to God who has enlighten me throughout my life. I would like to thank my **Family** members and my **Close Friends** who have appreciated me and given moral support. I hope this book will be the perfect guide to gain thorough knowledge about the subject.

I sincere regards to all those who have helped me directly and indirectly completing this book on time.

Last but not the least, I would like to thank a lot to **Thakur Publication Pvt. Ltd.** and the whole team, especially **Mrs. Priyanka Singh** (Editor-in-Chief), **Mr. Raj Kumar Singh** (Sr. Finance Editor) and **Mrs. Sona Singh** (Marketing Coordinator), for facilitating the work and providing an opportunity to be a part of their organization.

- Dr. Mandakini R. Bhosale

I would like to express my deep gratitude towards my **Principal** and the **Head** of our **MBA Department** for their thorough support, guidance and encouragement throughout my research journey.

I would also extend my appreciation towards all my **Colleagues** who supported me in the entire process.

I wish to appreciate the extended support of my **Family** without which it wouldn't have been possible.

Finally, My cordial thank goes to **Thakur Publication Pvt. Ltd.** specially **Mrs. Priyanka Singh** (Editor-in-Chief), **Mr. Raj Kumar Singh** (Sr. Finance Editor) and **Mrs. Sona Singh** (Marketing Coordinator) for publishing this book on time, their contribution to this work is beyond description for which I am deeply thankful to them.

- Dr. Surinder Sethi

Syllabus

101 - MANAGERIAL ACCOUNTING

Unit	Contents	Sessions
1	**Basic Concepts:** Forms of Business Organization. Meaning and Importance of Accounting in Business Organization, Basic Concepts and Terms used in Accounting, Capital & Revenue Expenditure, Capital & Revenue Receipts, Users of Accounting Information. Accounting Concepts and Conventions, Fundamental Accounting Equation, Journal, Ledger and Trial Balance.	(4+2)
2	**Financial Statements:** Meaning of Financial Statements, Importance and Objectives of Financial Statements. Preparation of Final Accounts of Sole Proprietary Firm.	(7+2)
3	**Cost Accounting:** Basic Concepts of Cost Accounting, Objectives, Importance and Advantages of Cost Accounting, Cost Centre, Cost Unit, Elements of Cost, Classification and Analysis of Costs, Relevant and Irrelevant Costs, Differential Costs, Sunk Cost, Opportunity Cost, Preparation of Cost Sheet.	(8+2)
4	**Short Term Business Decision Techniques – Marginal Costing:** Meaning, Principles, Advantages and Limitations, Contribution, P/V Ratio, Break-Even Point (BEP), Cost Volume Profit (CVP) Analysis, Short Term Business Decisions– Product Mix Decisions, Make or Buy (Outsourcing) Decisions, Accept or Reject Special Order Decisions, Shutting Down Decisions.	(8+2)
5	**Exercising Control – Budgetary Control & Standard Costing: Budgetary Control:** Meaning of Budget and Budgeting, Importance, Advantages and Disadvantages, Functional Budgets–Raw Material Purchase & Procurement Budget, Cash Budget and Flexible Budget. **Standard Costing:** Meaning, Importance, Advantages and Disadvantages, Cost Variance Analysis. Material Variances – Material Cost Variance, Material Rate Variance, Material Usage Variance, Material Mix Variance and Material Yield Variance. Labour Variances –Labour Cost Variance, Labour Rate Variance, Labour Efficiency Variance, Labour Mix Variance, Labour Idle Time Variance and Labour Yield Variance.	(8+2)

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UNIT 1

Basic Concepts of Accounting

1.1. BUSINESS ORGANIZATIONS

1.1.1. Meaning of Business Organizations

A **business organisation** is structured and operated in an authorised form of ownership. These enterprises are established with a major concern of earning huge profits. A business can be operated by one or a group of persons either privately by the government or under other public undertakings in the public sector. The type of business is started and owned by a single person, and is known as sole proprietorship. Whereas, a business run by two or more persons can be referred to as cooperatives, joint stock company or partnership firms.

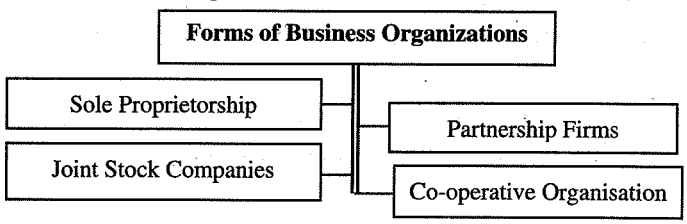
Decision-making is one of the most important skills a management needs to have in the different form of organisation. Management is required to make a large number of important decisions on regular basis. These decisions affect the eventual outcome of the business. These decisions also affect other stakeholders such as customers, workers and markets.

In **different form of organisation,** Business decision process involves the selection of most appropriate course of action out of various alternatives available. Decision-making impacts all the important aspects of business such as planning, controlling and organising.

The process of decision-making is influenced by many factors such as government regulation, market conditions, competition and other related aspects.

1.1.2. Forms of Business Organizations

Following are the major forms of business organizations:



1.1.3. Sole Proprietorship

A business which is owned by an individual and is not registered as a limited liability company with the state is known as a 'sole proprietorship'. It is the easiest form of enterprise ownership and one can be established easily without having much knowledge about it. **For example,** an entrepreneur working as a freelancer, a commission based salesman, independent contractor, a craftsman on a contract based job, etc. These entrepreneurs also called as sole proprietors.

According to L.H. Haney, "The individual proprietorship is the form of business organisation at the head of which stands an individual as one who is responsible, who directs its operations, and who alone runs the risk of failure".

According to James Stephenson, "A sole proprietor is a person who carries-on business exclusively by and for himself. He is not only the owner of the capital of the undertaking but is usually the organiser and manager and takes all the profits or responsibility for losses".

Features of Sole Proprietorship

Following are the distinctive features of sole proprietorship:

1) **Sole Ownership:** In this form, the business enterprise is solely owned by an individual. The capital required for the business is arranged from his personal funds or by borrowing funds from others.
2) **No Separate Legal Entity:** A sole proprietorship does not hold any separate legal entity from that of its owner. In the eyes of law, the owner and the business exist together. But the business will have no existence, if its owner dies or becomes insolvent. Both are considered as a single unit.
3) **Unlimited Liability:** The proprietor is the only person liable for all its debts. If the debts of the company are more and cannot be fulfilled by the assets then the proprietor can involve his personal property for clearing all dues.
4) **No-Profit Sharing:** A sole proprietor as an individual is entitled to all profits and all losses. There is no one to share the profits or losses with him. He is the only person to bear the entire losses of the company.
5) **No Legal Formalities:** No legal formalities are associated with the establishment or management of the sole proprietorship business. However, only a license is required for running the business.

1.1.4. Partnership Firms

Partnership is a contractual agreement between two or more persons who are willing to share the profits of a business which is owned by all or a single person acting for all. In general, it a business relationship between different persons having a common ownership or management of a business enterprise. The minimum number of persons required for a partnership is 2 and the maximum limit can be of 10 members (for banking industries) and 20 members (for other industries).

According to Section 4 of the Indian Partnership Act, 1932, "The relation between persons who have agreed to share profits of a business carried-on by all or any of them acting for all".

According to Kimball, "A partnership or firm as it is often called is, then a group of men who have joined capital or services for the prosecuting of some enterprise".

According to Haney, "Partnership is the relation between persons competent to make contract, who agree to carry-on a lawful business in common with a view to private gain".

Features of Partnership Firms

Following are the distinctive features of partnership firms:

1) **Two or More Persons:** At minimum two persons, and maximum 10 (in banking industry) and 20 (in other industries), are required for establishing a partnership firm. An individual cannot form a partnership with himself.
2) **Agreement or Contract:** There is an agreement or contract held between persons who are willing to share the business ownership. Usually, no legal problems are associated with these contracts or agreements as they are registered under specific terms and conditions.
3) **Lawful Business:** Partnership is an act of managing a business and earning profit, which is legally permitted by the government. Whereas, partnership formed for other purposes such as social or charitable work, or any illegal activity like black marketing, smuggling, etc. does not come under partnership law.
4) **Profit Sharing: Under the** partnership firm, all the profits are equally or proportionally shared between the partners as mentioned in the agreement. Likewise, losses are also shared in the same proportion among the partners.
5) **Mutual Agency:** Mutual agency is considered as a fundamental characteristic of partnership firms. As it is very important to have good faith, mutual trust and confidence between the partners.

1.1.5. Joint Stock Companies

A company legally owned by the shareholders and managed by an elected "Board of Directors", having limited liability in known as the 'Joint Stock Company'. In today's world, Joint Stock Company is the most suitable form of business ownership. It is considered as the only systematic and standard way of managing a business.

The Joint Stock Company is an association of persons in an organisation possessing a separate legal entity, perpetual succession with a unique name and a common seal. The capital of the company is divided into shares which are transferrable and it also has limited liability.

According to Prof. Haney, "Joint Stock Company is a voluntary association of individuals for profit, having a capital divided into transferable shares, the ownership of which is the condition of membership".

According to Chief-Justice Marshall, "A joint stock company is an artificial person invisible, intangible, and existing only in the eyes of law. Being a mere creature of law, it possesses only those properties which the charter of its creation confers upon it, either expressly or as incidental to its very existence".

Features of Joint Stock Companies

The distinctive features of joint stock companies are as follows:

1) **Artificial Person with a Separate Legal Entity:** A Joint Stock Company is intangible in nature and has no physical existence. It is an artificial person having an independent legal entity. It can purchase, sell or possess properties under its own seal. Various properties can be owned by Joint Stock Company and it can purchase or sell them under their seal. This company can conduct any other business and can sign an agreement with other businesses.
2) **Voluntary Association:** It is a voluntary association of individuals, which is established by the people themselves. It is not compulsory to follow any government rules and regulations while forming it.
3) **Perpetual Character:** This company exists permanently, it continuous until it is dissolved or sold off. It involves shareholders, who may join or leave the company or may change hands, as and when required.
4) **Common Seal:** Since the company has no physical existence, it is not liable to sign any document. For this purpose, the company has a common seal with a unique name on it, which is used for the purpose of signature. This seal is always kept safe and secure. There must be two directors present as a witness whenever the seal is used by the company on any document.
5) **Limited Liability:** In this business form, the shareholders or owners of the company have limited liability. This liability is limited up to the amount of capital invested by each shareholder.

1.1.6. Co-operative Organisation

A co-operative organisation is a legal entity which is owned and controlled by a group of persons having a common motive. There are several attributes of co-operatives like an elected Board of Directors, an administrative staff, indefinite life span, limited liability, etc. The owner of the company is liable to pay a fixed amount of fee to the co-operatives. The company's profit is shared by all the members in proportion to their contributions. This form of organisation is free to pay any kind of tax as they retain no profits.

It is an independent organisation which is owned and managed by its members or customers who buy the company's goods and services. Investors do not have any kind of role to play in a co-operative organisation.

The main purpose of co-operatives is to fulfil the needs of its members and owners. Unlike, a Joint Stock Company, it is not just about accumulating capital from the investors. This organisation follows a principle of 'one member, one vote', and each member has equal control over the cooperative.

The co-operative organisation stands on five pillars, i.e., equality, mutual trust, supervision, self-reliance and spontaneity. The main strength of an organisation is its cooperative spirit.

According to the International Labour Organisation, "A cooperative organisation is an association of persons, usually of limited means, who have voluntarily joined together to achieve a common economic end through the formation of a democratically controlled organisation, making equitable contributions to the capital required, and accepting a fair share of risks and benefits of the undertaking".

According to Indian Cooperative Societies Act, 1912, "A cooperative society has its objective, the promotion of economic interests of its members in accordance with cooperative principles".

Features of Co-operative Organisation

The distinctive features of co-operative organisation are as follows:

1) **Voluntary Association:** A co-operative organisation is a voluntary association of members. It is not at all mandatory to become its member.
2) **Legal Entity:** A co-operative organisation is an autonomous legal entity with a distinctive identity. After getting registered, it attains a common seal and a permanent succession.
3) **Equal Voting Right:** Regardless of the number of shares owned by the members of cooperative society, each member is eligible to vote only once. This confirms that a rich member or a member with more shares is not allowed to apply his own terms and conditions as everyone is treated equally.
4) **Service Objective:** Co-operatives are service-oriented organisations. They provide non-profit services primarily to their members. However, they can earn profits by rendering their services to outsiders other than its members.
5) **Economic Objective:** Along with service objectives, every co-operative organisation also has economic objectives. These objectives include delivering marketing services, providing land resources, distributing consumer goods, etc. **For example,** for an agricultural-based cooperative, economic objectives will be circulation of enhanced seeds, providing best fertilizers, low-priced credit facilities, etc.

Merits of Co-operative Organisation

The merits of co-operative organisations are as follows:

1) **Easy to Establish:** A co-operative organisation can be formed very easily. A group of ten individuals, with common interest, can start a cooperative organisation by getting it registered with the Registrar of Cooperative Societies.
2) **Open Membership:** Any individual having a common interest can become a member of a co-operative organisation. They can join or leave the organisation at their own will.
3) **Autonomous Control:** It is an autonomous body, where the members can elect a representative by casting their votes. A representative is responsible to run and administer the organisation.
4) **Limited Liability:** In a co-operative organisation, the members have a limited liability. This acts as an advantage for them, since it is only extended to the amount of capital invested by them.
5) **Government Support:** The support provides all type of assistance to co-operative organisations. This can be provided in form of capital funds, low rate loans, tax releases, subsidies in loan repayment, etc

1.2. BASIC CONCEPTS OF ACCOUNTING IN BUSINESS ORGANIZATION

1.2.1. Meaning and Definition of Accounting

Accounting is use as the language of business. Accounting involves the procedure of recording the financial transactions in the books of accounts which are helpful to its users for analysing and interpreting the financial soundness of business. The process of 'Accounting' involves the presentation as well as the interpretation of the financial results of a company's conduct with a view to make an assessment of its financial performance. Financial Accounting provides the requisite information necessary for taking investment decisions.

According to American Institute of Certified Public Accountants (AICPA), accounting is "The art of recording, classifying, and summarizing in a significant manner and in terms of money, transactions and events which are, in part, at least, of a financial character and interpreting the results thereof".

According to American Accounting Association (AAA), Accounting is, "The process of identifying, measuring and communicating economic information to permit informed judgments and decisions by users of the information".

According to R. N. Anthony, "Accounting system is a means of collecting, summarizing, analyzing, and reporting in monetary terms the information of the business".

1.2.2. Book-Keeping *Vs.* Accounting

Basis of Difference	Book-Keeping	Accounting
1) **Scope**	Book-keeping involves recording of various business transactions (financial only) and maintenance of books of accounts.	Accounting is beyond the recording and maintaining the books of accounts. It involves analysis and interpretation of the data provided through book-keeping, drawing conclusions therefrom and advising the management in taking decisions on the basis of the conclusions so arrived at.
2) **Stage**	It is the initial stage for recording of business transactions.	It is the next (secondary) stage for recording of business transactions.
3) **Objective**	Maintenance of proper record of all the business transactions in systematic manner.	Analysis and interpretation of business transactions with a view to arriving at some meaningful conclusion and facilitating the management in taking decisions.
4) **Nature**	Nature of book-keeping is mostly repetitive, routine and clerical.	Nature of accounting is analytical, interpretational and executive.
5) **Responsibility**	The responsibility of book-keeper is to record business transactions in an accurate manner.	An accountant is responsible for analysis and interpretation of a book-keeper's work.
6) **Knowledge Level**	As the book-keeper's job is routine in nature, it is not necessary for he/she to have an in depth knowledge as compared to an accountant.	An accountant is required to have expertise knowledge of accountancy as compared to a book-keeper. Further, he/she must have analytical skills as well.
7) **Supervision**	The job of a book-keeper does not include supervision and checking the work of an accountant.	An accountant's job includes supervision and checking of the work done by the book-keeper.
8) **Staff Involved**	The function of book-keeping in an organisation is performed by junior staff.	In an organisation, the function of accounting is performed by expert senior staff.

1.2.3. Features of Accounting

The 'Accounting' is characterised by following salient features:

1) **Monetary Transactions:** In 'Financial Accounting', only those business transactions, which can be expressed in monetary terms, are taken into account. Transactions of paramount importance from business point of view, which cannot be expressed in monetary terms, are out of the purview of the 'Financial Accounting'.

2) **Historical Nature:** Transactions which are likely to take place in future cannot be captured and recorded under the system of 'Financial Accounting'. Thus, only those transactions which have already taken place in past (i.e., are historical in nature) are taken into account under financial accounting.

3) **Legal Requirement:** 'Financial Accounting' is a statutory requirement. For a company, it is mandatory to have a proper record of all the monetary transactions taken place during the course of its business and prepare 'Financial Statements' like 'Profit and Loss Account', 'Balance Sheet', etc. at the year end. These 'Financial Statements' are subject to audit also, if the level of a company's business is beyond a prescribed threshold limit.

4) **External Use:** 'Financial Statements' prepared on the basis of 'Financial Accounting' is useful even for those stakeholders, who happen to be external to the business organisation, like customers, suppliers, lenders (banks and other financial institutions), investors, regulators, etc. They can assess the financial health of the organisation on the basis of 'Financial Statements' and thereby take a correct decision for investment purpose.

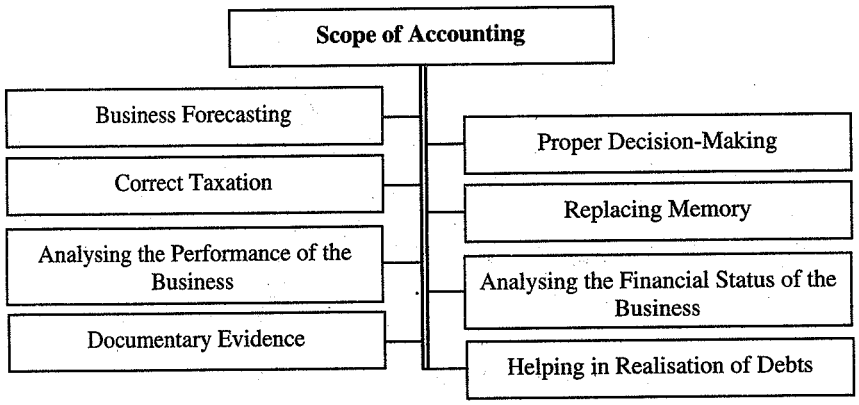
5) **Disclosure of Financial Status:** The 'Financial Statements' of a company reveal its performance during a particular period (Profit & Loss Account) and financial status/health (Balance Sheet) as on a particular date.

6) **Interim Reports:** 'Financial Statements' are the end result of the process of 'Financial Accounting'. However, they are not considered as 'Final Reports', as they are interim in nature.

7) **Financial Accounting Process:** The various steps involved in the process of 'Financial Accounting' vary according to the accounting policies adopted by various companies. Such accounting policies generally pertain to:
   i) Valuation of inventories, and
   ii) Calculation of depreciation on assets, for which a number of options are available for the companies.

1.2.4. Scope of Accounting

The scope of accounting is wide-ranging, and is evident from the following:



1) **Business Forecasting:** A business entity is interested in ascertaining its future level of business on the basis of its past as well as present level of business activities. Such forecasting enables it to chalk out strategies (like expansion, diversification etc.) in advance, relating to future plans with regard to the level of existing activities. **For example,** if a manufacturing unit engaged in the manufacturing of 'Sports Shoes' wants to go ahead for expansion and diversification of its activities into manufacturing of formal shoes, the 'Accounting' may prove to be extremely helpful for taking appropriate forecasting decision.

2) **Proper Decision-Making:** A company is frequently required to take important decisions regarding:
   i) Fixing the price of finished goods on the basis of their cost,
   ii) Investment level in new projects,
   iii) Increase in salary and wages of its employees/labours,
   iv) Distribution of dividends/bonus, etc.

   Such decisions need to be taken very carefully, as any wrong decision may land the company into financial trouble, coming out of which may not be that easy. An efficient and effective 'Accounting System' of the company may be of immense help in taking crucial decisions by the management.

3) **Correct Taxation:** A company has to fulfil its obligations with regard to liabilities on account of various government taxes, e.g. service tax, income-tax, sales tax, excise duty, customs, etc. The amount of such taxes is decided on the basis of financial results reflected through financial records. An appropriate 'Accounting System' prevailing in the company facilitates determination of accurate amount of each category of tax.

4) **Replacing Memory:** A business entity enters into a number of financial transactions, which are complex and diversified in nature. It is not possible for any business entity to memorise every single transaction. As each transaction is recorded in writing through the 'Accounting System', the need to memorise every single transaction is not required.

5) **Analysing the Performance of the Business:** Maintenance of a proper record of every business transaction through Accounting enables one to ascertain the performance of a company in terms of 'Income', 'Expenditure', 'Profitability', etc. in detail. Analysis of various sub-head items can also be carried out and necessary steps may be taken to improve the performance of a company.

6) **Analysing the Financial Status of the Business:** Financials (mainly 'Profit and Loss Account' and 'Balance Sheet') of a company prepared on the basis of various records of business transactions depict the status of the business. While the 'Profit and Loss Account' reflects the profit made or loss incurred during a particular financial year, the 'Balance Sheet' reflects the position of assets and liabilities of the company on a specific date.

7) **Documentary Evidence:** Records maintained through accounting have the legal sanctity too. They can be produced as a proof in the court of law to verify the necessary business claims. As the records of business transactions are based on documentary evidence and each and every entry is supported by authentic vouchers, they are accepted by the court of law as evidence.

8) **Helping in Realisation of Debts:** As part of accounting process, the personal ledger accounts of all the parties, with whom a company deals with, are prepared. These accounts reveal the accurate amount due from the debtors as well as the amount payable to the creditors. They may be forwarded to the debtors for confirmation of the entries and early payment of the debts. Timely payment to the creditors may also be ensured by the company.

1.2.5. Objectives of Accounting

Objectives of accounting have been summarised in the following points:

1) **To Keep Systematic Records:** During the conduct of business, a number of financial transactions take place. As it is not possible to memorise all the transactions, there is a need to record them. Maintenance of a proper and systematic record of all the business transactions is the main objective of 'Financial Accounting'.

2) **To Determine Profit and Loss:** Preparation of 'Profit and Loss Account' or 'Income and Expenditure Statement' by a business entity helps to achieve next objective of 'Financial Accounting', i.e., to determine the figure of profit earned or loss incurred during an accounting year.

3) **To Ascertain Financial Position:** One of the most important objectives of 'Financial Accounting' is to enable the management of a business entity to ascertain the financial position of the business, i.e., where the business stands, what is owned by it and what it owes. In order to achieve this objective, a balance sheet is prepared at the end of a financial year. The balance sheet depicts the statement of assets and liabilities of a business entity as on a particular date.

4) **To Communicate the Information:** 'Financial Accounting' is helpful to the management of a company, majorly in the decision-taking process by providing financial data and other vital information/facts to different stakeholders, *viz.*, owners, creditors, employees, investors, regulators, tax/other government authorities, etc. Thus, communicating the information of financial data is equally important, than other objectives of 'Financial Accounting'.

1.2.6. Functions of Accounting

The step-wise functions of accounting are in the following order:

1) **Identifying:** Identification of the transactions relating to the business directly from the root documents.

2) **Recording:** Once the business transactions are identified, the next step involves proper recording thereof in terms of money. Such recording of business transactions is carried-out in a book termed as 'Journal', which may be further sub-divided, for convenience sake, into various 'Subsidiary Books'.

3) **Classifying:** Various business transactions recorded in 'Journal' (or the subsidiary books) are further classified on the basis of their nature. Such classification of similar items is followed by their posting in another set of books termed as 'Ledger'. **'Journal'** entries are simply recording of various business transactions without any differentiation, whereas the **'Ledger'** posting is done after classifying each entry under appropriate head.

4) **Summarising:** The summarised information relating to the business transactions of an organisation is very useful for the stakeholders, both internal to the specific business as well as those who are external to the

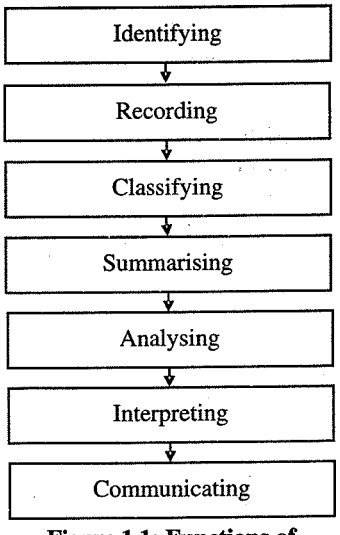


Figure 1.1: Functions of Accounting

business. Internal users are generally the management and employees of the organisation, whereas the external users are the creditors, investors, regulators, taxmen, labour unions, trade associations, etc.

5) **Analysing:** An in-depth analysis of financial statements, *viz.* 'Profit and Loss Account' and 'Balance Sheet' of an organisation helps in identifying the financial strengths and weaknesses of the organisation. Such analysis facilitates in overcoming the weaknesses and making utilisation of the strength for the betterment of the organisation.

6) **Interpreting:** Interpretation of the financial statements is helpful for the management in decision-making process and formulating future business strategy with regard to growth, expansion and diversification. Other stakeholders are also benefited from such data in taking decisions from their perspective.

7) **Communicating:** Communicating the duly summarised and analysed financial data to other stakeholders, enables them to interpret the same in their own way, taking their interests into consideration for taking decision-making at their level.

1.2.7. Importance/Advantages of Accounting

'Accounting' has, over a period of time, become an integral part of a business organisation, without which no business entity can survive. Broadly, the advantages of accounting are as follows:

1) **Helps in Management:** The information generated through the process of 'Accountancy' enables the management of a company to perform their job in an efficient manner by appropriate planning, monitoring and taking decisions to the advantage of the business organisation.

2) **Substitute to Memory:** 'Accounting' necessitates recording of all the business transactions in a scientific and classified manner. This eliminates the need to 'memorise all the transactions' entered into by a business organisation.

3) **Comparative Study:** The end products of 'Accounting' are financial statements in the form of 'Profit and Loss Account' and 'Balance Sheet', which allows the management of a company to compare the annual result of a year and initiate requisite actions, necessary for the growth of the business.

4) **Settlement of Taxation Liability:** Proper maintenance of books of accounts ensures hassle-free assessment of tax liabilities by various Government Departments, especially 'Income Tax', 'Service Tax' and 'Sales Tax' authorities.

5) **Evidence in Court:** A systematic record of all the business transactions may be produced as evidence in the court of law. Courts are inclined to accept such records as 'Good Evidence'.

6) **Sale of Business:** In case a businessman decides to dispose of his business, maintenance of proper and accurate books of accounts become handy in estimating the cost of the business set-up and finalisation of the 'Purchase Price' by the purchaser.

7) **Assistance to an Insolvent Person:** In the event of an individual becoming insolvent, he is required to explain a number of business activities undertaken during the conduct of business by him. This process becomes extremely smooth, if proper books of accounts were maintained by him.

1.2.8. Disadvantages of Accounting

Despite the advantages of accounting, its process also suffers from certain limitations, which may be considered as the disadvantages of accounting. They are as follows:

1) **Non-Monetary Items Overlooked:** Under the 'Accounting System' all the business transactions, which can be expressed in terms of money, are recorded. Business transactions, which are non-monetary in nature, are completely out of the purview of the existing 'Accounting System'.

2) **Original Cost:** The original cost is taken into consideration, while recording the details of fixed assets. The amount spent on them, which logically should be added to their acquisition cost, is not taken into account. As a result their true value is not reflected in the 'Balance Sheet' of the business entity.

3) **Possibility of Manipulation:** The 'Accounting System' envisages profit to be the only parameter to assess the performance of the company's management. This concept is illogical and need not be considered trustworthy, as certain major items especially those relating to Research and Development, Advertisement, etc. are excluded. Further, the possibility of certain manipulations cannot be ruled out.

4) **Bases on Estimates:** Accounting data recorded in the books of accounts are sometimes based on estimation instead of the actual. Such data need not be correct and reliable.

5) **Rule of Consistency:** Some of the companies, at times, fail to observe the basic tenets of accounting. The underlying principles are compromised in certain cases. This is especially true in the case of depreciation on fixed assets, as few companies tend to frequently change their policy relating to 'depreciation on fixed assets' from year to year. Inconsistency arising due to following the policy of 'Straight Line Method' (SLM) during one year and 'Written Down Value Method' (WDVM) during the next year results in failure in depicting correct picture of the company's performance.

1.2.9. Terms Used in Accounting

Various terms generally used in relation to accounting and a brief description thereof are discussed below:

1) **Assets:** Assets are items of property owned by a person or company, which have some value in terms of money and are available to meet debts or commitments. They are resources of a business organisation used for the conduct of its operations and fulfilling debt obligations. Assets can be expressed by following accounting equation:

Assets = Liabilities + Capital

Classification of Assets

i) **Fixed Assets:** Fixed assets are those assets which are acquired for the purpose of carrying out organisation's core business activity, *viz.* manufacturing of goods or providing services. They are not purchased with the intention of selling them and booking profit in the routine manner of business. Fixed assets may fall under one of the following categories:

a) **Tangible Fixed Assets:** Fixed assets which are physically visible and can be touched or felt. **Examples** of tangible fixed assets are land, building, plant & machinery, furniture, vehicles, etc.

b) **Intangible Fixed Assets:** Fixed assets which do have value in monetary terms, but are not visible and cannot be touched or felt. **Examples** of intangible assets are trademarks, copyrights, patents, goodwill of a company, etc.

ii) **Current Assets:** Current assets are "the cash and other assets that are expected to be converted into cash within a year". Current assets are important to business because they are the assets that are used to fund the day-to-day operations and pay-off the recurring expenses.

Examples of current assets are cash-in-hand, cash-at-bank, stock of raw material (RM)/Work in Progress (WIP)/finished goods (FG), sundry debtors, bills receivables, etc.

iii) **Fictitious Assets:** Fictitious assets are not the real assets as they do not have a physical form. They find a place on the asset side of the balance sheet on legal and technical basis. They are created by an accounting entry (included under assets in the balance sheet) that has no tangible existence or realisable value. They are also termed as 'Deferred Revenue Expenditures'. The purpose of creating a fictitious asset is to account for expenses (such as the preliminary expenses, loss incurred in connection with the issue of shares, advertising expenses), which cannot be placed under any normal account heading. Fictitious assets are required to be written off as early as possible against the firm's earnings.

2) **Liabilities:** Liabilities are the obligations or debts that a business entity has to pay the outsiders in terms of money or services at a future date. Thus, liabilities are the claims of those who are not owners of the business entity. They are expressed as:

Liabilities = Assets – Capital

Classification of Liabilities

i) **Current Liabilities/Short-term Liabilities:** Current liabilities or the short-term liabilities are those liabilities, which are required to be paid/discharged within a short span of time (generally within a year). **Examples** of short-term liabilities are trade creditors, salaries/wages payable, bank overdraft, bills payable, etc.

ii) **Long-term Liabilities:** Long-term liabilities of a company are obligations that extend beyond the current year, or the current operating cycle. Most commonly, these include long-term debts such as bonds and debentures issued by the company and term loans availed from a bank/financial institution.

3) **Capital:** The investment made by the owner of a business entity for the conduct of the business is termed as 'Capital'. From the perspective of the business entity, capital infused by the owner is a liability towards the owner. It is also referred to as **Owner's Equity,** (owner's claim against the assets) or **proprietorship** or **net worth.** Capital may be expressed as:

Capital = Assets – Liabilities

4) **Proprietor:** The owner of a business is known as the proprietor. He is the one who undertakes initiative to start a business by taking risk of infusing capital in the business.

5) **Drawings:** The money withdrawn from the business or the goods taken for personal use by the proprietor is referred to as 'Drawings'. When an asset belonging to the business is put to the personal or domestic use, it also tantamount to **'Drawing'. For example,** when a vehicle purchased for the business activities, is used for personal purpose or when a business premise is utilised for personal use, they are considered as 'Drawings'.

6) **Revenue:** The earnings of a business by making available services to its clients or by way of selling the products manufactured by it are termed as **'Revenue'.**

7) **Expenses:** The amount of money spent by a business organisation during the conduct of its business, manufacturing process or providing services, etc. are referred to as **'Expenses'.**

8) **Income:** The difference between the revenue earned and expenditure incurred during the conduct of a business is the **'Income'.**

Income = Revenue – Expenditure

9) **Losses:** If the income of a business organisation happens to be a negative figure (revenue expenditure is more than the revenue earned), it is considered as business losses incurred by the entity. In addition to the above, a business entity may be compelled to bear losses due to natural calamities like flood or storm and/or those caused by humans like theft or some unforeseen accidents (like fire). 'Losses' are obviously different from the 'Business Expenses' in the sense that while the former are unwarranted and unwelcome incidents, the latter are part of normal business activities undertaken consciously.

10) **Purchases:** Goods acquired, either on credit or on cash payment, for the own use of the business entity (including raw materials) or for resale thereof as part of the business operations are referred to as **'Purchases'.**

11) **Purchases Return:** Purchases made in excess of the required level or surplus purchases and purchases that have a defect are returned to the supplier/seller. Such returns are termed as **'Purchases Returns' or 'Returns Outward'.**

12) **Sales:** The total amount of goods and services extended by a business organisation to its clients/customers are known as **'Sales'.**

13) **Sales Return:** At client's/customer's end, when the goods sold/supplied by a business organisation become in excess of the required level, the surplus goods are returned by them. Such returns are termed as **'Sales Return' or 'Returns Inward'.** To arrive at the 'Net Sales' figure, 'Sales Return' figure is reduced from the 'Sales' figure.

14) **Stock/Inventory:** Inventory or stock refers to the goods and materials that a business holds for the ultimate purpose of resale (or repair). It has also been defined as "a complete listing of merchandise or stock on hand, work in progress, raw materials, etc., made each year by a business entity".

15) **Debtors (Accounts Receivable):** A 'debtor' is an entity that owes a debt to another entity. Such debt generally arises due to receipt of goods and/or services on credit in the normal course of business. When the counterpart of this debt arrangement happens to be a bank, the debtor is referred to as a borrower.

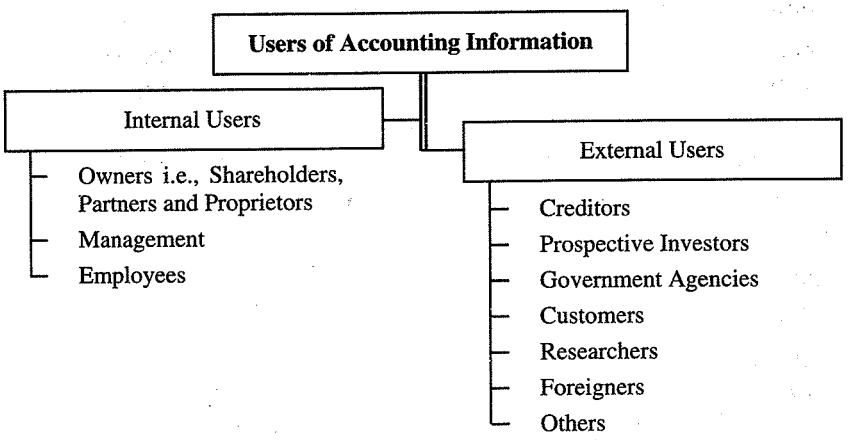
16) **Creditors (Accounts Payable):** 'Creditors' are the counterparties of 'Debtors'. They are the entities which supply/extend/provide goods and services on credit in the normal course of business. They are also referred to as **'Accounts Payable'.**

17) **Vouchers:** A voucher is the documentary evidence of a business transaction. The monetary value of a transaction is indicated in the voucher. This document may be in the form of 'Cash Receipt', 'Cash Memo', 'Invoice', 'Pay-in Slip' of a bank, etc. Vouchers are very useful for an 'Audit Trail'.

18) **Invoice:** Invoice is another document prepared at the time of a 'Sale Transaction' by the seller of goods. Important and crucial information of the buyer and seller, like their names and addresses, date of sale, quantity and selling price of goods, etc. are incorporated in the 'Invoice'.

1.2.10. Users of Accounting Information

Accounting information, especially 'Financial Statements' of a company are of immense utility not only for the management and owners, but other stakeholders, which are external to the business, viz. Government agencies, regulators, potential investors, potential business associates, etc. Such users of 'Accounting Information' may be grouped under two broad categories, i.e. 'External Users' and 'Internal Users'. The details of these users are discussed in the following points:



1) **Internal Users:** The internal users of 'Accounting Information' are as under:
   i) **Owners i.e., Shareholders, Partners and Proprietors:** They are the driving force behind the conduct of any business activity, as it is their ideas as well as the funds, which works for the efficient and convenient flow of business, its growth, development, diversification, etc. Some of the factors on their priority lists are enhanced turnover, improved profitability, better dividend distribution, healthy financial indicators, growth/expansion of business, better future prospects regarding earnings, etc. This sub-category of 'Internal Users' are contended with published annual results, like 'Profit and Loss Account', 'Balance Sheet' and other miscellaneous statements.
   ii) **Management:** The accounting data is useful for all the three tiers of a company's management, *viz.* 'Top Management' (Board of Directors), 'Middle Management' (Senior Executives) and 'Lower Management (Supervisory Staff). The data facilitates in finalisation of policy statement at 'Top Management' level, implementation of such policies at 'Middle Management' level and discharge of supervisory functions at 'Lower Management' level. Setting up of a 'Vision Document' and 'Objectives' of a company and monitoring thereof, becomes possible on the basis of inputs provided by accounting. Various budgetary tools and techniques relating to production, sales budget, cash budget and also master budget can be effectively utilised on the basis of statements and reports provided by accounting.
   iii) **Employees:** For a sustainable success and growth of a business entity, mutual understanding and cordial relationship between the employer and the employee is the basic requirement. Management of a company can have a wholesome industrial relationship (between the management on one side and the trade union on the other) through highly contended, committed and motivated employees. This can be achieved with the help of accounting information, which is capable of ensuring industrial harmony by averting various forms of labour unrest like, strikes for demand of better working conditions, wages/salaries, bonus, other benefits, etc.

2) **External Users:** The details of external users of 'Accounting Information', *viz.* customers, investors, regulators, Government agencies, creditors/suppliers, researchers, foreigners, etc. are discussed in the following paragraphs:
   i) **Creditors:** Suppliers of goods or raw materials (in the case of a manufacturing unit) extend short-term credit to their customers with the understanding that the entire payment would be made within a stipulated timeframe. Such kind of credit is termed as **'Trade Credit'**. Such trade creditors or the suppliers have concerns with regard to the timely receipt of their dues. Therefore, it is important for them to ascertain the 'Short-term Solvency' of their customers (to whom supplies have been made). The status of 'Liquidity' or 'Short-term Solvency' of a company can easily be found out with the help of 'Accounting Statements'. The creditors can frame their policy of extending short-term 'Trade Credit' on the basis of financial data.

ii) **Prospective Investors:** The prospective investor may include, besides individuals, banks, and insurance/finance/mortgage companies. The vital guiding factors behind any investment decision taken by a potential investor are:
   a) Safety of funds invested,
   b) Recurring returns in the form of dividends, and
   c) Possible capital gain.

   These factors can be determined through the study and analysis of financial data, *viz.* profitability, dividend policy, and the company's historical background for the last few years. The accounting statements also enable a potential investor to ascertain the viability of current projects, future growth and expansion plans, etc.

iii) **Government Agencies:** The accounting information is required by Government agencies, like tax-authorities, Company Law Board (CLB), Registrar of Companies (RoC), Securities and Exchange Board of India (SEBI), Ministry of Trade and Commerce (MoTC), etc. with a view to ensuring that a company has complied with all the guidelines, directions, rules and regulations issued by the respective Government agencies. **For example**, CLB may be interested in knowing whether all the provisions of Companies Act, 2013 have been complied with or not. Similarly, SEBI would be interested in ensuring that the interests of investors and public in general are protected.

iv) **Customers:** The interest of a customer or normal citizen lies in knowing the factors like monitoring mechanism prevailing in the company regarding production, selling and other expenses, which affect the market price of the finished goods. Accounting Statements are helpful in ascertaining such factors. Of late 'Customer Protection Groups' have started exercising some supervision over the business and industry and also creating awareness about the concept of 'Corporate Social Responsibility' (CSR).

v) **Researchers:** Analysis and interpretation of financial statements of a company by researchers may lead to some unexpected and surprising results and innovative ideas, which may be used by different entities in different ways. **For example**:
   a) The Government agencies may use such results/ideas to amend their existing policies and come out with the new ones,
   b) The regulators may use them to improve their regulatory regime, and
   c) Management of the company may use them as a guiding factor for reviewing their current policies.

   The accounting statements are, therefore, very important for the research scholars, as they serve two purposes, *viz.* research in accountancy, and improving the business standards and systems.

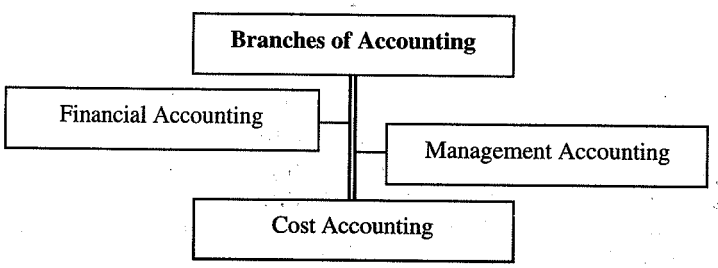
vi) **Foreigners:** Foreigners are interested in accounting data of a company because of their exposure in 'Joint Ventures' (JVs), 'Foreign Direct Investments' (FDIs), etc. Further, there are increasing number of cases of merger and acquisition of Indian Companies with Multi-National Companies (MNCs).

vii) **Others:** Entrepreneurs, trade associations, stock exchanges, media, political parties, etc. are some of the other stakeholders, who are keen on well-being of a business entity and its growth, etc. They also undertake the study of impact on financial and social business environment.

Thus, it is clear that the uses and benefits of accounting information are multi-dimensional and various agencies use them in their own manner.

1.2.11. Branches of Accounting

Several branches of accounting have developed with the passage of time. They can be broadly categorised as mentioned below:



1) **Financial Accounting:** It is that branch of accounting, which deals with the past data of the business, which is generally for one year and ends with the preparation of 'Financial Statements' like:
   i) 'Profit and Loss Account', indicating the profit earned or losses incurred during a particular period, and
   ii) 'Balance Sheet', indicating the financial position, i.e. assets and liabilities held as on the last day of the accounting period.

   The 'Financial Statements' are prepared on the basis of Generally Accepted Accounting Principles (GAAP) articulated by the accounting profession and other statutory provisions and accounting standards. Such statements are more useful for the stakeholders and outside the business entity, *viz.* taxmen, regulators, and other Government agencies.

2) **Management Accounting:** It is that branch of accounting, which is 'Forward Looking' and 'Futuristic' in nature rather than dealing with historical data and generally involves 'Cost Accounting' and 'Budgeting'. Its aim to provide inside reporting to the manager of a company. While preparing various statements under management accounting, it is not necessary to ensure compliance with the provisions of Generally Accepted Accounting Principles (GAAP) articulated by the accounting profession and other statutory provisions and accounting standards.

   The essence of 'Management Accounting' lies in the fact that it highlights the 'Control', 'Planning' and 'Decision-making' aspect of the accounting (and not the 'Stewardship' aspect of accounting). The framework of management accounting is specifically designed to serve the requirement of a particular company's management.

   Importance/Advantages of Management Accounting

   In the context of present day business environment, the role of 'Management Accounting' is very important, as may be seen from the following:

   i) **Increasing Efficiency:** Under the system of 'Management Accounting', each department is brought under the purview of the 'Performance Budgeting' exercise; targets are prescribed in advance and performance is monitored on an on-going basis. As the efficiency of all the departments is measured through the system, there is significant improvement in their performance. Better performance of departments results in overall improved efficiency of the organisation.

   ii) **Proper Planning:** The inputs and feedbacks received through the various techniques applied by the system of 'Management Accounting' are of immense help to the management of the organisation especially in the area of 'Forecasting' and 'Future Planning'.

   iii) **Measurements of Performance:** There are certain tools under the 'Management Accounting', through which the performance of departments of an organisation may be measured. They are:
      a) **Budgetary Control:** This tool forecast target for departments in advance, against which their performance is monitored. Any shortfall in achievement of the targets is treated as a negative performance of the individual / concerned department.
      b) **Standard Costing:** Under this technique, 'Standards' with regard to costing of each step involved in various activities is prescribed beforehand and all the 'Actual' costs incurred compare against that 'Standard Cost'. Any deviation in the 'Actual Cost' from the 'Standard Cost' is not considered favourable by the management.

   iv) **Maximizing Profitability:** Various techniques of the 'Management Accounting' focus on controlling the costs involved in different steps of production cycle. This leads to reduction in the cost of production and efficiency boosting in respect of each department. The final outcome is the 'Maximization of Profit' for the business organisation.

   v) **Improves Service to Customers:** Once the cost of production is curbed through the techniques of 'Management Accounting', the organisation finds itself in a position to reduce the prices of finished goods. Further, the quality of goods produced is also maintained because of the application of 'Standard Costing' technique. As a result, the organisation is able to supply good quality products to its customers at a fair price without compromising with its 'Profitability'. It is a 'Win Win' situation both for the enterprise and its customers.

vi) **Effective Management Control:** Various techniques and tools applied under the system of 'Management Accounting' acts as a major support for the management in discharging its core functions like 'Coordinating', 'Controlling', 'Planning', 'Forecasting' etc. The 'Standard Costing' technique, which envisages setting 'Standard' and comparing the same with the 'Actual' performance, is an example of 'Management by Exception'.

3) **Cost Accounting:** It is that branch of accounting, which deals with costing aspect of the business transactions. All the costs, i.e., present and future, are recorded, classified and reported under 'Cost Accounting'. It is an accounting process used to determine cost of production of a business entity by analysing input and fixed cost of production. It also captures the cost of operations of a business.

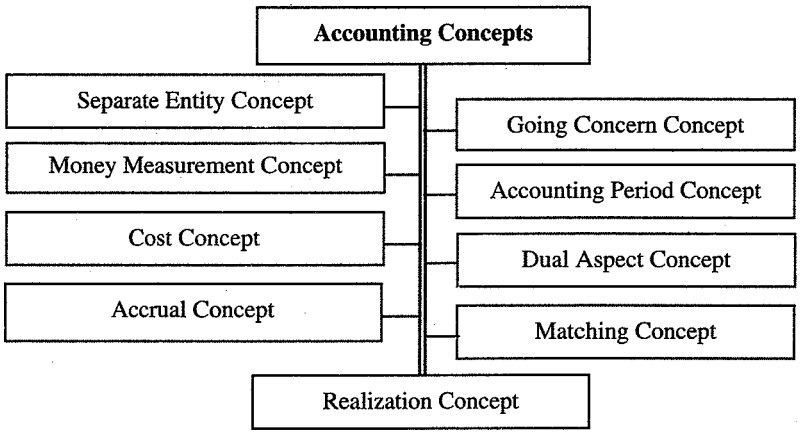
1.2.12. Difference between Management Accounting, Cost Accounting and Financial Accounting

Basis of Difference	Management Accounting	Cost Accounting	Financial Accounting
1) **Orientation**	Management Accounting is concerned with all the economic affairs from the management angle, irrespective of the fact whether the same is monetary or non-monetary in nature.	Cost Accounting is concerned with money as a measure of commercial activities of the business organisation.	Financial Accounting is concerned with funds as an economic resource, i.e., cash.
2) **Scope**	Management accounting covers accounting information system, which includes financial accounting as well as cost accounting, in addition to other facets of financial management. It facilitates the management in discharging its various functions especially the decision making and evaluation of its performance.	The scope of cost accounting includes the measurement of the economic performance of various cost centres. It provides appropriate data for the measurement of the economic performance of cost centres / cost units.	The scope of financial accounting deals with the financial aspect of a business entity. It is done by preparing Trading and Profit and Loss A/c for a specific period (accounting year), and Balance Sheet of a business organisation as of a specific date.
3) **Analysis of Performance**	Management accounting emphasises focus on various departments of the business organisation and submit reports regarding the performance in relation to the profitability of the business organisation.	Cost accounting focuses on collection, classification and analysis of cost data.	Financial accounting focuses on recording, classification, summarising and analysis of financial data.
4) **Time Factor**	Management accounting is futuristic in nature. It pays more attention on future operations, profitability, etc.	Cost accounting is concerned with the data pertaining to the past and current operations.	Financial accounts focus on past and present operations.
5) **Legal Compulsion**	Management accounting system is introduced in a business organisation with a view to facilitating the management in discharging some of its basic functions, e.g. decision making, capital budgeting, budgetary control, etc. However, there is no statutory compulsion of having a management accounting system.	Maintenance of records pertaining to the costing is also not compulsory under the law. Such records are maintained with a view to fulfilling the requirement of the management. However, the latest Companies' Act has made it mandatory to keep the cost records for some manufacturing industries.	Financial accounting is compulsory for all the companies due to legal provision.

1.3. ACCOUNTING CONCEPTS

The word **'Concept'** may be defined as "a general idea or understanding of thought". It is an idea of what thought is or how it works. **"Accounting concepts"** are the necessary assumptions, conditions or postulates upon which the accounting is based. They are developed to facilitate communication of the accounting and financial information to all the readers of Financial Statements, so that all readers can interpret the statements with the same meaning and context.

Following are the accounting concepts, which have been broadly accepted by accountants:



1.3.1. Separate Entity Concept

There is a presumption under 'Separate Entity Concept' that as far as accounting is concerned the 'owners of a business organisation' and the 'business organisation itself' are two independent and separate entities. The business transactions undertaken by the owner are altogether separate from the personal transactions undertaken by him. **For example**, the capital invested by the owner in a business is recorded as a liability for the business. Similarly, if any asset (including cash or goods) belonging to the business is taken by the owner for his personal use, it is not considered as business expenditure, instead it is treated as **'withdrawal'** or **'drawing'** by the owner. This is the cornerstone of 'Accounting Concepts'.

1.3.2. Going Concern Concept

Going concern is one of the fundamental assumptions in accounting on the basis of which financial statements are prepared. The assumption is that a business entity will continue to operate in the foreseeable future without the need or intention on the part of management to liquidate the entity and it will realise its assets and settle its obligations in the normal course of the business. In simple words, it means that every business entity has continuity of life and it will not be dissolved in the near future.

The assumption of going concern is the basis of all the financial transactions of a business entity like entering into long-term contracts with other parties, obtaining loans from banks/financial institutions, extending loans, investing in long-term securities, purchasing bonds/debentures, etc. This concept also enables a business entity to defer some of their costs like prepaid expenditure, closing stocks, etc. which are required to be charged against future incomes. Even the depreciation on fixed assets is charged on the assumption of continuation of the business. Fixed assets are shown in the balance sheet at acquisition cost less depreciation charged. They are not shown at their market value, due to the going concern concept. In the absence of this concept, it would not have been possible to classify assets into 'Fixed Assets' and 'Current Assets'. The income measurement techniques can be justified only on the basis of assumption of 'Going Concern Concept'.

1.3.3. Money Measurement Concept

In accounting, every transaction is recorded in terms of money, i.e. rupees and paise in India. Receipt of income, payment of expenses, purchase and sale of assets, etc., are monetary transactions and therefore are recorded in the books of accounts. The assumption under money measurement concept enables to have a common measure, in terms of money, for all the transactions, assets and liabilities, which facilitate the preparation of financial statements.

This concept would be better understood with the help of a few **examples.** A company, on a particular day, makes some purchases, *viz.* 10 office chairs, 50 meters of curtain cloth and 20 rims of A-4 paper, all of which have different measuring units. However, as all the transactions can be measured in terms of money, they can be easily recorded in the books of accounts.

On the other hand, an event of a machinery breakdown in a manufacturing unit cannot be recorded in the books of accounts as it does not have a monetary value. However, the expenditure incurred for the repair of the machinery can be measured in monetary terms and hence can be recorded. Similarly, the quality of service, health of the owner or the morale of work-force of a company cannot be measured in terms of money and therefore cannot be recorded in the books of accounts.

1.3.4. Accounting Period Concept

An accounting period is the interval of time at the end of which the financial statements are prepared to ascertain the financial performance of a company. This is known as accounting period concept. Although the "going concern" concept emphasises the continuing nature of an organisation, it is necessary to review its performance. The preparation of financial statements (balance sheet and profit & loss account) at periodic intervals (known as accounting period) helps in taking timely corrective action and developing appropriate strategies. The accounting period is generally of twelve months (which may be a calendar year or a financial year), although it may be for three months or six months as well in case of new startup. Preparation of financial statements also serves some other purposes like calculation of profit, tax calculation, submission of reports to regulators and other Government agencies, etc.

1.3.5. Cost Concept

As per this concept, cost of an asset is recorded in the books of accounts at the price paid to acquire it (including overheads like transportation and installation charges, if any) and not at the market price. Fixed assets, e.g., land, building, plants, machinery, furniture, fixtures, etc. are taken in the record at the price paid for acquiring them, which is also termed as 'Historical Cost'. However, the cost of assets recorded at the time of purchase may be systematically reduced through depreciation.

This concept may be better described through an **example**. If a piece of land is purchased to set-up a factory for ₹50,000 in the books of accounts this would be recorded at ₹50,000 only. After sometime there may be an increase in the market value of the land, but in the books of account, the cost would continue to be shown at ₹50,000 only. Further, at the end of the financial year the land would be subject to depreciation and the reduced cost would be taken over as the opening balance for the next financial year. This will be repeated year after year and the cost would be shown at a lower level despite appreciation in the market price of the land.

1.3.6. Dual Aspect Concept

This is the fundamental concept of accounting. This concept follows from the Entity Concept. All entities own certain assets. Such assets are acquired through contributions of those who have provided the funds for the purpose. Funds are made available either through the surplus of the entity or loans. Logically such providers of funds are claimants to the assets. At any point of time, the assets will be equal to the claims. It may thus be concluded that each and every transaction, a business entity enters into, has a dual effect. With every increase in the money owed to others, there has to be an equal increase in assets or loss. This concept supports the accounting equation, i.e.,
Assets = Owner's Funds (Capital and Reserves) + Liabilities Or Owner's Funds = Asset – Liabilities

This concept may be understood in a better way by an **example**. A business is started by Mr. 'X' with an investment of ₹2,50,000. This transaction has two aspects, *viz.* 'Capital Account' and 'Asset Account'. While the amount lying in 'Capital Account' (₹2,50,000) is a liability for the business (which it owes to Mr. 'X'), whereas the amount lying in 'Asset Account' (₹2,50,000) is an asset for the business.

1.3.7. Accrual Concept

Under the cash system of accounting, the incomes and expenditures are recorded only if there is actual receipt or payment in cash, irrespective of the accounting period to which they belong. But, under the accrual concept, occurrence of claims and obligations in respect of incomes or expenditures, assets or liabilities, diminution in values, etc., are recorded even though actual receipts or payments of money may not have taken place. This

concept depicts that incomes and expenses need to be recognised as and when they are earned and incurred respectively, notwithstanding the fact whether the money is actually received or paid in this regard. This concept may be understood in a better way by an **example**. If a company dealing in medicines gets supply of medicine worth ₹20,000 on March 29, 2014 (during the year 2013-14) and the payment is made on April 2, 2014 (during the year 2014-15), under the accrual system, the expenditure though incurred during 2014-15 needs to be recognised for the year 2013-14 (the year during which the supply was made).

1.3.8. Matching Concept

The matching concept is an outcome drawn from the accrual concept. It emphasises that the revenue earned and the expenditure incurred must belong to the same accounting period. To ascertain the surplus or deficit made by a business entity during an accounting period, it is necessary that the costs incurred are matched with the revenue earned by the entity during that accounting period. Therefore, once the revenue is realised or expenditure incurred, they need to be allocated to the relevant accounting period. The matching concept is different under 'Cash System' and 'Accrual Systems' of accounting:

1) In 'Cash System' of accounting only actual receipts and payments are recognised for any accounting year. Revenues are recognised when realised in cash and expenditures are recognised when actually paid in cash. The period during which the goods were supplied or services were rendered does not matter. The cheque received or issued is as good as the cash realised or cash paid.
2) In 'Accrual System' of accounting, revenues are recognised for the accounting period during which the sale is made or service is rendered irrespective of the fact whether realised in cash or not. Similarly, expenditures are recognised for the accounting period during which they are incurred (on due basis), whether paid in cash or not. This process of recognising and relating expenses with revenues is called **matching concept**.

1.3.9. Realisation Concept

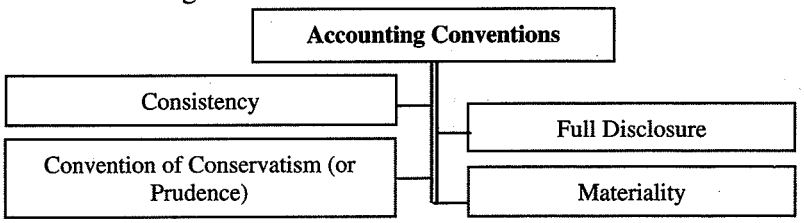
According to this concept, revenue needs to be accounted for only when it is actually realised or it has become certain that the revenue will be realised. However, in order to recognise revenue, actual receipt of cash is not necessary. What is important is that the organisation should be legally entitled to receive the amount for the services rendered or the goods sold. Revenue is said to have been realised when cash has been received or right to receive cash on the sale of goods or services or both has been created. It means that selling goods is realisation, whereas receiving order is not. This concept may be understood in a better way by an **example**:

1) A telecommunication company sells talk-time through scratch cards. No revenue is recognised when the scratch card is sold, but it is recognised when the subscriber makes a call and consumes the talk-time.
2) A monthly magazine receives 1,000 subscriptions of ₹240 to be paid at the beginning of the year. Each month it recognises revenue worth ₹20,000 [(₹240 ÷ 12) × 1,000].

1.4. ACCOUNTING CONVENTIONS

'Accounting Conventions' are the guidelines that arise from the practical application of accounting principles. They are not a legally binding practice, rather they are generally accepted practices based on customs, and are designed to help accountants to overcome practical problems faced by them during the preparation of financial statements.

As customs change, so will accounting conventions.



1.4.1. Consistency

The convention of consistency emphasises that the accounting principles/practices followed by an entity should be consistently applied by it over the years so as to achieve compatibility. It facilitates comparison of financial performance of an entity from one accounting period to another.

This convention may be understood in a better way by an **example**:

1) There are a number of methods for the valuation of inventories, *viz.* 'First In First Out' (FIFO), 'Last In First Out' (LIFO), or 'Weighted Average Method'. If an organisation has adopted one method of valuation say 'Weighted Average Method' during one accounting year, the same method needs to be followed during the subsequent accounting years. Any change in the method of inventory valuation would distort the financial results in a substantial manner. In case a change is made, it should be disclosed.
2) Similarly, the method of providing depreciation, provision for bad and doubtful debts, change from 'Cash Basis' to 'Accrual Basis' are some of the examples where there are a number of options available to the business entities. But once a method is adopted, the same method should be continued in the forthcoming financial years, so as to maintain consistency. Any change, if at all necessitated, needs to be disclosed properly, as it would result in a difficulty for the stakeholders to compare the financial performances.

1.4.2. Full Disclosure

The term "disclosure" implies that there must be a sufficient exposure of information which is of material interest to all the stakeholders, *viz.* owners, creditors, lenders, investors, public, etc. The accounts and the financial statements of an entity should disclose full and fair information to the beneficiaries in order to enable them to form a correct opinion on the performance of such entity, which in turn would allow them to take informed and correct decisions.

This convention may be understood in a better way by an **example**:

1) If a business organisation has raised substantial amount of bank finance immediately after the balance sheet date but before the finalisation of its balance sheet, it needs to be disclosed appropriately in the financial statements. **For example,** if the balance sheet data is 31st March, 2014, and the business has availed finance on 25th April, 2014, then the amount of bank finance will be disclosed to the period subsequent i.e., 31st March, 2015.
2) The figures of the sundry debtors should necessarily be accompanied by the figure of Bad and Doubtful Debts (BDDs) and the provisions made against such BDDs.

1.4.3. Convention of Conservatism (or Prudence)

This convention means a cautious approach or policy of having a conservative approach. According to this convention, the anticipated profits need to be ignored but all anticipated losses need to be provided for in the books of accounts of an entity. In other words, all the prospective losses are taken into consideration, while no doubtful income is taken into consideration in recording of transactions by an entity.

There are numerous examples of the 'Convention of Conservatism', which are adopted and practiced by many of the business organisations. Some of them are stated in the following points:

1) Providing for the Bad and Doubtful Debts (BDD) in the anticipation of some debtors' inability to pay their debts.
2) Providing for 'Discount' on debtors.
3) Valuation of the 'Stock-in-hand' at the 'Market Price' or 'Cost Price', whichever is lower.
4) Creation of 'Investment Fluctuation Reserve'.
5) 'Capital Expenditures' of small amount (like crockery purchase) are charged as 'Revenue Expenditure'.

1.4.4. Materiality

The term material refers to the relative importance of an item or event. An item should be regarded as material, if there is a sufficient reason to believe that knowledge of it would influence the decision of informed creditors, lenders, investors, public and other stakeholders. The accounts and the financial statements should impart importance of all material information so that true and fair view of the state of affairs of the entity is given to its beneficiaries. Hence, keeping the convention of materiality in view, unimportant items are not disclosed separately and are merged with other items.

The concept of materiality is relative. What is material for a small company may not be material for a large company. The cost of small tools may be material for a small vehicle repair workshop, but the same may not be material for large manufacturers of vehicles. Similarly, the nature of a transaction is also important in deciding the 'Materiality'. A difference of ₹1,000 in cash would be considered 'Material', whereas the difference of the same amount (₹1,000) otherwise would be considered 'Immaterial'.

1.5. FUNDAMENTAL ACCOUNTING EQUATION

1.5.1. Introduction

The 'Accounting Equation' is the foundation of double entry system of accounting. It states that both debit and credit side should have an equal balance. The main focus of accounting equation is financing of all the assets either through borrowing or paying money. Thus, the accounting equation is:

Assets = Liabilities + Shareholders Equity (Capital)

A = L + C

Where, A = Assets, L = Liabilities, and C = Capital (Shareholders Equity)

The balance sheet is a complex display of the above equation, showing that the total assets of a company are equal to the total of liabilities and shareholders equity. Any purchase or sale by an accounting entity has an equal effect on both sides of the equation, or offsetting effects on the same side of the equation.

Assets are the resources or things a company owns, e.g., land, building, machinery, equipment, cash, debts receivable, inventory, investments, prepaid insurance, etc. It also includes intangible assets, like goodwill, trademarks, preliminary expenses, etc.

Liabilities are the obligations of a company or the amounts a company owes to others, e.g., borrowings, account payable, salaries/wages payable, interest payable, dues of Government authorities like income tax, sale tax, service tax, etc. Liabilities may be categorised under two heads:

1) Claims by creditors against a company's assets, and
2) A source of a company's assets (owners' funds).

Capital, also known as owner's equity or stockholders' equity, is the residual amount after deducting liabilities from the assets:

Assets – Liabilities = Owner's Equity or Capital

'Owner's Equity' or 'Capital' comprises of the 'Paid-up Capital' and the 'Profit' earned and accumulated over the years ('Net Income' not withdrawn or distributed as 'Dividend') kept either as it is or in the form of 'Reserves'. As the 'Accounting Equation' reveals the basic relationship between various constituents of the 'Balance Sheet', it is also referred to as the 'Balance Sheet Equation'. Balance Sheet of a company is statement of its 'Owner's Equity' and various items of its 'Assets' and 'Liabilities'. An event impacting any of the 'Balance Sheet' components ('Owner's Equity', 'Assets' or 'Liabilities') is termed as a 'Transaction'. Before recording a transaction in the books of accounts, it is necessary to identify two 'Heads of Accounts', which would be 'Credited' or 'Debited'.

1.5.2. Accounting Equation Based on Classification of Accounts

Types of Accounts	Meaning	Example
1) **Assets Accounts**	'Assets Accounts' of a company represents all the 'Tangible' as well as 'Intangible' asset items of the balance sheet.	Tangible Assets: Land, Building, Plant/Machinery, Cash/Bank balance, etc. Intangible Assets: Goodwill, Patents, Trademarks, etc.
2) **Liabilities Accounts**	'Liabilities Accounts' of a company are its financial obligations it owes towards outsiders.	Sundry Creditors, Outstanding Expenses (including tax liabilities), Bank Borrowings (Overdraft, Cash Credit or Term Loans), etc.
3) **Capital Accounts**	'Capital Accounts' of a company represents its 'Inside Liabilities', i.e. its liabilities towards owners.	Capital a/c, Drawings a/c, etc.
4) **Revenue Accounts**	'Revenue Accounts' represents the amount charged for goods sold or services rendered or allowing others to use enterprise resources yielding interest, royalty or dividend.	Sales a/c, Discount Received a/c, Dividend Received a/c, Royalty Received a/c, Interest Received a/c, etc.
5) **Expenses Accounts**	'Expenses Accounts' are the expenses incurred in the conduct of business in the normal course or losses suffered due to some unforeseen causes, like fire, flood, theft, etc.	Purchases a/c, Discount Allowed a/c, Royalty Paid a/c, Interest Payable a/c, Loss by Fire/Flood/Theft a/c, etc.

1.5.3. Procedure for Developing an Accounting Equation

Following steps play a major role in the development of an 'Accounting Equation':

Step 1) Identification of the variable-items of the balance sheet, i.e., Assets, Liabilities or Capital impacted by a transaction.

Step 2) Ascertaining the impact (increase or decrease) of the transaction on the variable-items of the balance sheet.

Step 3) Showing the impact of the transaction on the appropriate variable-items of the balance sheet and ensuring that both the sides of the balance sheet, *viz.* Assets and Liabilities remain matched. In other words, the total of 'Assets Side' of the balance sheet needs to be equal to the total of the 'Liabilities Side'.

Example 1: Show the effect of the following transactions on assets, liabilities and capital through accounting equation:

1) Started business with cash ₹1,50,000.
2) Rent received ₹10,000.
3) Invested in Shares ₹50,000.
4) Purchased goods on credit ₹25,000 from X.
5) Paid to X ₹25,000.
6) Salary paid ₹13,000.

Solution:

S. No.	Transactions	Assets	=	Liabilities	+	Capital
1)	Started business with cash ₹1,50,000	1,50,000	=	0	+	1,50,000
2)	Rent received ₹10,000	10,000	=	0	+	10,000
	New Equation	1,60,000	=	0	+	1,60,000
3)	Invested in Shares ₹50,000	(–) 50,000	=	0	+	0
		50,000	=	0	+	0
	New Equation	1,60,000	=	0	+	1,60,000
4)	Purchased goods on credit ₹25,000 from X	25,000	=	25,000	+	0
	New Equation	1,85,000	=	25,000	+	1,60,000
5)	Paid to X ₹25,000	(–) 25,000	=	(–) 25,000	+	0
	New Equation	1,60,000	=	0	+	1,60,000
6)	Salary paid ₹13,000	(–) 13,000	=	0	+	(–) 13,000
	New Equation	**1,47,000**	=	**0**	+	**1,47,000**

Example 2: Show the accounting equation on the basis of the following transactions:

1) Yunus started business with cash ₹90,000.
2) Purchased goods on credit ₹50,000.
3) Purchased furniture for cash ₹10,000.
4) Sold goods costing ₹20,000 for ₹40,000.
5) Sold goods costing ₹20,000 on credit for ₹42,000.
6) Bought goods worth ₹20,000 (₹15,000 paid in cash and balance on credit)
7) Withdrawal of cash for personal use ₹5,000.
8) Paid for rent ₹1,000.
9) Paid to creditors ₹40,000.
10) Paid for salaries ₹3,000.
11) Received from debtors ₹12,000.

Solution:

S. No.	Transactions	Assets	=	Liabilities	+	Capital
1)	Yunus started business with cash ₹90,000	90,000	=	0	+	90,000
2)	Purchased goods on credit for ₹50,000	50,000	=	50,000	+	0
	New Equation	1,40,000	=	50,000	+	90,000
3)	Purchase furniture for cash ₹10,000	(–) 10,000	=	0	+	0
		10,000	=	0	+	0
	New Equation	1,40,000	=	50,000	+	90,000
4)	Sold goods consisting ₹20,000 for ₹40,000	(–) 20,000	=	0	+	0
		40,000	=	0	+	20,000
	New Equation	1,60,000	=	50,000	+	1,10,000
5)	Sold goods consisting ₹20,000 on credit for ₹42,000	(–)20,000	=	0	+	0
		42,000	=	0	+	22,000
	New Equation	1,82,000	=	50,000	+	1,32,000
6)	Bought goods worth ₹20,000	20,000	=	0	+	0
	(₹15,000 paid in cash and balance on credit)	(–) 15,000	=	5,000	+	0
	New Equation	1,87,000	=	55,000	+	1,32,000
7)	Drawn for personal use ₹5,000	(–)5,000	=	0	+	(–)5,000
	New Equation	1,82,000	=	55,000	+	1,27,000
8)	Paid for rent ₹1,000	(–) 1,000	=	0	+	(–)1,000
	New Equation	1,81,000	=	55,000	+	1,26,000
9)	Paid to creditors ₹40,000	(–) 40,000	=	(–) 40,000	+	0
	New Equation	1,41,000	=	15,000	+	1,26,000
10)	Paid for salaries ₹3,000	(–) 3,000	=	0	+	(–) 3,000
	New Equation	1,38,000	=	15,000	+	1,23,000
11)	Received from debtors ₹12,000	(–)12,000	=	0	+	0
		12,000	=	0	+	0
	New Equation	**1,38,000**	=	**15,000**	+	**1,23,000**

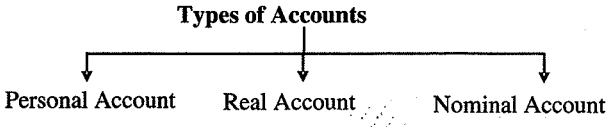
1.6. DOUBLE ENTRY SYSTEM

1.6.1. Meaning of Double Entry System

The 'Double Entry' system of accounting is based on the fact that every financial transaction has equal and opposite effect in at least two different accounts. It is used to satisfy the equation Assets = Liabilities + Equity, whereby each entry is recorded so as to maintain the relationship. Under double entry system dual aspect of a business transaction is recorded in terms of 'Debit and Credit'. The 'Double Entry System' of book-keeping is the underlying basis of the standard system practised by most of the accountants and the business enterprises to record 'Financial Transactions'. This system, also known as the Venetian method of accounting and was first introduced by an Italian mathematician, **Luca Pacioli**. This new system revolutionised the business and Pacioli was given a place in history as **"The Father of Accounting"**.

1.6.2. Types of Accounts

The debiting and crediting of the accounts are done on the basis of certain rules. These rules are called **rules of journalising**. They differ according to the type of the effected account. All the accounts may be classified into three categories. Details of all categories of accounts and the 'Rules of Journalising' applicable under each category are described below:



1.6.2.1. Personal Account

An account of a person which may be a natural person, an artificial person or representative group of persons is referred as 'Personal Account'. Examples of categories of personal account:

1) A **Natural Person/Individual:** Ram, Shyam, Sohan, or Mohan,
2) An **Artificial or Legal Person/Entity:** A business firm, company, organisation, e.g. bank, FI, Government Department,
3) A **Representative/Group of Persons**: Debtors, creditors, borrowers, lenders, etc.

Thumb Rule Governing the Personal Accounts
Debit the Receiver,
Credit the Giver.

Thumb rule governing a 'Personal Account' is very simple. Whenever a benefit is received by a person (natural, artificial/legal, or representative/group of persons), the account of that person is debited by the amount of the benefit received. Conversely, when a benefit is given by a person, his account is credited by the amount of benefit given.

1.6.2.2. Real Account

Accounts relating to properties or assets are known as the 'Real Accounts'. A separate account is maintained for each asset e.g., Cash, Inventory, Plant, Machinery, Land, Building, etc. Real accounts may be either tangible or intangible:

1) **Tangible Real Accounts:** These accounts represent assets and properties which can be seen, touched, felt, measured, purchased and sold, e.g., Land account, Building account, Machinery account, Cash account, Furniture account, Inventory account, etc.
2) **Intangible Real Accounts:** These accounts represent assets and properties which cannot be seen, touched or felt but they can be measured in terms of money. e.g., Goodwill accounts, Patents account, Trademarks account, Copyrights account, etc.

Thumb Rule Governing the Real Accounts
Debit What Comes In,
Credit What Goes Out.

The thumb rule governing 'Real Account' states that whenever some asset is added into the business by way of its purchase or otherwise, the account of that asset is debited. On the other hand, whenever some asset goes out of the business through its sales or otherwise, the account of that asset is credited.

1.6.2.3. Nominal Account

Accounts relating to income, expenses, gains and losses are known as 'Nominal Accounts'. These accounts are also known as 'Fictitious Accounts' as they do not represent any tangible asset. A separate account is maintained for each head or expense or loss and gain or income. Wages account, Rent account, Commission account, interest received account, etc. are some examples of nominal account.

Thumb Rule Governing Nominal Accounts
Debit all Expenses or Losses,
Credit all Income or Gains.

The thumb rule of 'Nominal Account' states that if some expense is incurred in connection with services availed or losses made, that expense account is debited. Conversely, if income or revenue is generated by extending some services, that income account is credited.

1.7. JOURNAL

1.7.1. Meaning and Definition of Journal

The term 'Journal', from old French and Latin origins, suggests daily activity (jour is French for day). Personal diaries and newspapers are also sometimes called **journals** for the same reason. In bookkeeping and accounting, 'Journal' is a book for recording transactions as and when they occur. Recording of this transaction is called as **journal entry**. Entering transactions into a journal is considered the first step in the accounting cycle. That is why the 'Journal' is also called the book **of original or prime entry**. The process of recording the business transaction (journal entry) is termed as **journalising**.

According to Eric. L. Kohler, "The journal is a book of original entry in which are recorded transactions not provided for in specialised journals."

According to Robert Anthony, "The journal is a chronological record of accounting transactions showing the names of the accounts that are to be debited or credited, the amounts of debits and credits, and any useful supplementary information about the transactions. It is analogous to a diary."

1.7.2. Proforma of Journal

Proforma of Journal

Date	Particulars	L.F.	Dr. (₹)	Cr. (₹)
(1)	(2)	(3)	(4)	(5)

1) **Date:** In the first column of the proforma, date of the transaction is entered.
2) **Particulars:** Second column of the proforma is meant for entering the names of the two 'Heads of Account', influenced by a transaction.
3) **Ledger Folio:** In the third column of the proforma, the page number of the ledger, on which the particular account appears in that book, is entered.
4) **Amount (Debit):** In column number four the amount to be debited is indicated.
5) **Amount (Credit):** In column number five the amount to be credited is indicated.

The fundamental principle to be remembered all the time is that **'left hand side is debit'** and the **'right hand side is credit'**. Debit (abbreviated form Dr.) means entering an amount of transaction on the left side of the concerned account and credit (abbreviated form Cr.) means entering an amount of transaction on the right side of the concerned account. Both debit and credit may indicate either increase or decrease, depending upon the nature of an account.

1.7.3. Examples of Journal Entries

A few examples of 'Journal' entries, furnished in the following paragraphs, would facilitate a better understanding of the concept:

1) **Started Business with a Capital of ₹1,00,000:** The two 'Heads of Account' involved in this transaction are:
   i) Cash Account, and
   ii) Capital Account.

   Whereas 'Cash Account' is a 'Real Account', 'Capital Account' is a 'Personal Account'. As per the rule of the 'Real Account' (*Debit what comes in and Credit what goes out*), the 'Cash Account' would be debited (increase in cash). Similarly, as per the rule of 'Personal Account' (*Debit the receiver and Credit the giver)* the 'Capital Account' would be credited (decrease in capital i.e., cash).

   The entry would be shown in the 'Journal' as following:

Cash A/c	Dr.	1,00,000	
To Capital A/c			1,00,000
(Being commencement of business)			

2) **Purchased Goods from Mr. Shyam for Cash ₹50,000/Cash Purchases ₹50,000:** The two 'Heads of Account' involved in this transaction are:
   i) Goods (Purchase) Account, and
   ii) Cash Account.

   As both the accounts are 'Real Accounts', as per the rule of the 'Real Account' (*Debit what comes in and Credit what goes out*), the 'Goods Account'/'Purchase Account' would be debited (increase in assets) and 'Cash Account' would be credited (decrease in cash).

   The entry would be shown in the 'Journal' as following:

Purchases A/c	Dr.	50,000	
To Cash A/c			50,000
(Being goods purchased for cash)			

3) **Purchased Goods of ₹50,000 from Mr. Somit on Credit:** The two 'Heads of Account' involved in this transactions are:
   i) Purchases Account, and
   ii) Mr. Somit Account.

Whereas Purchases Account is a 'Real Account', Mr. Somit account is a 'Personal Account'. As per the rule of the 'Real Account' (*Debit what comes in and Credit what goes out*), the 'Purchases Account' would be debited (increase in assets). Similarly, as per the rule of 'Personal Account' (*Debit the receiver and Credit the giver)* the 'Mr. Somit Account' would be credited (decrease in cash).

The entry would be shown in the 'Journal' as following:

Purchases A/c	Dr.	50,000	
To Mr. Somit A/c			50,000
(Being goods purchased on credit)			

4) **Goods of ₹1,000 Returned to Mr. Tarun:** The two 'Heads of Account' involved in this transaction are:
   i) Mr. Tarun (Supplier) Account, and
   ii) Purchase Return/Return Outward Account.

Whereas 'Mr. Tarun (Supplier) Account' is a 'Personal Account', 'Purchase Return' is a 'Real Account'. As per the rule of 'Personal Account' (*Debit the receiver and Credit the giver)* the 'Mr. Tarun Account' would be debited (decrease the supplier dues). Similarly, as per the rule of the 'Real Account' (*Debit what comes in and Credit what goes out*), 'Purchase Return' account would be credited (decrease in business assets).

The entry would be shown in the 'Journal' as following:

Mr. Tarun A/c	Dr.	1,000	
To Purchase Return A/c			1,000
(Being goods returned to Mr. Tarun)			

5) **Goods Sold to Mr. Shyam for Cash ₹40,000/Cash Sales of ₹40,000:** Two 'Heads of Account' involved in this transaction are:
   i) 'Sales Account' and
   ii) 'Cash Account'.

As both the accounts happen to be 'Real Accounts', as per the rule of the 'Real Account' (*Debit what comes in and Credit what goes out*), the 'Cash Account' would be debited (increase the cash) and 'Sales Account' would be credited (decrease the assets).

The entry would be shown in the 'Journal' as following:

Cash A/c	Dr.	40,000	
To Sales A/c			40,000
(Being goods sold for cash)			

6) **Sale of ₹40,000 Goods to Mr. Varun on Credit:** Two 'Heads of Account' involved in this transaction are:
   i) Mr. Varun Account and
   ii) Sales Account.

'Mr. Varun Account' is a 'Personal Account' and 'Sales Account' is a 'Real Account'. As per the rule of 'Personal Account' (*Debit the receiver and Credit the giver)* 'Mr. Varun Account' would be debited (increase in debtors). Similarly, as per the rule of the 'Real Account' (*Debit what comes in and Credit what goes out*) 'Sales Account' account would be credited (decrease in the assets).

The entry would be shown in the 'Journal' as following:

Mr. Varun A/c	Dr.	40,000	
To Sales A/c			40,000
(Being goods sold on credit)			

7) **Goods of ₹1,000 Returned by Mr. Tarun:** Two 'Heads of Account' involved in this transaction are:
   i) Mr. Tarun Account and
   ii) Sales Return Account (Return Inwards).

Whereas Mr. 'Tarun Account' is a 'Personal Account', 'Sales Return Account' is a 'Real Account'. As per the rule of 'Personal Account' (*Debit the receiver and Credit the giver)* 'Mr. Tarun Account' would be credited (decrease in the assets). Likewise, as per the rule of the 'Real Account' (*Debit what comes in and Credit what goes out*) 'Sales Return Account' account would be debited (increase in the assets).

The entry would be shown in the 'Journal' as following:

Sales Return A/c	Dr.	1,000	
To Mr. Tarun A/c			1,000
(Being goods returned by Mr. Tarun)			

8) **Machinery of ₹10,000 Sold:** In this transaction, two 'Heads of Account' involved are 'Machinery Account' and 'Cash Account', both of which are 'Real Accounts'. As per the rule of the 'Real Account' (*Debit what comes in and Credit what goes out*), the 'Cash Account' would be debited (increase in cash) and 'Machinery Account' would be credited (decrease in value of machinery).

The entry would be shown in the 'Journal' as following:

Cash A/c	Dr.	10,000	
To Machinery A/c			10,000
(Being machinery sold)			

9) **Paid ₹5,000 as Salary:** Two 'Heads of Account' involved in this transaction are:
   i) Salary Account and
   ii) Cash Account.

The 'Salary Account' is a 'Nominal Account and 'Cash Account' is a 'Real Account'. As per the rule of 'Nominal Account' (*Debit all the expenses or losses and Credit all the income or gains*), the 'Salary Account' would be debited (increase in expenses). Likewise, as per the rule of 'Real Account' (*Debit what comes in and Credit what goes out*) 'Cash Account' would be credited (decrease in cash).

The entry would be shown in the 'Journal' as following:

Salary A/c	Dr.	5,000	
To Cash A/c			5,000
(Being salary paid)			

10) **Cash Received from Mr. Mohan ₹990 and Discount Paid ₹10:** In this transaction, three 'Heads of Account' are involved, viz.:
   i) Mr. Mohan Account,
   ii) Cash Account, and
   iii) Discount Account.

'Mr. Mohan Account' is a 'Personal Account' i.e., debtor, 'Cash Account' is a 'Real Account' and 'Discount Account' is a 'Nominal Account'. As per the rule of 'Personal Account' (*Debit the receiver and Credit the giver),* 'Mr. Mohan Account would be credited (decreases the assets). Further, as per the rule of 'Real Account' (*Debit what comes in and Credit what goes out*) 'Cash Account' would be debited (increases the cash). Similarly, as per the rule of 'Nominal Account' (*Debit all the expenses or losses and Credit all the income or gains*), 'Discount Account' would be debited (increases a loss in the form of discount allowed).

The entry would be shown in the 'Journal' as following:

Cash A/c	Dr.	990	
Discount A/c	Dr.	10	
To Mr. Mohan A/c			1,000
(Being cash received from Mr. Mohan and discount allowed)			

11) **Cash Paid to Mr. Rohan ₹1,980 and Discount Received ₹20:** In this transaction, three 'Heads of Account' are involved, viz.:
   i) Mr. Rohan Account,
   ii) Cash Account, and
   iii) Discount Account.

'Mr. Rohan Account' is a 'Personal Account' i.e., creditor, 'Cash Account' is a 'Real Account' and 'Discount Account' is a 'Nominal Account'. As per the rule of 'Personal Account' (*Debit the receiver and Credit the giver),* 'Mr. Rohan Account would be debited (decrease in liability). Further, as per the rule of 'Real Account' (*Debit what comes in and Credit what goes out*) 'Cash Account' would be credited (decrease in cash). Similarly, as per the rule of 'Nominal Account' (*Debit all the expenses or losses and Credit all the income or gains*), 'Discount Account' would be credited (increase in revenue in the form of discount).

The entry would be shown in the 'Journal' as following:

Mr. Rohan A/c	Dr.	2,000	
To Cash A/c			1,980
To Discount Received A/c			20
(Being cash paid to Mr. Rohan and discount received)			

12) **Depreciation on Furniture of ₹2,000:** Two 'Heads of Account' involved in this transaction are:
   i) Depreciation Account, and
   ii) Furniture Account.

'Depreciation Account' is a 'Nominal Account' and 'Furniture Account' is a 'Real Account'. As per the rule of 'Nominal Account' (*Debit all the expenses or losses and Credit all the income or gains*), 'Depreciation Account' would be debited (increase the expenses) Further, as per the rule of 'Real Account' (*Debit what comes in and Credit what goes out*) 'Furniture Account' would be credited (decrease the assets).

The entry would be shown in the 'Journal' as following:

Depreciation A/c	Dr.	2,000	
To Furniture A/c			2,000
(Being depreciation on furniture)			

13) **Interest on Capital ₹2,000:** Two 'Heads of Account' involved in this transaction are:
   i) Interest on Capital Account, and
   ii) Capital Account.

While the 'Interest on Capital Account' is a 'Nominal Account', the 'Capital Account' is a 'Personal Account'. As per the rule of 'Nominal Account' (*Debit all the expenses or losses and Credit all the income or gains*), 'Interest on Capital Account' would be debited (increase in the expenses). Similarly, as per the rule of 'Personal Account' (*Debit the receiver and Credit the giver)* 'Capital Account' would be credited (increase in the capital).

The entry would be shown in the 'Journal' as following:

Interest on Capital A/c	Dr.	2,000	
To Capital A/c			2,000
(Being interest on capital paid of ₹2,000)			

14) **Withdrawn ₹1,000 by Proprietor for Personal Use:** In this transaction, two 'Heads of Accounts' involved are:
   i) Drawing Account, and
   ii) Cash Account.

While the 'Drawing Account' is a 'Personal Account', 'Cash Account' is a 'Real Account'. As per the rule of 'Personal Account' (*Debit the receiver and Credit the giver)* 'Drawing Account' would be debited (decrease in capital). Further, as per the rule of 'Real Account' (*Debit what comes in and Credit what goes out*) 'Cash Account' would be credited (decrease in the cash).

The entry would be shown in the 'Journal' as following:

Drawing A/c	Dr.	1,000	
To Cash A/c			1,000
(Being cash withdrawn by the proprietor)			

15) **Withdrawn from Bank ₹1,000 for Office Use:** Two 'Heads of Account' involved in this transaction are:
   i) Cash Account and
   ii) Bank Account.

Whereas 'Cash Account' is a 'Real Account', the 'Bank Account' is a 'Personal Account'. As per the rule of 'Real Account' (*Debit what comes in and Credit what goes out*), 'Cash Account' would be debited (increase in cash). Similarly, as per the rule of 'Personal Account' (*Debit the receiver and Credit the giver),* 'Bank Account' would be credited (decrease in bank account).

The entry would be shown in the 'Journal' as following:

Cash A/c	Dr.	1,000	
To Bank A/c			1,000
(Being cash withdrawn from bank for office use)			

Example 3: Journalise the following transactions:

Date	Particulars	₹
2019		
Jan. 1	Started business with cash	30,000
Jan. 2	Deposited into bank	10,000
Jan. 3	Goods purchased for cash	5,000
Jan. 4	Withdraw from bank for office use	1,000
Jan. 5	Credit sales to Sumit	1,500

Solution:

Journal Entries

Date	Particulars		L.F.	Dr. (₹)	Cr. (₹)
2019					
Jan. 1	Cash A/c	Dr.		30,000	
	To Capital A/c				30,000
	(Being the business started with cash)				
Jan. 2	Bank A/c	Dr.		10,000	
	To Cash A/c				10,000
	(Being cash deposited into bank)				
Jan. 3	Purchase A/c	Dr.		5,000	
	To Cash A/c				5,000
	(Being goods purchased for cash)				
Jan. 4	Cash A/c	Dr.		1,000	
	To Bank A/c				1,000
	(Being cash withdrawn from bank for office use)				
Jan. 5	Sumit's A/c	Dr.		1,500	
	To Sales A/c				1,500
	(Being goods sold to Sumit on credit)				
		Total		**47,500**	**47,500**

Example 4: Pass the necessary journal entries in the books of Mr. Raghav.

Date	Particulars	₹
2019		
Mar. 1st	Raghav started business with cash	15,000
	Plant and machinery	20,000
Mar. 2nd	Bought the furniture from Raj	2,000
Mar. 8th	Purchased goods	8,000
Mar. 12th	Sold goods	1,000
Mar. 16th	Purchased goods from Vaibhav	500
Mar. 18th	Sold goods to Rajesh	3,000
Mar. 20th	Returned damaged goods to Vaibhav	100
Mar. 27th	Received cheque from Rajesh	2,000

Solution:

Journal Entries

(In the books of Mr. Raghav)

Date	Particulars		L.F.	Dr. (₹)	Cr. (₹)
2019					
Mar. 1	Cash A/c	Dr.		15,000	
	Plant and Machinery A/c	Dr.		20,000	
	To Capital A/c				35,000
	(Being business started with cash and plant and machinery)				
Mar. 2	Furniture A/c	Dr.		2,000	
	To Raj A/c				2,000
	(Being furniture bought from Raj)				
Mar 8	Purchase A/c	Dr.		8,000	
	To Cash A/c				8,000
	(Being goods purchased for cash)				
Mar. 12	Cash A/c	Dr.		1,000	
	To Sales A/c				1,000
	(Being goods sold for cash)				
Mar. 16	Purchase A/c	Dr.		500	
	To Vaibhav A/c				500
	(Being goods purchased from Vaibhav)				
Mar. 18	Rajesh A/c	Dr.		3,000	
	To Sales A/c				3,000
	(Being goods sold to Rajesh)				
Mar. 20	Vaibhav A/c	Dr.		100	
	To Purchase Return A/c				100
	(Being damages goods are returned to Vaibhav)				
Mar. 27	Bank A/c	Dr.		2,000	
	To Rajesh A/c				2,000
	(Being cheque received from Rajesh)				
		Total		**51,600**	**51,600**

Example 5: Record the following transactions in the journal of Mr. Mathur:

Date	Particulars	₹
2019		
Jan. 1	Started business with cash	1,00,000
Jan. 8	Purchased goods with 20% trade discount and 5% cash discount	50,000
Jan. 12	Bought shares of Bhavna Ltd. and brokerage paid ₹25	1,500
Jan. 16	Purchased a motor car in exchange of goods	20,000
	and cash	30,000
Jan. 20	Sold goods to Mr. Yash on credit	60,000
Jan. 22	Purchased goods from Rohit	25,000
Jan. 26	Goods distributed as free samples	1,000
Jan. 29	Mr. Yash became insolvent and only 50% is recovered from his assets	
Jan. 30	Cash withdrawn by the proprietor for personal use	5,000
Jan. 31	Paid into bank	5,000

Solution:

Journal Entries

(In the journal of Mr. Mathur)

Date	Particulars		L.F.	Dr. (₹)	Cr. (₹)
2019					
Jan. 1	Cash A/c	Dr.		1,00,000	
	To Capital A/c				1,00,000
	(Being business started)				
Jan. 8	Purchase A/c	Dr.		40,000	
	To Cash A/c				38,000
	To Discount A/c				2,000
	(Being goods worth ₹50,000 bought for cash less 20% trade discount and 5% cash discount. Trade discount is not shown in the books but cash discount is shown)				
Jan. 12	Investment in Share A/c	Dr.		1,525	
	To Cash A/c				1,525
	(Being purchase 100 shares in Bhavna Ltd. @ ₹15 per share plus brokerage ₹25)				

Jan. 16	Motor Car A/c	Dr.		50,000	
	To Purchase (Goods) A/c				20,000
	To Cash A/c				30,000
	(Being purchase motor cash from goods and cash)				
Jan. 20	Mr. Yash A/c	Dr.		60,000	
	To Sales A/c				60,000
	(Being goods sold to Mr. Yash on credit)				
Jan. 22	Purchase A/c	Dr.		25,000	
	To Rohit A/c				25,000
	(Being goods purchased from Rohit)				
Jan. 26	Free Sample A/c	Dr.		1,000	
	To Purchase A/c				1,000
	(Being goods given as free sample)				
Jan. 29	Cash A/c	Dr.		30,000	
	Bad Debts A/c	Dr.		30,000	
	To Mr. Yash A/c				60,000
	(Being half the dues received from Mr.Yash and balance written-off as bad debts)				
Jan. 30	Drawing A/c	Dr.		5,000	
	To Cash A/c				5,000
	(Being cash used by the proprietor for personal use)				
Jan. 31	Bank A/c	Dr.		5,000	
	To Cash A/c				5,000
	(Being paid into bank)				
		Total		**3,47,525**	**3,47,525**

1.8. LEDGER

1.8.1. Meaning and Definition of Ledger

'Ledger' is defined as an accounting book (principle book of accounts), wherein all the transactions are recorded in a summarised and classified manner. It is the most important book of accounts. It may be maintained either in the form of a physical book into which credits and debits are posted, or an accounting computer programme wherein various credits and debits are entered. Under the general ledger of the double-entry bookkeeping system, each transaction is required to be entered on the left/debit side of one account and simultaneously on the right/credit side of another account. As the transactions recorded in the 'Ledger' are permanent in nature, it is referred to as the **'Book of Final Entry'**.

According to L.C. Cropper, "The book which contains a classified and permanent record of all the transactions of a business is called the Ledger".

The process of transferring various 'Debit' and 'Credit' entries from the 'Journal' to the 'Ledger' is called **'Ledger posting'**. For this purpose, separate 'Heads of Account' are opened in the 'Ledger' and all the 'Credit' and 'Debit' entries are taken from the 'Journals' and posted in the 'Ledger'. In the final stage, all the accounts are balanced accurately.

1.8.2. Proforma of Ledger

There are two types of forms for writing Ledger Accounts, viz.:

1) Horizontal form, and
2) Vertical or 'T' form.

Nowadays, vertical or 'T' form is in practice, which is in 'T' shape and also termed as **"T Account"**. The vertical or 'T' form are classified on the basis of the right and left side of ledger account. The left side of the "T account" is called the **debit side** and its right side is called the **credit side**.

The proforma of "T Account' is furnished below:

Date (1)	Particulars (2)	Journal Folio (3)	Dr. (₹) (4)	Date (1)	Particulars (2)	Journal Folio (3)	Cr. (₹) (4)

It may be seen from the above, that each side of the proforma, i.e. 'Debit Side' and 'Credit Side' has four columns to record the requisite details of each transaction, which are:

1) **Date:** The date on which a transaction is recorded.
2) **Particulars:** It relates to a subject matter of transaction.
3) **Journal Folio (J.F.):** In this column, a reference is made to the page number of the 'Journal'.
4) **Amount:** It is the column of money paid or received during the transaction.

An account is debited or credited in terms of the 'Thumb Rule' applicable in respect of each category of account (i.e., 'Personal', 'Real' or 'Nominal').

1.8.3. Difference between Journal and Ledger

Basis of Difference	Journal	Ledger
1) **Nature of Book**	It is termed as the 'Original Book of Account' or the 'Book of Prime Entry'.	It is termed as the 'Book of Final Entry' or the 'Book of Secondary Entry'.
2) **Basis for Preparation**	It is prepared on the basis of 'Source Records of Transactions'.	It is prepared on the basis of the entries recorded in the 'Journals'.
3) **Stage of Recording**	'Journal Entries' are first stage of recording a transaction.	'Posting in the ledger' is the second stage of recording a transaction.
4) **Object**	Journal is prepared with the objective of recording all business transactions in a chronological order.	Ledger is prepared with the objective of knowing the net impact of various transactions affecting a specific account.
5) **Balancing**	There is no concept of balancing a Journal.	All ledger accounts (except nominal accounts) are required to be balanced periodically.
6) **Narration**	Writing a narration is must in respect of each entry.	Ledger posting does not involve writing of any narration.
7) **Name of the Process of Recording Entries**	The process of entering a transaction in journal is termed as 'Journalising'.	The process of transferring a 'Journal Entry' in the ledger is termed as 'Posting'.
8) **Basis of Preparation of Final Accounts**	Journals are not the basis for the preparation of 'Final Accounts' directly.	Ledgers are the basis for the preparation of 'Final Accounts'.

1.8.4. Ledger Posting

The process of transferring various 'Debit' and 'Credit' items from the books of prime entry, viz. 'Journals' to the corresponding accounts in 'Ledgers' is known as **'Ledger Posting'**. Ledger is called the 'Principal Book of Account' because finally all ledger accounts are considered for the preparation of 'Final Accounts' and accounting analysis. The ledger sums up the total of every account which is finally transferred into the 'Balance Sheet' and 'Profit & Loss Account'.

Rules Regarding Ledger Posting

During the exercise of 'Ledger Posting' following steps and rules are required to be followed:

1) **Opening of Separate Accounts:** In a 'Ledger', separate accounts are required to be opened for various 'Personal', 'Real' and 'Nominal' accounts. Minimum two accounts are impacted by a business transaction. No account is opened twice. Once an account is opened, all transactions relating to that account, whether debit or credit, should be posted therein, with a view to ensuring that net result of that account may be ascertained at any point of time.

2) **Posting of the Journal Entry to Appropriate/Concerned Side of the 'Ledger':** If an account is debited in the journal, at the time of posting in the 'Ledger', it must be ensured that the posting is made on the debit side of the concerned account in the ledger. In the same way, if an account is credited in the journal, at the time of posting in the 'Ledger', it must be ensured that the posting is made on the credit side of the concerned account in the ledger.

   For example, if a provision of ₹10,000 has been made for salary, 'Salary Account' would be debited and 'Outstanding Salary Account' would be credited in the 'Journal'. At the time of 'Ledger Posting', it may be ensured that 'Salary' account in the ledger is debited with the amount (₹10,000) and 'Outstanding Salary Account' is credited with the same amount.

3) **Use of Word "To" and "By":** The word 'To' is placed before writing the name of an account on the debit side of ledger, which is credited in the journal, whereas the word 'By' is placed before writing the name of an account on the credit side of ledger, which is debited in the journal.

4) **Balance in Account:** The heavy side of the account (Debit or credit) is totalled, first. The same total is put on the total column of the other side in the account. Thus, the net result is achieved. If total of the debit side is more than the total of the credit side, the net result would be 'Debit'. Likewise, if the total of the credit side is more than the total of the debit side, the net result would be 'Credit'.

Example 6: Enter the following transaction in the journal of Neerose and post them to ledger for the year ending 2018:

Date	Particulars	₹
March 1	Neerose commenced his business with a capital	1,00,000
2	Bought machinery	30,000
3	Bought goods for cash from Ram	5,000
4	Sold goods for cash to Ramesh	4,000
6	Credit sales to Harish	3,400
7	Bought goods from Kamal on Credit	1,920
10	Paid insurance premium	3,000
11	Paid salary to clerk	3,500

Solution:

Journal Entries

Date	Particulars		L.F.	Dr. (₹)	Cr. (₹)
1 March	Cash A/c	Dr.		1,00,000	
	To Capital A/c				1,00,000
	(Being business commenced)				
2 March	Machinery A/c	Dr.		30,000	
	To Cash A/c				30,000
	(Being machinery purchased)				
3 March	Purchases A/c	Dr.		5,000	
	To Cash A/c				5,000
	(Being goods purchased for cash from Ram)				
4 March	Cash A/c	Dr.		4,000	
	To Sales A/c				4,000
	(Being goods sold to Ramesh for cash)				
6 March	Harish's A/c	Dr.		3,400	
	To Sales A/c				3,400
	(Being credit sale made)				
7 March	Purchases A/c	Dr.		1,920	
	To Kamal's A/c				1,920
	(Being goods purchased from Kamal on credit)				
10 March	Insurance Premium A/c	Dr.		3,000	
	To Cash A/c				3,000
	(Being insurance premium paid)				
11 March	Salary A/c	Dr.		3,500	
	To Cash A/c				3,500
	(Being salary paid)				

Ledger

Dr. Capital Account Cr.

Date	Particulars	J.F.	₹	Date	Particulars	J.F.	₹
2018				2018			
31 March	To Balance c/d		1,00,000	1 March	By Cash A/c		1,00,000
			1,00,000				**1,00,000**
				2018			
				1 April	By Balance b/d		1,00,000

Dr. **Cash Account** **Cr.**

Date	Particulars	J.F.	₹	Date	Particulars	J.F.	₹
2018				2018			
1 March	To Capital A/c		1,00,000	2 March	By Machinery A/c		30,000
4 March	To Sales A/c		4,000	3 March	By Purchases A/c		5,000
				10 March	By Insurance Premium A/c		3,000
				11 March	By Salary		3,500
				31 March	By Balance c/d		62,500
			1,04,000				**1,04,000**
2018							
1 April	To Balance b/d		62,500				

Dr. **Machinery Account** **Cr.**

Date	Particulars	J.F.	₹	Date	Particulars	J.F.	₹
2018				2018			
2 March	To Cash A/c		30,000	31 March	By Balance c/d		30,000
			30,000				**30,000**
2018							
1 April	To Balance b/d		30,000				

Dr. **Purchases Account** **Cr.**

Date	Particulars	J.F.	₹	Date	Particulars	J.F.	₹
2018				2018			
3 March	To Cash A/c		5,000	31 March	By Balance c/d		6,920
7 March	To Kamal's A/c		1,920				
			6,920				**6,920**
2018							
1 April	To Balance b/d		6,920				

Dr. **Sales Account** **Cr.**

Date	Particulars	J.F.	₹	Date	Particulars	J.F.	₹
2018				2018			
31 March	To Balance c/d		7,400	4March	By Cash A/c		4,000
				6 March	By Harish's A/c		3,400
			7,400				**7,400**
				2018			
				1 April	By Balance b/d		7,400

Dr. **Kamal's Account** **Cr.**

Date	Particulars	J.F.	₹	Date	Particulars	J.F.	₹
2018				2018			
31 March	To Balance c/d		1,920	7 March	By Purchases A/c		1,920
			1,920				**1,920**
				2018			
				1 April	By Balance b/d		1,920

Dr. **Harish's Account** **Cr.**

Date	Particulars	J.F.	₹	Date	Particulars	J.F.	₹
2018				2018			
6 March	To Sales A/c		3,400	31March	By Balance c/d		3,400
			3,400				**3,400**
2018							
1 April	To balance b/d		3,400				

Dr. **Insurance Premium Account** **Cr.**

Date	Particulars	J.F.	₹	Date	Particulars	J.F.	₹
2018				2018			
10 March	To Cash A/c		3,000	31 March	By Balance c/d		3,000
			3,000				**3,000**
2018							
1 April	To Balance b/d		3,000				

Dr. **Salary Account** **Cr.**

Date	Particulars	J.F.	₹	Date	Particulars	J.F.	₹
2018				2018			
1 March	To Cash A/c		3,500	31 March	By Balance c/d		3,500
			3,500				**3,500**
2018							
1 April	To Balance b/d		3,500				

1.9. TRIAL BALANCE

1.9.1. Meaning of Trial Balance

'Trial Balance' is a statement, in which the balances of all the ledgers are compiled into debit and credit columns. A company prepares a trial balance periodically, usually at the end of every reporting period. The general purpose of preparing a trial balance is to ensure that the entries in a company's bookkeeping system are mathematically correct. On the debit side of a 'Trial Balance' 'Asset' and 'Expense' accounts appear, whereas on its credit side 'Liabilities', 'Capital' and 'Income' accounts appear. All the debit balances and credit balances appearing in the 'Trial Balance' must be equal, provided:

1) All accounting entries are recorded accurately; and
2) All the ledger balances are correctly extracted.

1.9.2. Features of Trial Balance

The 'Trial Balance' of a company exhibits following salient features:

1) It is a list of all the 'Ledger' accounts (both revenue and capital) contained in the ledger of a business. The debit balances are listed in the 'Debit' column of the trial balance and the 'Credit' balances are listed in the credit column.
2) It can be prepared at any point of time during the entire accounting period.
3) The arithmetic correctness of 'Ledger Posting' may be checked and ascertained through the preparation of 'Trial Balance'.
4) Trial balance is not covered under 'Double Entry System' of accountancy. It is an instrument to check and test the arithmetical precision of the 'Ledger Postings' to some extent (all the errors are not disclosed by it).
5) Checking and testing of accuracy of accounts through 'Trial Balance' has some limitations. 'Trial Balance' tallying does not prove the correctness of all the accounts.

1.9.3. Proforma of Trial Balance

Trial Balance

(as on .....)

Particulars	L.F.	Dr. (₹) (Total or Balance)	Cr. (₹) (Total or Balance)

Following are the details about the trial balance proforma:

1) Trial balance proforma consists of four column i.e., particulars, ledger folio, debit amount and credit amount.
2) On the top of the trial balance proforma, the date on which it is going to be prepared is mentioned.
3) First column covers the various heads of accounts.
4) Second column mention the ledger folio number.
5) Heads of accounts that have debit balance are mentioned in third column.
6) Heads of accounts that have credit balance are mentioned in fourth column.
7) At the end the total of both debit and credit column should be matched.

1.9.4. Methods of Preparing Trial Balance

For the preparation of a 'Trial Balance' following methods are available:

1) **Traditional Method/Total Method:** The 'Traditional Method' or the 'Total Method' considers the sub-totals of each ledger account. It is prepared by presenting the sub-totals relating to each ledger account in the relevant columns of the 'Trial Balance'. In this method, the debit total of each account is recorded on the debit side and the credit total of each account is recorded on the credit side of the 'Trial Balance'. Thereafter the total of the 'Trial Balance' is taken. If both debits and credits sides matches, it is presumed that the books of accounts are arithmetically correct.

2) **Modern Method/Balance Method:** The 'Modern Method' or the 'Balance Method' considers only the balances of the ledger accounts. It is prepared by presenting the balance relating to each ledger account in the relevant column of the trial balance. In this method, the difference between the 'Debit Side' and 'Credit Side' of an account is taken in the 'Trial Balance' instead of taking the totals of both the sides. If the 'Debit Side' happens to be more than the 'Credit Side', it is termed as 'Debit Balance'. Both sides of the 'Trial Balance' also need to be equal in this method. 'Modern Method' of preparing a 'Trial Balance' is considered to be a better one and therefore preferred over the other methods.

3) **Total and Balance Method:** This method is a combination of 'Traditional Method' and the 'Modern Method' of preparation of the 'Trial Balance'. Four columns are made under this method. The first two columns are for the totals (traditional method) and last two columns are for balances (modern method). However, this method has not gained popularity.

Example 7: Prepare a Trial balance from the following ledger balances:

Particulars	₹
Capital	50,000
Sales	25,000
Opening stock	5,200
Debtors	2,500
Creditor	1,000
Purchases	20,000
Salaries	2,000
Rent	1,500
Cash	2,000
Insurance	300
Drawing	10,000
Plant	28,000
Bank	4,500

Solution:

Trial Balance

(as on.....)

Particulars	L.F.	Dr. (₹)	Cr. (₹)
Capital		–	50,000
Sales		–	25,000
Opening stock		5,200	–
Debtors		2,500	–
Creditors		–	1,000
Purchases		20,000	–
Salaries		2,000	–
Rent		1,500	–
Cash		2,000	–
Insurance		300	–
Drawings		10,000	–
Plant		28,000	–
Bank		4,500	–
Total		**76,000**	**76,000**

Example 8: Prepare a Trial balance of Monika as on 31.3.2019 from the following Ledger Balances:

Particulars	₹
Purchases	60,000
Reserve Fund	20,000
Sales	1,00, 000
Purchase Returns	1,000
Sales Returns	2,000
Opening Stock	30,000
Expenses	20,000
Outstanding Expenses	2,000
Bank Balance (Dr.)	5,000
Fixed Assets	50,000
Debtors	80,000
Creditors	30,000
Capital	94,000

Solution:

Monika
Trial Balance
(as on 31-3-2019)

Particulars	L.F.	Dr. (₹)	Cr. (₹)
Purchases		60,000	–
Reserve Fund		–	20,000
Sales		–	1,00,000
Purchase Return		–	1,000
Sales Return		2,000	–
Opening Stock		30,000	–
Expenses		20,000	–
Outstanding Expenses		–	2,000
Bank Balance		5,000	–
Fixed Assets		50,000	–
Debtors		80,000	–
Creditors		–	30,000
Capital		–	94,000
Total		**2,47,000**	**2,47,000**

Example 9: Prepare a trial balance of Mr. Mudit as on March 31st, 2019.

Particulars	₹	Particulars	₹
Capital	9,20,000	Machinery	1,20,000
Creditors	2,88,520	Furniture	16,400
Bills Payable	69,300	Debtors	1,56,000
Sales	12,18,500	Opening Stock	2,50,400
Provision for Doubtful Debts	14,200	Cash in Hand	9,880
Interest (Cr.)	4,400	Bad Debts	6,130
Building	7,00,000	Audit Fees	5,000
Cash at Bank	1,45,340	Travelling Expenses	4,250
Bills Receivable	58,440	Discount (Dr.)	6,200
Purchases	8,55,220	Sales Returns	2,850
Carriage Inwards	12,910	Investments	89,220
Carriage Outwards	8,000	General Expenses	60,850
		Insurance	7,830

Solution:

Mr. Mudit
Trial Balance
(as on March 31st, 2019)

Particulars	L.F.	Dr. (₹)	Cr. (₹)
Capital		–	9,20,000
Creditors		–	2,88,520
Bills Payable		–	69,300
Sales		–	12,18,500
Provision for Doubtful Debts		–	14,200
Interest (Cr.)		–	4,400
Building		7,00,000	–
Cash at Bank		1,45,340	–
Bills Receivable		58,440	–
Purchases		8,55,220	–

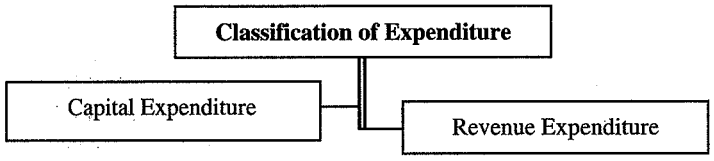
Carriage Inwards		12,910	–
Carriage Outwards		8,000	–
Machinery		1,20,000	–
Furniture		16,400	–
Debtors		1,56,000	–
Opening Stock		2,50,400	–
Cash in Hand		9,880	–
Bad Debts		6,130	–
Audit Fees		5,000	–
Travelling Expenses		4,250	–
Discount (Dr.)		6,200	–
Sales Returns		2,850	–
Investments		89,220	–
General Expenses		60,850	–
Insurance		7,830	–
Total		**25,14,920**	**25,14,920**

1.10. CAPITAL AND REVENUE EXPENDITURE

1.10.1. Meaning of Expenditure

Expenditure is the detail of the expenses and cost incurred in operating the business activities. The expenditure used for payment of cash or cash-equivalent for goods or services, or to pay off the current liabilities as per stated documents such as invoice, receipt, voucher, etc.

Hence, expenditure refers to those expenses of business which is used for getting the benefits like purchasing of goods and rendering of services. Expenditure incurred by an organisation can be classified into two categories as shown below:



1.10.2. Capital Expenditure

Expenditure incurred in connection with the following purposes is classified as 'Capital Expenditure':

1) Acquisition of physical assets such as land, building, plant, machinery or equipment. Investments made for undertaking new projects, etc. is also included in the 'Capital Expenditure.
2) Extension, improvement, or repairs of an existing fixed asset, and
3) Replacement of an existing fixed asset in a substantial manner.

'Capital Expenditure' is an expenditure, the benefits of which are reaped over an extended period of time, i.e. much beyond the current accounting period. It is incurred with a view to bringing improvements in output or earning competence.

Normally, following categories of expenditure are considered as part of 'Capital Expenditure':

1) **Expenditure Incurred in Connection with the Purchase, Receipt or Erection of a Fixed Asset:** In addition to the 'Purchase Price' of a fixed asset, expenditure incurred in connection with acquisition and making that fixed asset ready for use, are added to the cost of the fixed asset, classified as 'Capital Expenditure' and accordingly capitalised. However, the expenditure incurred, subsequent to the fixed asset having been made ready for use, is not capitalised.

2) **Expenditure Incurred in Connection with Improvement in the Efficiency of Fixed Assets:** An expenditure, which results in increased revenue generation capability of an existing fixed asset, needs to be treated as 'Capital Expenditure'. Such revenue generation capability is enhanced by:
   i) Lowering of cost, and/or
   ii) Increase in output.

3) **Expenditure Incurred in Connection with Obtaining Necessary Permission to Carry on Business:** Requisite expenditure incurred in connection with:
   i) Establishment of a business (like preliminary expenses for launching a company), and

ii) Obtaining necessary license/permission for setting up or carrying on a business from Government authorities/regulators, etc. is considered as 'Capital Expenditure'.

4) **Legal Expenses:** Expenses incurred in connection with legal cases pertaining to the protection of fixed assets, rights, goodwill, trademark, etc. are also treated as 'Capital Expenditure'.

Examples of Capital Expenditures

Following examples of 'Capital Expenditure' would further make the concept clearer:

1) Cost of:
   i) Fixed assets like land, building plant, machinery,
   ii) Equipments like computer, printer, fax-machine,
   iii) Furniture and fixtures,
   iv) Lease-hold properties,
   v) Motor vehicles, etc.
2) Cost of intangible assets, e.g. trademarks, patents, copyrights, goodwill, etc.
3) Cost of certain fictitious assets, like preliminary expenses.
4) Other miscellaneous costs, like:
   i) Cost of invention,
   ii) Cost of extension/expansion/erection,
   iii) Cost of increasing capacity of the existing fixed asset,
   iv) Legal charges,
   v) Stamp duty for conveyance or acquisition of property,
   vi) Architect's fees for supervision of construction,
   vii) Cost of experiments, and
5) Cost of standby equipments, etc.

1.10.3. Revenue Expenditure

'Revenue Expenditure' is an expenditure incurred on fixed assets in connection with their 'maintenance' rather than enhancing their earning capacity. These expenses are recurring in nature, incurred on a regular basis and the benefit from such expenses is obtained over a relatively short period of time (during the current accounting period).

Examples of Revenue Expenditures

Some examples of revenue expenditures are furnished below:

1) Expenses incurred in connection with day-to-day conduct of business, e.g. rent, salaries, wages, power and fuel, etc.
2) Expenses incurred in connection with maintaining of fixed assets, e.g. repairs and maintenance.
3) Expenses incurred in connection with purchase of raw material and goods to the extent of their utilisation level during the current year. The unutilised material needs to be considered as an asset, which may be shown in the 'Balance Sheet' at the year end.
4) Amount charged as depreciation of the fixed assets. A portion of the value of fixed assets is lost every year due to:
   i) Normal passage of time, and
   ii) Wear and tear.
   Such loss is reflected in reduced value of the asset after depreciation is charged to the 'Profit & Loss Account' at a certain rate in a certain manner. The depreciation is treated as 'Revenue Expenditure'.

Some other examples of revenue expenditures are as follows:

1) Interest paid on borrowed funds,
2) Taxes,
3) Loss arising out of sale of fixed asset,
4) Discounts and allowances,
5) Bad debts,
6) Loss of goods:
   i) In transit or at warehouse, and
   ii) By fire.
7) Fire insurance premium,
8) Distribution expenses, and
9) Administration expenses.

1.10.4. Difference between Capital Expenditure and Revenue Expenditure

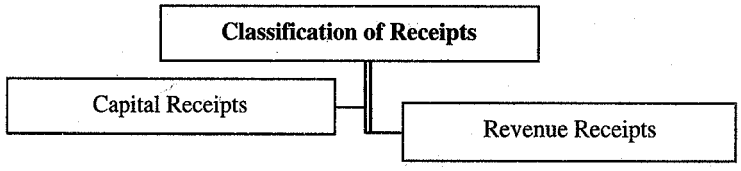
Basis of Difference	Capital Expenditure	Revenue Expenditure
1) **Nature**	Expenditure incurred in connection with the acquisition of physical assets such as land, building, plant, machinery or equipment, investments made for undertaking new projects, etc. is termed as 'Capital Expenditure. Extension, improvement, replacement or repairs of an existing fixed asset is also included.	'Revenue Expenditure' is an expenditure incurred on fixed assets in connection with their 'maintenance' rather than enhancing their earning capacity. These expenses are recurring in nature.
2) **Life of Assets**	Capital expenditure results in either acquisition of a new fixed asset or increase in the life of the existing fixed assets.	It does not result in either acquisition of a new fixed asset or increase in the life of the existing fixed assets.
3) **Cost**	It facilitates reduction in the cost of production and improvement in earning capacity.	It neither facilitates reduction in the cost of production nor improvement in earning capacity.
4) **Type**	'Capital Expenditure' is 'Non Recurring' in nature.	By nature, 'Revenue Expenditure' is a 'Recurring' one.
5) **Benefit Period**	'Capital Expenditure' is an expenditure, the benefits of which are reaped over an extended period of time, i.e., much beyond the current accounting period.	'Revenue Expenditure' is incurred on a regular basis and the benefit from such expenses is obtained over a relatively short period of time (generally during the current accounting period).
6) **Amount**	Amount involved in 'Capital Expenditure' is generally huge.	Amount involved in 'Revenue Expenditure' is relatively less.
7) **Examples**	Land & building, plant & machinery, furniture & fixture, etc.	Salary, wages, electricity, telephone charges, rent, etc.

1.11. CAPITAL AND REVENUE RECEIPTS

1.11.1. Meaning of Receipts

A receipt is a written document that provides information of the items or sum of money that has been acknowledged for the exchange of goods and services. The receipt gives the entitlement of the property which is obtained in exchange to it. Business basically raised funds from two types. It may raise the funds from long term receipts that mean the owner's capital which is either raised for the setup of business or introducing additional capital in business. The funds can also be raised from other investors in the form of mortgage receipts.

The other type of receipts is short term receipts which is actually a real profit of the business. These receipts might be commission received, cash from sale of goods, etc. There are two types of receipts are as follows:



1.11.2. Capital Receipts

The receipt which are non-recurring in nature and provides long term benefits to the business is known as **"Capital Receipts",** such as the capital invested by owner, loan from bank, income from sale of fixed assets etc. The treatment of capital receipts in the balance sheet is on the liabilities sides.

For example, Mr. Shyam as a proprietor introduced 1,00,000 capitals in the business. This capital amount is known as **capital receipts.** Similarly he sells the building for a sum of 50,000 this receipts is also a known as **capital receipts.** Capital receipts are shown during the preparation of the balance sheet.

1.11.3. Revenue Receipts

The receipt which are recurring in nature and provides short term benefits to the business and used for the operating of business activities (everyday needs) is known as **" Revenue Receipts",** such as sale proceeds of goods, interest received, commission received, rent received, dividend received etc. In simple words, revenue receipts are those receipts which are actual profit of the business and not needs to repay. Receipts like, income received from sale of goods or sale of old newspapers or scrap which have no book value are revenue receipts. These are shown while preparing the trading and profit and loss account.

1.11.4. Differences between Capital Receipts and Revenue Receipts

Basis of Differences	Capital Receipts	Revenue Receipts
1) Effect	The effect of capital receipts are long term effect that provides long term benefits to the business.	The effect of revenue receipts are short term effect that have current benefits to the business.
2) Occurrence	The nature of occurrence of capital receipts is non-recurring and irregular.	The nature of occurrence of revenue receipts is recurring and regular.
3) Visibility	It is treated as liability so it's shown in the Balance Sheet on the liability side.	It is treated as incomes so it's shown in profit and loss account on the credit side.
4) Generation of receipt	Capital receipt helps in generation of revenue receipts by owner investment. **For example,** capital is invested by the owner, business enjoyed the benefits as revenue receipt like sale proceeds of goods, assets etc.	A revenue receipt does not help in generation of capital receipt.
5) Increase or decrease of the value of asset or liabilities	The capital receipts either raises the value of liabilities or reduces the value of an asset. **For example,** additional capital invested by owner that creates the liability of the business.	A revenue receipt does not affect (neither increases nor decreases) the value of liability or asset.
6) Example	Usually the capital receipts incurred the expenses of revenue nature. **For example,** additional capital invested by owner that incurred the revenue expanses that payment of interest on capital of business.	Usually the revenue receipts incurred the expenses of capital nature. **For example,** purchase of building for a company is capital expenditure but rent received on building is a revenue receipt.

1.12. EXERCISE

1.12.1. Very Short Answer Type Questions

1) What do you understand by business decisions?
2) What is sole proprietorship?
3) What do you mean by debtor and creditor?
4) What do you mean by the terms drawings and capital?
5) What are the branches of accounting?
6) What do you mean by matching concept?
7) What do you mean by accounting equation?
8) What do you mean by journal?
9) What do you mean by trial balance?
10) What is expenditure?
11) What is receipt?

1.12.2. Short Answer Type Questions

1) What is the difference between book-keeping and accounting?
2) What are the functions of accounting?
3) Who are the users of accounting information?
4) Is there any similarity between Management Accounting and Financial Accounting?
5) Explain any five accounting concepts with suitable example.
6) What do you mean by accounting equation?
7) What is the rules of debit and credit in double entry system?
8) What do you mean by personal account and real account?
9) What are the rules regarding posting?
10) Differentiate between journal and ledger.
11) Explain the method of preparing trial balance.

1.12.3. Long Answer Type Questions

1) Discuss the forms of business organizations in detail.
2) What do you mean by accounting? Also discuss the scope and objectives of accounting.
3) Discuss the advantages and disadvantages of accounting in detail.
4) Discuss the different users of accounting information.
5) "Management Accounting provides immense help in management decision-making." Elaborate.
6) "Financial Accounting and Management Accounting are not same." Explain.
7) Explain the basic accounting concepts and conventions in financial accounting.
8) What do you mean by capital and revenue expenditure? Give the difference between capital expenditure and revenue expenditure.
9) What do you mean by capital and revenue receipts? Give the difference between capital receipts and revenue receipts.

1.12.4. Practical Questions

Accounting Equation

1) Show the Accounting Equation on the basis of the following transactions.
   i) Started business with cash₹70,000.
   ii) Purchased goods on credit of ₹18,000.
   iii) Purchased machinery by paying ₹20,000.
   iv) Payment made to creditors in full settlement ₹17,500.
   v) Depreciation on machinery ₹2,000.

   [Ans: Assets = ₹68,500; Liabilities = ₹9,500; Capital = ₹68,500]

2) Aman had the following transactions. Use the Accounting Equation to show their effect on his assets, liabilities and capital:
   i) Invested ₹15,000 in cash as capital.
   ii) Purchased furniture for cash ₹7,500.
   iii) Purchased a building for ₹15,000, giving ₹5,000 in cash and the balance through a loan.
   iv) Sold furniture costing ₹1,000 for ₹1,500.
   v) Purchased on old car for ₹2,800 cash.
   vi) Received cash as rent ₹3,600.
   vii) Paid cash ₹500 for loan and ₹300 for interest.
   viii) Paid cash for household expenses ₹300.
   ix) Received cash for dividend on securities ₹200.

   [Ans: Assets = ₹28,200; Liabilities = ₹9,500; Capital = ₹18,700]

3) Bitoo had the following transactions:
   i) Commenced business with cash of ₹50,000.
   ii) Purchased goods for cash ₹20,000 and on credit ₹30,000.
   iii) Sold goods for cash ₹40,000 (costing ₹30,000).
   iv) Rent paid ₹500.
   v) Rent outstanding ₹100.
   vi) Bought furniture ₹5,000 on credit.
   vii) Bought refrigerator for personal use ₹5,000.
   viii) Purchased building for cash ₹20,000.

   Use Accounting Equation to show the effect of the above transactions on the assets, liabilities and capital.

   [Ans: Assets = ₹89,500; Liabilities = ₹35,100; Capital = ₹54,400]

Journal

4) Pass journal entries for the following transactions:

Date	Particulars
2018	
June 1	Started the business with the owner investing ₹2,00,000 cash.
June 6	Purchased office supplies for ₹15,000 by paying ₹10,000 cash and putting the remaining ₹5,000 balance on credit.
June 8	Purchased office furniture by paying ₹50,000 cash.
June 16	Billed a customer ₹20,000 for services earned.
June 20	Provided services for ₹5,000 cash.
June 24	Withdrawal of ₹1,500 cash for office use.

5) Pass journal entries in the books of Mr. Arun. for the following transactions:

Date	Particulars
2018	
Dec 1	Started business with cash ₹1,00,000 building ₹20,000 and goods ₹30,000.
Dec 2	Purchased furniture by cash ₹12,000.
Dec 3	Purchased machinery on credit from XYZ company ₹25,000.
Dec 4	Purchased goods worth ₹18,000 by cash and ₹22,000 on credit from Vishal enterprises.
Dec 8	Sold goods for cash ₹15,000 and on credit to Vikram ₹30,000.
Dec 16	Received an amount of ₹16,000 from Vikram.
Dec 20	Paid to Vishal enterprises cash of ₹10,000.
Dec 28	Paid to XYZ company cash of ₹8,000.

Ledger

6) Journalise the following transactions in the books of Mr. Rai. Post them in the ledger and balance the various accounts opened in the ledger:

Date	Particulars	₹
2019		
April 2	Mr. Rai Started his Business with Cash	70,000
April 3	Deposited into bank	50,000
April 4	Purchased Goods For Cash From Mr. Karan	5,000
April 5	Bought Goods on Credit from Mr.Yusuf	6,000
April 6	Returned Goods to Yusuf	1,000
April 10	Sold Goods for cash to Mr. Aman	6,000
April 15	Sold Goods to Mr. Bitoo	6,000
April 16	Mr. Bitoo Returned Goods	1,000

April 17	Drew from Bank for Personal use	5,000
April 25	Paid to Mr. Yusuf in Full Settlement by Cheque	4,800
April 26	Received a Cheque from Mr. Bitoo in Full Settlement	4,900
April 27	Drew Cash from Bank for Office use	10,000
April 30	Drew cash for personal purposes	5,000
April 30	Paid Salaries to Staff	5,000
April 30	Issued a Cheque for ₹3,000 in Favour of Sr. Devi a Landlady towards Rent for April	
April 30	Withdrew Goods for Private use	1,000

Trial Balance

7) A book-keeper has submitted the following Trial Balance for your observation and suggestions and to correct the Trial Balance to agree. You are asked to prepare the Trial Balance correctly.

Particulars	₹	Particulars	₹
Land	60,000	Drawings	4,000
Plant and Machinery	80,000	Insurance Premium	800
Office Furniture	4,000	Sundry Creditors	31,000
Office Computer	8,000	Sales	2,20,000
Debtors	36,000	Capital	80,000
Purchases	1,00,000	Carriage Outward	900
Loan on Mortgage	40,000	Discount Allowed	400
Cash in Hand	600	Stock (opening)	16,300
Cash at Bank	34,000	Rent and Rates	2,400
Bad Debts	1,700		
Purchase Returns	1,600		
Trade Expenses	900		
Wages	15,000		
Interest on Loan	1,600		
Salaries	3,000		
Factory Power	3,600		
Discount Received	600		

[**Ans:** Total of Trial Balance ₹3,73,200]

8) Prepare a Trial Balance from the following:

Particulars	₹	Particulars	₹
Purchases	1,40,000	Sales Returns	16,000
Opening Stock	1,20,000	Sales	2,00,000
Trade Expenses	16,000	Cash	20,000
Purchases Return	8,000	Outstanding Expenses	8,000
Bank Overdraft	1,00,000	Debtors	1,12,000
Creditors	24,000	Building	2,00,000
Carriage Inward	8,000	Carriage Outward	12,000
Discount Received	20,000	Loan from Ajay	60,000
Capital	2,24,000	Salaries	50,000
Investment	20,000	Interest Received	20,000
Bank Loan	50,000	Miscellaneous Receipts	1,000
Printing and Stationery	500	Cleaning Expenses	500

[**Ans:** Total of Trial Balance ₹7,15,000]

9) From the following balances extracted from the books of a trader, prepare Trial Balance as on 31st March, 2019.

Particulars	₹
Cash in Hand	4,200
Cash at Bank	16,800
Bills Receivable	18,000
Bills Payable	16,000
Sundry Debtors	24,600
Sundry Creditors	32,400
Capital	50,000
Drawings	18,000
Sales	1,05,000
Purchases	75,000
Carriage Inward	2,700
Salaries	12,000
Advertisement	2,400
Insurance	1,600
Furniture	7,500
Stock	18,600
Office Rent	2,000

[**Ans:** Total of Trial Balance ₹2,03,400]

UNIT 2

Financial Statements

2.1. FINANCIAL STATEMENTS

2.1.1. Meaning and Definition of Financial Statements

'Financial Statement' (also referred to as the 'Financial Report') is a formal and structured record of the financial activities of a business entity. It reveals the financial health of an entity with regard to the profits earned during a specific period (Accounting Period), and position of assets and liabilities as on a specific date (Last date of the Accounting Period). It also gives an indication in respect of future prospects of the entity. Thus, the process of classifying the financial information assists in proper and regular presentation of financial health of the business entity.

According to John N. Myer, "The financial statements provide a summary of the accounts of a business enterprise, the balance sheet reflecting the assets, liabilities and capital as on a certain date and the income statement showing the results of operations during a certain period."

As per this definition financial accounts are the end products of the accounting process.

According to Robert N. Anthony, "Financial statements, essentially, are interim reports, presented annually and reflect a division of the life of an enterprise into more or less arbitrary accounting period – more frequently a year."

According to this definition, financial statements are a form of reports which helps management in decision-making process. Such reports are prepared after a fixed time period to enable management to carry out its day-to-day functions.

2.1.2. Nature of Financial Statements

The nature of financial statements is reflected through the following traits:

1) **Recorded Facts:** Various transactions recorded in the books of accounts relating to 'Cash in Hand', 'Bills Receivable', 'Bills Payables', 'Debtors', 'Creditors', 'Fixed Assets', 'Sales', 'Purchases', 'Wages', 'Salaries', 'Capital', etc. are termed as the 'Recorded Facts'. They are also referred to as 'Historical Data', as their recording is done on historical basis. 'Financial Statements' of a business organisation contain only historical data. Facts and events other than historical data are not reflected in the 'Financial Statements', inspite of their importance in the affairs of the business organisation.

2) **Accounting Conventions:** 'Accounting Conventions' relate to those accounting principles that have been in practice for long, some of which may be very conservative in nature, e.g. the conventions applied for the 'Valuation of Inventory', 'Allocation of Expenses between Capital and Revenue', Valuation of Assets', etc. Due to application of such conservative 'Accounting Conventions', the 'Financial Statements' may not reflect the true picture of an organisation's affairs. **For example,** as per the accounting convention, a provision is required to be made for 'Anticipated Losses', whereas the 'Anticipated Profit' is not taken into consideration.

3) **Postulates:** 'Postulates' are the presumptions made by a book-keeper or accountant at the time of following 'Accounting Conventions'. Some of the postulates are discussed in the following paragraphs:
   i) **Going Concern of the Business:** This postulate depicts valuation of assets at their cost value less depreciation. In the absence of such presumption, the value of assets would have been different, viz. at realisable value.

   ii) **Stability in the Value of Money Over a Period of Time:** This assumption results in not valuing the assets on the basis of change in the value of money by some of the accountants.

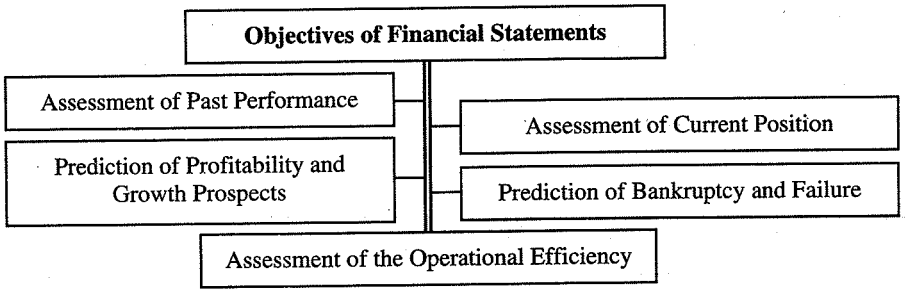
iii) **Realisation Postulate:** As per of this assumption, there exists a time-gap between the production of goods and their sale and the revenue is realised only after the sale of goods.

4) **Personal Opinion:** Personal opinion of an accountant is of paramount importance as it is the deciding factor at the time of the application of various conventions or postulates. Some of the examples, where the personal opinion of an accountant plays a significant role in such decisions related to the 'Method of Depreciating an Asset', 'Method of Valuation of Inventory', 'Making Provision for Doubtful Debts', 'Method of Amortisation of Fixed Assets, etc. Thus, the principle of consistency helps to judge the personal view of an accountant.

5) **Accounting Standards and Principles:** 'Accounting Standards and Principles' are of immense help while preparing the 'Financial Statements', as they ensure transparency and disclosure of vital information. Some of the Accounting Standards are mandatory for the business entity, e.g. Disclosure of Accounting Policies, Revenue Recognition, Accounting for Fixed Assets, etc.

2.1.3. Objectives of Financial Statements

Objectives of financial statements are enumerated in a nutshell as follows:



1) **Assessment of Past Performance:** Trend revealed through the various financial data, like 'Sales', 'Cost of Goods Sold', 'Operating Expenses', 'Net Income', 'Cashflow', etc. of an organisation's past performance may be a good pointer to its future performance. Prospective investors and current creditors rely on such past data to predict the future prospects of an organisation.

2) **Assessment of Current Position:** Through the exercise of financial statement analysis, the current status of an organisation with regard to the type of assets held by it and the nature of its liabilities, etc. can be ascertained.

3) **Prediction of Profitability and Growth Prospects:** The assessment and forecasting of the 'Earning Prospects' of a business organisation can be made through the exercise of 'Financial Statement Analysis'. It helps the prospective investors in comparing and choosing an investment out of the many investment opportunities available before them. It is also used by other stakeholders for various other purposes.

4) **Prediction of Bankruptcy and Failure:** 'Financial Statement Analysis' is a vital technique, through which the possibilities of an organisation going bankrupt in future or the chances of failure of a business can be predicted.

5) **Assessment of the Operational Efficiency:** 'Financial Statement Analysis' facilitates the assessment of the 'Operational Efficiency' of an organisation's management. It can be done through setting standards of performance on the basis of certain parameters and comparing them with the actual performance of the organisation. Any deviation between the 'Set Standards of Performance' and the 'Actual Performance' may be used as an indicator of 'Management Efficiency'.

2.1.4. Importance of Financial Statements

The importance of financial statements are as follows:

1) **Measurement of Short-Term Solvency:** An organisation's ability to pay its short term liabilities is referred to as its 'Short Term Solvency', assessment of which can be made through the 'Financial Statement Analysis'. If an organisation has a positive 'Short Term Solvency', its creditors would not hesitate in lending funds to it. On the other hand, if an organisation suffers from the lack of 'Short Term Solvency', the creditors would refrain from lending funds to it.

2) **Measurement of Long-Term Solvency:** An organisation's ability to repay its 'Long Term Liabilities', i.e. bonds and debentures issued by it and other secured liabilities, is referred to as its 'Long Term Solvency'. A company having such capability is said to be 'Solvent'. Thus, status of an organisation in terms of long term solvency is revealed through the analysis of financial statements.

3) **Measurement of Operating Efficiency:** Through the analysis of 'Financial Statements' of an organisation, the position with regard to its profitability can easily be ascertained. If a business organisation is incurring losses, the reasons thereof may also be found out through such analysis, and necessary corrective actions may be taken to curb losses and turn the loss incurring organisation into a profit earning organisation.

4) **Measurement of Profitability:** The existing earning capacity of an organisation as well as forecasting in respect of its future earnings can be gathered through the earning ratio analysis, inputs for which are provided by the 'Financial Statements'. 'Measurement of Profitability' is an important indicator, which helps in taking decision by the 'Investors' and 'Lenders' (Banks/FIs).

5) **Comparison of Inter-Firm Position:** The comparison of the financial position of one organisation with that of another organisation is an exercise undertaken through the 'Financial Analysis'. Such a comparative study is useful in deciding the course of action to be taken by 'Investors' and other 'Stakeholders'.

6) **Forecasting, Budgeting and Deciding Future Course of Action:** Through the 'Financial Statement Analysis', vital indicators of an organisation's financial health, e.g. 'Capital Base', 'Reserves and Provisions', 'Profitability', 'Working Capital Gap', 'Trend in Sales', etc. can be found out. Such indicators make a solid ground, on the basis of which the exercise of 'Forecasting' and 'Budgeting' may be undertaken. Future course of action to be taken for the betterment of the organisation may also be decided on this basis.

2.1.5. Limitations of Financial Statements

Although a lot of vital information may be gathered from the financial statements of a business organisation, it is characterised by certain inherent limitations, which are as follows:

1) **Interim Reports:** 'Financial Statements' of an organisation are basically considered 'Interim' in nature. The 'Profitability' during an 'Accounting Period' and 'Financial Position' as on a particular date, when compared to another period, can indicate the status at certain frequency during the lifetime of a business. However, the final outcome of conducting a business (in the form of strength, weaknesses, future prospects, etc.) is possible with accuracy only at the time of winding-up of the business.

2) **Discloses Only Quantitative Information:** The data revealed through the 'Financial Statements' is quantitative in nature, which can be measured in terms of money. Many other factors of a business, which are qualitative in nature, cannot be expressed in terms of money, and as such remain out of the purview of the 'Financial Statements'.

3) **Based on Historical Data:** Preparation of financial statements is done on the basis of past data, i.e. events that have already taken place. It does not give a foresight of the future events likely to take place and their impact on the profitability, financial health, etc. of a business organisation. Therefore, the financial statements are not of much help as far as the decision-making process is concerned.

4) **Based on Accounting Concepts and Conventions:** Preparation of financial statements is carried out on the basis of certain underlying presumptions as stated under 'Accounting Concepts/Conventions'. Some of such 'Concepts/Conventions' may be very conservative and unrealistic, rendering financial statements also unrealistic.

5) **Personal Opinion:** Personal opinion of an accountant is capable of influencing certain decisions taken during the preparation of financial statements. Such decisions generally relate to 'Choice of Depreciation Method', 'Mode of Amortisation of Fixed Assets', 'Treatment of Deferred Revenue Expenditures', 'Inventory Valuation', etc. At times, the personal opinion of an accountant may reflect a distorted picture of an organisation's performance.

6) **Window Dressing:** Resorting to the phenomena of 'Window Dressing', with a view to presenting a rosy picture of an organisation's affairs, is a common practice adopted by the management. Such an exercise creates a better, but false, image of the organisation's financial health, capabilities, etc. and the prospective investors and other stakeholders are misled.

2.2. FINAL ACCOUNTS OF SOLE PROPRIETARY FIRM

2.2.1. Introduction

'Final Accounts' are prepared with the objective of providing a precise summary of the profits made during a particular year and the financial position of the business as on a particular date.

'Final Accounts', also known as 'Financial Statements', are prepared in conformity with the provisions of the various statutory and regulatory frameworks, some of which are 'Income Tax Act', 'Companies Act', 'Partnership Act', 'Accounting Standards' (AS), 'Generally Accepted Accounting Principles' (GAAP), etc. Provisions of such acts are used for the valuation of various financial transactions with a view to standardise accounting information, which differs from one business organisation to another, depending upon their type.

As different business organisations are created and operated under different legal framework, the process of preparing their final accounts also differs in a marked manner.

2.2.2. Preparation of Final Accounts

In the case of a 'Trading Concern', final accounts are prepared in three distinct stages, which are as under:

1) **Trading Account:** Trading account is prepared to ascertain the gross profit or trading profit.
2) **Profit and Loss Account:** Profit and loss account reveals the net profit of the business organisation.
3) **Balance Sheet:** Balance sheet is a statement of assets and liabilities (financial position) of the organisation as on a specific date, normally the last date of the accounting period.

2.3. TRADING ACCOUNT

2.3.1. Meaning of Trading Account

The trading account of an organisation reveals its gross profit or loss. It reflects the result of buying and selling of goods and services, in which the business organisation deals during a trading period. While the trading account includes direct expenses associated with the purchases and sales of goods and services, the operating expenses of the business organisation are excluded. At the end of an accounting period, every business organisation is interested in knowing the result of its performance; whether the organisation has earned gross profit or incurred gross loss. This result (gross profit or loss) is discovered through the trading account.

'Gross Profit' is nothing but the excess of sales and closing stock over the opening stock, purchases and direct expenses, whereas 'Gross Loss' is the excess of opening stock, purchases and direct expenses over sales and closing stock.

2.3.2. Proforma of Trading Account

Trading Account
(for the period ending ----------)

Dr.			**Cr.**
Particulars	₹	**Particulars**	₹
To Opening Stock, or Stock in the Beginning, or		By Sales	
To Stock:		**Less:** Sales Return or	
Raw Material.....		Returns Inward	
Work in Progress....		By Closing Stock or	
Finished Goods		Stock at the End of the Year	
To Purchases		By Stock:	
Less: Purchases Return or Returns Outward		Raw Material	
To Carriage or Cartage or Carriage on Purchases		Work in Progress....	
To Octroi or Local Taxes		Finished Goods	
To Import Duty, Customs, Landing Charges, Clearing Charges		By Gross Loss – Transferred to	
To Wages or Productive Wage or Manufacturing		Profit & Loss A/c	
Wages or Direct Wages or Wages and Salaries		**(If debit side exceeds the credit side)**	

To Coal, Gas and Water To Fuel To Power or Motive Power To Manufacturing Expenses To Consumable Stores To Packing Charges To Manufacturing Expenses To Direct Expenses To Factory Expenses To Productive Expenses To Royalty To Gross Profit Transferred to Profit & Loss A/c **(If credit side exceeds the debit side)**			
	xxx		**xxx**

2.3.3. Items Appearing in Trading Account

Various items, which find a place on the 'Debit' side and those which find a place on the 'Credit' side of the trading account are discussed in the following paragraphs:

2.3.3.1. Items Appearing on Debit Side of the Trading Account

On the 'Debit' side of the trading account, following items find a place:

1) **Purchases:** Goods purchased by a business organisation for the purpose of resale are shown on the 'Debit' side of the trading account as 'Purchases'. Goods bought on cash payment as well as credit basis are included under the head 'Purchases'. Goods taken out for the personal use of the proprietors of the organisation are debited to their drawing account and reduced from the purchases.

2) **Purchases Returns:** A part of the goods purchased for the purpose of resale by an organisation, sometimes become surplus (and also due to certain other reasons), which is returned to the concerned suppliers. Such goods are referred to as 'Purchase Returns' or 'Returns Outwards' and shown as deduction from the purchases. Returns outwards account is always having credit balance.

3) **Opening Stock:** The amount shown as the opening stock of the current year's trading account is in fact the closing stock of the previous year's balance, which has been brought forward to the current year. Opening stock appears on the debit side of the trading account.

4) **Wages:** Wages are the monetary compensation paid by a business organisation to its workers for the work done by them in connection with the production of goods. If the manufacturing account is prepared by the business organisation, they are debited to that account. Otherwise, the trading account is debited.

5) **Carriage Inwards/Freight-In:** Carriage inwards, also termed as 'Freight-in' is the shipping and handling expenses incurred by a company, while receiving goods from its suppliers. These expenses are shown on the debit side of the trading account.

6) **Customs and Import Duty:** Customs and import duty are the statutory expenses incurred in connection with the import of goods from overseas markets. They appear on the debit side of the trading account.

7) **Royalty:** Royalty is defined as the payment made to the owner for the use of property, patents, copyrighted works, franchises or natural resources (like mines, tea-gardens, etc.). It is paid by those who make use of the assets for the commercial purposes. The mechanism of royalty payment has been evolved as a means to compensate the owner for the asset's use and has a legal backing. It is considered as a part of cost of production and as such debited to trading account.

8) **Gas, Electricity, Water and Fuel, etc.:** The expenses pertaining to gas, electricity, water and fuel, etc. are incurred in connection with the day-to-day operations (production of goods) of an organisation. They are considered as direct expenses and debited to the trading account.

9) **Packing Material:** Most goods manufactured in a factory need to be packed properly for their safe transportation to the customers place. The expenses incurred in this regard are considered as part of finished products and are debited to the trading account.

2.3.3.2. Items Appearing on Credit Side of the Trading Account

Credit side of the trading account contains the items listed below:

1) **Sales:** Sales of the goods, an organisation deals with (whether cash or credit), is shown on the credit side of the trading account. Sales of assets/items, the organisation does not deal with, are not included under sales.

2) **Sales Returns:** 'Sales Returns' or the 'Returns Inwards' include the goods supplied earlier to the customers and subsequently returned by them. The amount of the goods sold returned is deducted from the sales figure. Sales returned account has a debit balance.

3) **Closing Stock:** The stock of goods remained unsold at the year end is termed as 'Closing Stock'. The amount of such stock appears on the credit side of the trading account. Closing stock is not a part of ledger and is out of purview of the trial balance. Generally, the figure of the closing stock is given separately.

Example 1: Prepare the Trading Account from the following information for the year ending on 31st March, 2019:

Particulars	₹	Particulars	₹
Opening Stock	1,50,000	Freight Inward	3,000
Cash Sales	60,000	Cartage Inward	1,000
Credit Sales	12,00,000	Cash Purchases	50,000
Returns Outward	10,000	Credit Purchases	10,00,000
Wages and Salaries	4,000	Returns Inward	20,000
Carriage Inward	2,000		

Closing Stock as on 31.3.2019, ₹90,000 but its market value is ₹84,000.

Solution:

Trading Account

Dr. (for the year ending 31st March, 2019) **Cr.**

Particulars		₹	Particulars		₹
To Opening Stock		1,50,000	By Sales		
To Purchases			Cash Sales	60,000	
Cash Purchases	50,000		**Add:** Credit Sales	12,00,000	
Add: Credit Purchases	10,00,000		Total Sales	12,60,000	
Total Purchases	10,50,000		**Less:** Return Inward	20,000	12,40,000
Less: Return Outward	10,000	10,40,000	By Closing Stock		84,000
To Freight Inward		3,000			
To Wages and Salaries		4,000			
To Carriage Inward		2,000			
To Cartage Inward		1,000			
To Gross Profit		1,24,000			
		13,24,000			**13,24,000**

2.4. PROFIT AND LOSS ACCOUNT

2.4.1. Meaning of Profit and Loss Account

'Profit and Loss Account', also referred to as the **'Statement of Profit and Loss'**, is a financial statement, which summarises the revenues generated as well as costs and expenses made by a business organisation during an accounting period (generally quarterly or yearly). It highlights the net profit earned or net loss incurred as a result of the commercial activities undertaken by it, with the details of all the items of revenue gains, losses and operating expenses relating to that specific accounting period. After preparation of trading account the ledger balances of the remaining nominal accounts are required to be transferred to profit and loss account. Such transfers are affected strictly in accordance with the principle of double entry system, i.e.:

1) All expenses and losses (e.g., office and administration, selling and distribution, non-operating expenses, abnormal losses etc.) find a place on the debit side of the profit and loss account, and
2) All incomes and gains (e.g., operating, non-operating, abnormal gains, etc.) find a place on the credit side of the profit and loss account.

When during an accounting period, the revenues generated by a business organisation are in excess of expenses incurred by it, such excess amount is known as 'Profit', 'Net Income', or 'Net Earnings' of that organisation. On the other hand, if the expenses incurred exceed revenues generated, such excess amount is known as 'Loss' incurred by that organisation.

2.4.2. Proforma of Profit and Loss Account

Profit and Loss Account

Dr. (for the year ending .....) **Cr.**

Particulars	₹	Particulars	₹
To Gross Loss (Transferred from Trading Account)		By Gross Profit (Transferred from Trading Account)	
To Salaries or Salaries and Wages		By Interest Received, or	
To Rent, Rates and Taxes or Office Rent		Interest (Credit), or	
To Godown Rent or Storage, or Warehousing		Interest on Investment, or	
To Office Expenses, or Establishment		Interest on Fixed Deposits, or	
To Miscellaneous, or Sundry Expenses		Interest on Loans Advanced, or	
To Insurance		By Rent Received, or	
To Stationery		Rent (Credit)	
To Printing and Stationery		By Discount Received, or	
To Postage and Telegrams		Discount (Credit)	
To Telephone Expenses		By Commission Received	
To Law Charges, or Law Cost or Litigation Expenses, or Legal Charges		By Dividends Received	
To Repairs and Renewal, or Maintenance or Upkeep		By Profit from Sale of Assets	
To Distribution Expenses		By Refund of Tax	
To Traveling Expenses		By Compensation Received	
To General Expenses		By Income from Investment	
To Stable Expenses		By Apprenticeship Premium	
To Selling Expenses		By Difference in Exchange (Credit)	
To Carriage, or Freight Outward		By Interest on Drawings	
To Carriage on Sales		By Discount on Creditors	
To Indirect or Unproductive Wages		By Bad debts Recovered	
To Audit Fee		By Miscellaneous Receipts	
To Indirect Paid, or Interest (Debit) or Interest on Overdraft, or Interest on Loans Borrowed		By Appreciation or Increase in the Value of Assets	
To Discount Allowed, or Discount on Debtors		By Reserve for Bad and Doubtful Debts (Old reserve–if not treated at the debit side of P/L Account)	
To Bad Debts, or Bad Debts Written Off		By Net Loss–Transferred to Capital Account	
To Depreciation		**(If debit side exceeds credit side)**	
To Interest on Capital			
To Discounting Charges			
To Bank Charges, or Collection Charges			
To Export Charges			
To Administrative Expenses			
To Financial Expenses			
To Commission Paid			
To Advertisement			
To Charity and Donation			
To Sample Expenses			
To License Fee			
To Delivery Charges			
To Brokerage			
To Sales Tax Paid			
To Loss on Sale of Assets			
To Loss by Fire			
To Freight Outward			
To Commission			
To Net Profit-Transferred to Capital A/c			
(If credit side exceeds debit side)			
	XXX		XXX

2.4.3. Items Appearing in Profit and Loss Account

Items finding a place in the profit and loss account are listed in the following paragraphs:

1) **Salaries:** Salary payments to the employees of the business organisation are an item of expenditure and are debited to the 'Profit and Loss Account'. It is pertinent to note the difference between the items 'Salaries and Wages' and 'Wages and Salaries'. While the former item needs to be treated as 'Salaries' and as such debited to the 'Profit and Loss Account', the latter item needs to be treated as 'Wages' and as such debited to the 'Trading Account'.

2) **Rent:** The amount of rent may pertain to either 'Rent Paid' or 'Rent Received'. The former is an expenditure item, whereas the latter is an income item. Accordingly, the 'Rent Paid' is shown on the debit side of the 'Profit and Loss Account' and 'Rent Received' is shown on the credit side of the 'Profit and Loss Account'. Further, it is also noteworthy that the 'Rent' of the office, shop or showroom is an indirect expense and needs to be taken into 'Profit and Loss Account', whereas the 'Rent' of the factory needs to be taken to the 'Trading Account'.

3) **Discount:** Discount allowed by a business organisation may be put under either of the two categories:
   i) **Trade Discount:** 'Trade Discount' is availed of/allowed on bulk purchases and bulk sales. The amount of 'Trade Discount' is deducted from the amount of purchases/sales and the net amount is recorded in the books of accounts. As a result, it does not form part of the 'Trial Balance'.
   ii) **Cash Discount:** 'Cash Discount' is allowed by the suppliers/distributors with a view to encourage timely/prompt payments. When 'Cash Discount' is availed from the creditors/suppliers, it is a gain/income for the organisation and therefore, reflected on the credit side of its 'Profit and Loss Account'.

   Similarly, when the 'Cash Discount' is allowed to its customers, it is considered as an expense for the organisation and therefore, reflected on the debit side of its 'Profit and Loss Account'.

4) **Bad Debts:** A debt, the recovery of which is not possible, is considered as a 'Bad Debt' by the business organisation. This ultimately may prove to be a loss for the business and is shown on the debit side of the 'Profit and Loss Account'.

5) **Carriage Outwards or Freight Outwards:** 'Transportation Charges' paid on the goods sold by a business organisation are known as 'Carriage Outwards' or 'Freight Outwards'. It is an item of expenditure, and as such debited to 'Profit and Loss Account'.

6) **Drawings:** An amount in the form of 'Cash' or 'Goods' withdrawn/taken out from the business by the proprietor of the firm for personal use is referred to as the 'Drawings'. This item is debited to the 'Capital Account' of the proprietor, as it relates to the balance sheet.

7) **Income-Tax:** The 'Income Tax' paid by a business organisation gets different treatment on the basis of the constitution of the organisation as under:
   i) In the case of a 'Sole Trading'/'Proprietorship' firm, the income tax paid is required to be deducted from the 'Capital' ('Profit and Loss Account' need not be debited).
   ii) In the case of a 'Joint Stock Company', 'Income Tax' is debited to the 'Profit and Loss Account' as a 'Provision'.

8) **Loss by Fire, Theft, etc.:** These types of losses are considered as 'Abnormal Losses' and are charged (debited) to the 'Profit & Loss Account'.

9) **Loss or Gain on Sale of Fixed Assets:** Sale of its 'Fixed Assets' (like land and building, plant and machinery, furniture, etc.) by a business organisation is generally not undertaken in the normal course as part of the routine business transaction. However, when they do take place, the organisation may either incur losses or make a profit.

   They are treated as under:
   i) Losses on account of sale of a 'Fixed Asset' is considered as 'Revenue Loss' and therefore, needs to be debited to 'Profit and Loss Account'; and
   ii) Profit/Gain on account of sale of a 'Fixed Asset' being a 'Revenue Gain' needs to be credited to 'Profit and Loss Account'.

Example 2: Prepare Profit and Loss Account, from the following balances of Mr. Karan for the year ending 31.03.2019.

Particulars	₹	Particulars	₹
Office Rent	30,000	Salaries	80,000
Printing Expenses	2,000	Stationery	3,000
Tax & Insurance	4,000	Discount Allowed	6,000
Advertisement	36,000	Travelling Expenses	26,000
Gross Profit	2,50,000	Discount Received	4,000

Solution:

Profit and Loss Account of Mr. Karan
(for the year ending 31st March 2019)

Dr. Particulars	₹	Particulars	Cr. ₹
To Salaries	80,000	By Gross Profit	2,50,000
To Office Rent	30,000	By Discount Received	4,000
To Stationery	3,000		
To Printing Expenses	2,000		
To Tax & Insurance	4,000		
To Discount Allowed	6,000		
To Travelling Expenses	26,000		
To Advertisement	36,000		
To Net Profit (Transferred to Capital A/c)	67,000		
	2,54,000		**2,54,000**

2.5. BALANCE SHEET

2.5.1. Meaning and Definition of Balance Sheet

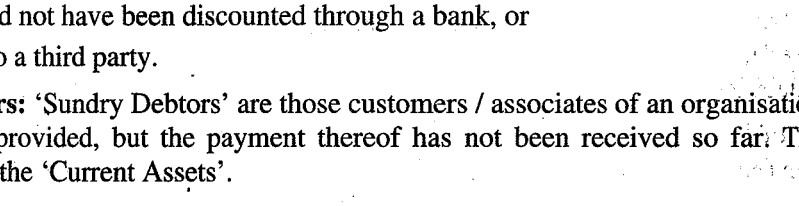
'Balance Sheet' of an organisation is a statement depicting what is owned by it (assets) and what it owes (liabilities) to others as on a specific date (generally the last date of an accounting period). It is the last stage of the accounting cycle (other stages being 'Trading Account', Profit and Loss Account', and 'Profit and Loss Appropriation Account') and is prepared at the end of the accounting period after the other parts of the final accounts are ready.

According to American Institute of Certified Public Accountants, "Balance Sheet is a list of balances in the assets and liability accounts. This list depicts the position of assets and liabilities of a specific business at a specific point of time".

According to Freeman, "A 'Balance Sheet' is an item-wise list of assets, liabilities and proprietorship of a business at a certain date".

2.5.2. Items of Balance Sheet

Following items generally find a place (either as an asset or a liability) in a balance sheet:



2.5.2.1. Items to be Shown on the Assets Side of a Balance Sheet

Asset side of a balance sheet generally contain the following items:

1) **Current Assets:** Current assets are defined as assets, which can be sold, consumed, or exhausted through the normal operations of a business within the current fiscal year (twelve months). Current assets include the following items:
   i) **Cash in Hand/at Bank:** Money held as cash either at the business premises or kept in a bank account maintained by the business organisation.
   ii) **Bills Receivable:** 'Bills of Exchange' drawn by an organisation and accepted by its customers are termed as 'Bills Receivable'. If they are yet to be matured for payment, they are treated as part of 'Current Assets' provided:
      a) They should not have been discounted through a bank, or
      b) Endorsed to a third party.
   iii) **Sundry Debtors:** 'Sundry Debtors' are those customers / associates of an organisation, to whom goods/ services were provided, but the payment thereof has not been received so far. They are one of the components of the 'Current Assets'.

iv) **Prepaid Expenses:** They are the business expenses, which have been paid in advance, and services in respect thereof are to be received in the future. They appear as part of 'Current Assets' on the right-side of the 'Balance Sheet'. 'Insurance Paid in Advance', and 'Rent Paid in Advance' are some of the examples of 'Prepaid Expenses'.

v) **Accrued Income:** Income which has been earned but yet to be received due to some reason or the other is referred to as 'Accrued Income' and considered as part of 'Current Assets'.

2) **Fixed Assets:** 'Fixed Assets' may be defined as assets, which are not consumed or sold during the normal course of business. They are not purchased for the purpose of reselling as part of the routine business. They are purchased and held as they are instrumental in the process of production of goods and services relating to the core business of an organisation. In accounting, 'Fixed' does not necessarily mean immovable; any asset expected to last, or be in use for, more than one year is considered a 'Fixed Asset'. 'Fixed Assets' may be classified into two categories:

i) **Tangible Fixed Assets:** Tangible fixed assets are those assets which can be perceived through the sense organs like eyes and skin. In other words, they can be seen and touched as they have a physical presence. **Examples** of tangible fixed assets are land and building, plant and machinery, fixture and furniture, motor vehicles, etc.

ii) **Intangible Fixed Assets:** Intangible fixed assets are those assets which cannot be perceived through the sense organs, as they do not have a physical presence. Some examples of 'Intangible FAs' are 'Goodwill', 'Trademarks', 'Patent', etc. which can neither be seen nor touched.

Valuation of the 'Fixed Assets' is carried out at the 'Cost Price' after providing for the 'depreciation' in accordance with the 'Accounting Policies' of the organisation.

3) **Investments:** Surplus funds available in the business at any given point of time are invested in appropriate financial instruments, which may be 'Government Dated Securities', 'Bonds/Debentures /Shares issued by Corporate, PSUs, or other bodies', etc. They are generally shown at 'Cost Price' in the 'Balance Sheet' of the organisation. However, if the 'Market Price' of the instrument is lower than the 'Acquisition/Cost Price' as on the date of 'Balance Sheet' preparation, appropriate provision needs to be made in the books of accounts and a 'Footnote' in this regard needs to be appended as 'Qualifications' to the 'Balance Sheet'.

2.5.2.2. Items to be Shown on the Liabilities Side of a Balance Sheet

Items appearing as 'Liabilities' on the 'Balance Sheet' of an organisation are listed below:

1) **Current Liabilities:** Current liabilities are defined as the liabilities, which are:
   i) Required to be paid / liquidated within a period of one year; and
   ii) Generally paid out of the 'Current Assets'.

   Following items are part of current liabilities:
   i) **Bank Overdraft:** Withdrawal of funds from a bank account maintained by an organisation beyond the balances held therein is termed as 'Overdrawing'. This is a short-term accommodation provided by a bank, referred to as 'Overdraft Facility', and needs to be liquidated at the shortest possible time.
   ii) **Outstanding Expenses:** The amount of expenses remaining unpaid, despite their having become due for payment, is known as 'Outstanding Expenses'. They are shown as 'Current Liabilities' and need to be paid at the earliest.
   iii) **Bills Payable:** 'Bills of Exchange' drawn by the creditors in their favour and accepted by an organisation are termed as 'Bills Payable' for that organisation. The amount lying in this account generally reflects the bills of exchange, which have not become due for payment.
   iv) **Sundry Creditors:** They may be the suppliers of 'Raw Materials' or providers of other goods and services. Under this 'Head of Account' the total of all balances lying in the accounts of various creditors is taken.

2) **Long-Term Liability:** 'Long-term Liabilities' are generally represented by those liabilities, which are:
   i) Not likely to become due for payment within a short period of time (within 12 months from the date of 'Balance Sheet');
   ii) Not current liabilities.

   Some **examples** of 'Long-term Liabilities' are debentures/bonds issued and term-loan, availed from a lending institution (banks/FIs).

3) **Capital:** It is the excess of total assets over total outside liabilities of an organisation. It is the amount of money contributed by the proprietors/partners/shareholders for starting the business. The profit earned during a year is added to, while the losses incurred/amount of drawings are reduced from this account.

4) **Drawings:** Any withdrawal made by the proprietor of a firm by way of 'Cash' or 'Kind' (goods) for his/her personal use is termed as 'Drawings'. It impacts the balance of the 'Capital Account', which gets reduced to the extent of the drawings. It is shown by way of deduction from 'Capital Account' in the balance sheet.

2.5.3. Proforma of Balance Sheet

The balance sheet is a list of all the resources owned by an organisation and shows that it is equal to the sum of all liabilities and the equity balance. There are two formats in which a balance sheet may be depicted, viz.

1) 'Horizontal Form' or 'Account Form', which is like 'T' - Account, listing all the assets on the right hand side and equity and liabilities on the left hand side, and
2) 'Vertical Form' or 'Report Form', which lists assets followed by liabilities and equity in vertical format.

2.5.3.1. Horizontal Form of Balance Sheet

This type of 'Balance Sheet 'may be better explained through the 'Grouping and Marshalling' of 'Balance Sheet'.

Grouping of Assets and Liabilities

Grouping of assets and liabilities involve the presentation/arrangement of various items, shown in a balance sheet, in proper groups. Items of similar nature are placed in the same group. **For example,** the balances lying in the accounts of all the customers to whom the business entity has supplied goods on credit are grouped under the heading 'Sundry Debtors'.

Marshalling of Assets and Liabilities

'Marshalling' of assets and liabilities involve the presentation / arrangement of various items, shown in a balance sheet, in a specific order. There are two orders followed in marshalling of assets and liabilities, which are explained in the following points:

1) **Assets can be Shown in Order of Liquidity**
   i) Assets are arranged according to the order of their liquidity. The most liquid asset (cash-in-hand) is placed first, whereas the least liquid asset (goodwill) is placed last;
   ii) Liabilities are arranged according to the order of their urgency of payment. The most urgent payment to be made (short-term creditors), is placed first, whereas the least urgent payment to be made (term loan) is placed last;
   iii) Generally, the 'Balance Sheets' of the 'Sole Proprietorship Firms', 'Partnership Concerns' 'Banking Companies' and 'Finance Companies' are prepared in the order of the liquidity.

Proforma of Order of Liquidity

A general proforma of a balance sheet in **order of liquidity** is shown below:

Balance Sheet of ....
(as on .....)

Liabilities	₹	Assets	₹
Current Liabilities		**Current Assets**	
Bank Overdraft		Cash in Hand	
Bills Payable		Cash at Bank	
Outstanding Expenses		Bill Receivable	
Sundry Creditors		Sundry Debtors	
Income Received in Advance		Prepaid Expenses	
		Accrued Income	
Long-term Liabilities		Closing Stock	
Loan		**Investment**	
		Fixed Assets	
Capital		Furniture and Fixtures	
Opening balance		Plant and Machinery	
Add: Net Profit		Building	
(**Less:** Net Loss)		Land	
Less: Drawings		Goodwill	
	XXX		**XXX**

2) **Assets and Liabilities can be Shown in the Order of Permanence**
   i) This order is just opposite to the 'Order of Liquidity' mentioned above. In this order, the assets find their place in the order of their permanence. The least liquid asset (goodwill) is placed first and the most liquid asset (Cash-in-hand) is placed last.
   ii) As regards the liabilities, the least urgent payment to be made (owners) finds its place first and the most urgent payment to be made (short-term creditors) finds its place in the last.
   iii) A company as defined under the Companies Act, 1956 needs to prepare its balance sheet according to the 'Order of Permanence'.

Proforma of Order of Permanence

A general proforma of a balance sheet in the **order of permanence** is shown below:

Balance Sheet of ....
(as on ....)

Liabilities	₹	Assets	₹
Capital		**Fixed Assets**	
Opening Balance		Goodwill	
Add: Net Profit		Land	
(**Less:** Net Loss))		Building	
Less: Drawings		Plant and Machinery	
		Furniture and Fixtures	
Long-term Liabilities		**Investment**	
Loan		**Current Assets**	
Current Liabilities		Closing Stock	
Income Received in Advance		Accrued Income	
Sundry Creditors		Prepaid Expenses	
Outstanding Expenses		Sundry Debtors	
Bills Payable		Bill Receivable	
Bank Overdraft		Cash at Bank	
		Cash in Hand	
	xxx		xxx

2.5.3.2. Vertical Form of Balance Sheet

The 'Vertical Form' or 'Report Form' of balance sheet depicts presentation of balance sheet in a statement form, which is more convenient and advantageous when compared with the 'Horizontal Form' or 'Accounts Form'. Most of the joint stock companies use this form in the preparation of their balance sheet.

'Vertical Form' of 'Balance Sheet' is advantageous in the following manner:

i) This is a simple form of 'Balance Sheet', which is easy to interpret even by an ordinary layman.
ii) Each items of 'Balance Sheet' are explained with clarity, e.g. 'Total Fixed Assets', 'Total Current Assets', 'Working Capital', etc. 'Sources of Funds' and 'Deployment of Funds' can also be understood through its help.
iii) The exercise of 'Ratio Analysis' and undertaking subsequent comparative study is also smooth under the 'Vertical Form' of 'Balance Sheet'.

Proforma Vertical Form of Balance Sheet

Balance Sheet
(as on.....)

Particulars	₹	₹
Current Assets:		
Stock-in-Trade	xxx	
Sundry Debtors	xxx	
Prepaid Expenses	xxx	
Accrued Income	xxx	
Bills Receivable	xxx	
Cash at Bank	xxx	
Cash-in-Hand	xxx	
Total Current Assets		xxx
Less: Current Liabilities:		
Sundry Creditors	xxx	
Bills Payable	xxx	
Bank Overdraft	xxx	

Outstanding Expenses	xxx	
Total Current Liabilities		xxx
Net Working Capital		xxx
Add: Fixed Assets:		
Goodwill	xxx	
Land and Building	xxx	
Plant and Machinery	xxx	
Furniture	xxx	
Investment	xxx	
Total Fixed Assets		xxx
Capital Employed		xxx
(Both owner's and outsiders)		
Less: Long-Term Liabilities		
Debentures	xxx	
Loans	xxx	
Total Long-Term Liabilities		xxx
Net Assets		xxx
Represented by:		
Owner's Capital	xxx	
Reserves and Surplus	xxx	
Shareholders Funds		xxx

Example 3: From the following balances taken from the books of Senthil & Co. prepare trading and profit & loss account for the year ending on 31st December 2018 and Balance sheet on that date:

Particulars	₹
Capital	35,000
Building	18,750
Machinery	9,250
Debtors	7,000
Salaries	1,110
Discount Allowed	200
Stock (1.1.18)	16,500
Bills Payable	5,000
General Expenses	800
Rent Paid	3,710
Proprietor's Drawings	650
Electric Charges	190
Carriage Inwards	850
Cash at Bank	3,000
Returns Outward	110
Sales	63,500
Purchases	46,850
Wages	2,500
Cash in Hand	1,800
Sundry Creditors	10,000
Return Inward	450
Closing Stock	18,210

Solution:

Trading and Profit and Loss Account

Dr. (for the year ending on 31st Dec. 2018) **Cr.**

Particulars		₹	Particulars		₹
To Opening Stock		16,500	By Sales	63,500	
To Purchases	46,850		**Less:** Return Inward	450	63,050
Less: Return outward	110	46,740	By Closing Stock		18,210
To Wages		2,500			
To Carriage Inwards		850			
To Gross Profit c/d		14,670			
		81,260			**81,260**
To Salaries		1,110	By Gross profit b/d		14,670
To General Expenses		800			
To Rent Paid		3,710			
To Electric Charges		190			
To Discount allowed		200			
To Net Profit (transferred to Capital Account)		8,660			
		14,670			**14,670**

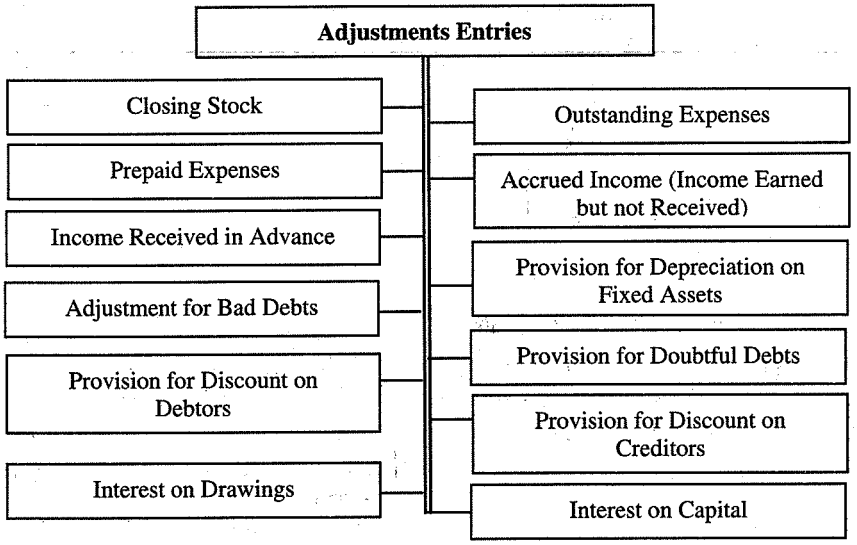
Balance Sheet
(as on 31st Dec. 2018)

Liabilities		₹	Assets	₹
Capital	35,000		Cash in Hand	1,800
Less: Drawing	650		Cash at Bank	3,000
	34,350		Debtors	7,000
Add: Net Profit	8,660	43,010	Machinery	9,250
Bills Payable		5,000	Building	18,750
Sundry creditors		10,000	Closing Stock	18,210
		58,010		**58,010**

2.6. ADJUSTMENTS ENTRIES

With a view to complying with the accrual concept of accounting, 'Income' and 'Expense' accounts need to be adjusted, for which certain journal entries are passed at the end of an accounting period. Such journal entries are known as **'Adjustment Entries'**. The main objective for passing such 'Adjustment Entries' is to ensure that incomes and expenses are charged to the appropriate/correct accounting periods. It is important to keep in mind that an 'Adjusting Entry' invariably involve either 'Income' or 'Expense' account. It is also noteworthy that the underlying transactions, which are recorded using adjusting entries, are spread over a wide range of time period.

The items, in respect of which adjustments are generally required to be made at the end of an accounting period for the preparation of the 'Trading & Profit and Loss Account', (with a view to ensuring compliance with the 'Accrual Basis of Accounting') are listed below:



1) **Closing Stock:** At the closing of an accounting period, the accumulation of unsold material, which consists of:
   i) Raw Material,
   ii) Semi-finished Goods, and
   iii) Finished Goods, are referred to as the 'Closing Stock'. It is valued at the 'Cost Price' or 'Market Price', whichever is lower.

   The 'Closing Stock' appearing in the 'Balance Sheet' of an organisation at the end of an 'Accounting Period', is required to be carried forward to the next 'Accounting Period', wherein it appears as the 'Opening Stock' in the 'Trial Balance' of that 'Accounting Period', from where it is transferred to the 'Trading Account'. The 'Trading Account' is debited and the 'Opening Stock' (last year's closing stock) is credited. The 'Stock Account' is closed in this manner.

   i) In case the value of the 'Closing Stock' is given in the 'Trial Balance', it will find a place only in the 'Balance Sheet', because it has already been taken into account at the time of calculating the cost of goods sold,

   ii) In case the value of the 'Closing Stock' is not shown in the 'Trial Balance' (it is given in adjustment), then it will find a place on the credit side of the 'Trading Account' also in addition to appearing on the assets side of the 'Balance Sheet'.

Following 'Adjustment Entry' is passed in respect of the 'Closing Stock':

Closing Stock A/c	Dr.
To Trading A/c	
(Being closing stock brought in books)	

Effects on Final Accounts

i) 'Closing Stock' will be displayed on the credit side of the 'Trading Account'; and

ii) It will appear on the asset-side of the 'Balance Sheet'.

Example 4: Closing Stock as at 31st March 2019 ₹100 appears outsides the Trial Balance. Accounts are closed on 31st March. Pass an adjusting entry and show how this will appear in final accounts.

Solution:

Journal Entry

Date	Particulars	L. F.	Dr. (₹)	Cr. (₹)
	Closing Stock A/c Dr.		100	
	To Trading A/c			100
	(Being closing stock brought to trading accounts)			

Trading Account

Dr. (for the year ended 31st March 2019) Cr.

Particulars	₹	Particulars	₹
		By Closing Stock	100

Balance Sheet

(as on 31st March 2019)

Liabilities	₹	Assets	₹
		Current Assets:	
		Closing Stock	100

2) **Outstanding Expenses:** These are the expenses, which have remained unpaid despite having become due for payment during the 'Accounting Period'. Some of the recurring expenses like 'Rent', 'Wages', 'Salaries', 'Interest', etc. are likely to remain unpaid at the end of any 'Accounting Period'. However, such expenses having remained unpaid, no entry in respect thereof might have been passed in the books of accounts.

If the 'Salary' for the last month of the 'Accounting Period', i.e., March has not been paid, the entry in this regard would also have not been passed in the books of accounts. Apparently, there should not be any need for 'Adjustment Entry'.

However, as such expenses have been accrued, in order to ensure compliance with the 'Accrual Principle of Accountancy' (and also with a view to arriving at the true picture of profitability), it is essential to take into consideration such outstanding expenses. The following entry, therefore, needs to be passed in the books of accounts:

Expense A/c	Dr.
To Outstanding Expense A/c	
(Being due expense remain unpaid)	

Effects on Final Accounts

i) 'Outstanding Expenses' would appear on the debit side of the 'Trading / Profit and Loss Account'; and

ii) As 'Outstanding Expenses' happen to be an item of 'Liability', they will be displayed on the liabilities side of the 'Balance Sheet'.

Example 5: Salary has been paid for 11 months from April 2018 to February 2019 @ ₹100 p.m. Salary for the month of March has not yet paid. Accounting year is financial year. Pass an adjusting entry and show how this will appear in final accounts.

Solution:

Journal Entry

Date	Particulars		L. F.	Dr. (₹)	Cr. (₹)
	Salary A/c	Dr.		100	
	To Outstanding Salary A/c				100
	(Being salary due remain upaid)				

Profit and Loss Account

Dr. (for the year ended 31st March 2019) Cr.

Particulars		₹	Particulars	₹
To Salary Paid	1,100			
Add: Outstanding salary	100	1,200		

Balance Sheet

(as on 31st March 2019)

Liabilities	₹	Assets	₹
Current Liabilities:			
Outstanding Salary	100		

3) **Prepaid (or Unexpired) Expenses:** These are the expenses, which have been incurred in advance, i.e., the payments in respect thereof have been made during the current 'Accounting Period', but the benefits of which would be availed during the next "Accounting Period'. Such expenses need to be adjusted for the value of the benefit, which would be received in the next 'Accounting Period'. The following entry, therefore, needs to be passed in the books of accounts:

Prepaid Expense A/c	Dr.
To Expense A/c	
(Being expense have been paid in advance)	

Effects on Final Accounts

i) 'Prepaid Expenses' are reduced from the particular head of expense on the debit side of the 'Profit and Loss Account'; and

ii) As they happen to be an item of 'Asset', they will be displayed on the assets side of the 'Balance Sheet'.

Example 6: On 1.10.2018, a Fire Insurance Premium of ₹720 paid on a policy which will expire on 30.9.2019. Accounts are closed on 31st March. Pass an adjusting entry and show now this will appear in final accounts.

Solution:

Journal Entry

Date	Particulars		L. F.	Dr. (₹)	Cr. (₹)
	Prepaid Insurance Premium A/c	Dr.		360	
	To Insurance Premium A/c				360
	(Being insurance premium paid in advance)				

Profit and Loss Account

Dr. (for the year ended 31st March 2019) Cr.

Particulars		₹	Particulars	₹
To Insurance Premium Paid	720			
Less: Prepaid Insurance Premium	360	**360**		

Balance Sheet

(as on 31st March 2019)

Liabilities	₹	Assets	₹
		Current Assets:	
		Prepaid Insurance Premium	360

4) **Accrued Income:** 'Accrued Income' is that income which has been earned / become due during the current 'Accounting Period', but has not been received before the end of that period. Such category of income needs to be shown on the credit side of the 'Profit and Loss Account'.

Some **examples** of 'Accrued Income' are 'Rent', 'Interest on Securities', etc., which have become due but yet to be received, for which following adjusting entry should be passed:

Accrued Income A/c	Dr.
To Income A/c	
(Being accrued income adjusted)	

Effects on Final Accounts

i) 'Accrued Income' is required to be taken as an additional income on the credit side of the 'Profit and Loss Account'; and

ii) Accrued income, being an asset, (debit balance) should be shown on the assets side of the 'Balance Sheet'.

Example 7: On 1.4.2018 a loan of ₹10,000 was given to Ram at a rate of interest of 12%, per annum. During the year, interest was received for 11 months from April to February Interest for the month for March has not yet been received – Accounting year is financial year. Pass an adjusting entry and show how this will appear in final accounts.

Solution:

Journal Entry

Date	Particulars		L. F.	Dr. (₹)	Cr. (₹)
	Accrued Interest A/c	Dr.		100	
	To Interest A/c				100
	(Being interest accrued)				

Profit and Loss Account

Dr. (for the year ended 31st March 2019) Cr.

Particulars	₹	Particulars		₹
		By Interest Received	1,100	
		Add: Accrued	100	1,200

Balance Sheet

(as on 31st March 2019)

Liabilities	₹	Assets	₹
		Investments:	
		12% Loan to Ram	10,000
		Current Assets:	
		Accrued Interest	100

5) **Income Received in Advance or Unearned Income:** This head represents the income received in advance during the current 'Accounting Period', although the same may not relate to the current 'Accounting Period' (may relate to the next 'Accounting Period'). Such income should not be taken as income and included in the 'Profit and Loss Account' for the current 'Accounting Period'. **For example,** Income (Rent) of ₹3,000 received in advance for the next six months @ ₹500 per month (relating to the next accounting period) needs to be deducted from the head 'Income Received' on the credit side of the 'Profit and Loss Account', and following adjusting entry may be passed:

Income A/c	Dr.
To Income Received in Advance A/c	
(Being Income Received in Advance)	

Effects on Final Accounts

i) 'Income Received in Advance' should be deducted from income on the credit side of 'Profit and Loss Account'; and

ii) It should be shown as a liability on the 'Liabilities' side of the 'Balance Sheet'.

Example 8: On 1.2.2019, a loan or ₹10,000 was given to Ravi at a rate of interest of 12% per annum and Accounting year is financial year. Interest was received for 3 months from February to April. Pass an adjusting entry and show how this will appear in final accounts.

Solution:

Journal Entry

Date	Particulars	L. F.	Dr. (₹)	Cr. (₹)
	Interest A/c Dr. To Interest Received in Advance A/c (Being unaccured interest received)		100	100

Profit and Loss Account

Dr. (for the year ended 31st March 2019) Cr.

Particulars	₹	Particulars		₹
		By Interest Received	300	
		Less: Advance Interest	100	200

Balance Sheet

(as on 31st March 2019)

Liabilities	₹	Assets	₹
Current Liabilities:		Investments:	
Advance Interest	100	Loan to Ravi	10,000

6) **Provision for Depreciation on Fixed Assets:** 'Fixed Assets' of a business organisation play a vital role in production of goods and contribute in a big way in the generation of revenue. However, over a period of time their 'Economic Life' gets depleted in a slow but steady manner, due to normal wear and tear. Such depletion is not directly reflected as an expense in the 'Profit and Loss Account'. There is a mechanism to show the depletion in the value of 'Fixed Assets' and provide 'Depreciation' on the 'Fixed Assets' in the 'Profit and Loss Account' every year as per the 'Depreciation Policy' of the organisation.

'Depreciation' is charged on 'Fixed Assets' to record the:

i) Reduction in their value on account of normal wear and tear due to business use, and

ii) Obsolescence effect. After the provision of 'Depreciation', the book value of the 'Fixed Assets' gets reduced. The amount, by which the value of the 'Fixed Assets' gets reduced, is reflected as an expense in the debit side of the 'Profit and Loss Account'. As the entry for 'Depreciation' is not passed in the normal course of maintaining books of accounts, it (Depreciation) does not find a place in the 'Trial Balance'.

Following steps are involved in passing entries at the time of providing 'Depreciation' on the 'Fixed Assets':

i) Charging depreciation on 'Fixed Assets'.

Depreciation A/c Dr. (With the amount of depreciation for the year)
To Appropriate Asset A/c
(Being depreciation charged on fixed asset)

ii) Transferring the 'Depreciation' amount to 'Profit and Loss Account'.

Profit and Loss A/c Dr. (With the amount of depreciation for the year)
To Depreciation A/c
(Being depreciation transfer to profit and loss account)

Effects on Final Accounts

i) Amount of 'Depreciation' is reduced from the value of the 'Fixed Assets' and in the 'Balance Sheet' and their reduced value is shown.

ii) Amount of 'Depreciation' is shown as an expense to the debit side of 'Profit & Loss Account'.

Example 9: In the trial balance on March 31, 2019 building appeared at a gross value at ₹1,20,000. The company has the practice of charging depreciation @ 10% on this asset.

Solution:

Journal Entry

Date	Particulars	L. F.	Dr. (₹)	Cr. (₹)
	Depreciation A/c Dr. To Building A/c (10% on ₹1,20,000) (Being depreciation provided on building @ 10%)		12,000	12,000
	Profit and Loss A/c Dr. To Depreciation A/c (Being depreciation transferred to profit and loss account)		12,000	12,000

Profit and Loss Account

Dr. (for the year ending March 31st, 2019) Cr.

Particulars	₹	Particulars	₹
To Depreciation on Building	12,000		

Balance Sheet

(as on March 31st, 2019)

Liabilities	₹	Assets		₹
		Building	1,20,000	
		Less: Depreciation @10%	12,000	1,08,000

7) **Adjustment of Bad Debts:** A debt, which has become difficult to recover despite all the efforts made for its recovery, is termed as 'Bad Debt'. It is an amount due from the customer of the business organisation, which he is not in a position to pay. Following entry are required to be passed in this regard:

Bad Debts A/c	Dr.
To Debtors A/c	
(Being sum of amount unrecovered from debtor)	

Effects on Final Accounts

i) It is reflected as a separate item on the debit side of the 'Profit & Loss Account', and

ii) Appears on the 'Assets' side of the 'Balance Sheet', by way of deduction from the 'Debtors'.

Example 10: ₹50,000 as extracted from the trial balance as on 31st March 2019. It includes the bad debts of Mr. Ramesh (debtor) ₹1,000 which could not be recovered from him. You are require to pass an adjusting entry and present these items in the final accounts.

Solution:

Journal Entry

Date	Particulars	L. F.	Dr. (₹)	Cr.(₹)
	Bad Debts A/c Dr.		1,000	
	To Mr. Ramesh A/c			1,000
	(Being ₹1,000 unrecovered from Ramesh and proved to be bad debts)			

Profit and Loss Account

Dr. (for the year ended 31st March 2019) Cr.

Particulars	₹	Particulars	₹
To Bad Debts	1,000		

Balance Sheet

(as on 31st March 2019)

Liabilities	₹	Assets		₹
		Current Assets:		
		Debtors	50,000	
		Less: Bad Debts	1,000	49,000

8) **Adjustment of Provision for Doubtful Debts:** During the conduct of a business, it is normal that some of the debtors fail to make payment. It is, therefore prudent on the part of the organisation to make general provision every year for a possible loss to the business, because of the 'Doubtful Debts'. Such a provision is created at a pre-decided percentage of net debtors (debtors less bad debts) as per the policy decided by the management. Following entries are required to be passed in this regard:

Profit and Loss A/c	Dr.
To Provision for Doubtful Debts A/c	
(Being Provision for Doubtful Debts charged)	

Effects on Final Accounts

i) It is reflected as a separate item on the debit side of the 'Profit & Loss Account'; and

ii) Appears on the 'Assets' side of the 'Balance Sheet', by way of deduction from the 'Sundry Debtors' (Net of Additional Bad Debts).

Example 11: From the trial balance data collected as on 31st March 2019. Pass the journal entry and prepare the necessary accounts.

Particulars	Dr. (₹)	Cr. (₹)
Sundry Debtors	50,000	
Bad Debts	1,200	

Adjustments:

i) The amount of ₹1,000 was not recovered from debtor and therefore it is treated as bad debts.

ii) Create provision of 5% for doubtful debts on debtors.

Pass the necessary journal entries and show the relevant accounts (including final accounts).

Solution:

Journal Entries

Date	Particulars		L.F.	Dr. (₹)	Cr. (₹)
	Bad Debts A/c	Dr.		1,000	
	To Debtor A/c				1,000
	(Being the amount of 1,000 was not recovered from debtor and proved to be bad)				
	Profit and Loss A/c	Dr.		2,200	
	To Bad Debts A/c				2,200
	(Being the bad debts amount transfer to Profit and Loss A/c)				
	Profit and Loss A/c	Dr.		2,450	
	To Provision for Doubtful Debts A/c				2,450
	(Being 5% provision made on doubtful debts on ₹49,000)				

Dr. Sundry Debtors Account Cr.

Particulars	₹	Particulars	₹
To Balance b/d	50,000	By Bad Debts A/c	1,000
		By Balance c/d	49,000
	50,000		**50,000**

Dr. Bad Debts Account Cr.

Particulars	₹	Particulars	₹
To Balance b/d	1,200	By Profit and Loss A/c	2,200
To Debtor (bad debt)	1,000		
	2,200		**2,200**

Dr. Provision for Doubtful Debts Account Cr.

Particulars	₹	Particulars	₹
To Balance b/d	2,450	By Profit and Loss A/c	2,450
	2,450		**2,450**

Profit and Loss Account

Dr. (for the year ended 31st March 2019) **Cr.**

Particulars		₹	Particulars	₹
To Bad Debts (Trial Balance)	1,200			
Add: Additional Bad Debts	1,000			
	2,200			
Add: Provision for Doubtful Debts	2,450	4,650		

Balance Sheet

(as on 31st March 2019)

Liabilities	₹	Assets		₹
		Current Assets:		
		Sundry Debtors	50,000	
		Less: Additional Bad Debts	1,000	
			49,000	
		Less: Provision for doubtful debts @ 5%	2,450	46,550

9) **Provision for Discount on Debtors:** Allowing discount by the seller to the buyer for making prompt payment of dues is a common business practice. 'Discount' is an incentive extended to the debtors with a view to encourage them for timely payment without delay. For allowing discount to the debtors / customers, a provision is required to be made in this regard, for which the following journal entry is passed:

Profit and Loss A/c	Dr.
To Provision for Discount on Debtor A/c	
(Being provision created for discount on debtors)	

The provision of discount should be extended to the 'Good and Disciplined Debtors' only. In other words, 'Bad Debts' and 'Provision for Bad & Doubtful Debts' need to be deducted from the 'Sundry Debtors', before making provision for 'Discount' for 'Good and Disciplined Debtors'.

Effects on Final Accounts

i) 'Provision for Discount on Debtors' would be reflected on the debit side of the 'Profit and Loss Account'; and

ii) Such provision will appear on the 'Assets' side of the 'Balance Sheet' by way of deduction from the 'Sundry Debtors'.

Example 12: Following are the extracts from the Trial Balance of a firm as at 31st March 2019:

Particulars	Dr. (₹)	Cr. (₹)
Sundry Debtors	50,000	
Bad Debts	1,200	
Discount	500	

Adjustments:

i) Provision for doubtful debts on debtors @ 5%.

ii) Provision for discount on debtors @ 2%.

iii) Additional discount of ₹1,000 are provided to debtor.

Pass the journal entries and prepare the relevant accounts showing the effects in final accounts.

Solution:

Journal Entries

Date	Particulars		L.F.	Dr. (₹)	Cr. (₹)
	Discount Allowed A/c	Dr.		1,000	
	To Sundry Debtors A/c				1,000
	(Being the additional discount allowed to debtors)				
	Profit and Loss A/c	Dr.		2,700	
	To Bad Debts A/c				1,200
	To Discount Allowed A/c				1,500
	(Being bad debts and discount allowed are transfer to P&L A/c)				
	Profit and Loss A/c	Dr.		2,450	
	To Provision for Doubtful Debts A/c				2,450
	(Being 5% provision for doubtful debts charged on debtors of (₹50,000 – 1,000@5%)				
	Profit and Loss A/c	Dr.		931	
	To Provision for Discount on Debtors A/c				931
	(Being 2% provision for discount on debtors made (i.e., ₹49,000– ₹2,450@2%)				

Profit and Loss Account

Dr. (for the year ended 31st March, 2019) Cr.

Particulars		₹	Particulars	₹
To Bad Debts (Trial Balance)		1,200		
To Provision for Doubtful Debts		2,450		
To Discount (Trial Balance)	500			
Add: Additional Discount	1,000	1,500		
To Provision for Discount on Debtors		931		

Balance Sheet

(as on 31st March, 2019)

Liabilities	₹	Assets		₹
		Current Assets:		
		Debtors	50,000	
		Less: Additional Discount	1,000	
			49,000	
		Less: Provision for Doubtful Debts @ 5%	2,450	
			46,550	
		Less: Provision for Discount @ 2%	931	45,619

10) **Reserve for Discount on Creditors:** 'Discount on Creditors' is just opposite of the 'Discount on Debtors'. In the case of 'Discount by Creditors' discount is allowed by the creditors for timely / prompt payment to them, as against the 'Discount on Debtors', wherein discount is extended to the debtors for timely / prompt payment by them. The amount of reserve for 'Discount on Creditors' is an item of gain, and as such shown on the credit side of the 'Profit and Loss Account' and is deducted from the amount of 'Sundry Creditors'.

The following journal entry needs to be passed:

Reserve for Discount on Creditors A/c	Dr.
To Profit and Loss A/c	
(Being reserve created on creditors)	

Effects on Final Accounts

i) Reserve for 'Discount on Creditors' will be reflected on the credit side of the 'Profit and Loss Account'; and

ii) It will appear on the 'Liabilities' side of the 'Balance Sheet' by way of deduction from 'Sundry Creditors'.

Example 13: From the Trial Balance as on 31st December 2018. Pass the journal entries and prepare relevant account showing the effects in final accounts.

Trial Balance
(as on 31st December 2018)

Particulars	Dr. (₹)	Cr. (₹)
Sundry Creditors		50,000
Discount		1,000
Reserve for Discount on Creditors	1,000	

Adjustments:

i) ₹500 additional discount received from creditors.

ii) Create 10% reserve for discount on creditors.

Solution:

Journal Entries

Date	Particulars		Dr. (₹)	Cr. (₹)
	Sundry Creditors A/c	Dr.	500	
	To Discount A/c			500
	(Being ₹500 additional discount received from creditors)			
	Discount A/c	Dr.	1,500	
	To Reserve for Discount on Creditors A/c			1,500
	(Being discount received transferred to reserve for discount A/c)			
	Reserve for Discount on Creditors A/c	Dr.	5,450	
	To Profit and Loss A/c			5,450
	(Being 10% reserve for discount on creditor are provided and further transfer to Profit and Loss A/c)			

Dr. Reserve for Discount on Creditors Account Cr.

Particulars	₹	Particulars	₹
To Balance b/d	1,000	By Discount A/c	1,500
To Profit and Loss A/c	5,450	By Balance c/d	4,950
	6,450		**6,450**

Profit and Loss Account
Dr. (for the year ending 31st December 2018) **Cr.**

Particulars	₹	Particulars		₹
		By Discount (Trial Balance)	1,000	
		Add: Additional Discount Received	500	
		Add: New Reserve	4,950	
			6,450	
		Less: Old Reserve	1,000	**5,450**

11) **Interest on Drawings:** 'Cash' and / or 'Goods' withdrawn by the owners of a business organisation for their personal or domestic use is referred to as the 'Drawings'. The business entity pays interest on the 'Capital' brought in by the proprietors for the running of the business. On the same analogy, the interest on the 'Drawings' made by the proprietors need to be / is charged by the business entity. Such interest amount is received by the business entity, and as such it is part of its income. In case of 'Drawing', it is desirable to debit 'Interest on Drawings' to 'Drawings Account' instead of 'Capital Account'.

Following 'Adjusting Entry' needs to be passed for 'Interest on Drawings':

Drawings A/c	Dr.
To Interest on Drawings A/c	
(Being interest on drawing charged)	

Effects on Final Accounts

i) 'Interest on Drawings', being an income for the business, is shown on the credit side of the 'Profit and Loss Account';

ii) It is deducted from the 'Capital Account' on the 'Liabilities' side of the 'Balance Sheet', because liabilities are reduced to that extent;

iii) However, if the item of 'Interest on Drawings' is shown in the 'Trial Balance', it indicates that the balance of 'Capital Account' has already been adjusted. In such a case, only 'Interest on Drawings' account is shown as income on the credit side of the 'Profit and Loss Account'.

Example 14: Capital is ₹2,00,000 and the total drawings are ₹10,000. The rate of interest is 6% on the drawings. The average amount of drawings on which interest is to be charged is ₹5,000. You are require to pass journal entry and present the data in final accounts.

Solution:

Journal Entry

Date	Particulars	L. F.	Dr. (₹)	Cr. (₹)
	Interest on Drawings A/c Dr.		300	
	To Profit and Loss A/c			300
	(Being interest on drawing charged) (interest @ 6% on drawings ₹5,000)			

Profit and Loss Account

Dr. (for the year ending....) Cr.

Particulars	₹	Particulars	₹
		By Interest on Drawings	300

Balance Sheet

(as on....)

Liabilities		₹	Assets	₹
Capital	2,00,000			
Less: Drawings	10,000			
	1,90,000			
Less: Interest on Drawings	300	1,89,700		

Drawings and Interest on drawings are reduced from Capital account.

12) **Interest on Capital:** 'Capital' of a firm is its proprietors' contribution for starting and running the business, for which interest is paid by the business entity to the proprietors. Such interest payment is a form of reward to the proprietors for having invested in the business. Following 'Adjustment Entry' is required to be passed in this regard:

Interest on Capital A/c	Dr.
To Capital A/c	
(Being interest charged on capital)	

Effect on Final Accounts

i) 'Interest on Capital' is an item of expenditure, and as such reflected on the debit side of the 'Profit and Loss Account'; and

ii) It is added in the 'Capital' on the liabilities side of the 'Balance Sheet'.

Example 15: The capital invested in a business is ₹10,000 and the rate of interest is 5%. You are required to pass journal entry and present the data in final accounts.

Solution:

Journal Entry

Date	Particulars	L. F.	Dr. (₹)	Cr. (₹)
	Interest on Capital A/c Dr. To Capital A/c (Being interest charged on capital @ 5%)		500	500

Profit and Loss Account

(for the year ending....)

Dr. Particulars	₹	Particulars	₹ Cr.
To Interest on capital	500		

Balance Sheet

(as on....)

Liabilities		₹	Assets	₹
Capital	10,000			
Add: Interest on capital	500	10,500		

2.7. SUMMARY OF ADJUSTMENTS ENTRIES

Adjustments	Journal Entry	Adjustment in a Trading and Profit & Loss Account	Adjustment in Balance Sheet
1) **Closing Stock**	Closing Stock A/c Dr. To Trading A/c	Posted at the credit side of Trading Account.	Shown at the assets side.
2) **Depreciation**	Depreciation A/c Dr. To Asset A/c	Posted at the debit side of P/L Account	Shown as deducted from concerned asset.
3) **Appreciation**	Asset A/c Dr. To Appreciation A/c	Posted at the credit side of P/L Account.	Shown as added to concerned asset.
4) **Outstanding Expenses**	Expenses A/c Dr. To Outstanding Expenses A/c	Added to the concerned expense at the debit side of P/L or Trading Account.	Shown at the liabilities side.
5) **Prepaid Expenses**	Prepaid expenses A/c Dr. To Expense A/c	Deducted from the concerned expense at the debit side of P/L Account.	Shown at the assets side.
6) **Accrued Income**	Accrued Income A/c Dr. To Income A/c	Added to the concerned income at the credit side of P/L Account.	**Shown** as the assets side.
7) **Unearned Income**	Income A/c Dr. To Unearned Income A/c	Deducted from concerned income from the credit side of P/L Account.	Shown at the liabilities side.
8) **Interest on Capital**	Interest on Capital A/c Dr. To Capital A/c	Posted at the debit side of P/L Account.	Added to capital at the liabilities.
9) **Interest on Drawings**	Drawings A/c Dr. To Interest on Drawings A/c	Posted at the credit side of P/L Account	Deducted from capital at the liabilities side
10) **Interest on Investment**	Investment A/c Dr. To Interest on Investment A/c	Posted at the credit side of P/L Account	Added to investment at the assets side
11) **Interest on Loans (Borrowed)**	Interest on loan A/c Dr. To Loan A/c	Posted at debit side of P/L Account.	Added to loan at the liabilities side.
12) **Interest on Loans (Advance)**	Loans A/c Dr. To Interest on loans A/c	Posted at the credit side of P/L Account.	Added to loan at the assets side.
13) **Bad Debts**	Bad debts A/c Dr. To Debtors A/c	Posted at the debit side of P/L Account.	Deducted from debtors at the assets side.
14) **Provision for Bad Debts**	P/L A/c Dr. To Provision for bad debts A/c	Posted at the debit side of P/L Account.	Deducted from debtors at the assets side.
15) **Provision for Discount on Debtors**	P/L A/c Dr. To Provision for discount on debtors A/c	Posted at the debit side of P/L Account.	Deducted from debtors at the assets side.
16) **Provision for Discount on Creditors**	Provision for Discounts on creditors A/c Dr. To Profit & Loss A/c	Posted at the credit side of P/L Account	Deducted from creditors at the liabilities side
17) **Accidental Loss of Stock**	Loss of stock A/c Dr. To Trading A/c or To Purchases A/c	Deducted from purchases or posted at the credit side of Trading Account and also at the debit side of P/L Account.	No Effect.

18) **Accidental Loss of Goods if Insured (Partial Admission of Claim)**	Insurance Co. A/c Dr. or Loss of stock A/c Dr. To Trading A/c	Insurance Co. Account will be shown with the amount of claim admitted by Co., loss of stock will be shown with amount of the claim not admitted at the credit side of Trading Account. Loss of stock will also be shown at the debit side of P/L Account.	Insurance Co. Account will be shown at the assets side.
19) **Loss of Assets by Fire (If not Insured)**	Loss by fire A/c Dr. To Asset A/c	Loss by fire Account will be shown at the debit side of P/L Account.	Loss will be deducted from the asset at the assets side.
20) **Accidental Loss of Assets (If Insured) (Partial Admission)**	Insurance Co. A/c Dr. or Loss by fire A/c Dr. To Assets A/c	Loss by fire Account will be shown at the debit side of P/L Account.	Insurance Co. Account will be shown at the assets side, loss by fire Account will be deducted from asset.
21) **Outstanding Manager's commission**	P/L A/c Dr. To Outstanding manager's commission A/c	Posted at the debit side of P/L Account.	Shown at the liabilities side
22) **Goods sent on sale on return basis treated as sales (cost of goods 6000 included in sales 8000)**	Sales A/c Dr. To Debtors A/c And Stock A/c Dr. To Trading A/c	₹8,000 will be deducted from sales and ₹6,000 will be added to closing stock at the credit side of trading Account	₹8000 deducted from debtors and ₹6000 added to stock at the assets side
23) **Goods taken by the proprietor for personal use**	Drawings A/c Dr. To Purchases A/c	Amount of goods taken by the proprietor will be deducted from purchases in Trading Account.	Amount of goods taken by the proprietor will be deducted from capital at the liabilities side.
24) **Goods Given as Charity**	Charity A/c Dr. To Purchases A/c	Deducted from purchase at the debit side of Trading Account.	Deducted from capital at the liabilities side.
25) **Goods Distributed as Free Samples**	Advertising A/c Dr. Or Free Samples A/c Dr. To Purchases A/c	Posted at the debit side of P/L Account. Also deducted from purchases Account	No effect.

2.8. MISCELLANEOUS PROBLEMS

Example 16: Following is the Trial Balance of ABC as on 31/03/2015. Prepare Trading Account, Profit and Loss Account for the year ended 31/03/2015 and prepare Balance Sheet as on that date.

Particulars	Debit Balance (₹)	Particulars	Credit Balance (₹)
Opening Stock	25,000	Creditors	25,000
Plant	1,64,000	Bills Payable	25,000
Rent	20,000	Capital	3,25,000
Salaries	30,000	Sales	4,00,000
Wages	20,000		
Carnage Inward	2,000		
Carriage Outward	3,000		
Factory Rent	5,000		
Purchases	2,25,000		
Insurance	1,000		
Furniture	80,000		
Debtors	2,00,000		
	7,75,000		**7,75,000**

Additional Information:

1) Closing stock valued ₹25,000 as per market value and as per book value ₹20,000.
2) Depreciation on furniture ₹8,000.

Solution:

Trading and Profit and Loss Account

Dr. (For the year ended 31st March 2015) Cr.

Particulars	₹	Particulars	₹
To Opening Stock	25,000	By Sales	4,00,000
To Purchases	2,25,000	By Closing Stock	20,000
To Wages	20,000		
To Carriage Inward	2,000		
To Factory Rent	5,000		
To Gross Profit c/d	1,43,000		
	4,20,000		**4,20,000**
To Rent	20,000	By Gross Profit b/d	1,43,000
To Salaries	30,000		
To Carriage Outwards	3,000		
To Insurance	1,000		
To Depreciation on Furniture	8,000		
To Net Profit	81,000		
	1,43,000		**1,43,000**

Balance Sheet

(As on 31st March 2015)

Liabilities		₹	Assets		₹
Capital	3,25,000		Plant		1,64,000
Add: Net Profit	81,000	4,06,000	Furniture	80,000	
Creditors		25,000	**Less:** Depreciation	8,000	72,000
Bills Payable		25,000	Debtors		2,00,000
			Closing Stock		20,000
		4,56,000			**4,56,000**

Example 17: Following is the Trial Balance and DEF as on 31/03/2010. Prepare Trading Account, Profit and Loss Account for the year ended 31/03/2010 and Balance Sheet as on that date.

Particulars	Debit Balance (₹)	Particulars	Credit Balance (₹)
Opening Stock	15,000	Bank Loan	50,000
Machinery	2,00,000	Capital	2,75,000
Insurance	1,000	Creditors	35,000
Printing and Stationery	5,000	Sales	1,86,500
Telephone Charges	500	Returns Outward	1,500
Wages	25,000		
Advertising	15,000		
Purchases	1,00,000		
Returns Inward	1,000		
Motor Van	30,000		
Debtors	1,50,000		
Motive and Power	5,500		
	5,48,000		**5,48,000**

Additional Information:

1) Closing stock valued at ₹5,000.
2) Wages outstanding ₹5,000 and prepaid insurance ₹500.

Solution:

Trading and Profit and Loss Account

Dr. (For the year ended 31st March 2010) Cr.

Particulars		₹	Particulars		₹
To Opening Stock		15,000	By Sales	1,86,500	
To Purchases	1,00,000		**Less:** Return Inward	1,000	1,85,500
Less: Return Outward	1,500	98,500	By Closing Stock		5,000
To Wages	25,000				
Add: Outstanding	5,000	30,000			
To Motive and Power		5,500			
To Gross Profit c/d		41,500			
		1,90,500			**1,90,500**

To Printing and Stationary		5,000	By Gross Profit b/d	41,500
To Telephone Charges		500		
To Advertising		15,000		
To Insurance	1,000			
Less: Prepaid Insurance	500	500		
To Net Profit		20,500		
		41,500		**41,500**

Balance Sheet
(As on 31st March 2010)

Liabilities		₹	Assets	₹
Capital	2,75,000		Machinery	2,00,000
Add: Net Profit	20,500	2,95,500	Motor Van	30,000
Bank Loan		50,000	Debtors	1,50,000
Creditors		35,000	Closing Stock	5,000
Outstanding Wages		5,000	Prepaid Insurance	500
		3,85,500		**3,85,500**

Example 18: The following balances were extracted from the books of Mohan on 31st December, 2018:

Particulars	₹	Particulars	₹
Capital	9,000	Purchases	15,000
Furniture	800	Carriage Outwards	200
Creditors	1,600	Salaries	2,000
Premises	13,000	Sales	18,000
Bad Debts	80	Rent Received	800
Cash	40	Discount Allowed	180
Drawings	900	Loan	4,000
Bank Overdraft	905	Reserve for Bad Debts	100
Debtors	1,500	Expenses	705

Adjustments:
1) Make provision for bad debts @ 3%.
2) Salary due ₹200.
3) Stock on 31-12-2018 ₹3,500.
4) Write-off 10% from furniture for depreciation.
5) Due from tenants rent ₹100.

Solution:

Trading and Profit and Loss Account
Dr. (for the year ending 31st Dec, 2018) **Cr.**

Particulars		₹	Particulars		₹
To Purchases		15,000	By Sales		18,000
To Gross Profit c/d		6,500	By Closing Stock		3,500
		21,500			**21,500**
To Salaries	2,000		By Gross Profit b/d		6,500
Add: Outstanding	200	2,200	By Rent Received	800	
To Discount Allowed		180	**Add:** Accrued Rent	100	900
To Bad Debts	80				
Add: Provision for Bad Debts	45				
Less: Old Reserve	100	25			
To Carriage Outward		200			
To Depreciation on Furniture		80			
To Expenses		705			
To Net Profit (Transferred to Capital A/c)		4,010			
		7,400			**7,400**

Balance Sheet
(as on 31st Dec, 2018)

Liabilities		₹	Assets		₹
Capital	9,000		Cash		40
Add: Net Profit	4,010		Closing Stock		3,500
	13,010		Debtors	1,500	
Less: Drawings	900	12,110	**Less:** Provision for Bad Debts	45	1,455
Creditors		1,600	Accrued Rent		100
Bank Overdraft		905	Premises		13,000
Outstanding Salary		200	Furniture	800	
Loan		4,000	**Less:** Depreciation	80	720
		18,815			**18,815**

Example 19: Prepare Trading & Profit and Loss account and Balance sheet from the following balances, relating to the year ending 31st March 2019.

Particulars	₹	Particulars	₹
Capital	10,000	Wages	5,000
Creditors	1,200	Bank	1,000
Returns Outward	500	Repairs	50
Sales	16,400	Stock	2,000
Bills Payable	500	Rent	400
Plant and Machinery	4,000	Manufacturing Expense	800
Sundry Debtors	2,400	Trade Expense	700
Drawings	1,000	Bad Debts	200
Purchases	10,500	Carriage	150
Return Inwards	300	Fuel and Power	100

Adjustments: 1) Closing Stock was valued at ₹1,450. 2) Depreciate Plant and Machinery by ₹400.
3) Allow 5% interest on capital. 4) A sum of ₹40 is due for repairs.

Solution:

Trading and Profit and Loss Account
(for the year ending 31st March, 2019)

Dr. Cr.

Particulars		₹	Particulars		₹
To Opening Stock		2,000	By Sales	16,400	
To Purchases	10,500		**Less:** Return Inwards	300	16,100
Less: Return Outwards	500	10,000	By Closing Stock		1,450
To Wages		5,000	By Gross Loss c/d		500
To Manufacturing Expenses		800			
To Carriage		150			
To Fuel and Power		100			
		18,050			**18,050**
To Gross Loss b/d		500	By Net Loss (Transferred to Capital A/c)		2,790
To Repairs	50				
Add: Outstanding Repair	40	90			
To Rent		400			
To Trade Expenses		700			
To Bad Debts		200			
To Depreciation on Plant and Machinery		400			
To Interest on Capital		500			
		2,790			**2,790**

Balance Sheet
(as on 31st March, 2019)

Liabilities		₹	Assets		₹
Capital	10,000		Plant and Machinery	4,000	
Less: Net Loss	2,790		**Less:** Depreciation	400	3,600
	7,210		Sundry Debtors		2,400
Less: Drawings	1,000		Bank		1,000
	6,210		Closing Stock		1,450
Add: Interest on Capital	500	6,710			
Creditors		1,200			
Bills Payable		500			
Outstanding Repairs		40			
		8,450			**8,450**

Example 20: From the following trial balance of Rajan, prepare trading and profit and loss account for the year ended 31.3.2016 and the Balance Sheet on at that date after taking into account the adjustments given below:

Trial Balance as on 31.3.2016

Dr. Particulars	₹	Cr. Particulars	₹
Drawings	7,600	Capital	2,90,000
Purchases	89,000	Sales	1,50,000
Sales Return	2,800	Purchases Return	4,500
Stock (1.4.2015)	12,000	Interest	3,500
Wages	8,000	Sundry creditors	12,000
Building	2,20,000	Bills payable	7,000
Freight	20,000		
Trade Expenses	2,000		
Advertisement	2,400		
Taxes and Insurance	1,300		
Sundry Debtors	65,000		
Bills Receivables	15,000		
Cash at Bank	12,000		
Cash in Hand	1,900		
Salaries	8,000		
	4,67,000		**4,67,000**

Adjustments:

1) Stock on 31.3.16 was valued at ₹15,000.
2) Insurance was prepaid of ₹400.
3) Salary outstanding ₹2000 and taxes outstanding ₹1300.
4) Depreciate Building at 2% P.A.

Solution:

Trading and Profit and Loss Account
(For the year ended 31st March 2016)

Dr. Particulars		₹	Cr. Particulars		₹
To Opening Stock		12,000	By Sales	1,50,000	
To Purchases	89,000		**Less:** Sales Return	2,800	1,47,200
Less: Purchase Return	4,500	84,500	By Closing Stock		15,000
To Wages		8,000			
To Freight		20,000			
To Gross Profit c/d		37,700			
		1,62,200			**1,62,200**
To Salaries	8,000		By Gross Profit b/d		37,700
Add: Outstanding	2,000	10,000	By Interest		3,500
To Trade Expenses		2,000			
To Advertisement		2,400			
To Tax and Insurance	1,300				
Less: Prepaid Insurance	400				
	900				
Add: Outstanding Taxes	1,300	2,200			
To Depreciation on Building		4,400			
To Net Profit (transferred to capital A/c)		20,200			
		41,200			**41,200**

Balance Sheet
(As on 31st March, 2016)

Liabilities		₹	Assets		₹
Capital	2,90,000		Cash in Hand		1,900
Add: Net Profit	20,200		Cash at Bank		12,000
	3,10,200		Building	2,20,000	
Less: Drawing	7,600	3,02,600	**Less:** Depreciation	4,400	2,15,600
Sundry Creditors		12,000	Sundry Debtors		65,000
Bills Payable		7,000	Bills Receivables		15,000
Outstanding Salary		2,000	Prepaid Insurance		400
Outstanding Taxes		1,300	Closing Stock		15,000
		3,24,900			**3,24,900**

Example 21: Following Trial Balance is prepared on 31.3.16 from Asha's book. Prepare Trading and Profit and Loss Account for the year ended 31.3.16 and the Balance Sheet as on that date after taking into account the adjustment given below:

Dr. **Trial Balance as on 31.3.2016** **Cr.**

Particulars	₹	Particulars	₹
Cash in hand	8,000	Sales	5,00,000
Cash at Bank	12,000	Creditors	40,000
Wages	20,000	Capital	3,20,000
Machinery	2,12,000	Commission	2,800
Bills Receivables	40,000		
Opening stock	84,000		
Purchases	3,30,000		
Sales Return	6,000		
Salaries	32,000		
Debtors	86,000		
Taxes and Insurance	22,000		
Bad debts	6,000		
Freight	4,800		
	8,62,800		**8,62,800**

Adjustments:

1) Closing stock ₹90,000.
2) Outstanding salaries ₹3,200 and wages ₹2000.
3) Prepaid insurance ₹900 and accrued commission ₹1200.
4) Charge 10% depreciation on machinery.

Solution:

Trading and Profit and Loss Account

Dr. (For the year ending 31st March 2016) **Cr.**

Particulars		₹	Particulars		₹
To Opening Stock		84,000	By Sales	5,00,000	
To Purchases		3,30,000	**Less:** Sales Return	6,000	4,94,000
To Wages	20,000		By Closing Stock		90,000
Add: Outstanding	2,000	22,000			
To Freight		4,800			
To Gross Profit c/d		1,43,200			
		5,84,000			**5,84,000**
To Salaries	32,000		By Gross Profit b/d		1,43,200
Add: Outstanding	3,200	35,200	By Commission	2,800	
To Taxes and Insurance	22,000		**Add:** Accrued Commission	1,200	4,000
Less: Prepaid Insurance	900	21,100			
To Bad Debts		6,000			
To Depreciation on Machinery		21,200			
To Net Profit (transferred to capital A/c)		63,700			
		1,47,200			**1,47,200**

Balance Sheet

(As on 31st March, 2016)

Liabilities		₹	Assets		₹
Capital	3,20,000		Cash in Hand		8,000
Add: Net Profit	63,700	3,83,700	Cash at Bank		12,000
Creditors		40,000	Machinery	2,12,000	
Outstanding Salary		3,200	**Less:** Depreciation	21,200	1,90,800
Outstanding Wages		2,000	Bills Receivables		40,000
			Debtors		86,000
			Accrued Commission		1200
			Prepaid Insurance		900
			Closing Stock		90,000
		4,28,900			**4,28,900**

Example 22: The following is the trial balance of Mr. Shah. Prepare the final account as on 31st March 2017.

Particulars	Dr. (₹)	Cr. (₹)
Capital		2,00,000
Land and Building	87,000	
Plant and Machinery	17,500	
Goodwill	20,000	
Drawings	22,600	
Cash in Hand	1,795	
Opening Stock	27,000	
Wages	10,000	
Purchases	69,000	
Carriage Inward	600	
Travellers Commission	6,000	
Insurance Premium	2,000	
Motor Car	3,000	
Carriage Outward	1,400	
Sales		94,000
Salaries	15,000	
Bank Charges	105	
RDD		1,500
Debtors	20,000	
Creditors		7,500
	3,03,000	**3,03,000**

Adjustments:

1) On 31st March 2017 the stock was valued at ₹46,000.
2) Depreciation on Plant and Machinery 10% and Motor car 20%.
3) Create a Reserve for Doubtful Debts at 10% on Debtors.
4) Insurance premium amounting to ₹800 is prepaid.
5) Outstanding salaries amount to ₹1,000.

Solution:

In the Books of Mr. Shah

Trading and Profit and Loss Account

(For the year ending 31st March, 2017)

Dr. Particulars		₹	Cr. Particulars	₹
To Opening Stock		27,000	By Sales	94,000
To Wages		10,000	By Closing Stock	46,000
To Purchase		69,000		
To Carriage Inward		600		
To Gross Profit c/d		33,400		
		1,40,000		**1,40,000**
To Salaries	15,000		By Gross Profit b/d	33,400
Add: Outstanding	1,000	16,000	By RDD	1,500
To Bank Charges		105		
To Carriage Outwards		1,400		
To Insurance Premium	2,000			
Less: Prepaid	800	1,200		
To Travellers Commission		6,000		
To Depreciation				
Plant and Machinery	1,750			
Motor Car	600	2,350		
To Reserve for Doubtful Debts on Debtors		2,000		
To Net Profit (Transferred to Balance Sheet)		5,845		
		34,900		**34,900**

Balance Sheet
(As on 31st March, 2017)

Liabilities		₹	Assets		₹
Capital	2,00,000		Cash in Hand		1,795
Add: Net Profit	5,845		Land and Building		87,000
	2,05,845		Plant and Machinery	1,7500	
Less: Drawing	22,600	1,83,245	**Less:** Depreciation @ 10%	1,750	15,750
Creditors		7,500	Goodwill		20,000
Outstanding Salaries		1,000	Motor Car	3,000	
			Less: Deprecation @ 20%	600	2,400
			Debtors	20,000	
			Less: Provision for Bad Doubtful Debt @ 10%	2,000	18,000
			Closing Stock		46,000
			Prepaid Insurance Premium		800
		1,91,745			**1,91,745**

Example 23: The following balances were extracted from the books of Mr. Ashok as on 31st March 2017. Prepare Trading and Profit and Loss Account for the year ended 31st March 2017 and the Balance Sheets as on that date after taking into account the following adjustments:

Particulars	Dr.(₹)	Cr. (₹)
Capital		3,30,000
Drawings	24,450	
Opening Stock	2,00,000	
Bills Receivable	25,000	
Purchases	2,75,000	
Sales		4,20,000
Bills Payable		60,000
Return Outwards		4,500
Return Inward	5,000	
Plant and Machinery	1,00,000	
Loose Tools	25,000	
Land and Building	25,000	
Sundry Debtors	1,25,000	
Sundry Creditors		1,40,000
Cash at Bank	77,550	
Salaries and Wages	50,000	
Repairs and Renewals	7,500	
Insurance	3,000	
Power and Fuel	3,500	
Printing and Stationery	2,000	
General Expenses	6,500	
	9,54,500	**9,54,500**

Adjustments:

1) Closing stock was worth ₹1,30,000.
2) Depreciation Land and Building 15% and Plant and Machinery by 5%.
3) Provide for Bad and Doubtful Debts at 5% on Debtors.
4) Insurance paid in advance to the extent of ₹750.
5) Revalue loose tools at ₹20,000.
6) Provide for outstanding expenses as follows:
   Salaries ₹2,500 and Wages ₹1,000.

Solution:

In the Books of Mr. Ashok

Trading and Profit and Loss Account

(For the year ending 31st March, 2017)

Dr. **Cr.**

Particulars		₹	Particulars		₹
To Opening Stock		2,00,000	By Sales	4,20,000	
To Purchases	2,75,000		**Less:** Return Inwards	5,000	4,15,000
Less: Return Outwards	4,500	2,70,500	By Closing Stock		1,30,000
To Power and Fuel		3,500			
To Gross Profit c/d		71,000			
		5,45,000			**5,45,000**
To Salaries and Wages	50,000		By Gross Profit b/d		71,000
Add: Outstanding Expenses on Salaries	2,500		By Net Loss (Transferred to balance Sheet)		20,750
Add: Outstanding Expenses on Wages	1,000	53,500			
To Repairs and Rewards		7,500			
To Insurance	3,000				
Less: Advance Paid	750	2,250			
To Printing and Stationery		2,000			
To General Expenses		6,500			
To Depreciation:					
Land and Building	3,750				
Plant and Machinery	5,000	8,750			
To Provision for Bad and Doubtful Debts on Debtor		6,250			
To Depreciation on Loose Tools (25,000 – 20,000)		5,000			
		91,750			**91,750**

Balance Sheet

(As on 31st March, 2017)

Liabilities		₹	Assets		₹
Capital	3,30,000		Cash at Bank		77,550
Less: Net Loss	20,750		Plant and Machinery	1,00,000	
Less: Drawing	24,450	2,84,800	**Less:** Depreciation @5%	5,000	95,000
Sundry Creditors		1,40,000	Loose Tools	25,000	
Bills Payable		60,000	**Less:** Depreciation	5,000	20,000
Outstanding Salaries		2,500	Land and Building	25,000	
Outstanding Wages		1,000	**Less:** Depreciation @15%	3,750	21,250
			Sundry Debtors	1,25,000	
			Less: PBDD	6,250	1,18,750
			Bills Receivable		25,000
			Closing Stock		1,30,000
			Prepaid Insurance		750
		4,88,300			**4,88,300**

Example 24: From the following Trial Balance of Ramesh as on 31-12-2018 prepare final accounts as on the above date:

Particulars	Dr. (₹)	Cr. (₹)
Capital		1,20,000
Drawings	15,000	
Bills Receivable	22,000	
Machinery	20,000	
Debtors and Creditors	60,000	58,000
Wages	39,000	
Purchases and Sales	2,52,000	3,55,000
Commission		5,500
Rent and Taxes	6,000	
Stock on 1-1-2018	90,000	
Salaries	10,500	
Travelling Expenses	2,000	
Insurance	600	
Repairs	3,400	
Bad Debts	3,500	
Furniture	9,000	
Returns	5,000	2,000
Cash in Hand and Bank	2,500	
	5,40,500	**5,40,500**

Adjustments:

1) Stock on hand on 31-12-2018 was ₹1,00,000.
2) Create 5% provision on debtors for doubtful debts.
3) Prepaid insurance amounted to ₹100.
4) Wages outstanding was ₹1,000.
5) Depreciate machinery by 5% and furniture by 10% p.a.

You are required to prepare Trading and Profit and Loss Account and the Balance Sheet as on 31st Dec., 2018.

Solution:

Trading and Profit and Loss Account of Ramesh
(for the year ended 31st December, 2018)

Dr. Particulars		₹	Cr. Particulars		₹
To Opening Stock		90,000	By Sales	3,55,000	
To Purchases	2,52,000		**Less:** Returns	5,000	3,50,000
Less: Returns	2,000	2,50,000	By Closing Stock		1,00,000
To Wages	39,000				
Add: Outstanding	1,000	40,000			
To Gross Profit c/d		70,000			
		4,50,000			**4,50,000**
To Salaries		10,500	By Gross Profit b/d		70,000
To Rent and Taxes		6,000	By Commission		5,500
To Travelling Expenses		2,000			
To Insurance	600				
Less: Prepaid Insurance	100	500			
To Repairs		3,400			
To Bad Debts	3,500				
Add: Provision for Doubtful Debts	3,000	6,500			
To Depreciation on:					
Machinery	1,000				
Furniture	900	1,900			
To Net Profit (Transferred to Capital A/c)		44,700			
		75,500			**75,500**

Balance Sheet of Ramesh
(as on 31st December, 2018)

Liabilities		₹	Assets		₹
Capital	1,20,000		Cash in Hand and Bank		2,500
Less: Drawings	15,000		Closing Stock		1,00,000
	1,05,000		Debtors	60,000	
Add: Net Profit	44,700	1,49,700	**Less:** Provision for Doubtful Debts	3,000	57,000
Creditors		58,000	Bills Receivable		22,000
Outstanding Wages		1,000	Prepaid Insurance		100
			Machinery	20,000	
			Less: Depreciation	1,000	19,000
			Furniture	9,000	
			Less: Depreciation	900	8,100
		2,08,700			**2,08,700**

Example 25: From the following balances extracted from the books of Mr. Amit, prepare Trading & Profit & Loss Account for the year ended 31.03.2019 and a balance sheet as on that date:

Particulars	₹
Stock on 1.04.2018	10,000
Purchases less returns	49,000
Wages	15,000
Loan from Shri. Aman	1,50,000
Capital	60,000
Sundry debtors	42,700
Reserve for bad debts (1.04.2018)	2,800
Rent and taxes	1,000

Salaries	8,000
General expenses	900
Discount allowed	500
Sundry Creditors	20,000
Sales	1,28,600
Plant and machinery	1,50,000
Furniture	18,000
Cash	2,300
Drawings	6,000
Sales returns	2,000
Cost of lease on 1.04.2018	56,000

Adjustments:

1) The closing stock was ₹9,800.
2) Write-off 10% depreciation on plant and 5% on furniture.
3) Write-off bad debts ₹700 and maintain Reserve for bad debts at 6% on debtors.
4) Write-off ₹6,000 in the current year from the cost of lease.

Solution:

Trading and Profit and Loss Account
(for the year ended 31.03.2019)

Particulars		₹	Particulars		₹
To Opening Stock		10,000	By Sales	1,28,600	
To Purchase Less Returns		49,000	**Less:** Sales Returns	2,000	1,26,600
To Wages		15,000	By Closing Stock		9,800
To Gross Profit c/d		62,400			
		1,36,400			**1,36,400**
To Salaries		8,000	By Gross Profit b/d		62,400
To Rent and Taxes		1,000			
To General Expenses		900			
To Discount Allowed		500			
To Bad Debts Written-off	700				
Add: New Provision for Bad Debts	2,520				
Less: Old Provision for Bad Debts	2,800	420			
To Depreciation:					
Plant	15,000				
Furniture	900	15,900			
To Lease written-off		6,000			
To Net Profit (transferred to Capital A/c)		29,680			
		62,400			**62,400**

Balance Sheet
(as on 31.03.2019)

Liabilities		₹	Assets		₹
Capital	60,000		Plant and Machinery	1,50,000	
Add: Net Profit	29,680		**Less:** Depreciation @ 10%	15,000	1,35,000
Less: Drawing	6,000	83,680	Furniture	18,000	
Loan from Shri. Aman		1,50,000	**Less:** Depreciation @ 5%	900	17,100
Sundry Creditors		20,000	Cost of Lease	56,000	
			Less: Lease written-off	6,000	50,000
			Sundry Debtors	42,700	
			Less: Bad debts written-off	700	
				42,000	
			Less: Provision for bad and doubtful debts	2,520	39,480
			Cash		2,300
			Closing Stock		9,800
		2,53,680			**2,53,680**

2.9. EXERCISE

2.9.1. Very Short Answer Type Questions

1) What do you understand by financial statements?
2) What are the stages of preparing final accounts?
3) What do you mean by Balance Sheet?

2.9.2. Short Answer Type Questions

1) Discuss the limitations of financial statements.
2) Give the proforma of profit and loss account.
3) Draw the proforma of vertical form of balance sheet.

2.9.3. Long Answer Type Questions

1) Explain the importance and objectives of financial statements.
2) What do you understand by trading account? Draw the proforma of trading account.
3) Explain the contents of profit and loss account.
4) Describe the items of balance sheet.
5) Give all the adjustments that are usually necessary, in the light of accrual basis of accounting, at the end of the accounting period for the preparation of the trading and profit and loss account.

2.9.4. Practical Questions

1) From the following trial balance of M/s XYZ and Sons prepare Trading and Profit and Loss A/c for the year ending March 31, 2019 and a Balance Sheet on that date:

Particulars	Dr. (₹)	Particulars	Cr. (₹)
Purchases	21,750	Sales	35,000
Discount allowed	1,300	Capital	13,000
Wages	6,500	Creditors	2,100
Salaries	2,000		
Travelling expenses	400		
Commission	425		
Carriage Inward	275		
Administrative expenses	105		
Trade expenses	600		
Interest	250		
Building	5,000		
Furniture	200		
Debtors	4.250		
Cash	7,045		
	50,100		**50,100**

Adjustments:

i) Stock on 31st March 2019 was ₹6,000.
ii) Depreciate building by 20%.
iii) Create a provision for bad debts @ 10% on debtors.
iv) Outstanding wages amount to ₹475.

[Ans: Gross Profit ₹12,000; Net Profit ₹5,495; and Total of Balance Sheet ₹21,070]

2) The following are the balances extracted from the books of ABC as on 31st December, 2018. From these balances, prepare his Trading and Profit and Loss Account and Balance Sheet as on that date:

Particulars	Dr. (₹)	Cr. (₹)
Opening Stock	12,000	
Purchases	40,000	
Sales		86,000
Discount		400
Sales Return	6,000	
Buildings	50,000	
Debtors	16,000	

Salaries	2,400	
Office Expenses	1,200	
Wages	10,000	
Purchase Return		4,000
Interest		800
Travelling Expenses	400	
Fire Insurance Premium	800	
Machinery	20,000	
Carriage on Purchases	700	
Commission	400	
Cash in hand	2,300	
Rent and Taxes	1,800	
Capital		62,000
Creditors		10,800
Total	**1,64,000**	**1,64,000**

Adjustments:

i) Closing Stock was valued at ₹16,000.
ii) Wages ₹2,000 and salaries ₹1,200 are outstanding.
iii) Rent for two months at the rate of ₹500 per month is outstanding.
iv) Depreciate Buildings by 5% and machinery by 10%.
v) Prepaid Insurance ₹200.

[Ans: Gross Profit ₹35,300; Net Profit ₹23,000; and Total of Balance Sheet ₹1,00,000]

3) The following Trial Balance was extracted from the books of M/s ABC on 31st December, 2018. You are required to prepare Trading and Profit and Loss Account for the year ending on 31st December, 2018 and a Balance Sheet as on that date:

Particulars	Dr. (₹)	Cr. (₹)
Capital		30,000
Drawings	5,000	
Debtors and Creditors	20,000	10,000
Loan from Bank		9,500
Interest on loan	300	
Cash in hand	2,000	
Provision for bad debts		700
Stock (1.1.2018)	6,800	
Motor vehicle	10,000	
Cash at bank	3,500	
Land and building	12,000	
Purchases and Sales	66,000	1,10,000
Returns	8,000	1,500
Carriage Outwards	2,500	
Carriage Inwards	3,000	
Salaries	9,000	
Rent and Insurance	3,000	
General expenses	6,900	
Bad Debts	500	
Discount		500
Bills Receivable and Payable	6,000	2,000
Rent Received		300
Total	**1,64,500**	**1,64,500**

Adjustments:

i) Stock on 31-12-2018 ₹7,000.
ii) Depreciation on Land and Building @ 2.5% and on Motor Vehicle @ 20% per annum.
iii) Salaries outstanding ₹200.
iv) Prepaid insurance ₹200.
v) Provision for bad debts is to be maintained at 5% on debtors.

[Ans: Gross Profit ₹34,700, Net Profit ₹10,700; and Total of Balance Sheet ₹57,400]

4) The following is the Trial Balance of Mr. Anil at 31st March, 2019 and it is desired to prepare final accounts showing the results of the transactions for the year:

Particulars	Dr. (₹)	Cr. (₹)
Capital Account		41,000
Plant and Machinery	51,000	
Office Furniture	2,600	
Stock on 1st April, 2018	48,000	
Motor Van	12,000	
Sundry Debtors	45,000	
Cash in Hand	400	
Balance at Bank	6,500	
Wages: Factory	1,50,000	
Office	14,000	
Purchases	2,13,500	
Sales		4,80,000
Bills Receivable	7,200	
Bills Payable		5,600
Sundry Creditors		52,000
Return Inward	9,300	
Provision for Bad Debts		2,500
Drawings	6,000	
Returns Outward		5,500
Rent	6,000	
Lighting	800	
Telephone	1,350	
Insurance	300	
Advertisement	6,350	
General Expenses	1,000	
Bad Debts	2,500	
Discount Allowed	6,500	
Discount Received		3,700
Total	**5,90,300**	**5,90,300**

The following adjustments are to made in the accounts:

i) Stock: 31st March, 2019 - ₹52,000
ii) Rent Due but not Paid: 31st March, 2019 - ₹2,000
iii) Lighting Due but not Paid: 31st March, 2019 - ₹300
iv) Insurance Paid in Advance - ₹100
v) Depreciation to be Written-Off:
Plant and Machinery @ 33-1/3%
Office Furniture @ 10%
Motor Van @ 33-1/3%
vi) The provision for Bad and Doubtful Debts has to be increased to ₹3,000
vii) Discounts at 2.5% on Debtors and Creditors are to be Provided

[Ans: Gross Profit ₹1,16,700; Net Profit ₹57,890; and Total of Balance Sheet ₹1,51,490]

5) ABC Manufacturing Co., Mumbai was registered with a nominal capital of ₹6,00,000 in Equity Shares of ₹10 each. The following is the list of balances extracted from its books on 31-3-2019.

Particulars	Dr. (₹)	Cr. (₹)
Furniture	7,200	
Calls-in-Arrears	7,500	
Plant and Machinery	3,00,000	
Premises	3,30,000	
Dividend (Interim)	37,500	
Stock (1-4-2018)	75,000	
Debtors	87,000	
Goodwill	25,000	
Cash in Hand	750	
Cash in Bank	60,900	
Purchases	1,85,000	

Preliminary Expenses	5,000	
Wages	84,865	
General Expenses	6,835	
Advertising	10,000	
Freight and Carriage	13,115	
Salaries	14,500	
Directors Fees	5,725	
Bad Debts	2,110	
Debenture Interest Paid	9,000	
Subscribed, Called-Up and Paid-Up Capital		4,00,000
Debenture (6)%		3,00,000
Profit and Loss A/c		14,500
Bills Payable		38,000
Creditors		50,000
Sales		4,15,000
General Reserve		25,000
Bad Debts Reserve (1-4-2018)		3,500
Loan		21,000
Total	**12,67,000**	**12,67,000**

Adjustments:

i) Depreciate Plant and Machinery by 5% Buildings by 2% and Furniture by ₹1,200.
ii) Write-off ₹1,000 from preliminary expenses.
iii) Provide for half years Debenture interest.
iv) RDD on 31-3-2019 should be equal to 1% on sales.
v) Outstanding Directors Fees ₹275.
vi) Goods of value of ₹1,500 were distributed as free samples during the year but no entry in this respect has been made.
vii) Closing stock ₹95,000.

Prepare Final Accounts.

[Ans: Gross Profit ₹1,53,520; Net Profit ₹70,125; and Total of Balance Sheet ₹8,82,900]

6) Following is the Trial balance of Mr. Kumar for the year ended 31st March 2019. Prepare Final Accounts.

Trial Balance

Particulars	Debit (₹)	Credit (₹)
12% Investments	35,000	
Stock as on 01st April 2018	30,400	
Purchases	98,700	
Sales		2,07,600
Carriage Inward	2,900	
Carriage Outward	5,600	
Salary	19,600	
Vehicle Expenses	8,400	
Printing and Stationery	6,870	
Insurance on Machinery	8,420	
Plant and Machinery	70,000	
Furniture and Fixtures	18,000	
Vehicles	25,000	
Debtors	36,000	
Creditors		46,700
Capital		12,5000
Commission Received		5,400
Loan		16,000
Rent paid	12,000	
Cash in hand	2,500	
Bank balance	24,000	
Interest on bank		2,690
Total	**4,03,390**	**4,03,390**

Adjustments

i) Closing Stock is valued at ₹22,420.
ii) Depreciation provided @ 10% on Plant and Machinery, @ 5% on Vehicles and @ 2.5% on Furniture and fixtures.
iii) Provide 5% on Debtors for doubtful debts.
iv) Outstanding salary is ₹4800. Investments are purchased on 11/10/2018.

[Ans: Gross Profit ₹89,600; Net Profit ₹31,845; and Total of Balance Sheet ₹2,24,345]

UNIT 3

Cost Accounting

3.1. BASIC CONCEPTS OF COST ACCOUNTING

3.1.1. Meaning and Definition of Cost

The total expenditure inclusive of money, labour and time incurred on goods and services is called **"Cost"**. In certain exceptional situations, the costs which are generally considered as expenses in the present scenario might not go into the record books.

The outcome of initiation and consumption of resources is denoted in monetary terms. The producer firms use it as an internal control process and it is not accessible to third parties.

The term 'Cost' is the money used for the purpose of producing something. Since it has been already utilized, it is not available for any other use. The cost also includes the money spent for acquiring something. From business point of view, the expenditure incurred for acquisition and repair purpose constitutes as "cost".

According to ICMA, London, "Cost is the amount of expenditure (actual or notional) incurred on, or attributable to, a specified thing or activity or cost unit".

According to AICPA Committee, "It is the amount measured in money or in cash expended or other property transferred, capital stock issued, services performed, or a liability incurred in consideration of goods or services received or to be received".

3.1.2. Meaning and Definition of Cost Accounting

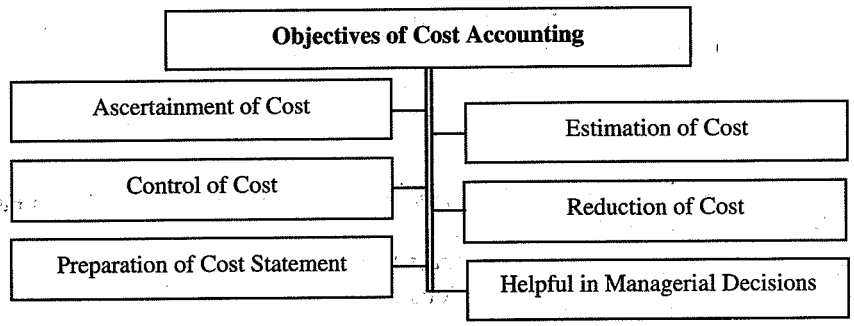
Cost Accounting may be defined as, "the process of accounting for cost". The process involves recording as well as controlling of various costs. This is a formal system of accounting which ascertains and controls costs related to various products and services.

According to the ICMA, London, cost accounting is "the process of accounting for cost which begins with the recording of income and expenditure and ends with the preparation of periodical statements and reports for ascertaining and controlling costs".

According to L.C. Cropper, "cost accounting means a specialized application of the general principles of accounting in order to ascertain the cost of producing and marketing any unit of manufacture or of carrying out any particular job or contract".

3.1.3. Objectives of Cost Accounting

Cost accounting serves various purposes which are as follows:



1) **Ascertainment of Cost:** The process of ascertainment of costs involves computation of costs which has been incurred. The costs may be ascertained for different products, services or activities using various costing methods. Determination of costs is important for the purpose of controlling them and for providing cost information to the top management for decision making. It should be ensured that the costs of a particular product or service are properly assessed.

2) **Estimation of Cost:** While costs are ascertained after it has been incurred, the estimation of costs is done before their actual incidence. For the purpose of estimation, an allowance is generally made for adjusting deviations as various costs such as material costs, labour costs and overheads tend to fluctuate. Cost estimation is useful for the following purposes:
   i) Cost estimation is important for the purpose of bidding and submitting quotations & tenders.
   ii) It is also useful for preparing budgets.
   iii) Cost estimation helps in evaluating projected performance.
   iv) Cost estimation is useful for the purpose of financial statement projection.
   v) It is important for cost control purpose.

3) **Control of Cost:** Cost control is related to controlling and reducing costs for the purpose of improving efficiency. Cost control can achieve this efficiency by full utilization of output at minimum cost. Two main techniques used for this purpose are Budgetary Control and Standard Costing. Under this process, various areas are identified where the efficiency may be increased by reducing wastage and costs. Various remedial measures may also be taken for cost control.

4) **Reduction of Cost:** The ending of cost control function is the stating of cost reduction function. It involves making actual and permanent reduction in the production costs of goods and services. It achieves the objective by eradicating non-essential or inefficient activities. Technological advances may render many activities non-essential. Similarly, brain storming sessions may be carried out to question standard activities. Various areas where costs may be controlled are product designing, administration, finance and production.

5) **Preparation of Cost Statement:** Different elements of total costs are shown in a cost statement. This statement shows total cost, unit cost, units produced and sold, closing stock in terms of units and value, etc. Production process and other processes should be constantly reviewed to maintain efficiency. Cost accounting presents information (such as production units and it total costs) with proper analysis for different time periods ranging from daily basis to yearly. Thus, a cost statement serves complete information tools for the management.

6) **Helpful in Managerial Decisions:** Cost accounting is framed in a manner to help the management. It provides information for making various decisions such as "product diversification", "fixing of selling prices" and "make or buy decisions". It is also helpful for evaluating a "product line", "factory shut down decision" and "determination of alternative methods of manufacture". Cost accounting is useful for the purpose of making various other decisions such as "shut down decision", "dropping a product line", "key factor areas" and "selection of profitable product-mix".

3.1.4. Importance/Advantages of Cost Accounting

Cost accounting has many advantages which are as follows:

1) **Aid to Management:** Cost accounting helps the management in taking various decision regarding different functions. The data provided by cost accounting helps the management in determining unprofitable activities. It also helps in measuring efficiency of various processes and functions.

   Cost accounting provides information which may be used for submitting tenders. It also aids in the process of framing policies. Elimination of wasteful activities leads to higher profits. A well-established cost accounting system helps in producing reliable data. The cost accounting systems serve the management in following ways:

   i) **Help in Eliminating and Turning Unprofitable Activities:** Cost accounting can help in disclosing profitable and unprofitable activities. The management can then decide to take appropriate measures for making unprofitable into profitable activities.

   ii) **Helpful in Comparison of Data:** Cost accounting also provides data for the purpose of making comparisons which enables to measure the efficiency of the business.

iii) **Provide Information on Cost Estimates:** Cost accounting is helpful for making appropriate bids and tender price quotations.

iv) **Helps in Managerial Decision Making:** Cost accounting helps in managerial decision making as it explains costs incurred and profits made for various lines of business.

v) **Helps in Locating Wasteful Activities:** Cost accounting helps in locating wasteful activities and management can take decision to control such wastes, leading to better profits.

vi) **Periodic View of Results:** Cost accounting helps the management in getting periodic view of results, i.e. (Profit or Loss) of operating activities without waiting for the end of the accounting period.

vii) **Provides Reliable Data:** Cost accounting follows established principles and thus provides reliable data.

2) **Advantage to Employee:** Cost accounting helps in reducing costs and increasing profitability with regard to cost incurred and production unit required in different department and processes. It is also helpful for the purpose of introducing incentive compensation system and bonus plans.

3) **Useful for Outsiders:** An efficient cost accounting system is good for outsiders (creditors, investors and bankers) as well. The data provided by cost accounting system helps them in evaluating financial soundness of the organization. Organizations with well-established accounting system are considered to be more efficient. This can help the firm in attracting more investment.

4) **Useful for Government and the Society:** Cost accounting helps organizations in reducing costs and increasing profits. It results in payment of higher taxes which helps the society and economy. Cost control helps in providing goods at lower costs. Cost accounting also helps in making national policies.

3.1.5. Disadvantages of Cost Accounting

Cost accounting has several shortcomings which are as follows:

1) **System is More Complex:** Establishing cost accounting processes is a complex system. It requires various forms and formulas for proper calculations and presentations. This system may be difficult to implement. The employees may also resist its implementation. The complexity of the system may make it difficult for a firm to implement it.

2) **Expensive:** Cost accounting system is not only complex but also involves outflow of substantial amount of funds which provide surety of increased wages for the workaholic employees. The ample of investment and work required in the process of analysis, allocation and absorption of overhead. Many small and medium scale organizations may not afford such expenses. It should be established that the benefits derived from such system are higher than the costs involved. The cost involved is one of the biggest limiting factors for a firm.

3) **Inapplicability of Costing Method and Technique:** Different organizations use different techniques and methods of cost accounting. These methods are based on the nature and requirements of different circumstances. It should be ensured that accounting procedures are in accordance with the requirements of the organization. Due to this reason, a firm needs to design its own accounting system to suit its procedures, making the process more difficult and expensive.

4) **Not Suitable for Small-Scale Units:** Cost accounting systems are generally not suitable for small scale units. For such units, traditional cost accounting methods are more suitable.

5) **Lack of Accuracy:** Cost accounting may not provide accurate information as it uses notional costs such as estimated costs and standard costs.

6) **No Social Accounting:** Cost accounting does not deal with social costs and thus provides one-sided view. It does not take cognizance of social duties of the firm.

7) **Frequent Reconciliation:** Cost accounting may show results different from financial accounting. In such cases, there is need to reconcile both the systems. Such reconciliations are resource intensive and may prove to be a limiting factor.

8) **Duplication of Work:** Cost accounting system is used in conjunction with financial accounting. This may lead to duplicity of efforts. For many firms, financial accounting serves these purposes in an economic manner.

3.1.6. Designing and Installing a Cost Accounting System

The system of cost accountancy is not like a 'Prefabricated Structure', which can be setup in any organisation. A cost accounting model, which is very successful in one business organisation, may not work efficiently (may even be a failure) in another organisation. In fact, each organisation needs to originate its own costing system depending upon a number of factors, like its size, nature of business, product-mix, market conditions, competitive edge, etc. Such a system is required to be developed with extreme caution and before setting up the same, it is essential to carry out the 'Cost Benefit Analysis' of such a system.

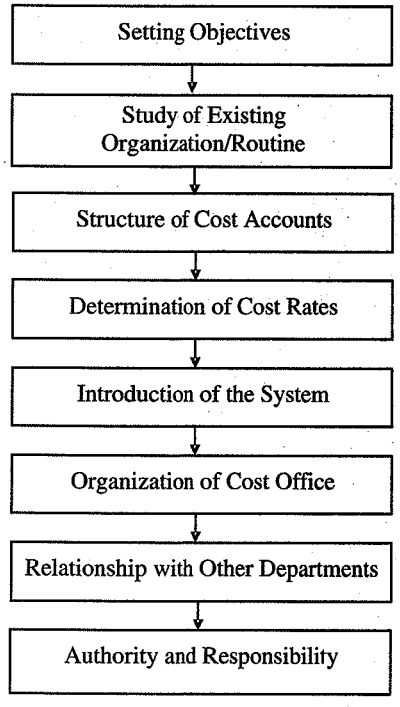
A costing system should be put in place by a business organisation at a time, when its need is experienced and the management holds a view that such a system would facilitate decision making process and the organisation would gain from such a system.

The cost accounting system helps the company in achieving the desired objectives by successfully mentoring the costs of the company. The system combines the relevant costs and allocates them to various costs objects like process, product, manufacturing, etc. by reporting the information of the individual costs. The system should be used and customized according to the company so as to utilize resources efficiently. The system helps in smooth flow of production and determining the profitability of the product.

Steps for Installing a Cost Accounting System

Following steps are necessary for putting a cost accounting system in a business organisation:

1) **Setting Objectives:** The specific goal, which the business organisation expects to achieve, through the system of cost accounting, needs to be spelt out clearly, so that the structure of the system may be erected accordingly. If the goal or objective happens to be only the cost determination, the structure would be a simple one. However, if the goal is manifold, including planning, budgeting, and decision making exercise by the management, then the structure would be more complex and sophisticated.



2) **Study of Existing Organization/ Routine:** Such a study is required to be undertaken in addition to the following:
   i) Nature of the business/operations carried out by the business organisation and the processes involved therein;
   ii) Blueprint of the factory, where the production of goods take place in general and manufacturing department in particular;
   iii) Level of responsibilities/authorities entrusted to different people within the organisation;
   iv) Maintenance of proper record pertaining to the working hours of the labourers, in addition to the calculation and payment of their wages;
   v) Prevailing system of issuing orders for manufacturing;
   vi) The arrangement for treating the wastage generated as a consequence of production of goods; and
   vii) Overheads involved in various categories of expenses like fixed, semi-variables and variables.

3) **Structure of Cost Accounts:** An appropriate cost accounting system needs to be designed and developed in respect of the entire manufacturing process. It is of paramount importance to ensure that such a structure is in synchronisation with the natural production line. Further, development of the structure should also be in tune with the gradual build-up of the costs involved at every stage of manufacturing process; both should be completed simultaneously.

4) **Determination of Cost Rates:** Before the determination of the cost rates, it is necessary to undertake a detailed study of the factory conditions. Decision in respect of the following activities is also required to be taken:
   i) Bifurcation of costs into direct and indirect;
   ii) Indirect costs need to be further classified into production, selling and administration;
   iii) Treatment of waste materials with a view to complying with statutory and regulatory requirements;

iv) Method of pricing issues;
v) Methods relating to the recovery of overheads; and
vi) Calculation of overheads rates.

A comprehensive cost accounting code needs to be finalized with a view to ensuring prompt classification of expenses as (a) source, and (b) cause.

5) **Introduction of the System:** For the successful implementation of the system, full cooperation from the various levels of person is a prerequisite. With this objective in mind, the details about the new system and implications thereof need to be shared with them. Further, the introduction of the new system should be in a gradual manner. Existing system should not be replaced with the new system in an abrupt manner.

6) **Organization of Cost Office:** Location of the cost office should ideally be in the proximity of the factory, which would ensure prompt rectification of various inconsistencies pertaining to the routing out documents / clearing up. Organisational set-up of the cost office should consist of following functional categories according to the responsibilities assigned:

S. No.	Category	Responsibilities
i)	Stores Accounts	a) Maintenance of stores ledger, i.e. posting of receipt of material and issue of stores; and b) Preparation of reports in respect of wastage of materials periodically.
ii)	Labour Accounts	a) Maintenance and evaluation of time sheets and job cards; and b) Preparation of payrolls and labour extracts.
iii)	Cost Accounts	Posting of all the cost accounts.
iv)	Cost Control	Preparation of various reports for the management on the basis of information available from all the above categories.

7) **Relationship with Other Departments:** The costing department should be self-sufficient and be given free hand without interference from any other department. They should be made to report only to the top management. Appropriate records are required to be maintained and suitable reporting system, as per the requirements of the management, should be put in place.

8) **Authority and Responsibility:** Authorities and responsibilities in respect of various functionaries of the costing department should be clearly spelt out without any ambiguity. This would ensure success of the system.

3.1.7. Cost Centre

Cost centres are important departments of a business organisation, which apparently do not make any direct contribution in the profitability, although they do contribute to the cost of running the organisation. **Examples** of cost centres are research and development departments, marketing departments, help desks, customer service/contact centres, etc. It is noteworthy that although the cost centres do not make any direct contribution to the revenue generation of a company, their indirect contribution can't be undermined. Years of hard work and efforts made by the research and development department may one day result in innovations, which may lead to huge revenue generation or heavy cost savings for the organisation. Similarly, customer service department may, over a period of time, help in image building of the company, which may ultimately attract more business for the company.

According to the Institute of ICMA, London, "Cost Centre is defined as a location, person or item of equipment (or group of them) in respect which costs may be ascertained and related to cost units for the purposes of cost control". It is the smallest segment of activity or area of responsibility for which costs are accumulated.

In manufacturing companies, cost centres are basically bifurcated into production and service cost centres. Production cost centres' main function is to carry out the production process. Service cost centres are supplementary functions provided to the production cost centres. **For example**, in a biscuit manufacturing company, making and packing relates to production cost centres and marketing, purchase stores are supplementary activity provided by service cost centres to production cost centres.

Classification of Cost Centres

Cost centres may also be classified as personal and impersonal as under:

1) **Personal Cost Centre:** It comprises of a person or group of persons.

2) **Impersonal Cost Centre:** It comprises of a location or an equipment (or group of location/equipment). Impersonal cost centres may be further divided into following two categories:
   i) **Operation Cost Centre:** A cost centre, where the machines and / or persons carry out similar operations.
   ii) **Process Cost Centre:** A cost centre, where a specific or continuous operation is carried out.

From the functional point of view, the cost centres may be classified into following types:

1) **Production Cost Centres:** Such cost centres consist of the departments, which are directly engaged in manufacturing activities. **For example,** cutting, assembling and finishing departments.

2) **Service Cost Centres:** These cost centres extend services to other departments. They are ancillary in nature and facilitate in manufacturing activities of production cost centres. Their contribution to the process of manufacturing is in an indirect manner. **For example,** personnel, cafeteria, maintenance department, steam plant, power plant, quality control department, etc.

3) **Service-cum-Production Cost Centres:** Such cost centres are also referred to as mixed cost centres. They are basically service cost centres, which may, at times, also undertake some production related work. **For example,** tools department producing dies and jigs as well as servicing tools.

Types of Cost Centres

Various types of cost centres are as follows:

1) **Engineered Costs Centres:** These type of costs include direct material and direct labour costs. These costs are incurred at the design stage. Upon finalization of design, engineered cost is proportional to the output. For engineered cost centres, the main focus is on productivity or efficiency. Cost drivers are identified and relationship between various costs and resources is identified. This helps in bringing more efficiency in operations.

2) **Discretionary Costs Centres:** Discretionary costs are decided in each budget. These costs are independent of output value. The main focus is on cost management and effectiveness. This is done by administering proper surveillance of budgets and measuring the performance.

3) **Administrative and Support Centres:** Administrative centres cover senior management and business unit management. Support centres are responsible to provide services to other centres. These type of units are difficult to manage as it is difficult to evaluate and quantify their functions. Another difficulty is related to the promotion of individual interest, which may hamper the process of attaining organizational goals. Budgets form an important part of evaluating the performance of administrative and support centres.

4) **R&D Centres:** R&D Centres are an important part of any modern organization. These centres may also be difficult to evaluate as it is difficult to quantify the performance of such centres. R&D teams are also generally not very commercially oriented. This department's budget is also highly subjective. Other reason is related to the subjectivity of the reports produced by the management. The expenditures stated in the reports are not objective and thus are not a proper base to measure the effectiveness of R&D operations.

5) **Marketing Centres:** Various costs incurred by these centres include advertising costs, sales promotion costs and training costs.

   While the output of such centres can be measured by checking revenue but the efficiency of various marketing efforts may be difficult to evaluate. Various assumptions involved in designing sales budget may be invalidated by external factors. Thus, the process of evaluating the performance of marketing centres is highly subjective.

3.1.8. Cost Unit

Cost unit may be defined as, "quantity of output for which costs are expressed or ascertained". A company may have more than one cost unit. A cost centre is defined as the organizational sub-unit for which costs are collected. Similarly, a unit of output for which separate cost is ascertained is known as cost unit.

For the purpose of effective analysis, it is important to identify proper cost unit. In certain cases, it may be necessary to select more than one cost unit. It is also important to consider the quality, quantity and other characteristics for the proper evaluation and selection of the cost unit.

According to the Institute of ICMA, London, cost unit is defined as "a unit of quantity of product, service or time in relation to which costs may be ascertained or expressed." It is a unit of quantity in terms of which costs may be computed.

Cost unit is important for expressing "Unit Cost" while cost centres are important for "Cost Collection and Cost Ascertainment".

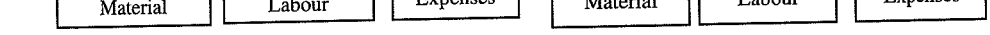
There are several factors which should be considered while selecting cost unit. **For example,** proper cost unit for transportation industry is Ton-Km. Such cost units may be composite or single units. Various factors which govern the selection of cost unit are nature of the business carried out by the firm, various requirements of costing system and information flow.

Following is the list of main cost units used in various industries.

Industry or Product	Cost Unit
Biscuit	Kilogram
Bread	Thousand loaves
Breweries	Barrel
Automobile	Number
Transport	Passenger-Kilometre, Tonne-Kilometre
Power & Electricity	Kilowatt-hour
Steel	Tonne
Brick	Thousand bricks
Coal, Cement	Tonne
Chemical	Litre, gallon, kilogram
Cotton Textile	K.G. of yarn or metre of cloth
Hospital	Patient-day
Hotel	Guest-day, Guest Room, etc
Gas	Cubic-foot or cubic-metre
Cigarettes	Thousand cigarettes

3.1.9. Elements of Cost

The elements of cost are shown as in **figure below**:



1) **Material:** Commodities or substances used for manufacturing products are called materials. Materials can be either Direct or Indirect.
   i) **Direct Material:** This material can be directly identified with the product. Direct materials can also be easily measured and thus charged directly to the product. **For example,** quantity of wood used for making a table can be identified directly to the product.

Following are the main characteristics of direct material:

a) All materials purchased specifically for a particular job, order or process.
b) Subsequent issues of materials from store for a particular job, order or process.
c) All the components are produced or purchased.
d) Material transferred among various processes.
e) Primary packing material such as cardboard boxes.

ii) **Indirect Material:** These materials are used for manufacturing a product but these are not direct materials as these cannot be directly tracked back to the product. These ancillary materials are known as indirect material. Main **examples** of indirect material are consumable stores and threads.

2) **Labour:** Like material, labour can also be direct or indirect.

i) **Direct Labour:** Direct labour includes employees and workers directly involved in the production of the goods. This type of labour can be identified with the goods or services. The compensation paid to direct labour is known as **direct wages**. **For example,** wages paid to a cook in a restaurant is included in the above category.

ii) **Indirect Labour:** Labour which is ancillary to the production of goods or services is known as indirect labour. The wages paid to such labour are known as indirect wages. Such costs cannot be traced back or identified with the product. Salaries paid to supervisors come under this category.

3) **Expenses:** Expenses may also be direct or indirect.

i) **Direct Expenses:** These are expenses which are incurred in the process of manufacturing of goods or rendering a service. It does not include direct wages and direct material costs. Main instances of such costs are Research & Development expenses. Following are some of the **examples** of direct expenses:

a) Hiring expenses for a machinery or an instrument for a specific product.
b) Costs associated with special drawings, layouts or designs.
c) Maintenance cost of specific equipment's.
d) Royalties related with production.

ii) **Indirect Expenses:** The expenses which are not directly identifiable with a particular product or service are categorized under this head. Following are the main **example** of such expenses:

a) Factory or production expenses,
b) Office and administrative expenses, and
c) Selling and distribution expenses.

4) **Overheads:** This category includes cost of indirect material, indirect labour and indirect expenses. Following are the main categories of overheads:

i) **Production or Factory Overhead:** This category shows the aggregate of indirect wages, indirect material costs and indirect expenses incurred for various manufacturing activities. Such overheads start with the supply of raw material and continue till the primary packing of finished goods.

Examples of production overhead are:

a) **Indirect Material Cost**
- Cost incurred for consumable stores and supplies like lubricating oil, etc.
- Cost of printing, postage & stationary used in production department.

b) **Indirect Labour Cost**
- Salary payable to supervisor, works manager and departmental superintendents.
- Contribution to ESI, P.F., leave pay, maternity pay, etc.

c) **Indirect Expenses**
- Rent, rates and taxes of factory building.
- Repairs, insurance and depreciation of factory building, plant and machinery and furniture.
- Factory telephone expenses.
- Lighting, heating and cleaning of factory.

ii) **Administration Overhead:** This category includes indirect material cost, indirect wages and indirect expenses incurred for strategic, control and administration purposes.

Following are the **some examples** of administrative overheads:

a) **Indirect Materials Cost**
- Printing, postage and stationery expenses incurred by administration department.
- Expenses incurred for dusters, brushes, etc., for cleaning.

b) **Indirect Labour Cost**
- Salary payable to managing director, whole time director, general manager, finance manager, accounts manager, secretary, legal manager and other staff working in administration department.
- Remuneration paid to internal and statutory cost and financial auditors, legal advisors.

c) **Indirect Expenses**
- Rent, rates and taxes of office building.
- Repairs, insurance and depreciation of office building, equipment, and furniture.
- Telephone expenses by administration department.
- Lighting, heating and cleaning of Administration office.

iii) **Selling Overhead:** This category shows the aggregate of all indirect material costs, indirect wages and indirect expenses incurred for generating and sustaining demand for product and services. It also includes the costs for fulfilling such demands.

Examples of selling overhead are:

a) **Indirect Materials Cost**
- Cost of printing, postage and stationery used in sales department.
- Cost of catalogues, list prices, etc.

b) **Indirect Labour Cost**
- Salary of sales director, sales managers, sales officers, salesmen and other staff working in sales department.
- Commission to selling agents.

c) **Indirect Expenses**
- Rent, rates and taxes of sales office and showroom.
- Repairs, insurance and depreciation of sales office building, equipment and furniture.
- Telephone expenses of sales office.
- Lighting, heating and cleaning of sales office.
- Advertising.
- Bad Debts.
- Debt collection charges.
- Travelling expenses of salesmen.
- Entertainment expenses on customer.

iv) **Distribution Overhead:** This category includes indirect material cost, indirect wages and indirect expenses incurred in packing finished products for dispatch and making them available to customers.

Examples of distribution overhead are:

a) **Indirect Materials Cost**
- Cost of printing, postage and stationery used by distribution office.
- Cost of secondary packaging of products.
- Cost of materials used in reconditioning of empty containers returned by customers for re-use.

b) **Indirect Labour Cost**
- Salary of staff attached to distribution office like, packers, dispatch staff.
- Salary of driver for distribution vehicle.

c) **Indirect Expenses**
- Rent, rates and taxes of distribution office/godown/ storage/ warehouse.
- Repairs, insurance and depreciation of distribution office building, equipment and furniture, delivery van.
- Telephone expenses of distribution office.
- Lighting, heating and cleaning of distribution office.
- Depreciation, repairs and running expenses of delivery vans.
- Freight and carriage outward.
- Insurance of finished stock in godown.

Example 1: Identify whether following expenses are factory/office and administration/selling and distribution overheads.

i) Show room rent
ii) Travelling expenses
iii) Cleaning charges
iv) General manager's salary
v) Storekeeper's wages
vi) Consumable stores
vii) Labour welfare expenses
viii) Sundry expenses
ix) Legal fees
x) Counting house salary

Solution:

Items	Types of Overheads
Show room rent	Selling and Distribution Overheads
Cleaning charges	Office and Administration Overheads
Storekeeper's wages	Factory Overheads
Labour welfare expenses	Factory Overheads
Legal fees	Office and Administration Overheads
Travelling expenses	Selling and Distribution Overheads
General manager's salary	Office and Administration Overheads
Consumable stores	Factory Overheads
Sundry expenses	Office and Administration Overheads
Counting house salary	Office and Administration Overheads

Example 2: Classify the following items into Factory Overheads, Administration Overheads and Selling and Distribution Overheads.

Loose tools	Income Tax	Postage and Telegram
Rent, Rate and Taxes	Land and Building	Delivery Charges
Indirect Material	Packing Expenses	Director's Fees
Warehouse Rent	Welfare Expenses	Overtime Wages
Advertising	General Expenses	Collection Charges
Property Tax on Building	Carriage Outward	Dividend Paid
Labour Expenses	Bad Debts	Unproductive Wages

Solution:

Items	Types of Overheads
Loose Tools	Factory Overheads
Rent, Rate and Taxes	Administration Overheads
Indirect Material	Factory Overheads
Warehouse Rent	Selling & Distribution Overheads
Advertising	Selling & Distribution Overheads
Property Tax on Building	Factory Overheads
Labour Expenses	Factory Overheads
Land and Building	Factory Overheads
Packing Expenses	Selling & Distribution Overheads
Welfare Expenses	Factory Overheads
General Expenses	Administration Overheads
Carriage Outward	Selling & Distribution Overheads
Bad Debts	Selling & Distribution Overheads
Postage and Telegram	Administration Overheads
Delivery Charges	Selling & Distribution Overheads
Director's Fees	Administration Overheads
Overtime Wages	Factory Overheads
Collection Charges	Selling & Distribution Overheads
Unproductive Wages	Factory Overheads

Note:

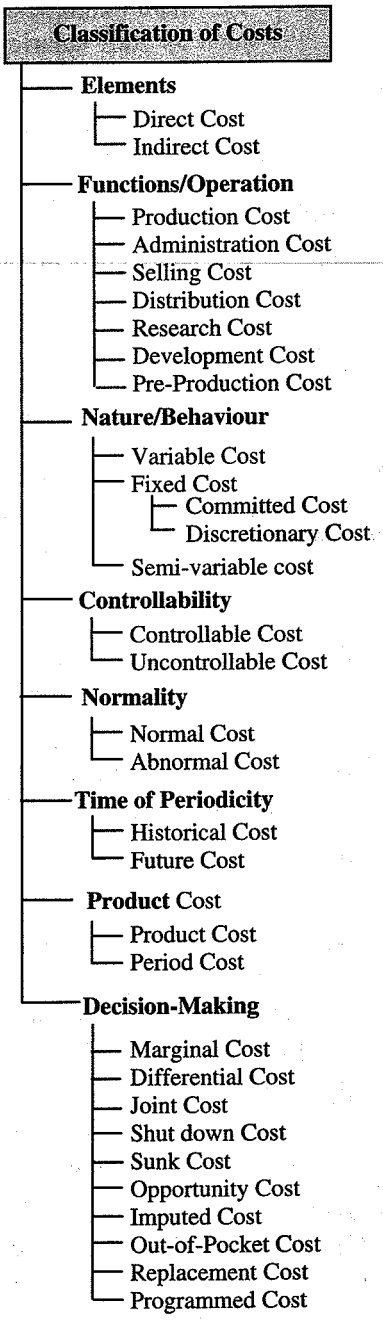
1) Property Tax on Building is assumed to be levied on factory building.
2) Income Tax and Dividend Paid are the financial nature of expenses. They are not classified under any overheads.

3.1.10. Classification and Analysis of Costs

There are several ways in which costs may be classified. Following are the main basis of cost classification:

1) **According to Elements:** The cost may be classified into a) Direct Cost, and (b) Indirect Cost according to elements, viz., Materials, Labour and Expenses.
   i) **Material:** Substances or commodities used for producing a product are called "Materials". Materials consumed may be direct or indirect.
      a) **Direct Material:** The materials which can be directly identified with the product are called "Direct Material". These materials can be easily measured and can be directly charged to the product. **For example,** steel used for producing a container.
      b) **Indirect Material:** These materials are used for production but cannot be directly identified with the product. These expenses are included in overhead category. Main **examples** of this category are consumable stores and stationary.
   ii) **Labour:** Like material, labour can also be divided into direct and indirect category.
      a) **Direct Labour:** Wages and salaries paid to workers and employees directly involved in the process of manufacturing goods is known as "Direct Labour". Wages paid to a machine operator is an **example** of direct labour.
      b) **Indirect Labour:** Wages and salaries payable to workers ancillary to production process are included in "Indirect Labour". The contribution of such workers cannot be directly traced to the product. **For example,** wages for store keepers is included in indirect labour category.
   iii) **Expenses:** Expenses may also be direct or indirect.
      a) **Direct Expenses:** These expenses are incurred in relation to a specific product. This category does not include direct material cost and direct wages. **For example,** cost of secret formula and special drawing are included in this category.
      b) **Indirect Expenses:** These expenses cannot be directly and completely allocated to specific product or cost centre. Office and administrative expenses are included under this category.

2) **According to Functions or Operations:** On the basis of functions, the costs may be classified as under:
   i) **Production Cost:** Production cost include all such costs which are incurred directly for producing goods either manually or through machine. It includes direct material cost, direct labour costs, direct expenses and overhead expenses related to production.
   ii) **Administration Cost:** It include all costs associated with cost of formulating the policy and measures required for running the operations. However, these costs are not directly traceable to production, selling and distribution or research and development activities. Salaries paid to managerial personnel and workers are included in this category.
   iii) **Selling Cost:** This category includes cost of all the activities which aid in the process of selling the goods. It includes sales promotion costs, consumer services costs and selling organizations. These costs help in increasing the revenue of the firm by ensuring customer satisfaction.
   iv) **Distribution Cost:** This cost includes packing expenses and other expenses related to making the product available for distributing and dispatch. It includes salary of dispatch clerks and running expenses of delivery trucks.



v) **Research Cost:** The research cost is incurred for the purpose of carrying out research in order to come up with new manufacturing processes or for improving the quality of the product.

vi) **Development Cost:** This is the cost of converting research into practical mode.

vii) **Pre-Production Cost:** These are the costs incurred in case of a new manufacturing unit. Costs associated with the launch of a new product also come under this category. These costs are treated as deferred revenue expenditure as these are matched against the cost of future production.

3) **According to Nature or Behaviour:** On the basis of nature, following are the main categories of costs:

i) **Variable Cost:** These costs change proportionally in response to change in volume of production. Direct material and direct labour costs are prime **examples** of variable cost and include fuel expenses, electricity expenses, etc.

ii) **Fixed Cost:** These costs do not vary with the change in level of production upto a certain level. These costs do change in the long run but remain fixed in the short term. Fixed costs can be further divided as follows:

a) **Committed Fixed Costs:** These costs arise due to possession of assets such as building and plants. The main examples are depreciation, insurance premium and rent. Basic organizational expenses such as staff salaries are also included in this category. Such costs generally cannot be reduced without having adverse impact on the long term performance of the concern.

b) **Discretionary Fixed Costs:** These costs are incurred on the basis of decisions made by the management. Following are the main features of such costs:

- These costs may arise out of regular decisions about the maximum outflow. Such decisions are generally taken on annual basis.
- These costs do not have clear correlation between inputs and outputs.

For example, advertising, executives training, health care and teaching expenses are included in this category.

iii) **Semi-Variable Cost:** These costs are partly fixed and partly variable. These do not bear linear relationship to the level of production. Telephone and electricity bills are **examples** of such costs.

4) **According to Controllability:** The costs may be divided as follows:

i) **Controllable Cost: CIMA states**, "a cost which can be influenced by the action of a specified member of an undertaking". These costs may be controlled by a specific authority within a specific period of time. These are variable costs and can be affected in a time period which is sufficiently long. It requires proper delegation of authority for effective cost control.

ii) **Uncontrollable Cost:** These costs cannot be controlled at any level of management. These costs are defined as the "cost which cannot be influenced by the action of a specified member of an undertaking".

5) **According to Normality:** On the normality basis the classification of costs is as under:

i) **Normal Cost:** These are the usual costs which are incurred for a given level of output under the given conditions. These costs are charged to costing profit and loss account.

ii) **Abnormal Cost:** These costs occur over and above normal costs and are not usual costs for a given level of production. These are also charged to costing profit and loss account.

6) **According to Time or Periodicity:** These costs may be classified as below:

i) **Historical Cost:** These are the actual costs which are determined after their occurrence. These costs may be calculated using financial records and accounts. However, these figures may not be appropriate for future projections during the times with frequent changes. These costs form the basis of financial accounting.

ii) **Future Cost:** These are the costs which are likely to be incurred in the future. This is an important category of costs as these can be easily controlled by the management. These costs are also important for control, projection, budgeting and appraisal.

7) **According to Association with Product:** On this basis, the costs may be classified as follows:

i) **Product Cost:** These costs can be directly traced back to a product. Product costs are the costs which are assigned to a particular product or a line of products. Under marginal costing, product cost would include variable manufacturing costs while in the case of absorption costing, product costs will include total manufacturing costs.

ii) **Period Cost:** These costs are incurred during a specified period of time and are not assigned to a product. The main examples of such costs are general and administrative expenses. These costs are charged against revenue of that particular period. While fixed factory overheads are treated as period costs under marginal costing, these may be considered product cost under absorption costing.

8) **According to Relevance to Decision Making and Control:** Following are the main categories of costs under this criteria:

i) **Marginal Cost:** It is the amount by which total costs change in response to increase or decrease in volume of output by one unit. In other words, this is the amount which could not be avoided if that particular unit was not produced.

ii) **Differential Cost:** Management is expected to compare the costs of alternatives proposal in order to make decisions. For this purpose, the costs which are common across different alternatives are ignored and only the differential is considered. Thus, a differential cost is the difference in cost among various alternatives. Incremental cost is the increase in cost due to the decision while decremental cost is the decrease in total cost.

iii) **Joint Cost:** These are the costs incurred upto the point till the products cannot be separately identified. Same raw material used for producing two different products is an **example** of joint cost. These costs are apportioned among concerned products.

iv) **Shut-down Costs:** These are the costs which are incurred even if the firm completely stops production. Office rent, insurance premium are **examples** of such costs. These costs are important to make decisions regarding continuing operations during 'off-season'. In general, it is decided to remain open as long as the revenue is sufficient to meet variable costs and a part of fixed costs.

v) **Sunk Cost:** These are the costs which have already been incurred and cannot be recovered. These costs are considered irrelevant for the purpose of decision making. This cost may be defined as, "an expenditure for equipment or productive resources which has no economic relevance to the present decision-making process". It may also be defined as, "the difference between purchase price and the salvage value of the asset". Sunk costs may also be non-incremental costs.

vi) **Opportunity Cost:** These are the economic resources which are required to be given up for accepting one proposal over another. **CIMA states**, "Opportunity cost is the value of a benefit sacrificed in favour of an alternative course of action". **For example**, a firm could sell its land for ₹5,00,000 but decided to retain it for setting up a factory. In such case, ₹5,00,000 is the opportunity cost.

vii) **Imputed/Notional Cost:** These are the costs which are implicit in the product and are not physically recorded in the accounts. These costs are hypothetical in nature and do not require actual cash outflow. Interest on capital is an **example** of such costs as no actual payment is required to be made. Although, no actual payments are made, but an estimation of earnings, had the funds been invested elsewhere is made.

These costs are important for decision making and forms a part of opportunity costs.

viii) **Out of Pocket Cost:** These are the costs which require an actual outflow of cash. It includes historical costs as well as future costs. Most of the expenses fall under this category. These costs are important for managerial decision making process and for determining the profitability of a project. Many fixed costs and most of the variable and semi-variable costs fall under this category.

ix) **Replacement Cost:** These costs need to be incurred for replacing a product with an identical one on the date of valuation. It excludes change in cost due to improvement in the product. This cost is different from the actual acquisition cost of the asset. It represents the outflow required for replacing an asset at present or at some point in future time.

x) **Programmed Cost:** These costs are incurred for long-term survival and have less importance to current processes. These costs are incurred at the discretion of management and can be controlled accordingly. Advertising cost is an **example** of such cost. Sales promotion expenses and research & development expenses also fall under this category.

3.1.11. Relevant Costs

Costs which differ from one option to another are called relevant costs. These costs are affected by a decision and are changed accordingly. Many costs such as differential costs, incremental costs, opportunity costs and out-of-pocket costs are included in this category.

Following are the two main features of relevant costs:

1) Relevant costs are future costs and are yet to be incurred. The costs such as sunk costs, which have already been incurred, are not relevant costs as these cannot be altered by any new decision.
2) Relevant costs are also incremental in nature. These costs may be avoided by taking appropriate decisions. Incremental costs are the changes in a cost among various alternatives.

3.1.11.1. Differential or Incremental Costs

The change in a cost item owing to change in decisions is termed as differential costs. Any cost which remains the same for different projects or alternatives is called irrelevant cost. **For example,** a piece of land may be used for constructing a plaza or mall. The cost of land is irrelevant for the purpose of decision-making as it would be incurred in either case.

For example, a company may decide to sell its products online using a website or through opening a shop. The costs associated with online selling are ₹10 million while the costs for opening a shop are ₹12 million. The difference of ₹2 million is relevant for making decisions. Differential costs are matched against differential revenue for the purpose of making decisions. If in the above, differential cost of ₹2 million is able to generate additional ₹5 million in sale, then it would be beneficial to sell offline than online.

3.1.11.2. Opportunity Costs

When a management chooses a specific course of action, it involves rejecting all other alternatives. The benefits which could have been obtained from the rejected options are known as **opportunity costs.** These costs are not recorded in the account books. However, these costs are important for the purpose of decision-making. Management is often faced with different alternatives, all with their pros and cons.

For example, Product A is expected to bring net profit of ₹50000 while Product B is likely to bring ₹40,000 in net profit. In such case, product A should be chosen. The net profit of ₹40,000 which has been forgone on account of rejecting product B is known as opportunity cost.

3.1.11.3. Out-of-Pocket Costs

These costs are the relevant expenses which are paid in cash. Such costs have direct impact on managerial decisions. This type of costs is associated with certain decisions and change with difference in decisions. **For example,** if a company needs to make decision between using own vehicles or public transport for delivering goods, then depreciated value of owned trucks is irrelevant as it is a sunk cost. However, other costs such as driver's wages and fuel costs are relevant as these are out of pocket cost.

3.1.12. Irrelevant Costs

These are the costs which do not affect the process of decision-making. These costs are not altered by any change in the alternative chosen. These costs are ignored while taking a decision. Sunk cost is a prime example of irrelevant cost. **For example,** a firm spent ₹5,00,000 on buying a machine. The machine has current book value of ₹3,00,000. It cannot be sold in the market on account of obsolescence. However, if it is modified then it can be sold for ₹1,00,000. The costs associated with modification are ₹40,000. In such scenario, cost of modification is a relevant expenditure. The expected sales value of ₹1,00,000 is also relevant. However, cost and book value of the machine are irrelevant costs as the actual cost is a sunk cost and will not change irrespective of decision made.

3.1.13. Sunk Costs

These are historical costs which have already been incurred. The costs were incurred for implementing past decisions and future decisions will have no impact on them. Purchase of fixed assets like building, machine, etc. are examples of such costs. These costs are irrelevant for the purpose of decision-making as future decisions do not have any impact on them.

A firm cannot change its past costs with any future decision. However, the firm may decide to sell their assets for recouping some of the investment made in them. Alternatively, they may decide to continue holding the assets. In such cases, the revenues derived will be matched against the costs incurred for them. Such costs which cannot be changed are known as sunk costs.

For example, Company A bought a machine for ₹5,00,000 with a working life of 10 years and nil scrap value. However, after a year, the management decides that the machine is not appropriate for their use. The machine is likely to result in ₹20,000 in operating expenses each year for next 4 years. It may be sold for ₹50,000 immediately.

In the above scenario, the relevant factors are ₹20,000 per annum in operating expenses and immediate sales price of ₹50,000. The money spent on purchasing the machine is not relevant as it is a sunk cost. The firm should make the decision based on relevant costs of ₹20,000 per annum for next 4 years and the immediate sales price. In this case, the expected costs are ₹80,000 for the lifespan of the machine while it can obtain ₹50,000 by selling the machine.

3.2. PREPARATION OF COST SHEET

3.2.1. Meaning of Cost Sheet

This is a statement to show the output during a particular period of time. It also shows the classification of costs. The data for the purpose of preparing this statement is collected from various sources. This sheet is useful for the purpose of analysing total cost of production and cost of sales. This statement shows per unit cost structure for various levels of production.

It is important to identify the stage of production and relevant price. Cost sheet is generally prepared in the case of manufacturing the single product. Cost sheet may be prepared for different time periods ranging from weekly to annually. It bifurcates the costs into various categories such as prime costs, cost of production and total costs. The main classifications used are prime cost, cost of production, works cost and cost of goods sold.

3.2.2. Types of Cost Sheet

Historical and estimated data is used for the purpose of preparing cost sheet:

1) **Historical Cost Sheet:** Actual costs are used for preparing this type of cost sheet. The statement showing costs after it has been incurred is known as historical cost sheet.
2) **Estimated Cost Sheet:** Estimated costs are used for preparing this cost sheet. It is prepared before the commencement of production. Such cost sheets are useful for various purposes such as bidding for tender. It is also used for estimating various components of cost. The estimates are made using present conditions and likely future conditions.

3.2.3. Proforma of Cost Sheet

Particulars	Total Cost (₹)	Cost Per Unit (₹)
Opening Stock of Raw Material		
Add: Purchase of Raw Material		
Carriage Inward		
Other Expenses on Purchase		
Less: Closing Stock of Raw Material		
Direct Material Consumed	–	
Add: Direct Labour/Wages		
Direct Expenses		
Direct Cost/Prime Cost	–	
Add: Factory/Manufacturing/Production/Work Overhead:		
Factory Rent		
Power and Fuel		
Haulage Charges		

Indirect Material		
Supervisor Salary		
Depreciation on Machinery		
Oil and Water Charges, etc.		
Factory Cost		
Add: Opening Work-in-Progress		
Less: Closing Work-in-Progress		
Factory/Manufacturing/Production/Work Overhead Cost		
Add: Office and Administration Overheads:		
Audit Fees		
Director Fees		
Legal Charges		
Depreciation on Furniture		
General expenses		
Postage, Telephone, Printing and Stationary		
Donation, Bank Charges, Insur ance of Office, etc.		
Office Cost/Cost of Production	–	
Add: Opening Stock of Finished Goods		
Less: Closing Stock of Finished Goods		
Cost of Goods Sold	–	
Add: Selling and Distribution Overheads:		
Trade Discount, Cash Discount Allowed, Brokerage, Commission, Sample Expenses, Branch Expenses, Delivery Van Expenses, Advertisement, Bad Debts, Free Gifts, After-Sale Service, Expenses of Catalogue and Price List, Driver Salary, etc.		
Total Cost	–	
Profit/Loss	–	
Sales	–	

3.2.4. Items Excluded from Cost Sheet

The following items are of financial nature and thus are not included while preparing a cost sheet:

1) Donation.
2) Cash discount.
3) Interest paid.
4) Income tax paid.
5) Dividend paid.
6) Preliminary expenses written-off.
7) Goodwill written-off.
8) Profit or loss on sale of fixed assets.
9) Transfer to reserves.
10) Provisions.

Example 3: Calculate material consumed, prime cost and works cost – Opening stock of raw material ₹40,000, closing stock of raw materials ₹50,000, purchase of raw material ₹4,00,000, Direct labour ₹3,00,000, Indirect labour ₹10,000, carriage outward ₹20,000, lubricant ₹10,000, Factory rent ₹60,000.

Solution:

Cost Sheet

Particulars		₹
Opening Stock of Raw Material	40,000	
Add: Purchase of Raw Material	4,00,000	
Less: Closing Stock of Raw Material	50,000	
Direct Material Consumed		3,90,000
Add: Direct Labour		3,00,000
Prime Cost		**6,90,000**
Add: Factory Overheads:		
Factory Rent		60,000
Lubricant		10,000
Indirect Labour		10,000
Works Cost		**7,70,000**

Example 4: Mr. A furnishes the following data relating to the manufacture of a Standard Product for the month of January 2019.

Particulars	₹
Materials	90,000
Direct Wages	60,000
Depreciation of Machinery	11,500
Power and Consumable Stores	12,000
Indirect Wages at Factory	15,000
Lighting of Factory	5,500
Cost of Rectification of Defective Work	3,000
Sale of Scrap	2,000
Office and Selling Overheads	39,000
Selling Price	3,16,000

Prepare the cost sheet.

Solution:

Cost Sheet
(For the Month of January 2019)

Particulars	₹
Materials	90,000
Direct Wages	60,000
Prime Cost	**1,50,000**
Add: Factory Overheads:	
Depreciation of Machinery	11,500
Power and Consumable Stores	12,000
Indirect Wages at Factory	15,000
Lighting of Factory	5,500
Cost of Rectification of Defective Work	3,000
	1,97,000
Less: Sale of Scrap	2,000
Work Cost/Production Cost	**1,95,000**
Add: Office and Selling Overheads	39,000
Cost of Sales	**2,34,000**
Add: Profit (Balancing figure)	**82,000**
Sales	**3,16,000**

Example 5: From the following information you are required to calculate prime cost, works cost, total cost of production, cost of sales and profit.

Particulars	₹	Particulars	₹
Direct Labour	1,00,000	Direct Expenses	1,50,000
Factory Overheads	2,00,000	Office Overheads	1,80,000
Direct Material	5,00,000	Sales	15,00,000
Selling and Distribution Overheads	50,000		

Solution:

Cost Sheet

Particulars	Total Cost (₹)
Direct Material	5,00,000
Direct Labour	1,00,000
Direct Expenses	1,50,000
Prime Cost	**7,50,000**
Add: Factory Overheads	2,00,000
Work Cost	**9,50,000**
Add: Office Overheads	1,80,000
Total Cost of Production	**11,30,000**
Add: Selling and Distribution Overheads	50,000
Cost of Sales	**11,80,000**
Add: Profit (Balancing figure)	3,20,000
Sales	**15,00,000**

Example 6: Prepare a cost sheet showing prime cost, works cost, total cost of production, cost of sales and profit from the following information related to Chennai Pvt. Ltd., Chennai.

Particulars	₹	Particulars	₹
Sales	5,00,000	Direct Wages	50,000
Direct Expenses	50,000	Factory Rent	20,000
Storekeeper Salary	10,000	Salary	1,00,000
Direct Material	2,00,000	Director's Fees	30,000
Showroom Rent	7,000		

Solution:

Cost Sheet

Particulars	Total Cost (₹)
Direct Material	2,00,000
Direct Wages	50,000
Direct Expenses	50,000
Prime Cost	**3,00,000**
Add: Factory Overheads:	
Factory Rent	20,000
Storekeeper Salary	10,000
Work Cost	**3,30,000**
Add: Office and Administration Overheads:	
Director's Fees	30,000
Salary	1,00,000
Total Cost of Production	**4,60,000**
Add: Selling and Distribution Overheads:	
Showroom Rent	7,000
Cost of Sales	**4,67,000**
Add: Profit (Balancing figure)	33,000
Sales	**5,00,000**

Example 7: From the following prepare cost sheet.

Particulars	₹
Sales	5,00,000
Direct Wages	1,10,000
Factory Rent	30,000
Direct Material	2,90,000
Showroom Rent	15,000
Direct Expenses	10,000
Printing and Stationery	5,000
Audit Fees	15,000

Solution:

Cost Sheet

Particulars	₹
Direct Material	2,90,000
Direct Wages	1,10,000
Direct Expenses	10,000
Prime Cost	**4,10,000**
Add: Factory Overheads:	
Factory Rent	30,000
Production Overhead/Cost	**4,40,000**
Add: Office and Administration Overheads:	
Printing and Stationery	5,000
Audit Fees	15,000
Total Cost of Production	**4,60,000**
Add: Selling and Distribution Overheads:	
Showroom Rent	15,000
Cost of Sales	**4,75,000**
Add: Profit (Balancing figure)	25,000
Sales	**5,00,000**

Example 8: From the following information you are required to calculate prime cost, works cost, total cost of production, cost of sales and profit.

Particulars	₹
Sales	1,00,000
Direct Material	40,000
Direct Expenses	2,000
Factory Overheads	7,000
Direct Wages	10,000
Selling and Distribution Overheads	2,000
Administrative Overheads	5,000

Solution:

Cost Sheet

Particulars	₹
Direct Material	40,000
Direct Wages	10,000
Direct Expenses	2,000
Prime Cost	**52,000**
Add: Factory/Work Overheads:	
Factory Overheads	7,000
Work Cost	**59,000**
Add: Official and Administrative Overheads:	
Administrative Overhead	5,000
Total Cost of Production	**64,000**
Add: Selling and Distribution Overheads	2,000
Cost of Sales	**66,000**
Add: Profit (Balancing figure)	34,000
Sales	**1,00,000**

Example 9: A factory produces a standard product. The following information is given to you from which you are required to prepare a cost sheet for January 2019:

Particulars	₹
Raw materials consumed	2,91,000
Direct wages	1,29,000
Other direct expenses	81,000
Factory overheads	80% of direct wages
Office overheads	10% of works cost
Selling & Distribution expenses	₹20 per unit sold
Units produced and sold during the month	10,000

Find the selling price per unit when the profit earned is at 20% on selling price.

Solution:

Cost Sheet

(For the Period of January, 2019)

Particulars	Total Cost (10,000 Units) (₹)	Per Unit (₹)
Raw Materials Consumed	2,91,000	29.10
Direct Wages	1,29,000	12.90
Other Direct Expenses	81,000	8.10
Prime Cost	**5,01,000**	**50.10**
Add: Factory Overheads:		
80% of Direct Wages	1,03,200	10.32
Work Cost	**6,04,200**	**60.42**
Add: Office and Administration Overheads:		
10% of Works Cost	60,420	6.04
Cost of Production	**6,64,620**	**66.46**
Add: Selling and Distribution Overheads:		
Selling and Distribution Expenses ₹20 per Unit (10,000× ₹20 p.u)	2,00,000	20.00
Cost of Sales	**8,64,620**	**86.46**
Add: Profit (20% on Selling Price)	2,16,155	21.61
Sales	**10,80,775**	**108.07**

Example 10: Mr. R & Sons furnishes the following data relating to the manufacture of a standard product during the month of April 2019.

Direct material consumed	₹20,000
Direct labour charges	₹10,000
Machine hour worked	₹1,000
Machine hour rate	₹6
Administration overheads	20% on works cost
Selling overhead	₹0.50 per unit
Unit produced	₹18,000
Units sold	17,000 at ₹4 per unit

You are required to prepare a cost sheet from the above, showing cost per unit sold and profit for the period.

Solution:

Cost Sheet
(For the Month of April 2019)

Particulars	Total Cost (18,000 units) (₹)	Cost Per Unit (₹)
Direct Raw Material Consumed	20,000	1.11
Direct Labour Charges	10,000	0.56
Prime Cost	**30,000**	**1.67**
Add: Factory Overheads (Machine hours × Machine-hour rate, i.e., 1000 × ₹6)	6,000	0.33
Factory Cost/Works Cost	**36,000**	**2.00**
Add: Administrative Overheads (20% on Works Cost)	7,200	0.40
Cost of Production	**43,200**	**2.40**
Less: Closing Stock of Finished Goods (1,000×2.40)	2,400	-
Cost of Goods Sold	**40,800**	**2.40**
Add: Selling Overheads (17,000×0.50)	8,500	0.50
Cost of Sales	**49,300**	**2.90**
Add: Profit (Balancing figure)	18,700	1.10
Sales	**68,000**	**4.00**

Working Note:

Closing Stock = Unit Produced – Unit Sold × Per Unit of Cost of Production
= 18,000 – 17,000 × 2.40 = ₹2,400

Example 11: Prepare Cost Sheet from the following information:

Particulars		₹
Raw materials consumed		40,000
Wages paid to labourers		10,000
Directly chargeable expenses		2,000
Oil and Waste		100
Wages of Foreman		1,000
Storekeepers Wages		500
Electric Power		200
Lighting:		
Factory	500	
Office	200	700
Rent:		
Factory	2,000	
Office	1,000	3,000
Repairs and Renewals:		
Factory Plant	500	
Machinery	1,000	
Office premises	200	1,700
Depreciation:		
Office Premises	500	
Plant and Machinery	200	700
Consumable stores		1,000

Manager's salary	2,000
Director's fees	500
Office printing and Stationery	200
Telephone charges	50
Postage and Telegrams	100
Salesmen's Commission and Salary	500
Travelling Expenses	200
Advertising	500
Warehouse charges	200
Carriage outward	150

Solution:

Cost Sheet

Particulars			Total Cost (₹)
Raw Material Consumed		40,000	
Wages Paid to Labours		10,000	
Directly Chargeable Expenses		2,000	
Prime Cost			**52,000**
Add: Factory Overheads:			
Indirect Material:			
Oil and Waste	100		
Consumable Stores	1,000	1,100	
Indirect Labour:			
Wages of Foreman	1,000		
Storekeepers Wages	500	1,500	
Indirect Expenses:			
Electric Power	200		
Factory Lighting	500		
Factory Rent	2,000	2,700	
Repairs and Renewals:			
Factory Plant	500		
Machinery	1,000	1,500	
Depreciation:			
Plant and Machinery		200	7,000
Factory or Work Cost			**59,000**
Add: Office and Administrative Overheads:			
Indirect Materials:			
Office Printing and Stationery		200	
Indirect Labour:			
Manager's Salary	2,000		
Director's Fees	500	2,500	
Indirect Expenses:			
Office Lighting	200		
Office Rent	1,000		
Telephone Charges	50		
Postage and Telegrams	100	1,350	
Repairs and Renewals:			
Office Premises		200	
Depreciation:			
Office Premises		500	4,750
Cost of Production			**63,750**
Add: Selling and Distribution Overheads:			
Indirect Labour:			
Salesmen's Commission and Salary		500	
Indirect Expenses:			
Travelling Expenses	200		
Advertising	500		
Warehouse Charges	200		
Carriage Outward	150	1,050	1,550
Cost of Sales			**65,300**

3.2.5. Treatment of Stock

Various type of stocks such as raw material, work-in-progress and finished goods.

1) **Stock of Raw Materials:** It requires information about the opening stock, purchase and closing stock of raw material. Following format may be used for calculating raw material consumed.

Particulars	₹
Opening Stock of Raw Materials	xxx
Add: Purchases of Raw Materials	xxx
	xxx
Less: Closing Stock of Raw Materials	xxx
Cost of Raw Materials Consumed	xxx

2) **Stock of Work-in-Progress:** Work-in-Progress are the units which have been partly worked on but are not finished yet. Work-in-Progress may be valued at prime cost or works cost basis. However, in most of the cases, works cost basis is preferred. Following format is used for the purpose of adjusting opening and closing stock if it has been valued at works or factory costs.

Particulars	₹
Prime Cost	
Add: Factory Overheads Incurred	xxx
Add: Work-in-Progress (Beginning)	xxx
	xxx
Less: Work-in-Progress (Closing)	xxx
Factory or Manufacturing or Works Cost	xxx

3) **Stock of Finished Goods:** Cost of goods sold may be calculated as below if opening and closing stock of finished goods is provided:

Particulars	₹
Cost of Production	xxx
Add: Opening Stock of Finished Goods	xxx
	xxx
Less: Closing Stock of Finished Goods	xxx
Cost of Goods Sold	xxx

Example 12: In a factory 20,000 units of product 'A' were manufactured in the month of March 2018. From the following figures obtained from the costing records. Prepare a cost sheet showing cost per unit.

Particulars	₹
Opening stock	5,000
Raw materials	55,000
Closing stock of finished goods	1,000
Closing stock of raw materials	10,000
Direct wages	25,000
Factory overhead	40,000
Office overhead	20,000
Material returned to seller	4,000

Solution:

Cost Sheet
(For March 2018)

Particulars		Total Cost (20,000 Units) (₹)	Cost per Unit (₹)
Opening Stock	5,000		
Add: Purchases of Raw Materials	55,000		
	60,000		
Less: Closing Stock of Raw Material	10,000		
	50,000		
Add: Direct Wages	25,000		
Prime Cost		**75,000**	**3.75**
Add: Factory Overhead		40,000	2.00
Factory Cost/Works Cost		**1,15,000**	**5.75**
Add: Office Overhead		20,000	1.00
Cost of Production		**1,35,000**	**6.75**
Less: Closing Stock of Finished Goods		1,000	0.05
Cost of Goods Sold		**1,34,000**	**6.7**
Less: Material Returned to Seller		4,000	0.2
Cost of Sales		**1,30,000**	**6.5**

Example 13: Following information is collected from the costing records of a factory. Prepare Statement of Cost:

Particulars	₹
Sales	4,29,000
Direct Wages	72,600
Indirect Wages	5,940
Rent of Office Building	11,880
Rent of Factory Building	19,800
Depreciation on Furniture	7,960
Depreciation on Machinery	6,600
Carriage Inward	3,360
Carriage Outward	1,320
Advertisement Expenses	7,920
Salary of Office Staff	19,600
Sale Commission	9,200
Motive Power	11,550
Dividend Paid	40,000
Direct Expenses	5,610
Work in Progress (1-1-2019)	26,730
Work in Progress (31-3-2019)	14,520
Stock of Raw Material (1-1-2019)	52,800
Stock of Raw Material (31-3-2019)	23,100
Stock of Finished Goods (1-1-2019)	14,850
Stock of Finished Goods (31-3-2019)	19,140
Purchase of Raw Material (per month)	99,400

Solution:

Statement of Cost

Particulars		Total Cost (₹)
Direct Material		
Add: Opening Stock of Raw Material		52,800
Purchase of Raw Material (99,400 × 3 month)		2,98,200
Carriage Inward		3,360
Less: Closing Stock of Raw Material		23,100
Direct Material Consumed		**3,31,260**
Add: Direct Wages		72,600
Direct Expenses		5,610
Prime Cost		**4,09,470**
Add: Work/Factory Overhead:		
Indirect Wages	5,940	
Rent of Factory Building	19,800	
Depreciation on Machinery	6,600	
Motive Power	11,550	
Factory Cost	**43,890**	
Add: Opening Stock of Work-in-Progress	26,730	
Less: Closing Stock of Work-in-Progress	14,520	56,100
Factory Overhead Cost/Work Cost		**4,65,570**
Add: Office and Administration Overheads:		
Depreciation on Furniture		7,960
Rent of Office Building		11,880
Salary of Office Staff		19,600
Office Cost/Cost of Production		**5,05,010**
Add: Opening Stock of Finished Goods		14,850
Less: Closing Stock of Finished Goods		19,140
Cost of Goods Sold		**5,00,720**
Add: Selling and Distribution Overheads:		
Carriage Outward		1,320
Advertisement Expenses		7,920
Sale Commission		9,200
Cost of Sales		**5,19,160**
Less: Loss (Balancing Figure)		**90,160**
Sales		**4,29,000**

Note: Dividend paid is excluded from cost sheet.

Example 14: The following figures are extracted from the cost books of Ashoka Company Ltd. for the year ended 31st March, 2019. Prepare a cost statement.

Particulars	₹
Stock on 1st April, 2018:	
Raw materials	72,000
Work-in-progress	40,000
Finished goods (1,000 units)	40,000
Stock on 31st March, 2019:	
Raw materials	55,600
Work-in-progress	50,000
Finished goods (2,000 units)	?
Purchase of raw materials	3,10,000
Direct wages	2,30,000
Indirect wages	20,000
Rent, Rates and Insurance (2/3 factory, 1/3 office)	66,000
Carriage Inwards	8,600
Salaries (2/4 factory, 1/4 office, 1/4 selling department)	80,000
General Expenses	6,250
Manager's Salary (2/3 factory, 1/3 office)	24,000
Carriage Outwards	8,000
Electricity (50% factory, 25% office, 25% selling department)	25,000
Sales	9,75,000

Production for the period is 16,000 units, selling and distribution expenses ₹4 per unit sold.

Solution:

Cost Sheet
(For the year ending 31st March, 2019)

Particulars	Total Cost (₹)
Opening Stock of Raw Material	72,000
Add: Purchase of Raw Material	3,10,000
Carriage Inwards	8,600
Less: Closing Stock of Raw Material	55,600
Direct Material Consumed	**3,35,000**
Add: Direct Wages	2,30,000
Prime Cost	**5,65,000**
Add: Factory Overheads:	
Indirect Wages	20,000
Rent, Rates and Insurance (2/3)	44,000
Salaries (2/4)	40,000
Manager's Salary (2/3)	16,000
Electricity (50%)	12,500
Factory Cost	**6,97,500**
Add: Opening Stock of Work-in-Progress	40,000
Less: Closing Stock of Work-in-Progress	50,000
Factory Overhead Cost/Work Cost	**6,87,500**
Add: Office and Administration Overhead:	
Rent, Rates, and Insurance (1/3)	22,000
Salaries (1/4)	20,000
General Expenses	6,250
Manager's Salary (1/3)	8,000
Electricity (25%)	6,250
Office Cost/Cost of Production	**7,50,000**
Add: Opening Stock of Finished Goods	40,000
Less: Closing Stock of Finished Goods	88,235
Cost of Goods Sold	**7,01,765**
Add: Selling and Distribution Overheads:	
Carriage Outward	8,000
Selling and Distribution Expenses (16,000 × 4)	64,000
Salaries (1/4)	20,000
Electricity (25%)	6,250
Total Cost	**8,00,015**
Add: Profit (Balancing figure)	1,74,985
Sales	**9,75,000**

Working Note:

$$\text{Closing Stock of Finished Goods} = \frac{\text{Cost of Production}}{\text{Total Units}} \times \text{Units of Closing Stock of Finished Goods}$$

$$= \frac{7,50,000}{18,000 - 1,000} \times 2,000 = ₹88,235$$

Units Produced = Units Sold + Closing Stock of Finished Goods – Opening Stock of Finished Goods
= 16,000 + 2,000 – 1,000
= 17,000 units

3.3. EXERCISE

3.3.1. Very Short Answer Type Questions

1) Define cost.
2) What do you mean by cost accounting?
3) What is cost control?
4) What is research cost?
5) What do you understand by cost ascertainment?

3.3.2. Short Answer Type Questions

1) Discuss the relevant and irrelevant costs.
2) What are the elements of costs?
3) Discuss the relevant and irrelevant costs.
4) What is differential and opportunity costs?
5) Explain sunk costs with suitable example.
6) Explain in brief classification of cost as per elements.
7) What are the types of cost sheet?
8) List few items which are excluded from a cost sheet.
9) Write short note:
   i) Cost Centre
   ii) Cost Unit

3.3.3. Long Answer Type Questions

1) What is cost accounting? Discuss its advantages and disadvantages in detail?
2) State in brief objectives of cost accounting.
3) "A good system of costing serve as a means of control over expenditure and helps to secure economy in manufacture." Discuss.
4) Explain the various types or classifications of cost?
5) What is cost sheet? What is the purpose of a cost sheet?
6) Prepare a specimen cost sheet while use your imaginary figures.

3.3.4. Practical Questions

1) The pen manufacturing company discloses the following information for the period ended 31st March 2019.

Particulars		₹
Stock on 1st April 2018		28,000
Work in Progress:		
Materials	10,000	
Labour	3,600	
Expenses	1,600	15,200
Stock on 31st March 2019		36,800
Raw Material working in Progress:		
Material	7,600	
Labour	3,000	
Expenses	1,040	11,640
Purchase of raw materials		88,000
Direct Labour		30,000
Duty on raw material Purchase		3,120
Carriage inward		4,000

Find the cost of production for the year ending 31st March 2019.

[Ans: Cost of Production is ₹1,19,880]

2) Prepare a cost sheet for January 2019 from the information provided for a product.

Particulars	₹
Raw Materials Consumed	91,000
Direct Wages	29,000
Other Direct Expenses	11,000
Factory Overheads	80% of Direct Wages
Office Overheads	10% of Works Cost
Selling and Distribution Expenses	₹2 per Unit Sold
Units Produced and Sold during the Month	10,000

There was no stock or work-in-progress either at the beginning or at the end of the period.

Find the selling price per unit on the basis that profit mark-up is uniformly made to yield a profit of 20% of the selling price.

[Ans: Cost of Sales = ₹1,89,620; Profit = ₹47,405**]**

3) Prepare Cost Sheet for the Period 30-6-2018:

Particulars	₹
Raw Materials Consumed	25,000
Wages	10,000
Direct Expenses	1,250
Consumable Stores	250
Supervisor's Wages	1,000
Wages paid to Shop-helper	300
Electric Power (Factory)	400
Electric Power (Office)	250
Rent (Factory)	2,500
Rent (Office)	1,000
Repairs and Renewals on:	
Plant and Machinery	2,500
Renovation of Office Building	500
Depreciation on Plant and Machinery	250
Depreciation on Office Building	100
Manager's Salary	1,500
Telephone Charges	100
Printing and Stationery	200
Postage and Telegram	75
Director's Fees	400
Advertising	400
Travelling Expenses	150
Salesman's Salary and Commission	500
Warehouse Rent	450
Delivery Van Expense	500
	49,575

[Ans: Cost of Sales is ₹49,575**]**

4) The following information has been obtained from the records of Ajay Co. Ltd., Pune for the year ended 31-3-2019, prepare cost sheet.

Particulars	₹
Stock on 1-4-2018:	
– Raw Materials	40,000
– Work-in-Progress	10,000
– Finished Goods	1,00,000

Stock on 31-3-2019:	
– Raw Materials	50,000
– Work-in-Progress	14,000
– Finished Goods	1,50,000
Purchases of Raw Materials	2,60,000
Productive Wages	1,26,000
Discount on Issue of Debentures	16,000
Freight Inward	30,000
Factory on Cost	64,000
Direct Chargeable Expenses	4,000
Defective Raw Materials Returned	10,000
Administrative Overheads	40,000
Value of Sales	6,00,000
Selling and Distribution Overheads	30,000

[Ans: Cost of Sales is ₹6,00,000]

5) A company produces a standard product and provides you the following information for the year ended 31st March 2019:

Particulars	₹
Opening Stock	10,000
Purchases	85,000
Closing Stock	4,000
Direct Wages	20,000
Other Direct Expenses	10,000

Additional Information:

i) Factory overhead 100% of direct wages.
ii) Office overhead 10% of works cost.
iii) Selling expenses ₹2 per unit.
iv) Finished Goods:
Opening Stock 1000 units (₹16,000)
Produced during the year 10,000 units
Closing Stock 2000 units

Prepare cost sheet for the year ending 31st March, 2019.

[Ans: Cost of Sales is ₹1,58,080]

UNIT 4

Short Term Business Decision Techniques

4.1. MARGINAL COSTING

4.1.1. Meaning and Definition of Marginal Costing

The term **'marginal cost'** is derived from an important economic concept of **'margin'**. However, economic concept of 'margin' is different from accounting concept. As an economic concept, marginal cost is the expenses incurred for producing an additional unit of a product, while accounting concept of margin cost implies the total cost by aggregating prime cost and variable cost. Thus, it includes all costs except fixed cost.

Marginal costing is not a costing method like process costing or job costing. It is a distinct technique and is primarily used for the purpose of managerial decision-making. Marginal costing may be used for measuring the profitability of various product lines or cost centres. This technique is helpful in bifurcating costs into distinct categories such as "variable costs" and "fixed costs". Variable cost constitutes marginal cost of the product.

The **Institute of Cost and Management Accountants, London,** has defined Marginal Costing as "The ascertainment of variable costs and of the effect on profit of changes in volume or type of output by differentiating between fixed costs and variable costs".

'Variable Costing' is another term used for **Marginal Costing**. It denotes the amount by which total cost change when there is an increase or decrease in the production volume by one unit. It equates to the total variable cost of a unit.

4.1.2. Principles of Marginal Costing

The followings are the principles of marginal costing:

1) In the course of time, fixed asset will be equal to the volume of sales and production (if the level of activity is under the relevant range). In case of the selling of an extra products and services the following will happens:
   i) Revenue increases with the increase in the value of sold product.
   ii) Costs increases by per unit increase in variable cost.
   iii) Profit increases by the increase in the additional items with the amount of contribution gained.
2) The profits decreases with the amount of contribution gained from an item when the sales volume decreases by one item. Thus, the amount of profit is totally depends on the total contribution.
3) Since the fixed is based on the time period and will not affects by increase and decrease in volume of sales therefore the amount of profit should be based on an examining of total contribution. So it is not appropriate to charge units of sale with a share of fixed costs.
4) When the manufacturer produces a unit of product, it incurred additional cost i.e., the variable production costs. Fixed cost are not influenced with any additional increase in unit of production or increase in the total output.

4.1.3. Scope of Marginal Costing

The wide scope of marginal costing can be explained in the following points:

1) **Profit Forecasting:** Marginal costing has a wide role in forecasting the future profits of the company. Also, the percentage of profit in future years can be ascertained. Hence, a flexible budget can be prepared by using this information.

2) **Price Determination:** Company decides the prices of their products after analysing various critical factors. One of the critical factors in price determination is marginal costing. For a company, it is essential to alter their product prices in order to maintain a competition in the market. Marginal costing is thus important in analysing the necessary changes in prices of the company's products and services.
3) **Facilitates Decision-Making:** Marginal costing helps in executing cost-volume-profit analysis that facilitates in decision-making for the company. Several important decisions related to production, volume, manufacturing, research are taken by the company in which the cost-volume-profit analysis is required.
4) **Profit Planning:** The company plans the profit when the price determination and profit forecasting is done. The planning for a particular profit is done with reference to the marginal costing and to the cost-volume-profit analysis. Since, cost-volume-profit analysis supports in decision-making, it also assists in profit planning:
   i) The analysis assists in the estimation of income at a specific level of sales.
   ii) The change in the level of profit due to the change in the sales volume can also be ascertained.
   iii) The analysis assists in the execution of the profit planning, as it provides the information related to inter-dependence of cost, profit and volume.

4.1.4. Advantages of Marginal Costing

Following are the main advantages of marginal costing:

1) **Simple to Use and Understand:** Marginal costing technique is simple and convenient to use. Since this technique excludes fixed costs and uses only variable costs, the cost statements are easy to understand and implement.
2) **Avoids Under-Absorption of Overheads:** This technique does not use the process of allocation, absorption or apportionment of fixed costs and thus reduces the issues related to under-absorption of fixed cost.
3) **Aids in Production Planning:** Variable costs per unit are constant in nature and remain same regardless of the level of activity. The use of these costs simplifies the process of production management.
4) **No Over-Valuation of Stocks:** Marginal costing does not carry forward fixed costs in the form of inventory valuation and thus avoids the possibility of over-inflating profit by inflated closing stock valuation.
5) **Facilitates Calculation:** Marginal costing helps in the calculation of various important factors such as key factor, make or buy decision, pricing decision, optimal sales mix, etc.
6) **Helps to Management:** Marginal costing helps the management in taking key decisions. These decisions may relate to shut down, designing of optimal production and sales mix and performance evaluation.
7) **Useful for Cost Control:** Only variable or marginal costs can be controlled while fixed costs cannot be controlled. Marginal costs help in cost control measures by segregating costs into fixed and variable categories.
8) **Profit Planning:** Marginal costing helps in profit planning by providing various tools such as graphs, break-even charts and Cost Volume Profit (CVP) analysis.

4.1.5. Limitations/Disadvantages of Marginal Costing

Marginal costing has following disadvantages:

1) **Segregation of Costs:** Marginal costing works by dividing costs into fixed and variable categories. However, in many cases, such segregation is not clear. **For example,** expenses related to payment of bonus to employees.
2) **No Cognizance of Fixed Overhead:** Marginal costing ignores fixed overheads and thus is unrealistic.
3) **Not Designed for Contract Costing:** Marginal costing cannot be used for job or contract costing purpose.
4) **Unrealistic Assumptions:** Marginal costing tends to make unrealistic assumptions while making analysis.
5) **Apportionment of Fixed Costs:** Marginal costing requires calculation of separate break-even points for different products. This leads to the problem of apportionment of fixed expenses.
6) **Under or Over-Absorption:** Marginal costing may lead to the problem of sub-optimal absorption of variable overheads.
7) **Unable to Fix Selling Prices:** Marginal costing ignores the fixed component of cost, thus making it unsuitable for calculating selling price in the long run.
8) **More Suitable for Short-Term:** Since marginal costing ignores fixed costs, it is more suitable for short-term decision-making and profit planning.

4.2. COST VOLUME PROFIT (CVP) ANALYSIS

4.2.1. Meaning and Definition of Cost Volume Profit (CVP) Analysis

Cost Volume Profit (CVP) analysis is used for establishing the relationship between cost, volume and profit associated with a product. There are many factors which have impact on profitability of the business; however, the most important determinants are the cost of production, sales volume and selling price.

According to Herman C. Heiser, "The most significant single factor in profit planning of the average business is the relationship between the volume of business, costs and profits". The CVP relationship is an important tool used for the profit planning of a business.

CVP analysis uses costs, volume and profit figures. These factors are interrelated as sales price impacts profit of a concern and in turn are determined by the costs incurred for production. The costs are determined by volume of production which is impacted by the expected sales volume. CVP analysis endeavour to analyse the relationship between change in volume and change in costs.

4.2.2. Features of CVP Analysis

Cost volume profit analysis has following features:

1) It studies the behaviour of cost with regard to production or sales volume.
2) It shows the impact of change in cost and volume of output on profit volume.
3) It forecasts the profit for a given volume or value of sales.
4) It projects the amount and quantity of production or sales to be achieved for a set profit level.
5) It helps to calculate value and volume of sales required for reaching break-even point.

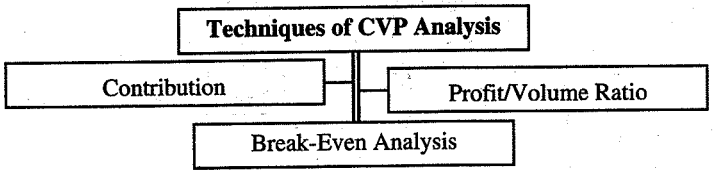
4.2.3. Assumption of CVP Analysis

CVP Analysis uses following assumptions:

1) Total costs comprises of fixed costs and variable costs.
2) Variable Costs tend to change with the level of output while Fixed Costs remain same regardless of output volume.
3) Selling Price per Unit remains same for different quantity of sales.
4) The business sells only one type of product. In case of multi-product business, the sales mix remains the same throughout.
5) Volume of production equals sales volume and thus, there is no opening or closing stock.
6) The efficiency and productivity level remains constant.
7) Material Price, wages rates and such other factors remain same for all the levels of production.

4.2.4. Techniques of CVP Analysis

CVP analysis uses sales volume, selling price, variable cost per unit and total fixed costs. The main techniques for CVP analysis are:



4.2.5. Contribution

Contribution is also known as "marginal cost of sales". It is calculated by deducting variable costs from sales. Other terms used for contribution are **"Gross Margin"** and **"Contribution Margin"**. It denotes the excess of sales over variable costs.

Contribution can be represented as:

Contribution = Sales – Variable (Marginal) Cost

Or Contribution (per unit) = Selling Price per unit – Variable (or Marginal) cost per unit

Or Contribution = Fixed Costs + Profit (– Loss)

Or Contribution = Sales × P/V Ratio

4.2.5.1. Uses of Contribution

Contribution is useful for following purposes:

1) Contribution is useful for making the calculation easy of net income and for break-even analysis.
2) Contribution margin is used for calculating break-even point, desired income sales and product line decision. It is also helpful for pricing and bonus decisions.
3) Contribution explains the relationship between sales and profit.
4) Contribution Income Statement is used for the purpose of calculating contribution margin. This statement groups fixed and variable costs.

4.2.5.2. Advantages of Contribution

Following are the main advantages of contribution:

1) It helps managers in fixing sales price.
2) It is helpful in calculating break-even point.
3) It is helpful in determining suitable product mix.
4) It is helpful in determining most efficient production method.
5) It can be used for making make or buy decisions.
6) It can be used for making decisions like, adding and dropping of product line.

Marginal Cost Equation

Marginal cost equation exhibits the relationship between contribution, fixed cost and profit. It explains that the excess of sales over the variable cost is the contribution towards fixed cost and profit.

The marginal cost equation can be derived as follows:

Sales – Variable Cost = Contribution
Or Sales = Variable Cost + Contribution
Or Sales = Variable Cost + Fixed Cost ± Profit/Loss
Or Sales – Variable Cost = Fixed Cost ± Profit/Loss
Or $S - V = F \pm P$

where,

'S' = Sales
'V' = Variable Cost
'F' = Fixed Cost
'P' = Profit/Loss

The marginal cost equation helps in determining the value of fourth factor if any three factors are known.

Example 1: From the following information find out the amount of profit earned during the year using the marginal costing technique:

Particulars	₹
Fixed cost	2,00,000
Variable cost	15 per unit
Selling price	18 per unit
Output level	80,000

Solution: The marginal cost equation is:

Sales – Variable Cost = Fixed Cost + Profit
Sales = 80,000 × ₹18 = ₹14,40,000
Variable Cost = 80,000 × ₹15 = ₹12,00,000
Fixed Cost = ₹2,00,000
Profit = ?
Sales – Variable Cost = Fixed Cost + Profit
14,40,000– 12,00,000 = 2,00,000 + Profit
2,40,000 = 2,00,000 + Profit
Profit = 2,40,000 – 2,00,000
Profit = ₹40,000

4.2.6. P/V Ratio (Profit-Volume Ratio)

The profit-volume ratio is also known as the **'Contribution Ratio'** or **'Marginal Ratio'**. It shows the relation between contribution to sales and can be expressed as under:

$$\text{P/V Ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100$$

Since, Contribution = Sales – Variable Cost = Fixed Cost + Profit, P/V ratio can also be expressed as:

$$\text{P/V Ratio} = \frac{\text{Sales} - \text{Variable Cost}}{\text{Sales}}, \text{ i.e., } \frac{S-V}{S}$$

Or $$\text{P/V Ratio} = \frac{\text{Fixed Cost} + \text{Profit}}{\text{Sales}}, \text{ i.e., } \frac{F+P}{S}$$

In case, when the profit and sales amount of two different years are provided in the question, then the following formula of P/V Ratio is used:

$$\text{P/V Ratio} = \frac{\text{Change in Profit or Contribution}}{\text{Change in Sales}}$$

P/V ratio remains constant irrespective of level of output. P/V ratio may be used in conjunction with margin of safety for proper analysis.

P/V ratio can be expressed in percentage form by multiplying it by 100. **For example,** P/V ratio for a product with selling price of ₹50 and variable cost of ₹40 may be shown as below:

$$/ \quad = \frac{50-40}{50} \times 100 = \frac{10}{50} \times 100 = 20\%$$

P/V ratio is useful for analysing the profitability of a business concern. It carries out the analysis by studying the relationship between contribution and sales. In the above example, every ₹100 of sales gives the contribution of ₹20 towards fixed expenses and profit. Higher P/V ratio denotes more profit and *vice versa*.

Desired Profit using P/V Ratio

P/V ratio can be used for calculating break-even point, profit at a certain level of sales and sales volume required for achieving certain level of profit.

Following formula is used for this purpose:

$$\text{P/V Ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100 \quad \text{Or, P/V Ratio} = \frac{\text{Fixed Cost + Profit}}{\text{Sales}}$$

Or, $$\text{Sales} = \frac{\text{Fixed Cost + Desired Profit}}{\text{P/V Ratio}} = \frac{F+P}{\text{P/V Ratio}}$$

4.2.6.1. Improvements of P/V Ratio

P/V ratio depends on contribution margin of a product. Following measures can be taken to improve the margin of a product:

1) Increase in sale price.
2) Reducing marginal cost by optimum usage of 3M's, i.e., Men, Material and Machines.
3) Focusing on sale of products with relatively better P/V ratio.

4.2.6.2. Significance/Uses of P/V Ratio

P/V ratio is useful for the following purposes:

1) For determining variable cost for any volume of sales,
2) For calculating margin of safety and break-even point,
3) Calculating profit or loss for a given sales volume,
4) Determining sales volume for achieving desired profit,

5) Determining selling price,
6) Determining most profitable line or lines of products,
7) Designing optimal sales-mix, and
8) P/V ratio may also be used for comparing sales methods, companies and individual factories, etc.

4.2.6.3. Limitations of P/V Ratio

Following are the main limitations of P/V ratio:

1) P/V ratio assumes linear relationship between costs and volumes. This may not hold true in real life.
2) P/V ratio requires classification of costs into variable and fixed categories. This process may turn complicated.
3) This analysis is useful for short-term analysis as fixed costs tend to change in long run.

Example 2: Find out (i) P/V ratio, (ii) Fixed Cost (iii) Sales volume to earn a Profit of ₹45,000. From the data given below:

Sales	₹1,20,000
Profit	₹30,000
Variable Cost	60%

Solution: Sales = ₹1,20,000

$$\text{Variable Cost} = 60\% = 1{,}20{,}000 \times \frac{60}{100} = ₹72{,}000$$

Contribution = Sales –Variable Cost = 1,20,000 – 72,000 = 48,000

i) $$\text{P/V Ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100 = \frac{48{,}000}{1{,}20{,}000} \times 100 = 40\%$$

ii) Contribution = Fixed cost + Profit
or 48,000 = Fixed cost + 30,000
or Fixed cost = 48,000 – 30,000 = ₹18,000

iii) $$\text{Sales} = \frac{\text{Fixed Cost} + \text{Desired Profit}}{\text{P/V Ratio}} = \frac{18{,}000 + 45{,}000}{40\%} = \frac{63{,}000 \times 100}{40} = ₹1{,}57{,}500$$

Proof:

Particulars	₹
Sales	1,57,500
Less: Variable Cost (60%)	94,500
Contribution	63,000
Less: Fixed Cost	18,000
Profit	**45,000**

Example 3: The sales turnover and profit during two years were as follows:

Year	Sales (₹)	Profit (₹)
2018	1,60,000	20,000
2019	1,80,000	25,000

You are required to calculate:
i) P/V ratio.
ii) Sales required to earn a profit of ₹60,000.
iii) Profit when sales are ₹1,40,000.

Solution:

i) $$\text{P/V Ratio} = \frac{\text{Change in Profit}}{\text{Change in Sales}} \times 100 = \frac{5{,}000}{20{,}000} \times 100 = 25\%$$

ii) Sales required to earn a profit of ₹60,000

$$\text{P/V Ratio} = \frac{\text{Fixed Cost} + \text{Profit}}{\text{Sales}}$$

$$\frac{25}{100} = \frac{\text{Fixed Cost} + 20{,}000}{1{,}60{,}000}$$

$$\frac{1{,}60{,}000 \times 25}{100} = \text{Fixed Cost} + 20{,}000$$

$40{,}000 - 20{,}000 = \text{Fixed Cost}$

Fixed Cost = ₹20,000

$$\text{Desired Sales} = \frac{\text{Fixed Cost} + \textit{Desired}\ \text{Profit}}{\text{P/V Ratio}} = \frac{20{,}000 + 60{,}000}{25/100} = \frac{80{,}000 \times 100}{25} = ₹3{,}20{,}000.$$

iii) Profit when sales are ₹1,40,000

$$\text{Sales} = \frac{\text{Fixed Cost} + \text{Profit}}{\text{P/V Ratio}}$$

Sales × P/V ratio = Fixed Cost + Profit

$$1{,}40{,}000 \times \frac{25}{100} = 20{,}000 + \text{Profit}$$

$35{,}000 = 20{,}000 + \text{Profit}$

Profit = 35,000 – 20,000 = ₹15,000

4.2.7. Break-Even Analysis

Cost-Volume-Profit analysis is also known as **Break-Even analysis**. The latter term may be used in narrow or broad sense. In a broad view, break-even analysis studies the relationship between costs, volume and profit for different levels of production and sales.

In the narrow sense, the term break-even analysis determines the level of operation which gives no profit and no loss. At this point, the total revenue is equal to total costs.

According to Matz, Curry and Frank, "A Break-even analysis indicates at what level costs and revenue are in equilibrium".

4.2.7.1. Assumptions of Break-Even Analysis

Break-even analysis makes following assumptions:

1) All components of cost can be divided into fixed and variable categories,
2) Total variable cost changes proportionately in response to change in production level,
3) Fixed cost remains same for all output levels,
4) Selling price per unit does not change with change in output quantity,
5) Cost is influenced only by production volume,
6) General price level does not change,
7) Sales mix remains the same and does not undergo change, and
8) Production and sales are inter-related.

4.2.7.2. Significance of Break-Even Analysis

Break-even analysis is important due to following reasons:

1) It is important to understand the relationship between costs, profits and volume for efficient forecasting of profit.
2) Break-even analysis helps in setting up flexible budget for showing costs for different output levels.

3) Break-even analysis helps in the process of performance evaluation.
4) Break-even analysis helps in the process of price fixation by showing the effect of different price structure on profits and costs.
5) Break-even analysis helps in determining the amount of overhead costs to be charged to product costs for different operation levels.
6) Break-even analysis is also useful for making various short-term decisions such as designing sales mix.

4.2.7.3. Limitations of Break-Even Analysis

Break-even analysis has following limitations:

1) Break-even analysis may give wrong results for different changes,
2) Segregating costs into fixed and variable parts is a difficult procedure,
3) Total fixed costs tend to change after certain point of activity volume,
4) It is difficult to predict sales mix in case of continuous change in demand,
5) Significant difference in opening and closing stock may affect results of the analysis, and
6) Break-even relationship changes in response to selling price and other factors such as wage rates and material prices.

4.2.8. Break-Even Point (BEP)

Break-even point denotes the quantity of sales where the firm neither earns any profit nor incurs any loss. At this volume level, sales revenue is equal to the total cost. Following equations elucidates the concept of break-even point:

Break-Even Sales = Fixed Cost + Variable Cost

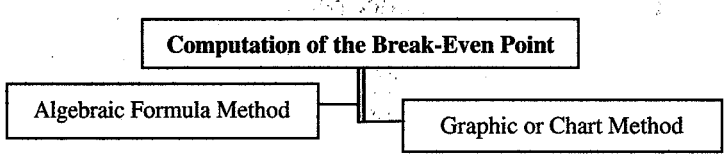
Break-even point is also known as **Balancing Point, Equilibrium Point, Critical Point or No-Profit No-Loss Point.**

According to Keller and Ferrara, "The break-even point of a company or a unit of a company is the level of sales income which will be equal to the sum of its fixed costs and variable costs".

According to Charles T. Horngren, "The break-even point is that point of activity (sales volume) where total revenues and total expenses are equal; it is the point of zero profit and zero loss".

4.2.8.1. Computation of the Break-Even Point

The break-even point can be calculated using algebraic formula method or graphic method.



4.2.8.2. Algebraic Formula Method for Computing the Break-Even Point

- Following concepts are used for calculating break-even point:

1) **Break-Even Point in Units:** At Break-Even Point, contribution margin equals fixed costs. Following formula is used for the calculation of break-even point in terms of units:

$$\text{Break - Even Point (units)} = \frac{\text{Fixed Cost}}{\text{Selling Price per Unit} - \text{Variable Cost per Unit}} = \frac{\text{Fixed Cost}}{\text{Contribution per Unit}}$$

2) **Break-Even Point in Terms of Budget – Total or Money Value:** For calculating break-even point in terms of total value, following formula is used:

Total Sales = Total Fixed Cost + Total Variable Cost

Or $S = F + V$ (where, S = Sales, F = Fixed Cost and V = Variable Cost)

Or $S - V = F$

Or $\frac{S-V}{S-V} = \frac{F}{S-V}$ (Dividing both Sides by S – V)

Or $1 = \frac{F}{S-V}$

Or $S \times 1 = \frac{F \times S}{S-V}$ (Multiplying both sides by S)

Hence, Break-Even Sales (in Rupee) $= \frac{\text{Fixed Cost}}{\text{Sales} - \text{Variable Cost}} \times \text{Sales} = \frac{\text{Fixed Cost}}{\text{Contribution}} \times \text{Sales}$

With the use of P/V Ratio,

$$\text{Break-Even Point} = \frac{\text{Fixed Cost}}{\text{P/V Ratio}}$$

where, $\text{P/V Ratio} = \frac{\text{Contribution}}{\text{Sales}}$ and Fixed Cost = Contribution – Profit.

3) **Break-Even Point as a Percentage of Estimated Capacity:** Break-Even Point can be calculated by dividing break-even sales by the total capacity sales. For example, if a firm's Break-Even Point is at 1,00,000 units and its total estimated capacity is at 2,00,000 units, then Break-Even Point is at 50% of capacity. Following formula may be used for this purpose:

$$\text{B.E.P (as \% age of capacity)} = \frac{\text{Fixed Cost}}{\text{Total Contribution}}$$

Example 4: From the given data, you are required to calculate the break-even point in units and in sales value:

Output	4,000 units
Selling price per unit	₹35
Variable cost per unit	₹25
Total fixed cost	₹30,000

Solution:

$$\text{Break Even Point (in units)} = \frac{\text{Fixed Cost}}{\text{Selling Price per unit} - \text{Variable Cost per unit}}$$

$$= \frac{30,000}{35-25} = \frac{30,000}{10} = 3,000 \text{ units.}$$

B.E.P (in sales value) = B.E.P (in units) × Selling price per unit = 3,000 × 35 = ₹1,05,000

or

$$\text{Break Even Point (in sales value)} = \frac{\text{Fixed Cost} \times \text{Sales}}{\text{Sales} - \text{Variable Cost}}$$

Fixed Cost = ₹30,000 (given)
Sales = 4,000 units × 35 per unit = ₹1,40,000
Variable Cost = 4,000 units × 25 per unit = ₹1,00,000

$$\text{Hence, B.E.P. (in sales value)} = \frac{30,000 \times 1,40,000}{1,40,000 - 1,00,000} = \frac{30,000 \times 1,40,000}{40,000} = ₹1,05,000$$

Example 5: XYZ Manufacturing, incurred following cost for a biscuit pack.

Fixed Factory Overheads Cost	₹70,000
Fixed Selling Overheads Cost	₹20,000
Variable Manufacturing Cost per unit	₹16
Variable Selling Cost per unit	₹4
Selling Price per unit	₹25

Calculate:

i) Break-even point in terms of sales value and in units.

ii) Number of units that must be sold to earn a profit of ₹95,000.

Solution:

i) $\text{Break-Even Point} = \dfrac{\text{Fixed Cost}}{\text{Selling Price per unit} - \text{Variable Cost per unit}}$

Variable Cost per unit = ₹16 + 4 = ₹20

Total Fixed Cost = ₹70,000 + 20,000 = ₹90,000

$\therefore$ B.E.P $= \dfrac{90,000}{25-20} = 18,000 \text{ units}$

B.E.P (in sales values) = B.E.P (in units) × Selling Price (Per unit)

B.E.P. (in sales values) $= 18,000 \times 25 =$ ₹ 4,50,000

ii) Number of units that must be sold to earn profit of ₹ 95,000

$$= \frac{\text{Fixed Cost + Profit}}{\text{Selling Price per unit} - \text{Variable Cost per unit}} = \frac{90,000 + 95,000}{25-20} = \frac{1,85,000}{5} = 37,000 \text{ units.}$$

Example 6: The fixed costs amount to ₹60,000 and the percentage of variable costs to sales is given to be $33\frac{1}{3}\%$. If at 100% capacity sales are ₹3,50,000, find out the break-even point and the percentage sales when it occurred. Determine profit at 75% capacity.

Solution: Percentage of Variable Cost to Sales $= 33\frac{1}{3}\%$, i.e., $\dfrac{100}{3}$

$\therefore$ Percentage of Contribution to Sales $= 100 - \dfrac{100}{3} = \dfrac{200}{3}$

$$\text{P/V Ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100 = \frac{200}{3} \times \frac{1}{100} \times 100 = \frac{200}{3} \text{ or } 66\frac{2}{3}\%$$

$$\text{Break-even Sales} = \frac{\text{Fixed Cost}}{\text{P/V Ratio}} = \frac{60,000}{66\frac{2}{3}\%} \text{ or } \frac{60,000}{200} \times 300 = ₹90,000.$$

100% Capacity Sales = ₹3,50,000

Hence, B.E.P. occurs at $\dfrac{90,000}{3,50,000} \times 100 = 25.71\%$ capacity.

Profit at 75% Capacity

At 100% Capacity Sales are ₹3,50,000

$\therefore$ 75% Capacity Sales $= 3,50,000 \times \dfrac{75}{100} =$ ₹2,62,500

Total Contribution at 75% capacity $= 2,62,500 \times \dfrac{200}{3} \times \dfrac{1}{100} = 1,75,000$

Less: Fixed Expenses	60,000
Profit at 75% Capacity	₹1,15,000

4.2.8.3. Graphical Method of Break-Even Analysis or Break-Even Chart/Graph

The break-even point can also be computed graphically. A break-even chart is a graphical representation of marginal costing. The break-even charts 'portrays a pictorial view of the relationships between costs, volume and profits'. It shows the break-even point and also indicates the estimated profit or loss at various levels of output. The break-even point as indicated in the chart is the point at which the total cost line and the total sales line intersect.

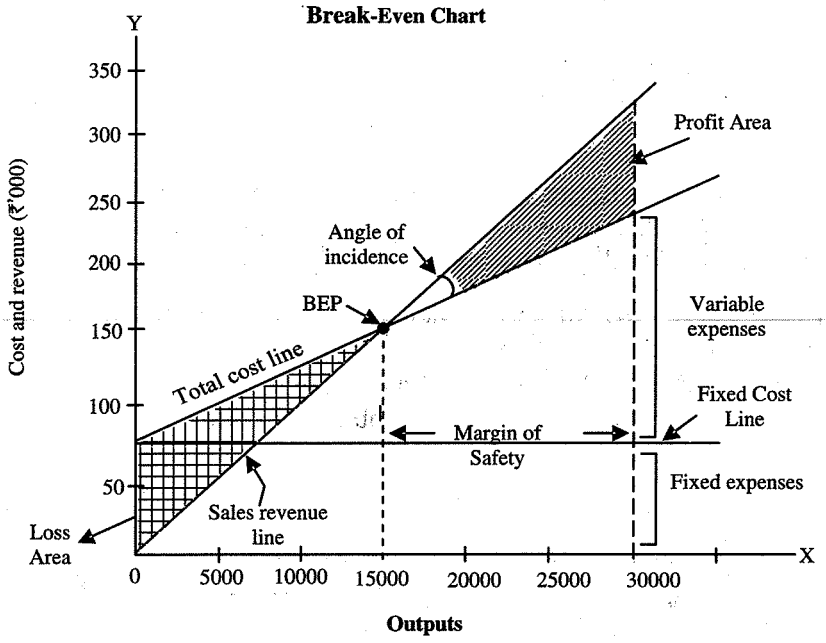
Construction of Break-Even Charts

Break-Even chart can be drawn using fixed cost line, total cost line and sales line. Following procedure is used for this purpose:

1) Horizontal axis shows the scale for production while vertical axis is used for costs and revenue.
2) Fixed cost is plotted on the vertical axis and fixed cost line is drawn, parallel to horizontal axis and passing through this point.
3) Variable cost is plotted, starting from the fixed cost line and these points are joined, giving total cost line. Alternatively, total cost for different levels is obtained and are plotted starting from horizontal axis.
4) Plot any sales volume and draw the sales line by joining the point and zero.

There are three methods of drawing a break-even chart/graph.

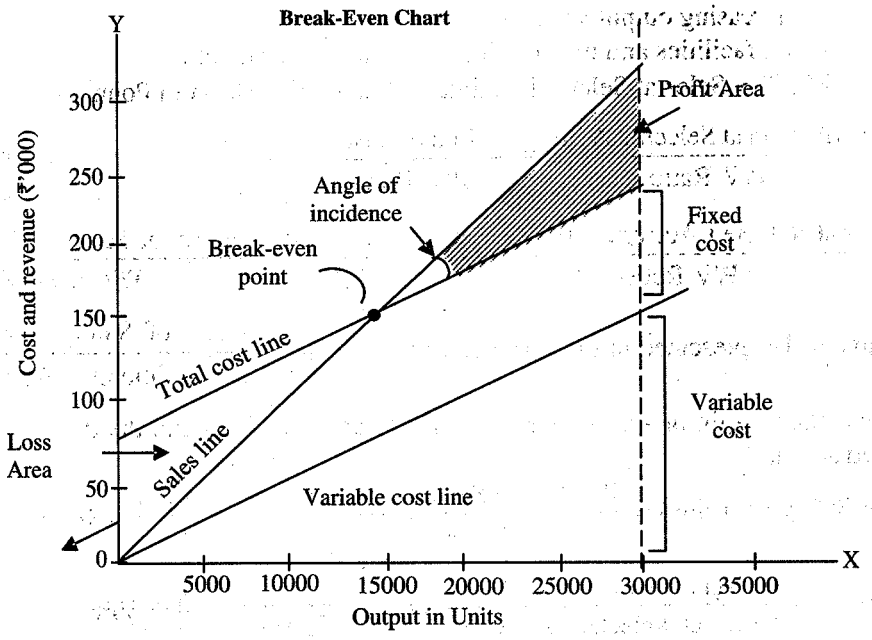
1) **First Method:** Under this method following steps are taken to draw the break-even chart:



i) Y-axis is used for representing costs and sales revenue.
ii) Since fixed expenses remain constant, the line is drawn parallel to X-axis.
iii) The variable costs for different volumes of production are plotted above the fixed cost line. The variable cost line meets fixed cost line at zero level of activity.
iv) Sales or revenue line is drawn by plotting sales value at different levels of output.
v) Break-even point is denoted by the intersection of revenue line and total cost line.
vi) A perpendicular line is drawn from the break-even point to determine the number of units to be produced for break-even.
vii) The sales revenue for break-even point is calculated by plotting a perpendicular line from the point of intersection of sales and cost line.
viii) The area below the break-even point denotes loss as total sales is less than the total cost while the area above the break-even point shows profit.

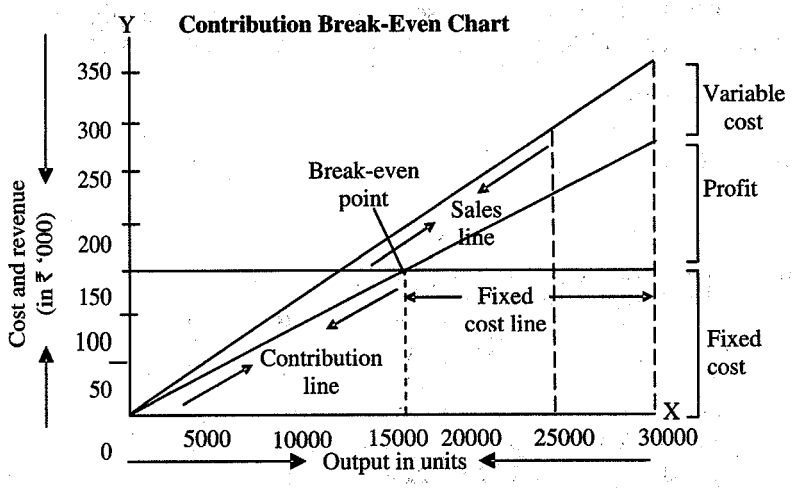
2) **Second Method:** The break-even chart can be plotted by drawing variable cost line and the total cost line. This method is more useful for managerial decision making purpose.
   i) The variable costs at various levels of sales are plotted.
   ii) Marginal contribution is depicted by the way of difference between sales and variable cost lines.
   iii) This type of graph shows the recovery of fixed costs for various levels of production.

   This method can be customised by showing different elements of fixed and variable costs such as variable factory overhead costs and labour costs.



3) **Third Method – Contribution Break-Even Chart:** This type of chart does not show total cost line. However, it uses contribution line which is drawn from the origin and goes up with increase in the output level. Fixed cost line is parallel to the x-axis and sales line is shown as usual.

The Break-even point is shown by the intersection of contribution line and fixed cost line. Since at this point, total contribution equals total fixed costs; there is neither any profit nor any loss. This chart presents clear representation of contribution at different levels of output.



4.2.9. Margin of Safety (MOS)

Margin of Safety is the excess of actual sales over the break-even sales. It may also denote excess of budgeted sales over break-even sales.

Margin of Safety = Total Sales –Break-Even Point (in Sales)

Margin of Safety is calculated at a selected level of production. It involves finding the difference between sales/production and the Break-Even sales or production. Higher the Margin of Safety, the better is for business.

High Margin of Safety shows that the business can survive substantial fall in its revenue without undergoing losses. However, businesses with lower Margin of Safety need to be more careful. In case of unsatisfactory Margin of Safety, following measures can be taken:

1) Margin of Safety may be boosted by increasing the selling price. However, proper attention should be paid to the price elasticity of demand.
2) Curtailing fixed costs.
3) Controlling variable costs.

4) Changing sales mix. Increasing output volume.
5) Modernising production facilities and undertaking cost effective practices.
6) Margin of Safety (MOS) = Sales at Selected Activity – Sales at Break-Even Point.

$$MOS = \frac{\text{Contribution at Selected Activity}}{\text{P/V Ratio}} - \frac{\text{Fixed Cost}}{\text{P/V Ratio}}$$

$$MOS = \frac{\text{Profit at Selected Activity} + \text{Fixed cost}}{\text{P/V Ratio}} - \frac{\text{Fixed Cost}}{\text{P/V Ratio}} = \frac{\text{Profit at Selected Activity}}{\text{P/V Ratio}}$$

This Margin of Safety is also presented in ratio or percentages as, $= \frac{\text{Margin of Safety (Sales)}}{\text{Sales at Selected Activity}} \times 100$

Example 7: Calculate the margin of safety when the profit earned is ₹300 and the P/V ratio is 50%. Sales of ₹2,000 is the selected activity.

Solution: Margin of Safety (in value) $= \frac{\text{Profit at Selected Activity}}{\text{P/V Ratio}} = \frac{300 \times 100}{50} = ₹600$

Margin of Safety (in %) $= \frac{\text{Margin of Safety (Sales)}}{\text{Sales at Selected Activity}} \times 100 = \frac{₹600}{₹2,000} \times 100 = 30\%$

4.2.10. Angle of Incidence

The angle formed by sales line and total cost line at break-even point is called **"Angle of Incidence"**. This angle shows the rate at which the firm generates profits above the break-even point. Larger angle is considered better as it shows higher profitability while narrower angle shows the converse. The management should endeavour to increase its angle of incidence and correspondingly its margin of safety.

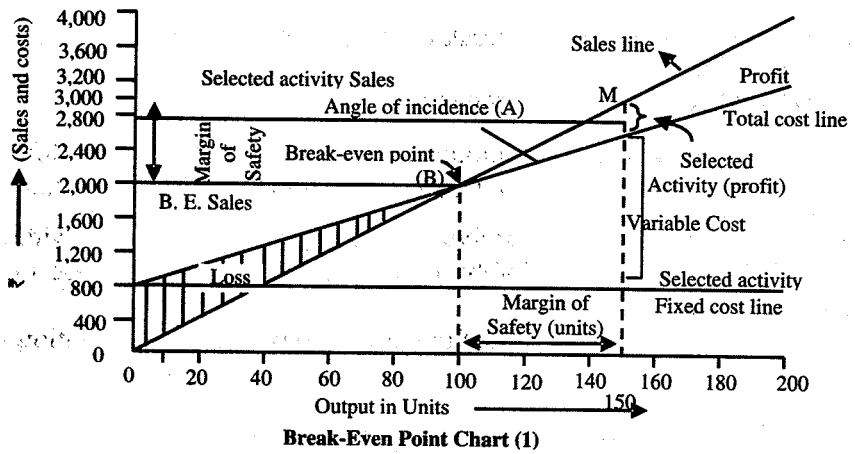
Example 8: A company produces a single article and sells at ₹20 each. The marginal cost of production is ₹8 each and total fixed cost of the concern is ₹600 per annum. Assuming the output at 40, 80,120 and 200.

Construct a break-even chart and show:
1) Break-Even Point;
2) Margin of Safety at sales ₹2,000;
3) Angle of Incidence;
4) Increase in selling price if the break-even point is reduced to 100 units.

Solution: A break-even chart is prepared by obtaining the information at these levels:

Output	**40** (₹)	**80** (₹)	**120** (₹)	**200** (₹)
Sales	800	1600	2,400	4,000
Fixed cost	600	600	600	600
Variable cost	320	640	960	1,600
Total	**1,720**	**2,840**	**3,960**	**6,200**

Fixed cost line, total cost line, and sales line are drawn one after another.



Break-Even Point Chart (1)

This chart shows clearly the break-even point, margin of safety and angle of incidence:

1) **Break-Even Point:** This is the point at which the sales line and the total cost line intersect. Here B is the break-even point equivalent to a sale of ₹2,000 or 100 units.
2) **Margin of Safety:** This is the difference in sales or units of production from the break-even point. Thus margin of safety at M is sales of (₹3,000 – ₹2,000), i.e., ₹1,000 or 50 units of production.
3) **Angel of Incidence:** This is the angle formed by the sales line and the total cost line at the break-even point. A large angle of incidence shows a high rate of profit being made.

   At 80 units the total cost (from the table) = ₹2,840 = Sales value of 100 units.

   Hence, selling price for break-even at 80 units = ₹2,840/100 = ₹28.4 per unit.
4) Increase in selling price is ₹8.40 or 10% over the original selling price of ₹20 per unit.

4.3. MISCELLANEOUS PROBLEMS

Example 9: The sales turnover and profit during two years were as follows:

Year	Sales (₹)	Profit (₹)
2011	1,40,000	15,000
2012	1,60,000	20,000

You are required to calculate:

1) P/V Ratio
2) Sales required to earn a profit of ₹40,000
3) Profit when sales are ₹1,20,000.

Solution:

1) Calculation of P/V Ratio $= \frac{\text{Change in Profit}}{\text{Change in Sales}} \times 100$

$$= \frac{20,000 - 15,000}{1,60,000 - 1,40,000} \times 100 = \frac{5,000}{20,000} \times 100 = 25\%$$

2) Calculation of sales required to earn a profit of ₹40,000

$$\text{Sales} = \frac{\text{Fixed Cost} + \text{Desired Profit}}{\text{P/V Ratio}}$$

Fixed Cost = Sales × P/V Ratio – Profit

$$= 1,40,000 \times \frac{25}{100} - 15,000 = 20,000$$

$$Sales = \frac{20,000 + 40,000}{25\%} = ₹2,40,000$$

3) Calculation of profit when sales are ₹1,20,000

Profit = (Sales × P/V Ratio) – Fixed Cost

= (1,20,000 × 25%) – 20,000 = 30,000 – 20,000

= ₹ 10,000

Example 10: From the following information:

Break-even Sales	₹1,00,000
Fixed Cost	₹25,000

Calculate:

1) P/V ratio
2) Profit when sales are ₹1,50,000
3) Sales to earn a profit of ₹50,000 and margin of safety at the level.
4) If the selling price is reduced by 10%, what will be new break-even point?

Solution:

1) **P/V Ratio** = Fixed cost/Break-even Sales = 25,000/1,00,000 = 25%
2) **Profit when sales are ₹1,50,000**
   Profit = (Sales × P/V Ratio) – Fixed Cost = (1,50,000 × 25%) – 25,000 = ₹12,500
3) **Sales to earn a profit of ₹ 50,000**
   Sales = Fixed Costs + Profit/P/V Ratio = 25,000 + 50,000/25% = ₹3,00,000
   Margin of Safety = Profit/P/V Ratio = 50,000/25% = ₹2,00,000
4) **If the selling price is reduced by 10%, New Break-Even Point**

Assuming the following data:

Particulars	Old (₹)	New (₹)
Selling Price	100	90
Less: Variable Cost	75	75
Contribution	25	15

Break-Even Point (in sales) = Fixed Cost/P/V Ratio
New, P/V Ratio = 15/90 = 1/6 or, 16.667%
Break-Even Point (in sales) = 25,000/1/6 = ₹1,50,000

Example 11: KLM Industries furnishes following information for the level of output of 1,000 units for the year 2010.

Selling Price per Unit	=	₹100
Variable Cost per Unit	=	₹40
Total Fixed Cost	=	₹30,000

Find:
1) P/V Ratio
2) BEP in units
3) BEP in sales
4) The amount of sales required to earn profit of ₹42,000 (assuming fixed cost in total remains same).

Solution:

1) $$\text{P/V Ratio} = \frac{Contribution}{Sales} \times 100$$

Contribution = Sales – Variable Cost= (1000 × 100) – (1000 × 40) = 1,00,000 – 40,000= 60,000

$$\text{P/V Ratio} = \frac{60,000}{1,00,000} \times 100 = 60\%$$

2) $$\text{BEP (in Units)} = \frac{\text{Fixed Cost}}{\text{Selling Price Per Unit} - \text{Variable Cost per Unit}} = \frac{30,000}{100-40} = 500 \text{ Units}$$

3) $$\text{BEP (in Sales)} = \frac{\text{Fixed Cost} \times \text{Sales}}{\text{Sales} - \text{Variable Cost}} = \frac{30,000 \times 1,00,000}{1,00,000 - 40,000} = ₹50,000$$

4) $$\text{Amount of Sales} = \frac{\text{Fixed Cost} + \text{Desired Profit}}{\text{P/V Ratio}} = \frac{30,000 + 42,000}{60\%} = ₹1,20,000$$

Example 12: The following information is furnished by JANTA Limited, Noida for the year 2018-19:
1) Sales Price ₹100 per unit
2) Number of Units sold 80,000
3) Direct Material Cost ₹40 per unit
4) Direct Labour Cost ₹20 per unit
5) Overheads ₹20 per unit [40% are variable]

You are required to find out:
1) P/V Ratio
2) Break-Even Point and
3) Margin of Safety

Solution:

1) $$P/V\ Ratio = \frac{Contributi\ on\ per\ Unit}{Selling\ Pr\ ice\ per\ Unit} \times 100$$

Contribution per Unit = Selling Price per Unit – Variable Overhead per Unit
Variable Overhead per Unit = Direct Material Cost + Direct Labour Cost + Variable Overhead
= 40 + 20 + 8 = ₹68 per unit
Contribution per Unit = 100 – 68 = ₹32

$$P/V\ Ratio = \frac{32}{100} \times 100 = 32\%$$

2) $$Break-Even\ Point\ (in\ Units) = \frac{Fixed\ Cost}{Contribution\ per\ Unit} = \frac{(20 \times 60\%)12 \times 80,000}{32} = \frac{9,60,000}{32}$$
= ₹30,000 units

$$Break-Even\ Point\ (in\ value) = \frac{Fixed\ Cost}{P/V\ Ratio} = \frac{9,60,000}{32} \times 100 = ₹30,00,000$$

3) Margin of Safety = Actual Sales – B.E.P. Sales = (80,000 × 100) – 30,00,000
= 80,00,000 – 30,00,000 = ₹50,00,000

Example 13: From the following given data:

Sales Price	₹400 per unit
Variable Cost	₹300 per unit
Fixed Cost	₹17,50,000

You are required to calculate:
1) Break-even point
2) Selling price per unit if break-even point is brought upto 20,000 units; and
3) Selling price per unit if break-even point is brought down to 15,000 units.

Solution:

1) **Calculation of Break-even Point**

$$BEP = \frac{Fixed\ Cost}{Contribution\ per\ unit}$$

Contribution per unit = Sales Price per Unit – Variable Cost per Unit
= 400 – 300 = 100

$$BEP = \frac{17,50,000}{100} = 17,500\ units$$

2) **Selling Price per unit if break-even point is brought upto 20,000 units.**

$$BEP = \frac{Fixed\ Cost}{Sales\ Pr\ ice\ per\ Unit - Variable\ Cost\ per\ Unit}$$

$$20,000 = \frac{17,50,000}{Contribution\ per\ Unit}$$

$$Contribution\ per\ Unit = \frac{17,50,000}{20,000} = 87.50\ per\ unit$$

Sales Price per Unit = Variable Cost per Unit + Contribution per Unit
= 300 per unit + 87.50 per unit
= ₹387.50 per unit

3) Selling price per unit if break-even point is brought down to 15,000 units.

$$BEP = \frac{\text{Fixed Cost}}{\text{Contribution per unit}}$$

$$15{,}000 = \frac{17{,}50{,}000}{\text{Contribution per unit}}$$

Contribution per Unit = ₹116.67 per unit

Sales Price per Unit = Variable cost per unit + Contribution per unit
= 300 per unit + 116.67 per unit
= ₹416.67 per unit.

Example 14: From the following information you are required to calculate:
1) Contribution
2) Fixed cost
3) Break-even point (in sales)
4) Margin of safety as percentage of sales.

Particulars	2018	2019
Sales	3,00,000	4,00,000
Total cost	2,80,000	3,60,000
Profit	**20,000**	**40,000**

Solution:

1) **Contribution = Sales – Variable Cost**

$$\text{Variable Cost to Sales Ratio} = \frac{\text{Change in Cost}}{\text{Change in Sales}} \times 100 = \frac{80{,}000}{100{,}000} \times 100 = 80\%$$

Variable Cost is 80% of the total sales.

Variable Cost for 2018 = 3,00,000 × 80% = ₹2,40,000
Variable Cost for 2019 = 4,00,000 × 80% = ₹3,20,000
Contribution for 2016 = 3,00,000 – 2,40,000= ₹60,000
Contribution for 2017 = 4,00,000 – 3,20,000= ₹80,000

2) **Fixed Cost = Total Cost – Variable Cost**
Fixed Cost for 2018 = 2,80,000 – 2,40,000 = ₹40,000
Fixed Cost for 2019 = 3,60,000 – 3,20,000 = ₹40,000

3) **Break Even Point** $= \frac{\text{Fixed Cost}}{\text{P/V Ratio}}$

$$\text{P/V Ratio} = \frac{\text{Change in Profit}}{\text{Change in Sales}} \times 100 = \frac{20{,}000}{1{,}00{,}000} \times 100 = 20\%$$

$$\text{Break Even Point (in Value)} = \frac{\text{Fixed Cost}}{\text{P/V Ratio}} = \frac{40{,}000}{20\%} = ₹2{,}00{,}000$$

4) **Margin of Safety as Percentage of Sale** $= \frac{\text{Margin of Safety}}{\text{Actual Sales}} \times 100$

Margin of Safety (in value) for 2018 = Actual Sales – Break-even Sales
= 3,00,000 – 2,00,000 = ₹1,00,000

$$\text{Margin of Safety (in Percentage)} = \frac{1{,}00{,}000}{3{,}00{,}000} \times 100 = 33\frac{1}{3}\%$$

Margin of Safety (in value) for 2019 = Actual Sales – Break-even Sales
= 4,00,000 – 2,00,000 = ₹2,00,000

$$\text{Margin of Safety (in Percentage)} = \frac{2,00,000}{4,00,000} \times 100 = 50\%$$

Example 15: The sales turnover and profits during two years were as follows:

Year	Sales	Profit
2016	45,00,000	5,00,000
2017	50,00,000	7,00,000

You are required to calculate:
1) P/V Ratio.
2) Fixed cost.
3) Break Even Point (₹)
4) Sales to Earn Profit of ₹10,00,000.
5) Margin of Safety.

Solution:

1) **Calculation of P/V Ratio**

$$P/V \text{Ratio} = \frac{\text{Change in Profit}}{\text{Change in Sales}} \times 100 = \frac{2,00,000}{5,00,000} \times 100 = 40\%$$

2) **Calculation of Fixed Cost**
Fixed Cost = Sales × P/V Ratio – Profit = 45,00,000 × 40% – 5,00,000 = ₹13,00,000

3) **Calculation of Break-Even Point (₹)**

$$\text{B.E.P (in Sales)} \frac{\text{Fixed Cost}}{P/V \text{Ratio}} = \frac{13,00,000}{40\%} = ₹32,50,000$$

4) **Calculation of Sales to Earn Profit of ₹10,00,000**

$$\text{Sales to Earn Profit of ₹10,00,000} = \frac{\text{Fixed Cost} + \text{Desired Profit}}{P/V \text{Ratio}}$$

$$= \frac{13,00,000 + 10,00,000}{40\%} = \frac{23,00,000}{40\%} = ₹57,50,000$$

5) **Calculation of Margin of Safety (MOS)**
MOS (For 2016) = Actual sales –Break-even point sales = 45,00,000 – 32,50,000 = ₹12,50,000
MOS (For 2017) = Actual sales – Break-even point sales = 50,00,000 – 32,50,000 = ₹17,50,000

Example 16: Himmatlal Co. Ltd., Pune provides the following information for the year 2014-15:
Sales – 15,000 units
Fixed Cost – ₹34,000
Sales Value – ₹1,50,000
Variable Cost – ₹6 per unit

You are required to calculate:
1) P/V Ratio, BEP in ₹ and Margin of Safety.
2) Also calculate and comment on the effect of the following:
   i) Decrease of 10% in selling price.
   ii) Increase of 10% in variable cost.

Solution:

1) **Calculation of P/V Ratio, BEP in ₹ and Margin of Safety**

$$P/V \text{ Ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100$$

Contribution = Sales – Variable Cost = 1,50,000 – (15,000 × 6) = 60,000

$$\text{P/V Ratio} = \frac{60,000}{1,50,000} \times 100 = 40\%$$

$$\text{Break-even Point (BEP in ₹)} = \frac{\text{Fixed Cost}}{\text{P/V Ratio}} = \frac{34,000}{40\%} = ₹85,000$$

$$\text{Margin of Safety (in value)} = \frac{\text{Profit}}{\text{P/V Ratio}}$$

Profit = Sales – (Variable Cost + Fixed Cost)= 1,50,000 – (90,000 + 34,000)= 1,50,000 – 124,000 =26,000

$$\text{Margin of Safety (in value)} = \frac{26,000}{40\%} = ₹65,000$$

2) **i) New P/V, BEP and Margin of Safety, if selling price is decrease by 10%.**

Original Sales = 1,50,000 – 15,000 = 1,35,000

$$\text{P/V Ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100$$

Contribution = Sales – Variable Cost = 1,35,000 – (13,500 × 6) = 54,000

$$\text{P/V Ratio} = \frac{54,000}{1,35,000} \times 100 = 40\%$$

$$\text{BEP} = \frac{\text{Fixed Cost}}{\text{P/V Ratio}} = \frac{34,000}{40\%} = ₹85,000$$

$$\text{Margin of Safety (in value)} = \frac{\text{Profit}}{\text{P/V Ratio}}$$

Profit = Sales – (Variable Cost + Fixed Cost)

= 1,35,000 – (81,000 + 34,000) = 1,35,000 – 1,15,000 = ₹20,000

$$\text{Margin of Safety (in value)} = \frac{20,000}{40\%} = ₹50,000$$

ii) New P/V, BEP and Margin of Safety, if variable cost increased by 10%.

Original Sales = 1,50,000

$$\text{P/V Ratio} = \frac{\textit{Contribution}}{\textit{Sales}} \times 100$$

Contribution = Sales – Variable Cost

= 1,50,000 – (15,000 × 6.6) = ₹51,000

$$\text{P/V Ratio} = \frac{51,000}{1,50,000} \times 100 = 34\%$$

$$\text{BEP} = \frac{\text{Fixed Cost}}{\text{P/V Ratio}} = \frac{34,000}{34\%} = ₹1,00,000$$

$$\text{Margin of Safety (in value)} = \frac{\text{Profit}}{\text{P/V Ratio}}$$

Profit = Sales – (Variable Cost + Fixed Cost)

= 1,50,000 – (99,000 + 34,000) = 1,50,000 – 1,33,000 = ₹17,000

$$\text{Margin of Safety (in value)} = \frac{17,000}{34\%} = ₹50,000$$

Example 17: The turnover and profits of the two years were as follows:

Particulars	Sales Units	Profit/Loss (₹)
I Year	8,000	15,000 Loss
II Year	10,000	15,000 Profit

The Selling Price per Unit is ₹120.

Calculate:

1) P/V Ratio
2) Fixed Cost
3) Sales at Break-Even Point
4) The number of units to be sold to earn a profit of ₹45,000.
5) Profit when sales are 30,000 units.

Solution:

1) **P/V Ratio** $= \frac{\text{Change in Profit}}{\text{Change in Sales}} \times 100 = \frac{15,000-(-15,000)}{12,00,000-9,60,000} \times 100 = \frac{30,000}{2,40,000} \times 100 = 12.5\%$

2) **Fixed Cost = Contribution – Profit**

Contribution = Sales × P/V Ratio

Contribution I Year = 9,60,000 × 12.5% = ₹1,20,000

Contribution II Year = 12,00,000 × 12.5% = ₹1,50,000

Fixed Cost I Year = 1,20,000 – (– 15,000) = ₹1,35,000

Fixed Cost II Year = 1,50,000 – 15,000 = ₹1,35,000

3) **Sales at Break-even Point** $= \frac{\text{Fixed Cost}}{\text{P/V Ratio}} = \frac{1,35,000}{12.5\%} =$ ₹10,80,000

4) **Number of units to be sold to earn a profit of ₹45,000**

$= \frac{\text{Fixed Cost + Desired Profit}}{\text{P/V Ratio}} = \frac{1,35,000+45,000}{12.5\%} =$ ₹14,40,000

5) **Profit when sales are 30,000 units**

Profit = Contribution – Fixed Cost

= (P/V Ratio × Sales) – Fixed Cost

= (12.5% × 36,00,000) – 1,35,000 = 4,50,000 – 1,35,000 = ₹3,15,000

Sales = 30,000 units × 120 per unit = ₹36,00,000

Example 18: The sales and profits during the two years were as follows:

Particulars	Sales (₹)	Profit (₹)
I Year	50,00,000	5,00,000
II Year	70,00,000	9,00,000

Assuming that the Cost Structure and Selling Prices remain the same in the two years, calculate:

1) Profit Volume Ratio
2) Break-Even Point (in Sales)
3) The Sales required to earn a profit of ₹8,00,000
4) Margin of Safety in II Year
5) Profit when sales are ₹40,00,000

Solution:

1) **Profit Volume Ratio** $= \frac{\text{Change in Profit}}{\text{Change in Sales}} \times 100 = \frac{4,00,000}{20,00,000} \times 100 = 20\%$

2) **Break-Even Point (in Sales)** $= \dfrac{\text{Fixed Cost}}{\text{P/V Ratio}}$

Fixed Cost = Contribution – Profit

Contribution = Sales × P/V Ratio

I Year = $50,00,000 \times \dfrac{20}{100} = ₹10,00,000$

II Year = $70,00,000 \times \dfrac{20}{100} = ₹14,00,000$

Fixed Cost = 10,00,000 – 5,00,000 = ₹5,00,000

or = 14,00,000 – 9,00,000 = ₹5,00,000

$\therefore$ Break-even Point (in Sales) $= \dfrac{5,00,000}{20\%} = ₹25,00,000$

3) **Sales required to earn a profit of ₹8,00,000**

Desired Sales $= \dfrac{\text{Fixed Cost} + \text{Desired Profit}}{\text{P/V Ratio}} = \dfrac{5,00,000 + 8,00,000}{20\%} = ₹65,00,000$

4) **Margin of Safety (II Year)** = Actual Sales – Break-Even Point Sales

= 70,00,000 – 25,00,000= ₹45,00,000

5) **Profit when Sales are 40,00,000**

Required Profit = (Sales × P/V Ratio) – Fixed Cost $= \left(40,00,000 \times \dfrac{20}{100}\right) - 5,00,000 = ₹3,00,000$

Example 19: The turnover and profits during the two periods were as follows:

	Sales (₹)	Profits (₹)
Period-I	40 lac	4 lac
Period-II	60 lac	8 lac

Assuming that the cost structure and selling prices remain the same in the two periods.

Calculate:
Profit Volume Ratio, BEP sales, the sales required to earn profit of ₹10 lac, margin of safety in period II.

Solution:

1) **Profit Volume Ratio** $= \dfrac{\text{Change in Profit}}{\text{Change in Sales}} \times 100 = \dfrac{4,00,000}{20,00,000} \times 100 = 20\%$

2) **Break-Even Point (in Sales)** $= \dfrac{\text{Fixed Cost}}{\text{P/V Ratio}}$

Fixed Cost = Contribution – Profit

Contribution = Sales × P/V Ratio

I Year = $40,00,000 \times \dfrac{20}{100} = ₹8,00,000$

II Year = $60,00,000 \times \dfrac{20}{100} = ₹\ 12,00,000$

Fixed Cost = 8,00,000 – 4,00,000 =₹ 4,00,000

or = 12,00,000 – 8,00,000 =₹ 4,00,000

$\therefore$ Break-Even Point (in Sales) $= \dfrac{4,00,000}{20\%} = ₹20,00,000$

3) **The sales required to earn profit of ₹10,00,000**

$$\text{Desired Sales} = \frac{\text{Fixed Cost} + \text{Desired Profit}}{\text{P/V Ratio}} = \frac{4,00,000 + 10,00,000}{20\%} = ₹70,00,000$$

4) **Margin of Safety (Period-II)** = Actual Sales – Break-Even Point(in Sales)
= 60,00,000 – 20,00,000 = ₹40,00,000

Example 20: The following figures of sales and profits for two months are available in respect of a concern:

Particulars	Sales (₹)	Profit (₹)
1st Month	1,50,000	20,000
2nd Month	1,60,000	24,000

You are required to find-out:
1) P/V ratio
2) Fixed cost
3) Break-even point
4) Profit on estimated sale of ₹1,50,000
5) Sales required to earn a profit of ₹40,000

Solution:

1) **Calculation of P/V Ratio**

$$\text{P/V Ratio} = \frac{\text{Change in Profit}}{\text{Change in Sales}} \times 100 = \frac{24,000 - 20,000}{1,60,000 - 1,50,000} \times 100 = \frac{4,000}{10,000} \times 100 = 40\%$$

2) **Calculation of Fixed Cost**

$$\textbf{Fixed Cost} = \text{Sales} \times \text{P/V Ratio} - \text{Profit} = 1,50,000 \times \frac{40}{100} - 20,000 = ₹40,000$$

3) **Calculation of Break-Even Point**

$$\textbf{B.E.P. (in Sales)} = \frac{\text{Fixed Cost}}{\text{P/V Ratio}} = \frac{40,000}{40} \times 100 = ₹1,00,000$$

4) **Calculation of Profit at an Estimated Sale of ₹1,50,000**
Profit = (Sales × P/V Ratio) – Fixed Cost = (1,50,000 × 40%) – 40,000 = ₹20,000

5) **Calculation of sales required to earn a profit of ₹40,000**

$$\text{Sales to earn a profit of ₹40,000} = \frac{\text{Fixed Cost} + \text{Desired Profit}}{\text{P/V Ratio}}$$

$$= \frac{40,000 + 40,000}{40\%} = \frac{80,000}{40} \times 100 = ₹2,00,000$$

Example 21: Following information of Zee Co. Ltd. are given.

Cost Element	Variable Cost (% of Sales)	Fixed Cost (₹)
Direct material	32	–
Direct labour	28	–
Factory overheads	12	2,00,000
Distribution overheads	5	60,000
Administration overheads	3	68,000

Budgeted sales are ₹19,00,000

You are required to calculate:

1) The break-even sales volume
2) The profit if actual sales;
   i) Decreased by 10%, and
   ii) Increased by 8% from budgeted sales.
3) The sales required to earn a profit of ₹1,50,000.

Solution:

Statement Showing Contribution and Profit on Budgeted Sales

Particulars		₹
Sales		19,00,000
Less: Variable Cost (80%) of Sales		15,20,000
Contribution		3,80,000
Less: Fixed Cost:		
Factory Overheads	2,00,000	
Distribution Overheads	60,000	
Administration Overheads	68,000	3,28,000
Profit on Budgeted Sales		**52,000**

1) B.E.P (in sales value) $= \frac{\text{Fixed Costs}}{\text{Total Contribution}} \times \text{Sales} = \frac{3,28,000}{3,80,000} \times 19,00,000 = ₹16,39,999.99$ or 16,40,000

2) i) **Profit if 10% Sales Decreased from Budgeted Sales:**

Budgeted Sales		19,00,000
Less: 10% of Budgeted Sales		1,90,000
		17,10,000
Less: Total Cost:		
Variable Cost (80% of ₹17,10,000)	13,68,000	
Fixed Cost	3,28,000	16,96,000
Profit		**14,000**

ii) **Profit if 8% Sales Increased of Budgeted Sales:**

Budgeted Sales		19,00,000
Add: 8% of Budgeted Sales		1,52,000
		20,52,000
Less: Total Cost:		
Variable Cost (80% of ₹20,52,000)	16,41,600	
Fixed Costs	3,28,000	19,69,600
Profit		**82,400**

3) **The Sales Required to Earn a Profit of ₹1,50,000**

$$\text{Sales for Desired Profit} = \frac{\text{Fixed Cost} + \text{Desired Profit}}{\text{P/V Ratio}}$$

$$\text{P/V Ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100 = \frac{3,80,000}{19,00,000} \times 100 = 20\%$$

$$\text{Sales for Desired Profit} = \frac{3,28,000 + 1,50,000}{20} \times 100 = ₹23,90,000$$

4.4. FORMULAS AT GLANCE IN MARGINAL COSTING

Measures in Terms of Units/Quantity

1) **Break-Even Point**

$$\text{Break-even Point} = \frac{\text{Fixed Cost}}{\text{Contribution per Unit}}$$

$\therefore$ Contribution per Unit = Selling Price per Unit – Variable Cost per Unit

2) **Units Sold to Earn a Desired Profit**

$$\text{Sales (Units)} = \frac{\text{Fixed Cost} + \text{Desired Profit}}{\text{Contribution per Unit}}$$

3) **Margin of Safety**

M/S = Total Units Sold – Units Sold at BEP

4) **Profit** = (Sales × Contribution per Unit) – Fixed Cost

Measures in Terms of Figure/Amount

1) $\text{Sales} = \frac{\text{Contribution}}{\text{P/V Ratio}}$ Where, Contribution = Fixed Cost + Profit

2) **Break-Even Point**

$$\text{Break-even Point (BEP)} = \frac{\text{Fixed Cost}}{\text{P/V Ratio}}$$

$$\therefore \text{P/V Ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100 \text{ Or } \frac{\text{Change in Profits}}{\text{Change in Sales}} \times 100$$

3) **Sales to Earn a Desired Profits**

$$\text{Sales} = \frac{\text{Fixed Cost} + \text{Desired Profits}}{\text{P/V Ratio}}$$

4) **Margin of Safety**

M/S = Total Sales – Sales at B.E.P Or

$$\text{M/S} = \frac{\text{Profits}}{\text{P/V Ratio}}$$

5) **Profit/Present Profits** = (Sales × P/V Ratio) – Fixed Cost

Where, Fixed Cost = Contribution – Profit

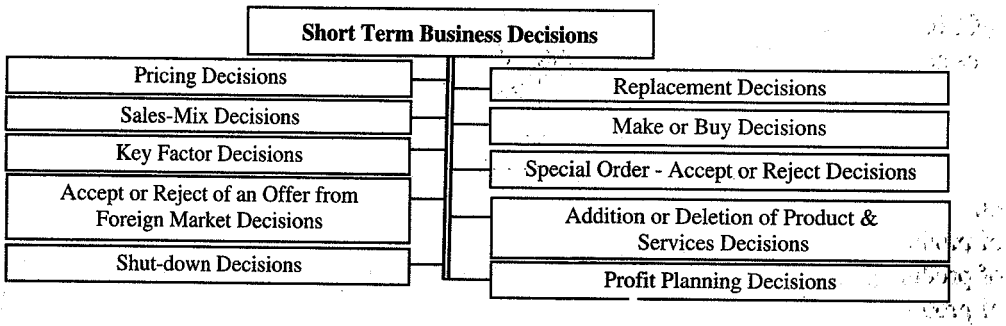
Equation: Sales – Variable Cost = Contribution – Fixed Cost = Profit

Note: Measures in terms of quantity or figure is to be calculated as per the information given in the question.

4.5. SHORT TERM BUSINESS DECISIONS

Managerial decision making follows different process than personal decision making as it requires more focus on systematic and specialised way of operations. Decision making is a difficult process. The main obstacles in the process are inadequate information, conflicting motives and changing circumstances. A solid decision making process is integral to good management as it helps in optimum allocation of resources and accomplishment of objectives.

Marginal costing techniques may be used for making following managerial decisions:



4.5.1. Product Mix/Sales-Mix Decisions

The firms producing more than one product face the issue of deciding sales-mix. This issue arises as different products yield different contributions.

The management should endeavour to design the sales mix which yields highest possible profit. Such ratio of quantities is known as **"Optimum Sales Mix"** and it can be designed using contribution per unit. The product with highest contribution should constitute highest proportion and vice versa.

Example 22: A company manufactures and sale three products Biscuits, Cookies and Brownie. All the three products are from the same set of machines. Production is limited by machine capacity. From the data below, indicate priorities for products with a view to maximising profits:

Particulars	Products		
	Biscuits	Cookies	Brownie
Raw Material Cost Per Unit (₹)	12.25	17.25	22.25
Direct Labour Cost Per Unit (₹)	3.50	3.50	3.50
Other Variable Cost Per Unit (₹)	2.00	2.50	3.60
Selling Price Per Unit (₹)	35.00	40.00	45.00
Standard Machine Time Required Per Unit in Minutes	40	22	30

Solution: Statement Showing the Contribution and Ranking of the Product

Particulars	Products		
	Biscuits	Cookies	Brownie
Selling Price Per Unit	₹35.00	₹40.00	₹45.00
Less: Variable Cost Per Unit	₹17.75	₹23.25	₹29.35
Contribution Per Unit (A)	₹17.25	₹16.75	₹15.65
Standard Machine Time Required Per Unit in Minutes (B)	40	22	30
Contribution Per Minute (A/B)	₹0.43	₹0.76	₹0.52
Ranking	III	I	II

Working Note: Calculation of variable cost per unit:

Particulars	Products		
	Biscuits (₹)	Cookies (₹)	Brownie (₹)
Raw Material Cost	12.25	17.25	22.25
Direct Labour Cost	3.50	3.50	3.50
Other Variable Cost	2.00	2.50	3.60
Total Variable Cost Per Unit	**17.75**	**23.25**	**29.35**

Example 23: Present the following information to show clearly to management.

1) The marginal product cost and the contribution per unit.
2) The total contribution and profits resulting from each of the following mixtures.

Particulars	Product T	Product Q
Direct material per unit (₹)	12	10
Direct wages per unit (₹)	4	3

Fixed expenses ₹1,000

Variable expenses are allotted to the products as 100% of direct wages.

Particulars	Product T	Product Q
Sales Price per unit (₹)	25	20

Sales Mixtures:

1) 100 units of product T and 200 of Q.
2) 150 units of product T and 150 of Q.
3) 200 units of product T and 100 of Q.

Solution:

Marginal Cost Statement

Particulars	Product T (₹)	Product Q (₹)
Selling Price (per unit)	25	20
Less: Marginal/Variable Cost:		
Direct Material	12	10
Direct Wages	4	3
Variable Expenses (100% of Wages)	4	3
	20	16
Contribution (per unit)	**5**	**4**

Sales Mixtures

100 Units of T and 200 of Q		150 units of T and 150 of Q		200 units of T and 100 of Q	
Particulars	**₹**	**Particulars**	**₹**	**Particulars**	**₹**
Contribution		Contribution		Contribution	
T (100 × 5)	500	T (150 × 5)	750	T (200 × 5)	1,000
Q (200 × 4)	800	Q (150 ×4)	600	Q (100 × 4)	400
Total contribution	**1,300**	**Total contribution**	**1,350**	**Total contribution**	**1,400**
Less: Fixed expenses	1,000	**Less:** Fixed expenses	1,000	**Less:** Fixed expenses	1,000
Profit	**300**	**Profit**	**350**	**Profit**	**400**

As sales mixture (3), i.e., 200 units of T and 100 units of Q gives the maximum profit, it is more profitable.

4.5.2. Make or Buy (Outsourcing) Decisions

A firm may buy or manufacture components and spare parts. Marginal costing techniques may be used to determine which course of action is more beneficial for the firm. For this purpose the marginal cost of manufacturing is compared with the purchase price from the market. The firm may choose to manufacture the product if its marginal cost is lower than the purchase price. Other factors such as capacity utilisation should also be taken into account while making such decision. In case of unused capacity, only variable costs should be compared with market price while if the firm is working at full capacity the opportunity costs should also be taken into account.

Factors Influencing Make or Buy Decision

Following factors should be kept in mind while making make or buy decisions:

1) **Plant Capacity:** Sub-contracting is generally advisable when the business is expanding or the sales are of seasonal nature.
2) **Profit Maximisation:** The firm may decide to outsource the production of less profitable products.
3) **Specialisation:** The products requiring specialised techniques may be better bought from outside.
4) **Nature of Product:** The products should be manufactured internally if their consistent supply is required.
5) **Secrecy:** It is better to manufacture the parts if the firm wants to maintain its trade secrets.

Example 24: TPCL Ltd., is producing 30,000 units of a component at a cost of ₹60 per unit. Fixed cost at this level of output is ₹20 per unit. 40% of fixed cost is avoidable. TPCL Ltd. can buy the component from outside at ₹40 per unit. Should the company produce or buy?

Solution:

Total Cost if Production is Continued		Total Cost if Component is Purchased from Outside	
Particulars	**₹**	**Particulars**	**₹**
Fixed cost (20 × 30,000)	6,00,000	Total Fixed Cost	6,00,000
Total variable cost (60 – 20) × 30,000	12,00,000	**Less:** Avoidable Fixed Cost (6,00,000 × 40%)	2,40,000
		Unavoidable fixed cost	**3,60,000**
		Purchase Price (₹40 × 30,000)	12,00,000
Total Cost	**18,00,000**	**Total Cost**	**15,60,000**

Conclusion:
Thus, total cost is less if the component is purchased from outside. Hence, the company should buy the component from outside. Also price paid ₹12,00,000 is less than variable cost plus avoidable fixed cost, i.e., ₹12,00,000 + ₹2,40,000 = ₹14,40,000. Therefore, purchasing from outside is profitable.

Example 25: With the help of the following data, advice a manufacturer whether to buy an item from the market or to produce in factory.

Particulars	Proposed (Make) (₹)
Sales	18,00,000
Costs: Variable	11,00,000
Fixed	6,00,000
Capital required	14,00,000

If the product is purchased from the market, the price would be ₹10,00,000 and fixed cost would be ₹5,00,000.

Solution:

Statement of Cost and Profitability

Particulars	Make (₹)	Buy (₹)
Sales (S)	18,00,000	18,00,000
Less: Variable Costs	11,00,000	10,00,000
Contribution (C)	7,00,000	8,00,000
Less: Fixed Costs	6,00,000	5,00,000
Profit (P)	**1,00,000**	**3,00,000**
P/V Ratio $= \frac{\text{Contribution}}{\text{Sales}} \times 100$	38.89%	44.44%
Percentage of profit on sales (P/S×100)	5.55%	16.67%

Conclusion:
From the above statement, buying of the item is better than to make, because the profit percentage on sales is more than making the product in the factory.

Working Note: Total costs would be reduced by ₹2,00,000 and by the same amount the profit would also increase.

P/V ratio will also increase by 5.55% and the percentage of profit on sales will also increase from 5.55% to 16.67%.

4.5.3. Accept or Reject Special Order Decisions

Sometimes a decision regarding acceptance of a special order is to be made. This, however, depends on the availability of spare capacity. The contribution available from such an order helps in making such a decision.

Example 26: Trishul Limited receives a special order from Quiral Private Limited for supply of 30,000 units of a product that usually sells for ₹10 per unit. Quiral Pvt. Ltd. offers ₹9 per unit for this product. Trishul Pvt. Ltd. incurs ₹6 per unit in variable costs to manufacture each item, plus ₹2 per unit for variable administrative cost.

Total fixed manufacturing costs are ₹1,60,000. Other fixed cost amounts to ₹80,000 per year. Productivity capacity is 2,10,000 units annually and sales volume through normal sales outlets will be about 1,60,000 units of this year.

Draw the marginal cost statement showing profitability and also advice whether to accept or reject the offer.

Solution:

Marginal Cost Statement

Particulars	Per Unit (₹)	Current Year Capacity (1,60,000 Units) (₹)	Proposed Additional Output (30,000 Units) (₹)	Total Capacity 100% (2,10,000 Units) (₹)
Sales	10.00	16,00,000	2,70,000	21,00,000
Less: Marginal cost:				
Variable cost	6.00	9,60,000	1,80,000	12,60,000
Variable admn. cost	2.00	3,20,000	60,000	4,20,000
	8.00	12,80,000	2,40,000	16,80,000
Contribution	**2.00**	**3,20,000**	**30,000**	**4,20,000**
Less: Total Fixed Cost		2,40,000	-	2,40,000
Profit/Loss		**80,000**	**30,000**	**1,80,000**

Note:

Total Fixed Cost = Total Fixed Manufacturing Costs + Other Fixed Cost
= 1,60,000 + 80,000 = ₹2,40,000

Conclusion:

If the order is accepted, revenue will increase by ₹2,70,000 (₹9 × 30,000). The incremental costs will only be ₹2,40,000 (8 × 30,000) - the variable cost of producing the additional units. Therefore, the company will gain ₹30,000 (2,70,000 – 2,40,000) by accepting the special order.

4.5.4. Accept or Reject of an Offer from Foreign Market Decisions

Any special order receives from existing or from new customer for supply of goods or services in the domestic or foreign market should be carefully evaluated by a business concern. Generally, such orders offer lower than normal selling price and are produced using idle capacity. In such cases, no additional fixed cost is incurred. If the quoted price is higher than the marginal cost of the product then the order should be accepted. However, additional costs such as special insurance charges, freight or packing costs should be taken into account. The net profit from the special order is reflected after deducting the additional cost from the contribution amount.

Example 27: A company sells 6,000 units of a product in a domestic market @ ₹25 per unit. The marginal cost of production is ₹15 per unit and the fixed costs are ₹15,000. He wants to sell another 6,000 units in the foreign market @ ₹17 per unit with additional cost of ₹1 per unit to export. Fixed costs are same. Advise the company whether it should try to sell in the foreign market.

Solution:

Particulars	Domestic Market (6,000 Units) (₹)	Additional Sale in Foreign Market (6,000 Units) (₹)	Total (12,000 Units) (₹)
Sales	1,50,000	1,02,000	2,52,000
Less: Marginal Cost	90,000	96,000	1,86,000
Contribution	60,000	6,000	66,000
Less: Fixed Cost	15,000	–	15,000
Profit	45,000	6,000	51,000

Note: Marginal costs for foreign market	= 6,000 ×15	=	₹90,000
Additional cost for export	= 6,000 × 1	=	₹6,000
			₹96,000

Conclusion:

1) It is advisable for the company to sell in the foreign market.
2) Additional sale of 6,000 units in foreign market gives an additional profit of ₹6,000 without incurring any extra costs.

4.5.5. Shutting Down Decisions

In case of firms operating on loss for sometime, the management needs to take decision about its shutdown. Shut-down decision can be put into two categories:

1) **Complete Shut-Down:** The firm decides to shut-down without any intention of reviving the business. This decision is taken in the following situations:
   i) When selling price goes below the variable cost.
   ii) The demand for the output is low and is not expected to be picked up in the future.

   In case of complete shut-down, the company saves on the fixed cost of running the firm.

2) **Partial or Temporary Shut-Down:** Such decisions are temporary in nature and the management intends to start production in future. The firm continues to incur fixed expenses such as salaries and remuneration. The savings from such shut down should be carefully evaluated.

   The shutdown decision should only be taken if the savings are substantial as reviving a business is a tedious process. Such decisions may also entail additional costs such as extra payments for repairing the equipment, training of the new employee, etc. Thus, the savings and costs should be considered for making a final decision.

Example 28: The cost per unit of the three products M, S, T of a company are given below:

Particulars	Product		
	M (₹)	S (₹)	T (₹)
Direct Material	30	20	20
Direct Labour	20	16	14
Variable Expenses	10	12	8
Fixed Expenses	8	8	6
	68	**56**	**48**
Profit	26	16	14
Selling Price	**94**	**72**	**62**
No. of Units Produced	**12,000**	**6,000**	**10,000**

Production arrangements are such that if one product is dropped down, the production of the others can be raised by 50%. The directors propose that 'T' should be drop down because the product has the lowest selling price. Show analysis of the data indicating whether the proposal should be accepted.

Solution:

Statement Showing Contribution Per Unit

Particulars	Product		
	M (₹)	S (₹)	T (₹)
Selling Price	94	72	62
Less: Marginal Cost			
Direct Material	30	20	20
Direct Wages	20	16	14
Variable Expenses	10	12	8
Contribution Per Unit	**34**	**24**	**20**

From the above statement we can conclude that product 'M' earns maximum contribution per unit. Although a limiting factor has not been specified, there is a condition which states that the dropping of a product would lead to the increase in the manufacture of remaining products by 50%. Consequently, before dropping or discontinuing a product, management should take into consideration incremental contribution arising from dropping product 'M' or product 'S' or product 'T'. This is shown below:

Statement Showing Product Discontinuance Analysis

Particulars	Alternatives					
	I		II		III	
	Discontinue Product 'M' (₹)		Discontinue Product 'S' (₹)		Discontinue Product 'T' (₹)	
Additional Contribution from Remaining Products	S (₹24 × 3,000 units)	72,000	M (₹34 × 6,000 units)	2,04,000	M (₹34× 6,000 units)	2,04,000
	T (₹20 × 5,000 units)	1,00,000	T (₹20 × 5,000 units)	1,00,000	S (₹24 × 3,000 units)	72,000
		1,72,000		**3,04,000**		**2,76,000**
Less: Loss of Contribution from Discontinued Product	M (₹34 × 12,000 units)	4,08,000	S (₹24 × 6,000 units)	1,44,000	T (₹20×10,000 units)	2,00,000
Incremental Contribution		**(2,36,000)**		**1,60,000**		**76,000**

The analysis reveals that alternative 'I' namely discontinuance of product 'M' generates the lowest incremental contribution. Therefore, discontinuing product 'T' would result in a sub-optimal decision. Accordingly, management is advised to discontinue product 'M' in order to optimise the company's profit and reject the proposal to discontinue product 'T'.

4.5.6. Pricing Decisions (Single Product and Multi Product Pricing)

Determining of selling price is a critical managerial decision. Various factors such as market conditions and regulations also impact this decision. Following are the various circumstances under which management may have to take such decisions:

1) Pricing under normal conditions,
2) During stiff competition,
3) During trade depression,
4) For accepting special bulk orders,
5) For accepting additional orders utilising idle capacity, and
6) For accepting foreign orders and exploring new markets.

Fundamental Principles of Pricing Decisions

Decision making is a continuous process in a company and the managers take different decisions from which the most important decisions are related to pricing. Listed below are some of the major pricing decisions taken by the managers:

1) Price setting for an innovative or a new product.
2) Price determination of the products traded under private labels.
3) Tackling with price strategy opted by the competitors in the market.
4) Pricing bids in conditions of open and sealed biddings.

Example 29: (Single Product Pricing): The product has variable cost of ₹8 per unit and fixed cost of ₹60,000. Selling price is ₹9 per unit, and 30,000 units can be sold at this price. Is it agreeable to continue production?

Solution:

Variable Cost ₹(30,000 × 8)	₹2,40,000
Add: Fixed Cost	60,000
Total Cost	₹3,00,000
∴ Cost per unit = ₹(3,00,000 ÷ 30,000) = ₹10	

Although the selling price does not cover the total cost, the production should not be discontinued as it will minimize loss. Loss will be ₹60,000 if production is discontinued but if production continues the loss will be as follows:

Sales ₹(30,000 × 9)	₹2,70,000
Less: Total Cost	3,00,000
Loss	(30,000)

Thus, by continuing production, loss is reduced by ₹30,000.

Example 30: (Multi Product Pricing): Pesso Goods Limited, manufactures pesticides. The accounts of the company show an expected profit of ₹15,00,000 from the manufacture of pesticides after charging fixed costs of ₹12,00,000. The pesticide is sold for ₹60 per cup and has a variable unit cost of ₹20.

Market fluctuations have the following responses to price changes:

Alternatives	Selling Price Reduced by	Quantity Sold Increase by
P	5%	10%
Q	7%	20%
R	10%	25%

Evaluate these alternatives and state which, on profitability consideration, should be adopted for the forthcoming year, assuming cost structure remains unchanged.

Solution:

Computation of Quantity Sold

Particulars	₹
Net profit	15,00,000
Add: Fixed cost	12,00,000
Contribution	**27,00,000**

Contribution per Unit = Selling Price – Variable Cost
= ₹(60 – 20) = ₹40
Unit Sold: ₹27,00,000/40 = 67,500 Units

Pesso Goods Limited
(Statement showing results for proposals)

Alternatives (1)	Reduction in Selling Price (2)	Rate (₹) (3)=60×(2)	Revised Contribution per Unit (₹) (4) = 40–(3)	Rate Increase in Sale (5)	Qty. (6)=67,500×(5)	Revised Sale Qty. (Units) (7)=67,500+(6)	Total Contribution (₹) (8)=(7)×(4)
P	5%	3.00	37.00	10%	6,750	74,250	27,47,250
Q	7%	4.20	35.80	20%	13,500	81,000	28,99,800
R	10%	6.00	34.00	25%	16,875	84,375	28,68,750

As per the above statement, we can conclude that alternative Q is more profitable than alternatives P&R. Thus, the company is advised to go for alternative Q for coming proposals.

4.5.7. Key or Limiting Factor Decisions

A key factor may be defined as, "a factor which may limit production or profit of a firm". Sale is the most common limiting factor. The firm's production is generally limited by its capacity to sale.

Other limiting factors may be materials, labour, capital or plant capacity. In such cases, the management may need to decide which products to be produced and in which quantity.

In the absence of any limiting factor, the firm will decide its production strategy on the basis of P/V ratio. However, if there are limiting factors, then the decision is made using contribution per unit of limiting factor.

In the presence of a limiting factor, the profit may be maximised by making a contribution analysis for that factor. **For example,** if labour is the limiting factor, then the labour hours required for each product must be calculated. The contribution is then expressed in terms of rupees per labour hours. The products with higher contribution per labour hour should be chosen for maximising profits.

Particulars	Product A	Product B	Product C	Total
Sales Units (Pcs.)	500	300	200	1000
Contribution (₹)	2500	1200	1400	5100
Labour hours utilised	6000	4800	2800	13600
Contribution per labour hour	0.42	0.25	0.50	

The above analysis shows that the firm can increase its profits by limiting the production of product B and increasing production of Product A and Product C. Similar analysis may be made for other limiting factors as well. In these situations where there are more than one limiting factors, the factor which restricts the activity level to the minimum should be regarded as the key factor. Maximum contribution may be generated by focusing on the product which makes the best use of the limiting factor. In such cases, profitability index may be used for calculating contribution per unit of limiting factor.

Example 31: The following information is obtained from TLC Co. Ltd. producing products M and N:

Particulars	Product M (₹)	Product N (₹)
Selling price	255	165
Direct material	90	90
Direct labour (₹5 per hour)	10 hrs.	6 hrs.

Variable overhead 50% of direct wages
Show the profitability statement of products considering labour as a key factor.

Solution: Formula used for determining profitability:

$$\text{Profitability} = \frac{\text{Contribution}}{\text{Labour Hrs.(Key or limiting Factors)}}$$

Statement Showing Contribution and Profitability

Particulars	Product M (₹)	Product N (₹)
Sales (S)	255	165
Less: Variable cost (V)		
Direct material	90	90
Direct wages (Rate per hour × No. of hours)	50	30
Variable overhead	25	15
	165	**135**
Contribution per unit (S –V)	**90**	**30**
Labour Hours	10 hours	6 hours
Profitability = Contribution/Labour hrs.	90/10= 9.00	30/6= 5.00

Conclusion: In the **above example**, it has been shown that the two products M & N had different revenue earning potentialities, which were exploited by several ways to increase total profit. Thus during labour shortage, Product M is more profitable than Product N.

Example 32: The following data is given:

Particulars	Product X (₹)	Product Y (₹)
Direct materials	50	70
Direct labour @ ₹3 per hour	15	40
Variable overhead @ ₹4 per hour	20	50
Selling price	300	320
Standard time	3 hours	4 hours

State which product you would recommend to manufacture when:

1) Labour time is the key factor,
2) Sales value is the key factor.

Solution:

Particulars	Product X (₹)	Product Y (₹)
Selling Price (S)	300	320
Less: Variable Cost		
Direct material	50	70
Direct labour	15	40
Variable overhead	20	50
Total Variable Cost (V)	**85**	**160**
Contribution (S – V)	215	160
Contribution per labour hour	215 ÷ 3 hrs = 72	160 ÷ 4 hrs = 40
Contribution per rupee of sales value	215 ÷ 300 = 0.72 paise	160 ÷ 320 = 0.50 paise

Conclusion:

1) Product X is recommended when labour time is the key factor because contribution per labour hour of product X is more than that of product Y.
2) When sales value is the key factor, product X is recommended because contribution per rupee of sales value of product X is more than that of product Y.
3) When sale quantity is the key factor, product Y is more profitable because its contribution per unit is less than that of product X.

4.5.8. Profit Planning Decisions

Variable costing techniques are useful for profit planning. This process involves planning of future operations for the achievement of maximum profits or for the maintenance of desired level of profits. Marginal costing is concerned with contribution margin.

It may be expressed as a percentage of sales. In such cases, it is known as **marginal ratio or contribution ratio**. It shows the amount of paisa obtained from every rupee of sales for the purpose of covering fixed costs and for generating profits.

In case of fixed production and selling capacity, short-run profits fluctuate with the change in variable costs, selling price and in volume and mix of products sold.

Example 33: The budgeted results for 'Y Co. Ltd.' include the following:

Sales	₹(in Lac)	Variable Cost as a Percentage of Sales Value (%)
Product 'P'	30	70
Product 'Q'	25	40
Product 'R'	45	60
Product 'S'	20	90
Product 'T'	33	70
	153	**66%**

Fixed overheads for the period are ₹50 lac.

You are required to prepare:

1) A statement showing the expected loss.
2) Recommend a change in the sales volume of each product which will eliminate the expected loss, assuming that the sale of only one product can be increased at a time.

Solution:

1)

Statement showing Contribution and Loss Expected ₹(in Lac)

Product	Sales	Variable cost		P/V Ratio	Contribution
		as a % of Sales Value	Amount		
(1)	(2)	(3)%	(4) = (2÷3) ₹	% (5) = (100% – 3)	(6) = (5-4) ₹
P	30	70	21	30	9
Q	25	40	10	60	15
R	45	60	27	40	18
S	20	90	18	10	2
T	33	70	23.10	30	9.9
Total Overheads					53.90
Less: Fixed overhead					50.00
∴Loss					**(3.90)**

Note: Contribution= Sales- Variable cost (30 – 21= 9; and so on)

2) Additional sales required to break-even is shown in the following statement assuming that the sale of only one product is increased at a time to generate an additional contribution of ₹3.90 lac (i.e., amount of loss).

$$\therefore \text{Additional Sales Required to Eliminate Expected Loss} = \frac{\text{Desired Contribution for Eliminating Loss}}{\text{P/V Ratio of Each Product}}$$

Particulars			Amount (₹in lac)
Product 'P'	=	₹3,90,000 / 30%	13
Product 'Q'	=	₹3,90,000 / 60%	6.5
Product 'R'	=	₹3,90,000 / 40%	9.75
Product 'S'	=	₹3,90,000 / 10%	39
Product 'T'	=	₹3,90,000 / 30%	13

4.6. EXERCISE

4.6.1. Very Short Answer Type Questions

1) Define marginal costing.
2) What do you mean by CVP Analysis?
3) What is contribution?
4) Define P/V ratio.
5) What is margin of safety?
6) What is angle of incidence?
7) What is profit planning?
8) What do you mean by make or buy decision?
9) List the types of decision making in marginal costing.

4.6.2. Short Answer Type Questions

1) Discuss the scope of marginal costing.
2) Explain the assumptions of CVP analysis.
3) Discuss the term contribution in detail.
4) What are the assumptions of break-even analysis?
5) What is the significance of break-even analysis?
6) What do you mean by break-even point? How it is calculated?

4.6.3. Long Answer Type Questions

1) Discuss the advantages and disadvantages of marginal costing.
2) Explain the applications and limitations of P/V ratio.
3) What is break-even analysis? Discuss its limitations?
4) What is break-even chart? Construct a break-even chart.
5) Explain the various types of short term business decisions in detail.

4.6.4. Practical Questions

CVP Analysis

1) From the following particulars:

Particulars	₹
Sales	50,000
Variable cost	30,000
Fixed cost	7,500

Calculate:
i) Break-Even Point
ii) P/V ratio
iii) Profit
iv) New Break-Even Point, if selling price is reduced by 10%.
v) New Break-Even Point, if variable cost increased by 10%.

[Ans: Break-Even Point = ₹18,750, P/V Ratio = 40%, Profit = ₹12,500, New Break-Even Point, if selling price is reduced by 10% = ₹22,500, New Break-Even Point, if variable cost increased by 10% = ₹22,058.8]

2) The following information is obtained for the year ended 31-3-19:

Particulars	₹
Sales (1,00,000 Units)	50,000
Marginal Cost	30,000
Fixed Cost	15,000

Calculate:
i) P/V Ratio
ii) Break-Even Point (Value)
iii) Sales to earn a profit of ₹7,500.
iv) Profit when sales amounted to ₹70,000.

[Ans: P/V Ratio = 40%, BEP (Sales Value) = 37,500 units, Sales to Earn a Profit of ₹7,500 = ₹56,250, Profit when Sales Amounted to ₹70,000 = ₹13,000]

3) From the following information:

Particulars	₹
Direct Material Cost	22,500
Direct Labour Cost	15,000
Variable Overhead Cost	7,500
Fixed Cost	17,000
Sales	75,000

Calculate:
i) P/V Ratio
ii) Break-Even Point
iii) Sales Required to earn Profit of ₹25,000
iv) The Profit made when Sales are ₹1,25,000

[Ans: P/V Ratio = 40%, Break-even Point = ₹42,500; Sales required to earn a profit of ₹25,000 = ₹1,05,000; The profit made when sales are ₹1,25,000 = ₹33,000]

4) Arjun Ltd. has furnished the following particulars relating to the year ending 31-3-2019:

Particulars	₹
Net Profit	2,00,000
Fixed Cost	6,00,000
Sales	20,00,000
Variable Cost	12,00,000

You are required to calculate:
i) P/V Ratio
ii) Break-Even Point
iii) Profit for Sales Volume ₹24,00,000
iv) Sales to earn profit of ₹4,00,000

[Ans: P/V Ratio = 40%; Break-Even Point = ₹15,00,000; Profit for Sales Volume ₹24,00,000 = ₹3,60,000; Sales required to earn a profit of ₹4,00,000 = ₹25,00,000]

5) Given:

Particulars	₹
Sales Price	40 per unit
Variable Manufacturing Cost	22 per unit
Variable Selling Cost	6 per unit
Fixed Factory Overheads	10,80,000 per year
Fixed Selling Costs	5,04,000 per year

Calculate:
i) Break-Even Point
ii) Sales required to earn a profit of ₹1,20,000.
iii) Sales required to earn a profit of 10% of sales.

[Ans: Break-even point = ₹52,80,000, Sales required to earn a profit of ₹1,20,000 = ₹14,20,000, Sales required to earn a profit of 10% of sales = ₹79,20,000]

6) The following figures are available from the records of Venus Enterprises as at 31st March, 2019:

Year	Sales (₹ in lac)	Profit (₹ in lac)
2018	300	60
2019	400	100

Calculate:
i) The P/V ratio
ii) Total fixed expenses;
iii) The break-even level of sales:
iv) Sales required to earn a profit of ₹180 lac;
v) Profit or loss that would arise if the sales were ₹560 lac.

[Ans: P/V Ratio = 40%; Total Fixed Expenses = ₹60 lakhs; Break-Even Level of Sales = ₹150 lakhs; Sales of ₹600 lac is required to earn a profit of ₹180 lac; Profit or Loss on the sales of ₹360 lac = ₹164 lakhs]

Sales Mix Decision

7) From the following information, suggest the best sales mix:

Particulars	X Per Unit (₹)	Y Per Unit (₹)
Selling Price	25	20
Direct Materials	8	6
Direct Wages	6	4

Variable overheads 150% of direct wages.
Fixed overheads ₹750.

Sales mix:
i) 250 units of X; 250 units of Y;
ii) 400 units of X, 100 units of Y;
iii) 400 units of Y only; and
iv) 150 units of X and 350 units of Y.

[Ans: Profit (1) ₹750; (2) ₹450; (3) ₹850; (4) ₹950; Mixture (4) is recommended]

8) Vishal Ltd. which produces three products, furnishes the following data for the year 2018-19:

Particulars	Products		
	Alfa	Beta	Gama
Selling Price Per Unit	₹100	₹75	₹50
Profit/Volume Ratio	10%	20%	40%
Maximum Sales Potential (Units)	40,000	25,000	10,000
Raw Material as % of Variable Cost	50%	50%	50%

The company uses the same raw material for all the three products. Raw material is in short supply and the company has a quota for supply of raw material of the values of ₹18,00,000 for the year for manufacture of its products to meet its sales. Total fixed cost is ₹6,80,000.

You are required to:

i) Determine a sales mix which will give the maximum overall profit keeping in view the short supply of raw material.

ii) Compute that maximum profit.

[Ans: i) Gama = 10,000 units; Beta = 25,000 units; Alfa = 20,000 units; ii) Maximum Profit = ₹95,000]

Make or Buy Decision

9) You are producer and seller of plastic products. You are producing 10,000 buckets at a fixed cost of ₹3,00,000 and variable cost of ₹50 per bucket, i.e., at an average cost of ₹80 per bucket. You can buy buckets from outside at a price of ₹65 per bucket. The capacity so vacated can be used for producing ₹7,000 drums at a variable cost of ₹40 per drum plus an additional fixed cost of ₹1,00,000.

Each drum can be sold @ ₹90 per drum. Will you continue producing buckets or instead produce drums and buy buckets from outside?

[Ans: Produce drums and buy buckets as it saves ₹1,00,000.]

10) X Ltd. manufactures component A-100 and the costs for the year 2019 when 50,000 units are produced is as follows:

Particulars	₹
Materials	2.50
Labour	1.25
Variable Overheads	1.75
Fixed Overhead	3.50
Total Cost	**9.00**

Component A-100 could be bought in for ₹7.75 and if so the production capacity utilised in 2019 would be unused. Assuming that there is no technical consideration, should component A be bought in or manufactured.

[Ans: The component should be manufactured and not bought because marginal cost of manufacture is ₹5.50 and the buying price is ₹7.75. The reason for this is that the fixed cost of ₹17,500 (50,000 units × ₹3.50) would continue and as the capacity is not used, the fixed overhead would not be absorbed into production. If component A-100 is bought, overall profit will fall by ₹1,12,500 being the difference between buying price and marginal cost of manufacture, i.e., (7.75 – 5.50 × 50,000)]

11) The management of a company finds that while the cost of making component part is ₹20. The same is available in the market at ₹18 with an assurance of continuous supply.

i) Give a suggestion whether to make or buy this part; and

ii) Give also your views in case the supplier reduces the price from ₹18 to ₹16.

The cost information is as follows:

Particulars	₹
Materials	7.00
Direct Labour	8.00
Other Variable Expenses	2.00
Fixed Expenses	3.00
Total	**20.00**

[Ans: (i) Variable cost ₹17, hence not profitable to buy; (ii) There is a saving of ₹1 per component, the offer may be accepted]

Accept or Reject Special Order

12) ABC Sports Ltd. is famous for specialised manufacture of quality chessboard sets. Presently, the company is working below its normal capacity of 1,000 units per month. The company sells its chessboards in the national market at ₹150 per unit. During April 2018, 600 units were sold which is the regular sales volume for each month all through the year. The unit cost of production is as follows:

Particulars	₹
Direct Material	60
Direct Labour	30
Factory Overhead	30
Selling and Administration Overhead	15

The company has received an export older on 20 April, 2018 for supply of 600 units to be dispatched by 30 June, 2018. However, the stipulated the price per unit is ₹100 only. The cost analysis indicates that the cost of direct material and direct labour that are to be incurred on the export order would be the same amount per unit as the regular line of production.

However, an amount of ₹2,000 will have to be incurred on special packing, labelling, packaging, etc. No additional factory selling or administrative overhead costs would be incurred in executing the export order since the firm is operating below normal capacity. Using differential cost analysis method, prepare the income statement to show whether the acceptance of the export order would be profitable to the company. Assumptions and comments, if any, may be given separately.

[Ans: Accepting the export order, the company's profit has increased by ₹4,000 for the period of 2 months covering May and June 2018. Hence, export order can be accepted.]

Operate or Shut Down Decision

13) G Ltd. produces and sells 95,000 units of 'X' in a year at its 80% production capacity. The selling price of product is ₹8 per unit. The variable cost is 75% of sales price per unit. The fixed cost is ₹3,50,000. The company is continuously incurring losses and management plans to shutdown the plant. The fixed cost is expected to be reduced to ₹1,30,000. Additional cost of plant shutdown is expected at ₹15,000.

Should the plant be shutdown? What is the capacity level of production of shutdown point?

[Ans: It is better to shutdown the plant. Capacity level of production is 86.31%]

UNIT 5

Budgetary Control & Standard Costing

5.1. BUDGETARY CONTROL

5.1.1. Meaning of Budget

A **budget** may be defined as a financial tool which is used for planning and for projecting future values, using past experience. It is a quantitative statement which states future course of action in a numerical form. It covers a definite period of future and presents written operational plan. A budget uses money and quantity to express its plan. It charts out the policy to be followed for the achievement of organisational objectives. The main goal of the budget is to achieve organisational goals in a specified manner.

According to CIMA Terminology, a budget is "A plan expressed in money. It is prepared and approved prior to the budget period and may show income, expenditure, and the capital to be employed. It may be drawn up showing incremental effects on former budgeted or actual figures, or be compiled by zero-based budgeting".

According to Crown and Howard, "A budget is a pre-determined statement of management policy during a given period which provides a standard for comparison with the results actually achieved".

Thus, a budget is a tool for financial administration and holds a prime position in financial planning. It is a financial statement and is used for correlating various operations in public finance. The statement of budget is put forward to the management for the approval and sanction. It provides forecast of future fiscal policies. Budgets are also used by governments to report their financial performance and to forecast their fiscal policies.

5.1.2. Meaning and Definition of Budgeting

Budgeting is the course of preparing, implementing and processing of budgets. It involves formulation of plans in numerical terms for a specified period of time in future. Budgeting attempts to prevent future issues and to resolve them. A business is required to evaluate its position in the current economic environment. It helps in establishing goals and objectives for all the levels in an organisation. Business transactions are required to be undertaken for the purpose of achieving these goals. Budgets are prepared for summarising the forecasted data. In nutshell, budgeting involves resource allocation. This tool can be used by a variety of business organisations including non-profit concerns.

Budgeting is a managerial tool and is used for the purpose of short-term planning and control. It denotes the complete cycle of designing, implementing and operating budgets. The main purpose of short-term budgets is to ensure that the plans are provided with adequate resources.

According to Shillonglow, "Budgeting is the preparation of comprehensive operating and financial plans for specific intervals of time".

According to William J. Vatter, "Budgeting is a kind of 'future tense' accounting in which the problems of future are met on paper before the transactions actually occur".

5.1.3. Meaning and Definition of Budgetary Control

Budgetary control involves determining budgeted figures for definite period of future, comparing actual figures with the budgeted ones and finding out the variances. This process helps in finding deviations and the reasons behind them. It aids in implementing corrective measures for improving the situation. It is an ongoing process and helps in planning and coordination. Budget is a tool by which implementation of budgeting processes are made to achieve the end result of budgetary control.

The first step is to prepare all the budgets. This is followed by the recording of actual results and their comparison. In case of discrepancies, the management is required to take remedial actions.

According to Brown and Howard, "Budgetary control is a system of controlling costs which includes the preparation of budgets, coordinating the department and establishing responsibilities, comparing actual performance with the budgeted and acting upon results to achieve maximum profitability".

According to CIMA, "The establishment of budgets relating the responsibilities of executives to the requirement of a policy, and the continuous comparison of actual with budgeted results either to secure by individual action the objective of that policy or to provide a basis for its revision".

5.1.4. Characteristics of Budgetary Control

Following are the main characteristics of budgetary control:

1) **Establishment of Budgets:** Budgets are prepared for each and every functional area in the business. These functional budgets are then merged to create a master budget, which is applicable throughout the organisation.

2) **Officers' Responsibility:** Every officer is given certain responsibilities and tasks. Such responsibilities and tasks contribute towards the accomplishment of organisational goal.

3) **Requirement of a Policy:** It is a policy statement and shows the future plans of the business. It also states a course of action for the implementation of such plans.

4) **Comparison of Actual Figures with Budget Figures:** Budgeting is also used as a control tool. It helps in setting up future plans, measuring actual performance and comparing it against budgeted figures. In case of deviations, the reasons are found out and are corrected.

5) **Revision of Policy:** In case of deviation from budgeted figures, several actions may be taken. Revision of policy may also be required in order to achieve the budgeted numbers. Therefore, the policies must be flexible to incorporate the corrective actions.

5.1.5. Objectives of Budgetary Control

Budgetary control has the following objectives:

1) **Basic Purpose:** A budgetary control system helps in planning, coordinating and controlling, which are its primary objectives.

2) **Cooperative Spirit:** Departmental and organisational budgets help in aligning different objectives. Budgetary control helps them in working towards a common goal.

3) **Maximum Profitability:** A budget is a planning tool and helps in increasing profit potential of the business.

4) **Centralised Control:** Budgets also help in facilitating control as it helps in delegation of authority and responsibility.

5) **Best Use of Resources:** Budgeting helps the organisation to stay on its course for the achievement of common objectives. It helps in making the best use of its resources.

6) **Coordination:** Budgeting helps in coordinating the activities of various parts of an organisation. It also coordinates various activities such as purchase, sales and administration.

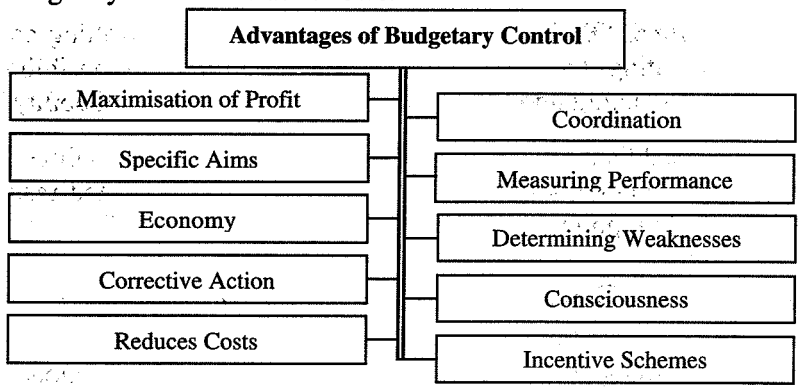
7) **Execution:** Budgeting also helps in overseeing execution of the plans. It helps in ensuring that the plans are being followed properly. It also takes prompt remedial actions if there are any deviations from the standards.

8) **Remedial Measures:** Budgetary controls are meant to take corrective measures as and when actual numbers deviate from the budgeted ones.

9) **Revision:** Budgetary controls also encompass making revisions if actual results are not in line with budgeted numbers.

5.1.6. Importance/Advantages of Budgetary Control

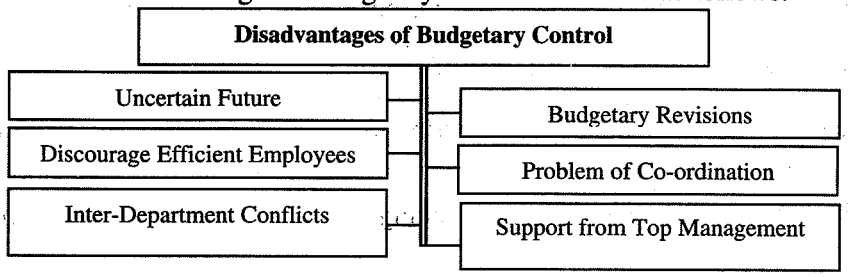
Various advantages of budgetary control are as follows:



1) **Maximisation of Profit:** The budgetary controls help the organisation in achieving maximum profits by properly planning and coordinating various activities. It also controls expenditure of capital and revenue nature. Thus, budgeting ensures that resources are used in the best possible manner.
2) **Coordination:** The working of different areas of an organisation is coordinated through the use of budgetary controls. Such coordination helps in the efficient achievement of organisational objectives. The success of budgeting depends on coordination.
3) **Specific Aims:** Budgetary controls help in setting up specific aims which are important for the purpose of planning and control. This also helps in setting up targets for various departments. Top management is responsible for setting up plans and policies to be followed.
4) **Measuring Performance:** Budgetary controls are important measuring tools as these help in setting up specific targets which help in measuring the performance. The actual performance is compared with the budgeted ones to find out the deviations. This helps in the implementation of 'management by exception' system.
5) **Economy:** Budgetary controls help in controlling expenditures as well. Systematic planning through budgeting is important for this purpose. The efficient use of resources is not only important for firms but has economy-wide impact. It also helps in reducing the wastage of precious resources and increases performance.
6) **Determining Weaknesses:** Comparison of actual performance helps in finding out weak spots. This leads to better performance as corrective measures can be taken for improving the performance.
7) **Corrective Action:** Budgetary control helps in timely identification of deviations. These deviations are required to be reported regularly to the management. In the absence of such controls, these deviations would be reported only at the end of the reporting period.
8) **Consciousness:** Budgetary control helps employees in getting acquainted with organisational goals and objectives. It also clearly demonstrates the targets that are required to be achieved.
9) **Reduces Costs:** Budgetary controls are also helpful in controlling costs which further helps in increasing profit by reducing the production cost. These controls help in designing product mix which is more profitable.
10) **Incentive Schemes:** Budgetary control helps in setting up of targets and measurement of actual performance. This facilitates the implementation of incentive schemes.

5.1.7. Disadvantages of Budgetary Control

There are several limitations/disadvantages of budgetary control which are as follows:



1) **Uncertain Future:** Budget involves future projections and future is uncertain. This greatly reduces the efficacy of budgets as future cannot be predicted with certainty. Any significant change in conditions may require change in budget as well. Such changes may increase the running costs of the organisation.

2) **Budgetary Revisions:** Since future is uncertain, the budgets may be required to be revised. However, frequent change in budgets is not desirable as it may affect the efficiency of employees. Such revisions are also expensive.

3) **Discourage Efficient Employees:** Budgetary control may demoralise efficient employees as it involves setting up of the targets. People tend to work towards the achievement of targets only and not beyond them. In such cases, budgetary control may curtail the performance instead of enhancing it.

4) **Problems of Coordination:** The coordination between various departments helps in proper implementation of budgetary control.

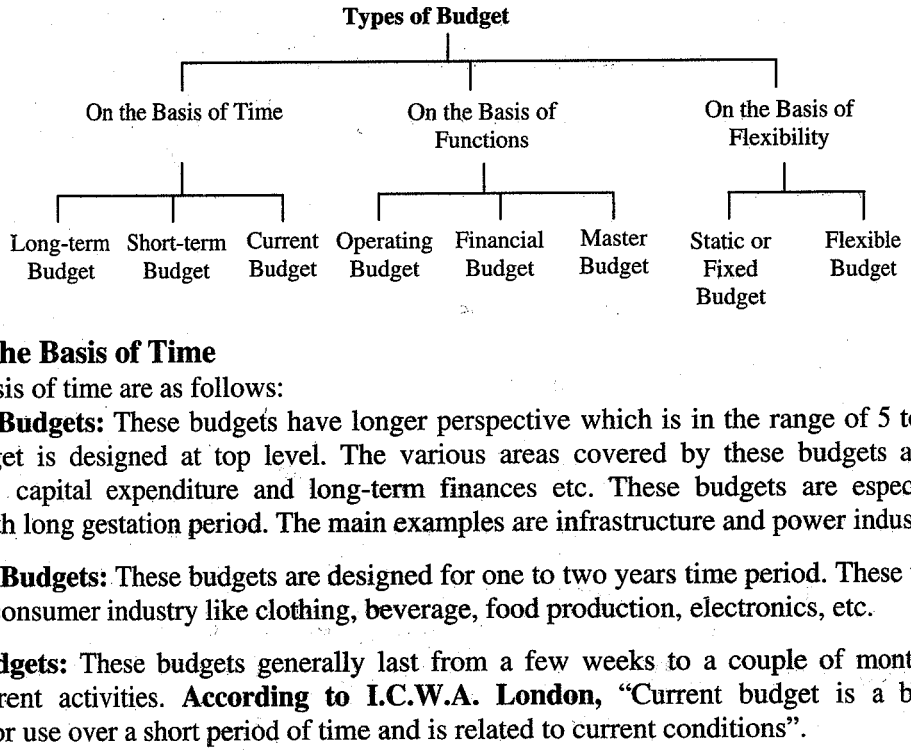
   The organisation may appoint a budgetary officer for the purpose of coordination. However, many organisations cannot bear additional employees which may impact the performance of budgetary control.

5) **Inter-Department Conflicts:** Budgetary control may lead to inter-departmental conflicts as they may fight for the limited resources. Different departments are provided with their own goals to be achieved and they may try to accomplish such goals at the expense of overall organisational objectives.

6) **Support from Top Management:** Budgetary control requires proper support from top management as it involves use of resources. Top management should cooperate with all the levels to ensure the success of budgetary control.

5.1.8. Types of Budget

Normally, budgets are classified according to their nature. The most commonly used budgets are:



5.1.8.1. On the Basis of Time

Budget on the basis of time are as follows:

1) **Long-Term Budgets:** These budgets have longer perspective which is in the range of 5 to 10 years. This type of budget is designed at top level. The various areas covered by these budgets are research and development, capital expenditure and long-term finances etc. These budgets are especially useful for industries with long gestation period. The main examples are infrastructure and power industry.

2) **Short-Term Budgets:** These budgets are designed for one to two years time period. These types of budgets are used for consumer industry like clothing, beverage, food production, electronics, etc.

3) **Current Budgets:** These budgets generally last from a few weeks to a couple of months. This budget involves current activities. **According to I.C.W.A. London,** "Current budget is a budget which is established for use over a short period of time and is related to current conditions".

5.1.8.2. On the Basis of Functions/ Functional Budget

Budget on the basis of functions/functional budget are as follows:

1) **Operating Budgets:** These budgets are related to various operations or activities of a firm. Different organisations have different budgets, depending on their organisational structure and processes. The most common operating budgets are as follows:
   i) **Material Budget:** This type of budget deals with raw materials, packing materials and components which are consumed for the purpose of production. It presents quantitative and monetary representation.

ii) **Production Budget:** This budget forms a part of the master budget. It involves determination of future production targets. This budget projects the number of units to be produced for each product.

iii) **Sales Budget:** This budget estimates the expected sales during the budget period. This budget is comprehensive in nature and forms the plan for sales department.

iv) **Labour Budget:** This budget is drawn in conjunction with production budget. Once the production quantity is budgeted, the next step is to calculate the requirement of labour for carrying out production. This budget also projects labour cost by multiplying budgeted labour hours with wage rate.

v) **Overhead Budget:** This budget is required for the preparation of Production Budget. This budget takes care of indirect expenses. It collates the information from various departments.

If a firm constructs its budget in the form of programmes or responsibility areas, it may involve following:

i) **Programme Budget:** This budget enumerates expected revenue and costs related to various projects or products. This budget shows revenue, cost and estimated profit of various programmes run by the organisation.

ii) **Responsibility Budget:** An operating budget designed for a responsibility area is known as responsibility budget. It shows the plans for people responsible for accomplishing them. It can also be used for control purpose. This budget can be used for evaluating the performance of various managers who are in charge of different programmes.

2) **Financial Budget:** This type of budget involves working capital, capital expenditure, cash receipts and disbursement and financial position of the business. Most commonly used financial budgets are as follows:
   i) Cash Budget,
   ii) Working Capital Budget,
   iii) Capital Expenditure Budget,
   iv) Income Statement Budget,
   v) Statement of Retained Earnings Budget, and
   vi) Budgeted Balance Sheet or Position Statement Budget.

3) **Master Budget:** A master budget is designed by merging several functional budgets. According to I.C.W.A. London, "The Master Budget is the summary budget incorporating its functional budgets". This budget is useful for top level management and is used for coordinating the activities of different departments. This type of budget can also be used as control measure.

5.1.8.3. On the Basis of Flexibility

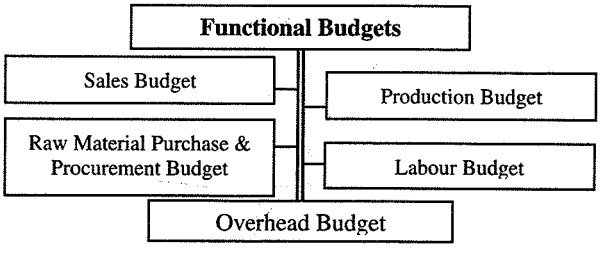
Budget on the basis of flexibility are as follows:

1) **Static or Fixed Budget:** The static budgets are designed for a particular level of activity. It is prepared before the beginning of the financial year. This budget is not adjusted in response to actual circumstances. There is generally a gap of more than 12 months between budgeted and actual figures.

2) **Flexible Budgets:** A flexible budget involves a series of budgets dealing with different levels of activities. As suggested by its name, the budget varies according to the level of activities. It classifies various costs into fixed, semi-fixed and variable cost categories.

5.2. FUNCTIONAL BUDGETS

Functional budgets relate to various functional activities of the business, sales, purchases, production. Functional budgets commonly prepared are:



5.3. SALES BUDGET

5.3.1. Meaning of Sales Budget

A sales budget makes projections regarding estimated sales for the budgeting period. It is an important tool and is also known as **"backbone of the enterprise"**. This budget forms the basis for other budgets such as production budget, materials budget and labour budget. The accuracy of these budgets depends on the accuracy of operating budget.

A sales budget shows potential sales in terms of units and value. This budget acts as a complete plan of action for sales department. Sales budget is prepared by the sales manager. The information for preparing this budget is collected from various internal and external sources.

5.3.2. Objectives of Sales Budget

A sales budget has following objectives:

1) **Planning:** Budgeting facilitates the attainment of marketing and sales objectives set by the organisation. Sales budget divides these objectives and reflects the same among products, customers and territories.
2) **Co-ordination:** A budget helps in coordinating revenue and expenditure. This also helps in controlling expenditure as various executives know the amount of money and other resources at their disposal.
3) **Control:** A budget can act as a control tool as it helps in evaluating executive's performance against the set standards. A sales budget clearly enumerates sales and costs goals.
4) **Evaluation:** Sales budget clearly enumerates the goals to be accomplished and makes the process of evaluation more efficient.
5) **Others:** Other objectives of sales budget are:
   i) To expand scope of the market, ii) To earn estimated profit, and iii) To conduct market research.

5.3.3. Advantages of Sales Budget

Following are the main advantages of a sales budget:

1) It sets sales targets to be achieved by the sales department.
2) This budget helps in proper allocation of resources amongst various products and territories for the efficient achievement of sales targets.
3) This budget also helps in controlling costs and thus maximising net profits.
4) It also helps in control process as it sets the standards and measures the actual performance against them.
5) It finds out problem areas where the performance is lacking.

5.3.4. Format of Sales Budget

Period / Area	Budgeted in the Current Period			Actual Sales			Budgeted for the Future Period		
	Units	Rate (₹)	Amount (₹)	Units	Rate (₹)	Amount (₹)	Units	Rate (₹)	Amount (₹)
Northern Area									
Product X									
Product Y									
Southern Area									
Product X									
Product Y									
Eastern Area									
Product X									
Product Y									
Western Area									
Product X									
Product Y									
Grand Total									

Example 1: ABC Manufacturing Co. produces and sells Biscuit, Chips and Cold drink.
The company bifurcates its market into Department A and Department B. Following are the sales figures for the previous year:

Products	Department A		Department B	
	Units	Unit Price (₹)	Units	Unit Price (₹)
1) Biscuit	3,000	1,100	2,000	1,100
2) Chips	3,500	1,600	3,000	1,600
3) Cold drink	2,000	1,500	2,500	1,500

Department A is expected to experience 20% increase in biscuit sales while chips sale is likely to be increased by 500 units in Department B. Cold drink sale is expected to be increased by 20% in both the regions. The company would keep biscuit and chips prices constant, though it will cut cold drink price by ₹100 in both Department A and Department B. Prepare quantitative cum-financial budget for sales in the next year.

Solution:

Sales Budget of ABC Manufacturing Company

Products	Biscuit			Chips			Cold drink			Total
	Units	Unit Price (₹)	Amount (₹)	Units	Unit Price (₹)	Amount (₹)	Units	Unit Price (₹)	Amount (₹)	Units
Department A	3,600	1,100	39,60,000	3,500	1,600	56,00,000	2,400	1,400	33,60,000	1,29,20,000
Department B	2,000	1,100	22,00,000	3,500	1,600	56,00,000	3,000	1,400	42,00,000	1,20,00,000
Total	**5,600**	**2,200**	**61,60,000**	**7,000**	**3,200**	**1,12,00,000**	**5,400**	**2,800**	**75,60,000**	**2,49,20,000**

Example 2: XYZ Ltd. sells two products which are shirt and t-shirt. Budgeted sales for the current year are given below:

Product	Division I	Division II
Shirt	300 at ₹110	400 at ₹110
T-shirt	220 at ₹100	200 at ₹100

Actual sales for the above period were as follows:

Product	Division I	Division II
Shirt	400 at ₹110	350 at ₹110
T-shirt	200 at ₹100	240 at ₹100

Following are various decisions taken by the company for future:
i) The company decides to increase the price of T-shirt by ₹20, due to its high demand.
ii) The shirt is likely to have better demand if its price is reduced by ₹10.

Following are the estimates for percentage increase in demand of the products:

Product	Division I (%)	Division II (%)
Shirt	10	25
T-shirt	5	10

Prepare a sales budget incorporating all the changes.

Solution:

Sales Budget of XYZ Ltd.

Division	Product	Budget for Future Period			Budget for Current Period			Actual Sales for Current Period		
		Qty.	Price (₹)	Amount (₹)	Qty.	Price (₹)	Amount (₹)	Qty.	Price (₹)	Amount (₹)
I	Shirt	330	100	33,000	300	110	33,000	400	110	44,000
	T-shirt	231	120	27,720	220	100	22,000	200	100	20,000
	Total	**561**	**220**	**60,720**	**520**	**210**	**55,000**	**600**	**210**	**64,000**
II	Shirt	500	100	50,000	400	110	44,000	350	110	38,500
	T-shirt	220	120	26,400	200	100	20,000	240	100	24,000
	Total	**720**	**220**	**76,400**	**600**	**210**	**64,000**	**590**	**210**	**62,500**
Total	Shirt	830	100	83,000	700	110	77,000	750	110	82,500
	T-shirt	451	120	54,120	420	100	42,000	440	100	44,000
	Total	**1,281**	**220**	**1,37,120**	**1,120**	**210**	**1,19,000**	**1,190**	**210**	**1,26,500**

5.4. PRODUCTION BUDGET

5.4.1. Meaning of Production Budget

Once sales budget is prepared, the management is required to design its production budget. This budget is also a part of master budget and deals with the forecast of level of production for the budgeted period. A production budget is an endeavour to determine the number of units to be produced during the budgeted period.

Production budget is designed to ensure that there are enough units available for sales. It also ensures that no over-production is carried out. This budget takes into account the quantity of opening inventory. A production budget takes opening inventory into account to determine the units to be produced for the period under planning.

5.4.2. Advantages of Production Budget

Following are the main advantages of production budget:

1) This budget leads to best possible utilisation of equipments like plant and machinery.
2) It also ensures proper use of labour.
3) This budget helps in maintaining optimum level of inventory to ensure that there is no stock out or over-accumulation of goods.
4) It offers production of finished goods in accordance to delivery schedules.
5) It helps in proper routing and scheduling of labour.
6) It reduces cost by bringing uniformity in production process.
7) It helps in framing purchase budget which is helpful for reducing purchase costs.

5.4.3. Factors Considered while Preparing a Production Budget

Following are the salient factors to be considered while preparing a production budget:

1) **Sales:** Sales requirements such as quantity, product specifications, customer demands and other related variables.
2) **Inventory Policies:** Inventory policies pertaining the stock size of work-in-progress and finished goods required to be maintained at different point of time.
3) **Availability of Production Resources:** Factor considered in the availability of production resources, such as plant capacity, expansion of plant, plant improvement, availability of manpower.
4) **Key Factor:** Key factor such as labour, wages, etc. puts major restriction on the production process.
5) **Plant Maintenance:** Plant maintenance standards expected during the budget period.
6) **Technological Obsolescence:** The rate of technological obsolescence also reflects the production efficiency.
7) **Production Control:** Production control procedures, their effectiveness and flexibility.
8) **Time Lag:** Production budget get affected due to the time lag between production and sale.
9) **Time Involved:** The major considerations are given to the time involved in the production process.
10) **Changeover:** With the change in production of one product to another, related costs also changes.
11) **Economic Batch Quantity:** Economic batch quantity of production.
12) **Cost and Production Run:** The production budget effects the relation between cost and production run.

5.4.4. Format of Production Budget

A format of a detailed production budget is given as under:

.....Co. Ltd.
Production Budget
(January, February and March 20XX)

Particulars	January	February	March
Sales in Quantity (as per sales budget)	xxx	xxx	xxx
Add: Desired Inventory at the End	xxx	xxx	xxx
Total Quantity Required	xxx	xxx	xxx
Less: Stock at Beginning	xxx	xxx	xxx
Quantity to be Produced	**xxx**	**xxx**	**xxx**

Example 3: ABC Ltd. expects to sell 1,00,000 units of product A in the first fiscal quarter, 1,20,000 units in the second quarter, 1,45,000 units in the third quarter and 1,50,000 units in the fourth quarter and 1,25,000 units in the first quarter of the following year. The closing stock is 24,000 units in the first fiscal quarter, 29,000units in the second quarter, 30,000 units in the third quarter and 25,000 units in the fourth quarter. The opening stock at the beginning of the first quarter of the current year was at 15,000 units. The company plans to have closing inventory equal to one-fifth of the sales for the next fiscal quarter. Prepare the Production budget for the year.

Solution:

Production Budget of ABC Ltd.

Particulars	First Quarter (Units)	Second Quarter (Units)	Third Quarter (Units)	Fourth Quarter (Units)
Sales Budget	1,00,000	1,20,000	1,45,000	1,50,000
Add: Desired stock/inventory at the end	24,000	29,000	30,000	25,000
Total	1,24,000	1,49,000	1,75,000	1,75,000
Less: Stock at beginning	15,000	24,000	29,000	30,000
Quantity to be Produced	**1,09,000**	**1,25,000**	**1,46,000**	**1,45,000**

Example 4: The following information has been made available from the records of precision tools Ltd. for the six months of 2018 (and the sales of January 2019), in respect of production X:

i) The units to be sold in different months are:
July 2018 = 1,100; August 2018 = 1,100; September 2018 = 1,700;
October 2018 = 1,900; November 2018 = 2,500; December 2018 = 2,300;
January 2019 = 2,000

ii) There will be no work in progress at the end of any month.

iii) Finished units equal to half the sales of the next month will be in stock at the end of every month (including June 2018).

You are required to prepare production budget for the six months of 2018.

Solution:

Production Budget

(For the six months ending Dec. 2018)

Particulars	July (₹)	Aug. (₹)	Sep. (₹)	Oct. (₹)	Nov. (₹)	Dec. (₹)	Total (₹)
Estimated Sales	1,100	1,100	1,700	1,900	2,500	2,300	
Add: Closing Stock	550	850	950	1,250	1,150	1,000	
	1,650	1,950	2,650	3,150	3,650	3,300	
Less: Opening Stock	550	550	850	950	1,250	1,150	
Production	**1,100**	**1,400**	**1,800**	**2,200**	**2,400**	**2,150**	**11,050**

Note: As per the adjustment given in the question, closing stock will be half of units sold of the following month and closing stock of previous month will be opening stock of the following next month.

5.5. RAW MATERIAL PURCHASE & PROCUREMENT BUDGET

5.5.1. Meaning of Material Budget

Material budget shows the quantity and cost of materials required for production. It deals with the direct material requirements for the budgeted output. This budget does not include indirect materials as such materials are treated as a part of factory overheads. It involves forecasting total quantity and cost of materials to be used for production process.

Material budget shows the total quantities of each item of raw materials and its components. It also shows the total costs of such materials. The budget also takes cognizance of opening inventory and various organisational policies regarding the inventory. The budget can be presented for different months when the inventory is required. This type of budget involves forecasting of quantity as well as prevailing price level. Various steps need to be followed for the purpose of drafting a material budget.

5.5.2. Objectives of Material Budget

A material budget serves following objectives:

1) Determining the quantity and value of the raw material required for the budgeted production.
2) To ensure smooth production process.
3) To reduce the cost of production by curtailing the lag time.
4) To aid the process of drawing purchase schedules at the desired time.
5) To minimise the wastage of material.

5.5.3. Factors Considered while Preparing a Material Budget

Following factors need to be kept in mind for preparing the material budget:

1) Raw material required for producing budgeted output.
2) Previous records are to be used for calculating the percentage of raw materials to total cost of production. This percentage may be used as the basis for computing requirement of raw material for the budgeted output.
3) Required opening and closing inventory balances should be considered.
4) Time lag between ordering the raw material and its receipt should also be taken into account.
5) If the availability of the raw material is seasonal in nature then this factor should be considered while drawing the budget.
6) Attention must be paid to the price trend in the market.

5.5.4. Advantages of Material Budget

A material budget has following advantages:

1) This budget is helpful for estimating material prices during the production period.
2) It is helpful in analysing quantity of material required for the budgeted period.
3) It makes the process of inventory planning smoother.
4) It allows for longer purchase lead time.
5) Material budget helps to minimise cost incurred in transportation.
6) It helps in ensuring better supplier relationship.
7) Material budget helps in better coordination between the departments.
8) It also ensures smooth and continuous operations.

5.5.5. Format for Materials Budget

Material Budget

(For the year .....)

Particulars	Kinds of Raw Material (in Units)	
	A	B
Units of Raw Material Required as per Production Budget:		
For Product: A	XXX	XXX
B	XXX	XXX
Add: Desired Stock of Raw Material at the End of Year	XXX	XXX
Less: Available Stock of Raw Material at the Beginning of the Year	XXX	XXX
Material Budget Physical	XXX	XXX
Standard Price of Raw Material (P.V.) (in ₹)	XXX	XXX
Material Cost Budget (in ₹)	**XXX**	**XXX**

Example 5: The sales director of a manufacturing company expects to sell 50,000 units of product A.
The production manager presented following figures:
Product A needs raw materials, X and Y. Each unit of the product requires 2 units of X and 1 unit of Y. The estimated opening balances of the next year are:
Product A - 12,000 units, X - 15,000 units, Y - 10,000 units.
The firm wants following closing balances at the end of the next year are:
Product A - 10,000 units, X - 11,000 units, Y - 14,000 units.
Prepare a Material Purchase Budget for the next year.

Solution:

Material Purchase Budget (Quantitative)

Particulars	Product A (units)	Material	
		X (Units)	Y (Units)
Production Budget	50,000	1,00,000	50,000
Add: Opening Balances	12,000	15,000	10,000
	62,000	1,15,000	60,000
Less: Closing Balances	10,000	11,000	14,000
Estimated Sales of Product	52,000	-	-
Estimated Purchase of Materials	-	1,04,000	46,000

Example 6: A Ltd. expects to have following inventory as on the beginning:

Type of Material	Quantity	Price (₹)	Amount (₹)
A	1,200 units	5	6,000
B	40 tons	40	1,600
C	4,000 kg.	0.50	2,000
D	1,500 gallons	1.20	1,800
E	20,000 units	0.25	5,000
			16,400

Following statement shows the consumption of different types of raw material for various departments:

Material	Departments			
	I	II	III	IV
A (units)	8,000	-	5,000	2,000
B (tons)	20	10	-	5
C (kg)	-	1,000	1,200	2,500
D (gallons)	500	1,500	-	-
E (units)	11,000	12,000	3,500	1,500

Desired Closing Inventory	
A	1,500 units
B	30 tons
C	2,000 kg.
D	1,600 gallons
E	21,000 units

Prepare the Purchase Budget, assuming that all prices will remain constant.

Solution:

Purchase (Material) Budget

Material	Units	Opening Stock	Expected Consumption	Closing Stock	Purchase Requirement	Price (₹)	Budget (₹)
A	Units	1,200	15,000	1,500	15,300	5.00	76,500
B	Tons	40	35	30	25	40.00	1,000
C	Kg	4,000	4,700	2,000	2,700	0.50	1,350
D	Gallons	1,500	2,000	1,600	2,100	1.20	2,520
E	Units	20,000	28,000	21,000	29,000	0.25	7,250
							88,620

Purchase Requirement = Closing Stock − Opening Stock + Expected Consumption

5.6. LABOUR BUDGET

5.6.1. Meaning of Labour Budget

Labour budget presents "the estimate of direct labour" in order to achieve the targets set under "production budget". This budget is important as it involves an active factor of production. It enumerates the potential requirement for various types of labour for the budget period. It also calculates the estimated cost of labour. This budget is important as it involves an active factor of production.

Labour budget is drafted on the basis of "production budget". It involves determination of standard time by using time and motion study. Requirement of labour for each department or cost centre is identified by converting the calculated hours into labour requirements. These hours are multiplied with wage rate to determine direct labour cost.

Departmental Labour Budget

Cost Centre: Product	Output Quantity	Month: Standard Hours
A B C		

Workers	Number	Hours	Rate of Wages	Direct Labour Cost	Total
Male: Skilled Semi-skilled Unskilled **Female:** Skilled Semi-skilled Unskilled					
Total:					

On the above basis, the manpower requirement is designed by the Production Manager. Requirement of indirect workers is also included. Both are added to reach total manpower requirement.

Note: Labour costs associated with indirect workers such as clerical staff and supervisors are included in appropriate budgets under Overheads.

5.6.2. Advantages of Labour Budget

The advantages associated with labour budget are as follows:

1) It shows the requirement of direct and indirect labour.
2) The personnel department can make plans about its recruitment and training programme. It helps in reducing labour turnover.
3) It forms the basis for preparing cash budget and manufacturing cost budget.
4) It helps in determining labour cost and rate per unit.
5) It helps in optimal production schedule, reducing idle hours as less as possible.
6) Labour budget acts as a cost control measure and helps in cost minimisation and profit maximisation.

Example 7: Prepare a labour cost budget for the next year from the following information of Paras Industries Ltd.:

1) **Production Forecast**

Production X	–	50,000 units
Production Y	–	40,000 units
Production Z	–	20,000 units

2)

Kinds of Labour Required (Per Unit)	Product		
	X	Y	Z
Skilled	2 hours	3 hours	3 hours
Semi-skilled	3 hours	3 hours	4 hours
Unskilled	4 hours	5 hours	3 hours

3) **Labour Rate (Per Hour)**
Skilled – ₹5; Semi-skilled – ₹4; Unskilled – ₹3

Solution:

Labour Cost Budget of Paras Industries Ltd.
(for the year ending .....)

Particulars	Kinds of Labour			
	Skilled	**Semi-Skilled**	**Unskilled**	**Total**
1) Labour Hours Required for Budgeted:				
Product X	1,00,000	1,50,000	2,00,000	4,50,000
Product Y	1,20,000	1,20,000	2,00,000	4,40,000
Product Z	60,000	80,000	60,000	2,00,000
Total Labour Hours	**2,80,000**	**3,50,000**	**4,60,000**	**10,90,000**
2) Labour Rates (₹)	5	4	3	–
3) Labour Cost for Budgeted Production:				
Product X	5,00,000	6,00,000	6,00,000	17,00,000
Product Y	6,00,000	4,80,000	6,00,000	16,80,000
Product Z	3,00,000	3,20,000	1,80,000	8,00,000
Total Labour Cost or Labour Cost Budget	**₹14,00,000**	**₹14,00,000**	**₹13,80,000**	**₹41,80,000**

Example 8: The direct labour hour requirements of three of the products manufactured in a factory, each involving more than one labour operation, are estimated as follows:

Direct Labour Hours per Unit (in minutes)

Operations	Products		
	1	**2**	**3**
Operation 1	18	42	30
Operation 2	-	12	24
Operation 3	9	6	-

The factory worked 8 hours per days in a week. The budget quarter is taken as 13 weeks, and during a quarter lost hours, due to leave and holidays and other causes, are estimated to be 124 hours.

The budgeted hourly rates for the workers manning the operation 1, 2 and 3 are ₹2.00; ₹2.50 and ₹3.00 respectively.

The budgeted sales for the products during the quarter are:
Product 1 9,000 units
Product 2 15,000 units
Product 3 12,000 units

There is a carryover of 5,000 Units of Product 2 and 4,000 units of product 3 and it is proposed to build up a stock at the end of the budget quarter as follows:
Product 1 1,000 units
Product 3 2,000 units

Prepare a man-power budget for the quarter showing for each operation, (i) direct labour hours, (ii) direct labour cost, and (iii) the number of workers.

Solution:

Quarterly Man-Power Budget

Operations	Hourly Rate	Product 1		Product 2		Product 3		Total		No. of Workers
		Direct Labour Hrs.	Cost (₹)	Direct Labour Hrs.	Cost (₹)	Direct Labour Hrs.	Cost (₹)	Direct Labour Hrs.	Cost (₹)	
1	2.00	3,000	6,000	7,000	14,000	5,000	10,000	15,000	30,000	30
2	2.50	-	-	2,000	5,000	4,000	10,000	6,000	15,000	12
3	3.00	1,500	4,500	1,000	3,000	-	-	2,500	7,500	5

Working Notes:

Production Budget

Particulars	Product (units)		
	1	2	3
Budgeted Sales	9,000	15,000	12,000
Add: Closing Stock	1,000	-	2,000
	10,000	15,000	14,000
Less: Opening Stock	-	5,000	4,000
Production Budget	**10,000**	**10,000**	**10,000**

Total Hours Available (per man in a quarter)

Total hours (8 hours × 6 days × 13 weeks) = 624
Less: Hours lost due to leaves etc. 124
Total hours available per man 500

i) **Direct Labour Hours**

Product 1

$$\text{Operation 1} = \frac{18 \times 10{,}000}{60} = 3{,}000, \quad \text{Operation 3} = \frac{9 \times 10{,}000}{60} = 1{,}500$$

Product 2

$$\text{Operation 1} = \frac{42 \times 10{,}000}{60} = 7{,}000, \quad \text{Operation 2} = \frac{12 \times 10{,}000}{60} = 2{,}000, \quad \text{Operation 3} = \frac{6 \times 10{,}000}{60} = 1{,}000$$

Product 3

$$\text{Operation 1} = \frac{30 \times 10{,}000}{60} = 5{,}000, \quad \text{Operation 2} = \frac{24 \times 10{,}000}{60} = 4{,}000$$

ii) **Direct Labour Cost**

Product 1

Operation 1 = 3,000 hours × ₹2 = ₹6,000, Operation 3 = 1,500 hours × ₹3 = ₹4,500

Product 2

Operation 1 = 7,000 hours × ₹ 2 = ₹14,000, Operation 2 = 2,000 hours × ₹2.50 = ₹5,000
Operation 3 = 1,000 hours × ₹ 3.00 = ₹3,000

Product 3

Operation 1 = 5,000 hours × ₹ 2.00 = ₹10,000, Operation 2 = 4,000 hours × ₹ 2.50 = ₹ 10,000

iii) **Number of Workers/Men Required**

$$\text{Number of Men Required} = \frac{\text{Direct labour hours required}}{\text{Total available hours per man}}$$

$$\text{Operation 1} = \frac{15{,}000}{500} = 30 \text{ men}, \quad \text{Operation 2} = \frac{6{,}000}{500} = 12 \text{ men}, \quad \text{Operation 3} = \frac{2{,}500}{500} = 5 \text{ men}$$

5.7. OVERHEAD BUDGET

5.7.1. Meaning of Overhead Budget

Overhead budget is used for preparing production budget. It enumerates all the indirect expenses which are incurred in the production process. These expenses include production overheads, office and administration overheads and selling and distribution overheads. These overheads are illustrated separately in the budget.

5.7.2. Steps in Preparation of Overhead Budget

Following are the steps needed to be undertaken for the preparation of overhead budget:

1) The expenditure should be bifurcated into:
   i) Fixed Costs, and ii) Variable Costs.

   The semi-variable costs should be appropriately divided into the above categories. "Fixed costs" remain constant irrespective of the production level while "variable costs" are expected to fluctuate with change in production level.

2) Various expenses should be apportioned and allocated among different departments on the basis of set policies. It helps in controlling various overheads.
3) Following types of overheads should prepare their own overheads budget as these expenses require different treatments:
   i) Factory overhead budget,
   ii) Administration overhead budget, and
   iii) Selling and Distribution overhead budget.

   Owing to their inherent natures, different overheads need to be treated differently. Production overheads vary with the quantum of production while selling and distribution overheads are more closely related to sales quantum.
4) For an effective control of overheads, it is important to segregate different overheads as these overheads cannot be subjected to the same measure of control. The overhead budgets should be prepared carefully for the purpose of controlling costs.

Example 9: The costs of manufacture in a factory in a particular year are as follows:

Particulars		₹
Direct Labour		40,000
Direct Material		1,00,000
Factory Overhead:		
Fixed	30,000	
Variable	50,000	80,000

The estimation of costs for the next year envisages the following changes:
i) The average rate for direct labour will go up from ₹4 to ₹4.50 per hour;
ii) The production efficiency will decrease by 5%;
iii) Direct labour hours will increase by 10%; and
iv) The purchase price per unit of direct materials, and of the other materials and services included in the overhead will remain unchanged.

Draw up a budget and compute a factory overhead rate, the overheads being absorbed on a direct wages basis.

Solution:

Overhead Budget

Particulars	Original Budget (₹)	Budget for Next Year (₹)
Direct Labour Cost	40,000	52,105
Direct Material Cost	1,00,000	1,10,000
Prime Cost	**1,40,000**	**1,62,105**
Factory Overhead: Fixed	30,000	30,000
Variable	50,000	55,000
Total Cost	**2,20,000**	**2,47,105**

Working Notes:

i) **Direct Labour Cost:** The production efficiency will go down by 5%. As such more labour to the extent of $\frac{100}{95}$ would be required to meet the production.

Secondly, direct labour hours will increase by 10% to have greater production. The present labour hours are 40,000 ÷ ₹4 = 10,000 hrs.

So $10{,}000 \times \frac{100}{95} \times \frac{110}{100} = 11{,}579$ hrs @ ₹4.50 per hour = ₹52,105.

ii) **Direct Material:** Due to increase in labour hours by 10%, it has been assumed that production will increase by $\frac{110}{100}$. So Material Cost will be ₹ $1{,}00{,}000 \times \frac{110}{100} = ₹1{,}10{,}000$.

iii) **Variable Overheads:** On the analogy of 10% increase in production, the Variable Factory Overheads shall be ₹$50{,}000 \times \frac{110}{100} = ₹55{,}000$.

5.8. CASH BUDGET

5.8.1. Meaning and Definition of Cash Budget

Cash budget is presented in the statement form consisting the cash receipts and cash payments of different months. This budget provides an estimate of cash receipts and cash payment expected to occur during the budgeted period.

This budget ensures that the business does not face liquidity issues. There should be enough money to meet various obligations. At the very same time, the cash should not be lying around idle. With the help of cash budget, a business can maintain proper amount of cash.

According to Guthmen and Dougal, "Cash budget is an estimate of cash receipts and disbursements for a future period of time".

According to James C. Van Horne, "A cash budget is a forecast of future cash receipts and cash disbursements over various intervals of time".

According to S.C. Kuchhal, "Cash budget is a schedule to record cash inflows and cash outflows over a period with a view to locate the timing and magnitude of cash surplus and shortage".

Cash Budget presents "future projections about the cash required and the expected sources from which funds would be received". Cash Budget can be prepared for various durations ranging from weekly to annually depending on the requirements of the business.

5.8.2. Characteristics of Cash Budget

Cash budget has following characteristics:

1) A cash budget shows anticipated cash receipts and payments.
2) It is prepared to pre-determine future time period.
3) It is expressed in monetary terms, i.e. absolute figure.
4) A cash budget shows financial budget of the enterprise.
5) It offers a preview of future plans and policies of the organisation.

5.8.3. Advantages of Cash Budget

Following are the main advantages of cash budget:

1) **Assists in Establishing Future Cash Requirement:** Cash budget assists in estimating future inflows and outflows of cash for operating business activities.
2) **Assists in Making Plans:** Cash budgets are helpful in designing realistic plans for achieving the set target of the business.
3) **Investment of Surplus Funds:** A cash budget helps in establishing the requirement for cash. In case, the organisation has excess cash lying around, it can invest these funds in more profitable avenues.
4) **Assists in Cash Control and Maintaining Liquidity:** Cash budget may be used as a control tool. It also helps in ensuring proper liquidity in the business.
5) **Availing Cash Discount:** Cash budget also helps in availing cash discount during the transaction taking place.
6) **Decision Regarding the Form of Borrowing:** If budget shows that there is a requirement to borrow funds then the firm can plan in advance about various avenues it can approach.

5.8.4. Disadvantages of Cash Budget

Following are the shortcomings of cash budget:

1) **Causes Distortions:** Cash budgets may lead to ambiguities as cashflows are independent of profit. Various items of cash inflows such as sales proceeds of an asset do not represent operational activities of the business.
2) **Manipulation:** Cash budgets can be easily manipulated. It may give misleading results if big pay-outs or pay-ins are scheduled on different days. If the business is making losses, such situation may be covered by delaying the payments. Accounts can be presented with false accounts by making a huge payout just before the end of the period. Such transactions offer erroneous results. Similarly, cashflows may be negatively impacted by making early payments.

3) **Reliance on Estimates:** Budgeting requires forecast and hence involves estimates. It uses previous year figures for the purpose of estimates. Such estimates may or may not match with the actuals, limiting the effectiveness of cash budget.

4) **Non-Financial Factors:** Though cash budget deals with cash, but it gets affected by various non-financial factors as well. **For example,** a product may generate poor revenue but may have good impact on the company's goodwill and brand loyalty in the market. In such cases, reliance on sales budget will lead to erroneous decision of discontinuing the product.

5.8.5. Items in Cash Budget

Cash receipt and cash payment has various sources. They are as follows:

Sources of Cash Receipts	Sources of Cash Payments
1) Cash sales, 2) Collection from debtors, 3) Sale of fixed assets, 4) Issue of share and debenture, 5) Loans and borrowing, 6) Interest and dividend, and 7) Miscellaneous receipts.	1) Cash purchase, 2) Payment to creditors, 3) Labour cost, 4) Factory overhead, 5) Loan repayment, 6) Purchase of market securities, 7) Interest and dividend, 8) Tax payment, 9) Payment for wages and salaries, rent and other expenses, and 10) Miscellaneous expenses.

5.8.6. Format of Cash Budget

Cash Budget
(For the period.....)

Particulars	Jan (₹)	Feb (₹)	March (₹)
Opening Cash Balance			
Add: Estimated Cash Receipts:			
i) Cash Sales			
ii) Cash Collection from Debtors			
iii) Interest Received from Investments			
iv) Cash Inflow on Issue of New Securities			
v) Raising of Loans			
vi) Sale of Assets			
vii) Dividend			
Total Cash Receipts			
Less: Estimated Cash Payments:			
i) Payment for Cash Purchases			
ii) Payment to Sundry Creditors for Credit Purchases			
iii) Payment for Wages and Salaries			
iv) Payment for other Administrative Expenses			
v) Payment in the Nature of Capital Expenditure			
vi) Loan Repayment			
vii) Dividend Payment			
viii) Payment of Interest on Loan			
Total Cash Payments			
Closing Cash Balance (Surplus/Deficit)			
Total Cash Receipts – Total Cash Payments			

Example 10: Use following information for preparing cash budget for the months of June, July and August 2018:

1) Opening cash balance in June ₹8,000.
2) Cash sales for June ₹25,000; July ₹20,000 and August ₹30,000.
3) Wages payable ₹8,000 every month.
4) Interest receivable ₹1500 in the month of July.
5) Purchase of furniture for ₹10,000 in August.
6) Cash Purchases for June ₹12,000; July ₹10,000 and August ₹11,000.

Solution:

Cash Budget
(For the period June to August 2018)

Particulars	June (₹)	July (₹)	August (₹)
Opening Cash Balance	**8,000**	**13,000**	**16,500**
Add: Estimated Cash Receipts:			
Cash Sales	25,000	20,000	30,000
Interest Received	-	1,500	-
Total Cash Receipts	**33,000**	**34,500**	**46,500**
Less: Estimated Cash Payments:			
Cash Purchases	12,000	10,000	11,000
Payment of Wages	8,000	8,000	8,000
Purchase of Furniture	-	-	10,000
Total Cash Payment	**20,000**	**18,000**	**29,000**
Closing Cash Balance (Surplus/Deficit) (Total Cash Receipts – Total Cash Payments)	**13,000**	**16,500**	**17,500**

Note: The closing cash balance in June will be the opening cash balance for the month of July.

Example 11: Using the following information, draw a budget for three months starting from October 2018:
1) Opening cash balance in October ₹4,000.
2) Cash Sales October: ₹35,000; November ₹30,000; December; ₹25,000.
3) Credit purchases – September ₹20,000; October ₹14,000; November ₹15,000; December ₹14,000.
4) Suppliers allow one month period of credit.
5) Dividend receivable in November is ₹5,000.
6) Advance tax payable in October is ₹4,000 payable.
7) Sale of an old asset during the month of October is ₹10,000.

Solution:

Cash Budget
(For the period October to December 2018)

Particulars	October (₹)	November (₹)	December (₹)
Opening Cash Balance	4,000	25,000	46,000
Add: Estimated Cash Receipts:			
Cash Sales	35,000	30,000	25,000
Dividend Received	-	5,000	-
Sales of Asset	10,000	-	-
Total Cash Receipts	**49,000**	**60,000**	**71,000**
Less: Estimated Cash Payments:			
Payment to Suppliers	20,000	14,000	15,000
Advance Income Tax Payable	4,000	-	-
Total Cash Payment	**24,000**	**14,000**	**15,000**
Closing Cash Balance (Surplus/Deficit) (Total Cash Receipts – Total Cash Payments)	**25,000**	**46,000**	**56,000**

Note: Since suppliers offer one month credit, the purchases made in September will be payable in October and so on.

Example 12: Using following information, draw a cash budget for the months of April, May and June 2018:

Month	Credit Sales	Credit Purchases	Office Expenses
February	55,000	20,000	7,000
March	45,000	35,000	6,000
April	50,000	15,000	8,000
May	50,000	30,000	6,000
June	60,000	45,000	7,000

Other Information:

1) Opening cash balance ₹4,000.
2) Office expenses are paid in the same month.
3) Suppliers allow two months of credit period.
4) Dividend receivable in the month of May is ₹2,000.
5) Customers are given one month of credit.
6) Interest payable in June is ₹1,500.

Solution:

Cash Budget
(For the period April to June, 2018)

Particulars	April (₹)	May (₹)	June (₹)
Opening Cash Balance	4,000	21,000	32,000
Add: Estimated Cash Receipts:			
Cash Receivable from Customers	45,000	50,000	50,000
Dividend Received	-	2,000	-
Total Cash Receipts	**49,000**	**73,000**	**82,000**
Less: Estimated Cash Payments:			
Payments to Suppliers	20,000	35,000	15,000
Office Expenses	8,000	6,000	7,000
Interest Payable	-	-	1,500
Total Cash Payment	**28,000**	**41,000**	**23,500**
Closing Cash Balance (Surplus/Deficit) (Total Cash Receipts – Total Cash Payments)	**21,000**	**32,000**	**58,500**

Example 13: From the following prepare cash budget for 3 months for January. February and March, 2018.

Month	Total Sales (₹)	Materials (₹)	Wages (₹)	Overheads	
				Production (₹)	Selling & Distribution (₹)
January	40,000	42,000	8,000	6,400	1,600
February	44,000	28,000	8,800	6,600	1,800
March	48,000	28,000	9,200	6,600	1,600
April	52,000	24,000	9,200	6,800	1,800
May	56,000	24,000	9,600	7,000	1,800
June	60,000	32,000	9,600	7,200	2,000

Other Information:

1) Cash balance on January 1, 2018 was ₹20,000.
2) A new machine is to be installed at ₹60,000 on credit, to be repaid by two equal instalments in March and April, 2018.
3) Period of credit allowed by suppliers 1 month.
4) Period of credit allowed to customers 1 month.
5) Delay in payment of overheads 1 month.
6) Delay in payment of wages 1 month.
7) Assume cash sales to be 50% of total sales.

Solution:

Cash Budget
(For the Period January to March)

Particulars	January (₹)	February (₹)	March (₹)
Opening balance	20,000	40,000	24,000
Add: Estimated Cash Receipts:			
Amount Received from Customers	–	20,000	22,000
Cash Sales	20,000	22,000	24,000
Total Cash Receipts	**40,000**	**82,000**	**70,000**

Less: Estimated Cash Payments:			
Amount Paid to Purchase of Machinery			30,000
Amount Paid to Creditors	–	42,000	28,000
Payment of Overheads:			
Production Overheads	–	6,400	6,600
Selling Overheads	–	1,600	1,800
Payment of Wages	–	8,000	8,800
Total Cash Payments	**–**	**58,000**	**75,200**
Closing Balance (Surplus/Deficit) (Total Cash Receipts – Total Cash Payments)	**40,000**	**24,000**	**–5200**

Example 14: A newly started company wishes to prepare cash budget from January. Prepare a cash budget for the first six months from the following estimated revenue and expenses:

Month	Total Sales (₹)	Purchases (₹)	Wages (₹)	Production Overheads (₹)	Selling and Distribution Overheads (₹)
January	20,000	20,000	4,000	3,200	800
February	22,000	14,000	4,400	3,300	900
March	28,000	14,000	4,600	3,400	900
April	36,000	22,000	4,600	3,500	1,000
May	30,000	20,000	4,000	3,200	900
June	40,000	25,000	5,000	3,600	1,200

Cash balance on 1st January was ₹10,000. A new machinery is to be installed at ₹20,000 on credit, to be repaid by two equal instalments in March and April.

Sales commission @ 5% on total sales is to be paid within a month following actual sales.

₹10,000 being the amount of 2nd call may be received in March. Share premium amounting to ₹2,000 is also obtainable with the 2nd call.

Period of credit allowed by suppliers – 2 months.

Period of credit allowed by customers – 1 months.

Delay in payment of wages $^1/_2$ month and overheads 1 month.

Assume cash sales to 50% of total sales.

Solution:

Cash Budget

(For the period January to June)

Particulars	January (₹)	February (₹)	March (₹)	April (₹)	May (₹)	June (₹)
Opening Cash Balance	10,000	18,000	29,800	27,000	24,700	33,100
Add: Estimated Cash Receipts:						
Second Call Money Received	–	–	10,000	–	–	–
Share Premium Received			2,000			
Amount Received from Customers	–	10,000	11,000	14,000	18,000	15,000
Cash Sales	10,000	11,000	14,000	18,000	15,000	20,000
Total Cash Receipts	**20,000**	**39,000**	**66,800**	**59,000**	**57,700**	**68,100**
Less: Estimated Cash Payments:						
Amount Paid to Purchase of Machinery	–	–	10,000	10,000	–	–
Sales Commission Paid	–	1,000	1,100	1,400	1,800	1,500
Amount Paid to Creditors	–	–	20,000	14,000	14,000	22,000
Payments of Overheads (Production + Selling and Distribution)	–	4,000	4,200	4,300	4,500	4,100
Payment of Wages	2,000	4,200	4,500	4,600	4,300	4,500
Total Cash Payments	**2,000**	**9,200**	**39,800**	**34,300**	**24,600**	**32,100**
Closing Balance (Surplus/Deficit) (Total Cash Receipts –Total Cash Payment)	**18,000**	**29,800**	**27,000**	**24,700**	**33,100**	**36,000**

Working Note:

Calculation of Sales Commission:

January = 20,000 × 5/100 = ₹1,000 **February** = 22,000 × 5/100 = ₹1,100 **March** = 28,000 × 5/100 = ₹1,400

April = 36,000 × 5/100 = ₹1,800 **May** = 30,000 × 5/100 = ₹1,500 **June** = 40,000 × 5/100 = ₹2,000

Calculation of Wages:

January = 4,000 × ½ = ₹2,000

February = Balance of wages left in previous month + Half of total wages of current month

$$= 2{,}000 + \frac{4{,}400}{2} = ₹4{,}200$$

March = Balance of wages left in previous month + Half of total wages of current month

$$= 2{,}200 + \frac{4{,}600}{2} = ₹4{,}500$$

April = Balance of wages left in previous month + Half of total wages of current month

$$= 2{,}300 + \frac{4{,}600}{2} = ₹4{,}600$$

May = Balance of wages left in previous month + Half of total wages of current month

$$= 2{,}300 + \frac{4{,}000}{2} = ₹4{,}300$$

June = Balance of wages left in previous month + Half of total wages of current month

$$= 2{,}000 + \frac{5{,}000}{2} = ₹4{,}500$$

Example 15: A company expects to have opening balance of bank of ₹15,000 on 1st May, 2018. Prepare cash statement for the month of May, June and July using the following information:

Month	Sales (₹)	Purchases (₹)	Wages (₹)	Office Expenses (₹)	Factory Expenses (₹)	Selling Expenses (₹)
March	52,000	32,000	5,000	5,000	5,500	3,500
April	54,000	31,000	7,500	5,000	6,500	3,500
May	50,000	25,000	6,000	5,000	7,000	4,500
June	60,000	44,000	8,000	5,000	8,500	4,000
July	96,000	42,000	8,500	5,000	7,000	3,500

Other Information:

1) Cash sales account for 25% of total sales, while remaining amount is collected in the following month.
2) Suppliers offer two month's credit.
3) Wages and all other expenses are paid in the following month.
4) Dividend to shareholders and bonus to workers of ₹15,000 and ₹25,000 respectively is payable in the month of June.
5) 70,000 is payable in the month of May for the purchase of plant.
6) Income tax of ₹15,000 is payable in June.

Solution:

Cash Budget

(For the period May to July, 2018)

Particulars	May (₹)	June (₹)	July (₹)
Opening Bank Balance	15,000	– 56,500	1,12,500
Add: Estimated Cash Receipts:			
Cash Sales (25% of sale)	12,500	15,000	24,000
Debtors (75% of previous month sale)	40,500	37,500	45,000
Total Cash Receipts	**68,000**	**– 4,000**	**1,81,500**

Less: Estimated Cash Payments:			
Creditors	32,000	31,000	25,000
Wages	7,500	6,000	8,000
Office Expenses	5,000	5,000	5,000
Factory Expenses	6,500	7,000	8,500
Selling Expenses	3,500	4,500	4,000
Dividend to Shareholders	–	15,000	–
Bonus to Workers	–	25,000	–
Purchase of Plant	70,000	–	
Income Tax	–	15,000	–
Total Cash Payment	**1,24,500**	**1,08,500**	**50,500**
Closing Cash Balance (Surplus/Deficit) (Total Cash Receipts – Total Cash Payments)	**– 56,500**	**1,12,500**	**1,31,000**

5.9. FLEXIBLE BUDGET

5.9.1. Meaning and Definition of Flexible Budget

A flexible budget incorporates a number of budgets at different production output/levels. A flexible budget takes into account various circumstances which may lead to change in actual production level. It recognises different types of costs viz. fixed, semi-fixed and variable costs. This concept helps in making these budgets more realistic and close to actual life situations.

According to C.I.M.A., flexible budget is, "A budget which, by recognising the difference in behaviour between fixed and variable costs in relation to fluctuations in output, turnover or other variable factors such as number of employees, is designed to change appropriately with such fluctuations".

For the purpose of designing a flexible budget, a company should study the impact of change in circumstances on its final production. Such circumstances may relate to the installation of new machinery or change in the production policies.

5.9.2. Features of Flexible Budget

Following are the main features of a flexible budget:

1) **Covers a Range of Activities:** Future events are uncertain and managers attempt to minimise the impact of such uncertainties. A flexible budget shows expected results at various different levels of activities while a fixed budget enumerates only one estimate. A flexible budget helps the managers in obtaining information about activities at different levels.

2) **Dynamic in Nature:** Flexible budgets help managers in adjusting the plans according to change in situation. It can accommodate change in level of activity. This type of budget is helpful in decision-making process.

3) **Facilitate Performance Measurement:** These budgets are helpful in efficiency measurement and thus play a major role in performance reporting.

5.9.3. Advantages of Flexible Budget

Following are the main benefits accruing from a flexible budget:

1) A flexible budget helps to establish future plan for any level of activity as it is possible to make predictions within the relevant range.

2) A flexible budget can be used for control purpose as it makes easy access to the performance of the managers in relation to the activities level achieved.

3) Flexible budgets can be used for the purpose of controlling overheads.

4) It also helps in ascertaining costs at various levels of activities. This further helps in fixing quotation prices.

5) It helps to ascertain the impact of different volume of activities on total revenue, costs and profits.

5.9.4. Disadvantages of Flexible Budget

Following are the main limitations of flexible budget:

1) It assumes costs to be linear and ignore various factors such as bulk discounts. However, in real life scenario, costs are not linear and hence such budgets lose their utility.
2) Flexible budgets assume continuing operations and do not take into account erratic behaviour of costs.
3) Flexible budgets divide costs into fixed and variable elements in an arbitrary fashion. Therefore, budgeted numbers do not bear close resemblance to actual costs.
4) Flexible budgets keep fixed costs at the same level for all activity levels, while in reality, even fixed costs change after crossing relevant output range.

5.9.5. Format of Flexible Budget

Following is a format of a flexible budget:

Flexible Budget
Normal Level of Activity.....
(period ending.....)

Particulars		Capacity 60% (₹)	Capacity 80% (₹)	Capacity 100% (₹)
Prime Cost				
Variable Overheads				
Marginal Cost	(A)			
Sales	(B)			
Contribution	(D) = (B – A)			
Fixed Cost	(C)			
Profit	(D – C)			

Example 16: Following are the budgeted expenses for a factory operating at 70% level of activity with 1,400 hours:

Variable	2,800
Semi-variable	3,200
Fixed	4,000

The semi-variable expenses go up by 15% between 80 and 90 percent level of activity, and by 20% above 90 percent activity. Prepare a flexible budget for 80, 90, and 100 percent level of activity, and calculate recovery rate per hour.

Solution:

Flexible Budget

Level of Activity	70%	80%	90%	100%
Budgeted Hours	**1400** (₹)	**1600** (₹)	**1800** (₹)	**2000** (₹)
Variable Expenses	2,800	3,200	3,600	4,000
Semi-variable Expenses	3,200	3,680	3,680	3,840
Fixed Expenses	4,000	4,000	4,000	4,000
Total Cost	**10,000**	**10,880**	**11,280**	**11,840**
Recovery Rate Per Hour = Total Cost / Budgeted Hours	$\frac{10,000}{1,400} = 7.143$	$\frac{10,880}{1,600} = 6.8$	$\frac{11,280}{1,800} = 6.267$	$\frac{11,840}{2,000} = 5.92$

Working Note:

1) A semi-variable expense between 80% and 90% of activity is: 3,200 + 15% of 3,200 = ₹3,680.
2) Semi-variable expenses above 90% level of activity is: 3,200 + 20% of 3,200 = ₹3,840

Example 17: A company had budgeted following expenses for producing 12,000 units.

Particulars	Per Unit (₹)
Materials	60
Labour	30
Variable overheads	25
Fixed overheads (₹1,20,000)	10
Direct variable overheads	6
Selling and distribution expenses (25% fixed)	24
Administrative expenses (₹60,000 rigid for all levels of production)	5
	160

Prepare a budget for the production of 10,000 units.

Solution:

Flexible Budget

Production	12,000 Units		10,000 Units	
Particulars	**Per Unit (₹)**	**Amount (₹)**	**Per Unit (₹)**	**Amount (₹)**
Production Cost:				
Materials	60.00	7,20,000	60.00	6,00,000
Labour	30.00	3,60,000	30.00	3,00,000
Variable Overheads	25.00	3,00,000	25.00	2,50,000
Direct Variable Overheads	6.00	72,000	6.00	60,000
Fixed overheads (1,20,000)	10.00	1,20,000	12.00	1,20,000
Selling and Distribution Expenses:				
Fixed	6.00	72,000	7.20	72,000
Variable	18.00	2,16,000	18.00	1,80,000
Administration Expenses	5.00	60,000	6.00	60,000
Total Cost	**160.00**	**19,20,000**	**164.20**	**16,42,000**

Working Note:

25% of Selling and Distribution Expenses (i.e. 24 ×25% = 6 ×12,000 = ₹72,000) are fixed and the remaining 75% of Selling and Distribution Expenses (i.e. 24 × 75% = 18×12,000 = ₹2,16,000) are variable expense. Therefore, the fixed selling and distribution expense for actual output of 10,000 units unit will be = ₹72,000 and variable selling and distribution expense for actual output of 10,000 units will be = 18×10,000 = ₹1,80,000.

Example 18: The expenses for the production of 6,000 units in a factory are given as follows:

Particulars	Per Unit (₹)
Materials	40
Labour	15
Variable Overheads	20
Fixed Overheads (₹42,000)	7
Administrative Expenses (10% variable)	15
Selling Expenses (10% Fixed)	10
Distribution Expenses (20% Fixed)	8
Total cost of sales per unit	**115**

You are required to prepare a budget for the production of 5,000 units.

Solution:

Flexible Budget

Production	Output 6,000 Units		Output 5,000 Units	
Particulars	**Per Unit (₹)**	**Amount (₹)**	**Per Unit (₹)**	**Amount (₹)**
Materials	40.00	2,40,000	40.00	2,00,000
Labour	15.00	90,000	15.00	75,000
Prime Cost	**55.00**	**3,30,000**	**55.00**	**2,75,000**

Factory Overheads:				
Variable Overheads	20.00	1,20,000	20.00	1,00,000
Fixed Overheads	7.00	42,000	8.40	42,000
Works Cost/Factory Cost	**82.00**	**4,92,000**	**83.40**	**4,17,000**
Administrative Expenses:				
Fixed Expenses	13.5	81,000	16.2	81,000
Variable Expenses	1.5	9,000	1.5	7,500
Cost of Production	**97.00**	**5,82,000**	**101.1**	**5,05,500**
Selling and Distribution Expenses:				
Selling Expenses:				
Fixed Expenses	1	6,000	1.2	6,000
Variable Expenses	9	54,000	9	45,000
Distribution Expenses:				
Fixed Expenses	1.6	9,600	1.92	9,600
Variable Expenses	6.4	38,400	6.4	32,000
Total Cost of Sales	**115.00**	**6,90,000**	**119.62**	**5,98,100**

Working Notes:

1) 10% of administrative expenses (i.e., 15 ×10% = 1.5 ×6,000 = ₹9,000) are variable and remaining 90% of administrative expenses (i.e., 15 ×90% = 13.5 ×6,000 = ₹81,000) are fixed expenses.

   Therefore, the fixed administrative expenses for the actual output of 5,000 units will be = ₹81,000 and variable administrative expenses for actual output of 5,000 units will be = 1.5 ×5,000 = ₹7,500.

2) 10% of Selling expenses (i.e., 10×10% = 1×6,000 = ₹6,000) are fixed and remaining 90% (i.e., 10×90% = 9 ×6,000 = ₹54,000) are variable expenses.

   Therefore, the fixed selling expenses for the actual output of 5,000 units will be = ₹6,000 and variable selling expenses for actual output of 5,000 units will be = 9 ×5,000 = ₹45,000.

3) 20% of distribution expenses (i.e., 8×20% = 1.6 ×6,000 = ₹9,600) are fixed. Therefore, remaining 80% (i.e., 8×80% = 6.4 ×6,000 = ₹38,400) are variable expenses.

Therefore, the Fixed distribution expenses for actual output of 5,000 units will be = ₹8,000 and the variable distribution expenses for actual output of 5,000 units will be = 6.4 ×5,000 = ₹32,000.

Example 19: The expenses for the production of 5,000 units in a factory are given as follows:

Particulars	Per Unit (₹)
Materials	50
Labour	20
Variable Overheads	15
Fixed Overheads (₹50,000)	10
Administrative Expenses (5% Variable)	10
Selling Expenses (20% Fixed)	6
Distribution Expenses (10% Fixed)	5
Total Cost of Sales per unit	**116**

You are required to prepare a budget for the production of 7,000 units.

Solution:

Flexible Budget

Production	5,000 Units		7,000 Units	
Particulars	**Per Unit (₹)**	**Amount (₹)**	**Per Unit (₹)**	**Amount (₹)**
Materials	50.00	2,50,000	50.00	3,50,000
Labour	20.00	1,00,000	20.00	1,40,000
Prime Cost	**70.00**	**3,50,000**	**70.00**	**4,90,000**

Factory Overheads:				
Variable Overheads	15.00	75,000	15.00	1,05,000
Fixed Overheads	10.00	50,000	7.14	50,000
Work Cost/Factory Cost	**95.00**	**4,75,000**	**92.14**	**6,45,000**
Administrative Expenses:				
Fixed Expenses	9.50	47,500	6.78	47,500
Variable Expenses	0.50	2,500	0.5	3,500
Cost of Production	**105.00**	**5,25,000**	**99.42**	**6,96,000**
Selling Expenses:				
Fixed Expenses	1.20	6,000	0.85	6,000
Variable Expenses	4.80	24,000	4.80	33,600
Distribution Expenses:				
Fixed Expenses	0.50	2,500	0.35	2,500
Variable Expenses	4.50	22,500	4.50	31,500
Total Cost of Sales	**116.00**	**5,80,000**	**109.92**	**7,69,600**

Example 20: Following is the budget for producing 1,00,000 units of the only product manufactured by ABC Ltd. for a costing period:

Particulars	₹
Direct Material	2.50 per unit
Direct Labour	0.80 per unit
Direct Expenses	0.20 per unit
Works Overheads (50% fixed)	2.40 per unit
Administration Overheads (75% fixed)	0.40 per unit
Selling Overheads (40% fixed)	0.25 per unit

The actual production for the said period was only 50,000 units. Prepare the revised schedule showing budgeted cost per unit.

Solution:

Revised (Flexible) Budget

Production	Output 1,00,000 units		Output 50,000 units	
Particulars	**Per Unit (₹)**	**Amount (₹)**	**Per Unit (₹)**	**Amount (₹)**
Direct Material	2.50	2,50,000	2.50	1,25,000
Direct Labour	0.80	80,000	0.80	40,000
Direct Expenses	0.20	20,000	0.20	10,000
Prime Cost	**3.50**	**3,50,000**	**3.50**	**1,75,000**
Works Overheads				
Fixed Overhead	1.2	1,20,000	2.4	1,20,000
Variable Overhead	1.2	1,20,000	1.2	60,000
Cost of Production	**5.90**	**5,90,000**	**7.10**	**3,55,000**
Administration Overheads:				
Fixed Overhead	0.3	30,000	0.6	30,000
Variable Overhead	0.1	10,000	0.1	5,000
Selling Overheads:				
Fixed Overhead	0.10	10,000	0.20	10,000
Variable Overhead	0.15	15,000	0.15	7,500
Total Cost of Sales	**6.55**	**6,55,000**	**8.15**	**4,07,500**

Working Notes:

1) 50% of works overheads (i.e., 2.40×50% = 1.20 ×1,00,000 = ₹1,20,000) are fixed. Therefore, remaining 50% (i.e., 2.40×50% = 1.20 ×1,00,000 = ₹1,20,000) are variable overhead.

   Therefore, the Fixed works overheads for actual output of 50,000 units will be = ₹1,20,000 and the variable works overhead for actual output of 50,000 units will be = 1.2 ×50,000 = ₹60,000.

2) 75% of administrative overhead (i.e., .40 ×75% = 0.30× 1,00,000 = ₹30,000) are fixed and remaining 25% of administrative overhead (i.e., .40 ×25% = 0.10× 1,00,000 = ₹10,000) are variable overheads.

Therefore, the fixed administrative overheads for the actual output of 50,000 units will be = ₹30,000 and variable administration overhead for actual output of 50,000 units will be = 0.1×50,000 = ₹5,000.

3) 40% of Selling overheads (i.e., .25×40% = 0.1 ×1,00,000 = ₹10,000) are fixed and remaining 60% (i.e., 0.25×60% = 0.15 ×1,00,000 = ₹15,000) are variable overhead.

Therefore, the fixed selling overheads for the actual output of 50,000 units will be = 10,000 and variable selling overhead for actual output of 50,000 units will be = 0.15 ×50,000 = ₹7,500.

Example 21: From the following information, prepare a flexible budget at 60% and 80% capacity.

Particulars	70% Capacity (₹)
Variable Overheads:	
Indirect Material	5,000
Indirect Labour	15,000
Semi-Variable Overheads:	
Electricity (60% Variable)	50,000
Repairs and Maintenance (35% Fixed)	5,000
Fixed Overheads:	
Salaries	10,000
Depreciation on Machine	14,000
Insurance	6,000
	1,05,000

Solution:

Flexible Budget

Particulars	Level of Capacity		
	60%	70%	80%
Variable Overheads:			
Indirect Material	4,286	5,000	5,714
Indirect Labour	12,857	15,000	17,143
Semi-variable Overheads:			
Electricity (60% Variable and 40% Fixed)	45,714	50,000	54,286
Repairs and Maintenance (35% Fixed and 65% Variable)	4,536	5,000	5,464
Fixed Overheads:			
Salaries	10,000	10,000	10,000
Depreciation on Machine	14,000	14,000	14,000
Insurance	6,000	6,000	6,000
Total Overhead Costs	**97,393**	**1,05,000**	**1,12,607**

Working Note:

1) **Calculation of Variable Overheads:**

$$\text{Indirect Material on 60\% Capacity} = 5,000 \times \frac{60\%}{70\%} = 4,286$$

$$\text{Indirect Material on 80\% Capacity} = 5,000 \times \frac{80\%}{70\%} = 5,714$$

$$\text{Indirect Labour on 60\% Capacity} = 15,000 \times \frac{60\%}{70\%} = 12,857$$

$$\text{Indirect Labour on 80\% Capacity} = 15,000 \times \frac{80\%}{70\%} = 17,143$$

2) Electricity amount to ₹50,000 at 70% capacity. Out of this 40% (i.e., ₹20,000) is fixed and 60% (i.e., 30,000) is variable. ₹20,000 will remain the same at all capacities while variable part will vary. Variable cost at 60% capacity will be $\left(\frac{30,000}{70}\times 60\right)$ ₹25,714 and at 80% capacity will be $\left(\frac{30,000}{70}\times 80\right)$ ₹34,286. So, the total expenditure will be (20,000 + 25,714) ₹45,714 and (20,000 + 34,286) ₹54,286 at 60% and 80% capacities respectively.

3) Repairs and maintenance amount to ₹5,000 at 70% capacity out of this 35% (i.e., 1,750) is fixed and 65% (i.e., 3,250) is variable. ₹1,750 will remain same at all capacity while variable part will vary. Variable cost at 60% capacity will be $\left(\frac{3,250}{70}\times 60\right)$ ₹2,786 and at 80% capacity will be $\left(\frac{3,250}{70}\times 80\right)$ ₹3,714. So, the total expenditure will be (1,750 + 2,786) ₹4,536 and (1,750 + 3,714) ₹5,464 at 60% and 8% capacity respectively.

Example 22: Prepare a flexible budget of a manufacturing company and show the forecast of profit at 75% and 100% capacity operations, from the given data of 50% capacity:

50% Capacity Worked	₹	Total (₹)
Fixed Cost		
Salaries	84,000	
Rent and Rates	56,000	
Depreciation	70,000	
Other Administrative Expense	80,000	2,90,000
Variable Cost		
Materials	2,40,000	
Labour	2,56,000	
Other expenses	38,000	5,34,000

Possible sales at various levels of working are:
75% Capacity – Sales ₹11,50,000
100% Capacity – Sales ₹15,25,000

Solution:

Flexible Budget

Particulars	Level of Capacity		
	50%	75%	100%
1) **Sales**	–	11,50,000	15,25,000
2) **Variable Expenses:**			
Material	2,40,000	3,60,000	4,80,000
Labour	2,56,000	3,84,000	5,12,000
Other Expenses	38,000	57,000	76,000
Total Variable Expenses	**5,34,000**	**8,01,000**	**10,68,000**
3) **Fixed Cost:**			
Salaries	84,000	84,000	84,000
Rent and Rates	56,000	56,000	56,000
Depreciation	70,000	70,000	70,000
Other Administrative Expense	80,000	80,000	80,000
Total Fixed Cost	**2,90,000**	**2,90,000**	**2,90,000**
4) **Cost of Sales (2 + 3)**	8,24,000	10,91,000	13,58,000
5) **Profit (1 – 4)**	–	59,000	1,67,000

Working Note:
Calculation of Variable Expenses:

Material on 75% Capacity $= 2,40,000 \times \frac{75\%}{50\%} =$ ₹3,60,000

Material on 100% Capacity $= 2,40,000 \times \frac{100\%}{50\%} =$ ₹4,80,000

Labour on 75% Capacity $= 2,56,000 \times \frac{75\%}{50\%} = ₹3,84,000$

Labour on 100% Capacity $= 2,56,000 \times \frac{100\%}{50\%} = ₹5,12,000$

Other expense on 75% Capacity $= 38,000 \times \frac{75\%}{50\%} = ₹57,000$

Other expense on 100% Capacity $= 38,000 \times \frac{100\%}{50\%} = ₹76,000$

5.10. STANDARD COSTING

5.10.1. Concept of Standard Costs

Standard Costs are pre-decided costs, which are incurred in an ideal condition of optimum operational efficiency and utilisation of resources; they are compared with the actual costs with regard to certain factors such as work, operation or activity.

According to Chartered Institute of Management Accountants (CIMA), London has defined Standard Cost as "A pre-determined cost which is calculated from management's standards of efficient operation and the relevant necessary expenditure."

According to Morton Backer and L.E. Jacobsen, Standard cost has defined as "the amount the firm thinks a product or the operation of a process for a period of time should cost, based upon certain assumed conditions of efficiency, economic conditions and other factors."

5.10.2. Meaning and Definition of Standard Costing

Standard Costing is a sophisticated technique, under which the standards are decided in advance and the actual costs are compared with such standard costs. Causes of variations, especially the unfavourable ones, are analysed and appropriate corrective measures are initiated so as to have the optimum efficiency in production.

According to Chartered Institute of Management Accountants, London has defined the Standard Costing as "The preparation and use of standard costs, their comparison with actual costs, and the analysis of variance to their causes, and points of incidence."

According to Institute of Cost and Works Accountants, London has defined Standard Costing as "An estimated cost, prepared in advance of production or supply correlating a technical specification of material, and labour to the price and wage rates estimated for a selected period of time, with an addition of the apportionment of overheads expenses estimated for the same period within a prescribed set of working conditions."

5.10.3. Objectives of Standard Costing

The broad objectives of standard costing are summarised in the following points:

1) **Promoting and Measuring Efficiency:** Standard Costing, besides enhancing the competence and performance, also acts as a tool to measure them. If the actual cost happens to be less than the standard cost, it indicates efficiency and competency; on the other hand, if the actual cost happens to be more than the standard cost, it is indicative of inefficiency and competency.

2) **Controlling and Reducing Costs:** While computing the Standard Cost, appropriate provisions are made with regard to normal wastage, normal breakdown, normal idle capacity, normal mistakes, etc. This exercise ensures proper monitoring as well as reduction in the cost.

3) **Simplifying Costing Procedure:** The exercise of Standard Costing is invariably specific to a product / process / job. For every product, or process, or job a separate exercise of standard costing is required to be undertaken. This is generally done by the professionals (cost and management accountants) after having a thorough discussion with the management and the technical specialists, which ensures smoothening of the entire process.

4) **Valuing Inventories:** Undertaking the valuation exercise in respect of stock and issue of material on the basis of standard costs results in substantial savings of time and energy in the maintenance of stores ledger. Computation of stocks on the basis of standard cost is done by multiplication of the quantity of stock in hand with the standard cost.

5) **Fixing Selling Price:** Selling price of a product may be fixed either on the basis of actual cost or on the basis of standard cost. It has been experienced that due to various reasons, there are lot of fluctuations in actual cost and as such the selling price cannot be fixed on the basis of actual cost, because price of a product needs to be generally stable and not volatile.

Thus, the preferred basis of fixing the price of a product is standard cost, to which a suitable margin is added.

5.10.4. Importance/Advantages of Standard Costing

Following advantages accrue from the standard costing:

1) **Measuring Efficiency:** Standard costing may be used as a measuring tool for the management to measure efficiency. The difference between the actual costs and standard costs reflects the level of performance of different cost centres.

2) **Formulation of Production and Price Policy:** Creation of production policies is facilitated by the standard costing. On the basis of prevailing conditions, the standards are fixed. Standard costs are helpful in deciding production plans and in determining prices of different products.

3) **Determination of Variance:** Variances are determined by comparing actual cost and standard cost. Such variations brings core areas of incompetence, which in turn enables the management to fix staff accountability for inefficiencies and initiate remedial steps promptly and ensure better performance in future.

4) **Reduction of Work:** Under the historical costing, a number of records are required to be regulated for the determination of costs, whereas under the standard costing minimal records are required to be maintained, resulting in the reduction of clerical job to a great extent. Management is provided with only relevant data and unnecessary information is skipped.

5) **Management by Exception:** A system, which envisages that every individual should be given a target to be achieved within a defined timeframe, is referred to as management by exception. Day-to-day supervision of various activities and individual performance is not required under this system.

6) **Facilitates Cost Control:** The twin objective of any costing system is cost control and cost reduction. Both the objectives are achieved in the system of standard costing. The standards are scrutinised and analysed on an ongoing basis with a view to have further improvement in efficiency. Causes of variance, whenever takes place, are thoroughly studied and analysed, which is followed by initiation of prompt remedial steps.

7) **Eliminating Inefficiencies:** There is a different manner in which the standards are set for manufacturing, administrative, and selling expenses. Improvised techniques are applied for setting the above standards. **For example,** the standard for manufacturing expenses are set on the basis of item and motion study for labour; and effective material control devices for materials, etc. Similarly, in the case of other expenses also appropriate studies would be required.

8) **Helpful in Taking Important Decisions:** Through the system of standard costing vital feedback is received by the management, which facilitates decision-making process. Specific problems posed by inflationary forces, rising prices, etc. can be sorted out with the aid of standard costing. It can be effectively used as a tool for providing incentive plans for employees.

5.10.5. Disadvantages of Standard Costing

The system of standard costing exhibits the following disadvantages:

1) **Fixation of Standards:** Fixation of realistic standards during the introduction of standard costing system is an extremely significant and challenging job. A wrong assessment of standards may lead to a faulty system. Fixation of standards on a realistic basis requires a technical expertise, which is difficult to implement.

2) **Frequent Technology Alterations:** Some of the industries are known for frequent technological alterations. Introduction of standard costing system in such industries may prove to be a costly affair, as the standards would need a frequent revision. Standard costing system, therefore, may not be suitable for such industries.

3) **Expensive Technique:** Expenses involved in setting up and continuing the system of standard costing are quite heavy. It is difficult for small-sized business organisations to afford such a costly system.

4) **Fixation of Responsibility:** In case of any lapse, the job of fixing staff accountability, *perse*, is a difficult one. In case of standard costing system, such accountability can be fixed only for the controllable variances and not for the uncontrollable variances. But to establish whether a variance is controllable or uncontrollable is also a tough exercise, due to the fact that the variances controllable at one point of time may become uncontrollable at another point of time.

5) **Analysis of Historical Events:** Variance analysis involves analysis of historical events, which cannot be changed. As such it is of limited use for the management.

5.10.6. Difference between Standard Costing and Budgetary Control

Basis of Difference	Standard Costing	Budgetary Control
1) **Concept**	Setting of standards is specific to a product or service.	Preparation of budgets is carried out for a business organisation as a whole.
2) **Basis**	The basis for setting standard costs is technical inputs.	The basis for preparation of budgets are: i) Historical data, and ii) Future projections.
3) **Scope**	Standards are fixed for expenditures only. In case of manufacturing departments, standards are fixed for various elements of cost i.e., material, labour and overheads.	The scope of budgetary control is broad-based and includes incomes, expenditures and other activities, etc.
4) **Emphasis**	Under the system of standard costing, the standards are fixed and the focus is to ensure that the variance between the standard cost and the actual cost remain minimal.	Under the budgetary control, expenditure limits are prescribed, beyond which the expenditure is not allowed. The focus is to ensure that the expenditure remains within the targeted limits.
5) **Objective**	Standard costs are fixed on the basis of norms prescribed by the management.	Budgets are prepared keeping in view the existing level of competency.
6) **Relationship**	Standard costing has some relationship with cost accounts.	Budgetary Control has some relationship with the financial accounts.
7) **Variance Analysis**	Computation of standard costing variances is carried out for various components of cost, viz. material, labour and overheads. Such variances are analysed on the basis of their causes.	Under the budgetary control, the variances are dealt with in a holistic manner. They may be computed either for various individual departments or for the business organisation in its entirety.

5.11. COST VARIANCE ANALYSIS

5.11.1. Meaning and Definition of Variance & Variance Analysis

Variance is the difference between the actual amount of costs or revenue and the expected amount. Variance Analysis compares actual amounts with the standard amounts and calculates efficiency. The analysis also explains the reason behind the difference. Variance analysis helps management in finding the causes for variation. It aids the process of decision making.

According to Terminology of CIMA, Variance is "The difference between planned, budgeted or standard cost and actual costs (similarly in respect of revenue)".

Sometimes the situation of **Zero Variance** also occurs when the actual results are just equal to planned results.

According to C.I.M.A. London, Terminology, variance analysis is "The process of computing the amount of variance and isolating the causes of variance between actual and standard".

In nutshell, variance analysis is used to measure the efficiency of the business by comparing actual and standard costs as well as revenue.

5.11.2. Use/Applications of Variance Analysis

Variance analysis has several uses which are as follows:

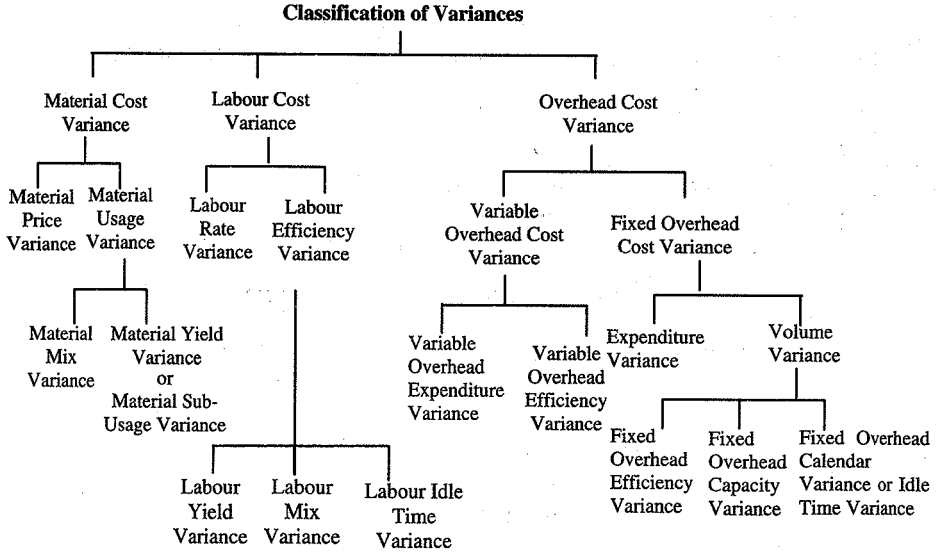
1) It is an important tool for cost control and cost reduction.
2) This analysis helps in exercising Management by exception.
3) Variance analysis is useful for planning purpose.
4) This analysis creates awareness about costs and revenue in the organisation.
5) Variance analysis helps the management in profit maximisation by bifurcating variance into controllable and uncontrollable costs.
6) Variance analysis helps management in taking timely action by detecting inefficiency at the earliest.

5.11.3. Classification of Variances

Variance analysis forms a major basis of Standard Costing. It explains the difference between actual and standard performance. Remedial actions are taken by appropriate authority when variances are unfavourable.

Purchase managers are held responsible for adverse Material Price Variance while negative Material Usage Variance is the responsibility of the production manager.

Following is the figure showing different types of variances:



5.12. MATERIAL VARIANCES

5.12.1. Material Cost Variance (MCV)

Direct materials cost variance is the difference between the standard cost of specific direct material and the actual cost of direct material used in the production process.

According to Terminology of CIMA defines Material Cost Variance as "The difference between the standard direct material cost of the actual production volume and the actual cost of direct material".

It is the total variance caused by the variations in prices of materials, usage or mix, yield and other causes. Material cost variances are sub-divided into:

1) Material price variance, and
2) Material usage variance.

Material cost variance is calculated by using following formula:

Materials Cost Variance = Standard Material Cost – Actual Material Cost

Standard Material Cost = Standard price per unit × Standard quantity of materials

Actual Material Cost = Actual price per unit × Actual quantity of materials

The variance is considered favourable, if the standard cost is more than the actual cost. If the actual cost is more than the standard cost, the variance will be considered unfavourable or adverse.

5.12.2. Material Rate/Price Variance (MPV)

Material price variance is the difference between the standard price of material and the actual price of material used for production.

Material price variance is stated as:
Materials Price Variance = Actual Quantity (Standard Price – Actual Price)

Following may be the various reasons behind material price variance:
1) Due to frequent variation in the market price of material,
2) Irregular and inefficient purchasing,
3) Change in the requirement of material quantity,
4) Change in the specification of materials, and
5) Change in the legal policy as a result of change in the amount of taxes and duties, etc.

5.12.3. Material Usage Variance (MUV)

It is also known as material quantity variance. It explains the difference between the standard quantity of material purchased for actual production and the actual quantity of material used.

Following is the formula used for calculating Material Usage Variance:
Materials Usage Variance = Standard Price (Standard Quantity – Actual Quantity)

Following are the reasons for material usage variances:
1) Surplus waste of material,
2) Mishandling of materials,
3) Providing wrong description of materials, and
4) Improper checking of material, etc.

Following are the main sub-categories of the Material Usage Variance:
1) Material Mix Variance, and
2) Materials Yield Variance.

5.12.4. Material Mix Variance

A product uses more than one type of material for its production. These materials are mixed in a specific ratio. Difference in actual mix and standard mix leads to Material Mix Variance.

Following formula is used for calculating Material Mix Variance:
Material Mix Variance = (Revised Standard Mix of Actual Input – Actual proportion) × Standard Price

The variance is calculated under two situations:

i) **When Actual Weight and Standard Weight of Mix are Equal**

Mix variance is calculated by using following formula:

= Standard cost of standard mix – Standard cost of actual mix.

= (Standard Price × Standard Quantity) – (Standard Price × Actual Quantity)

or, Standard unit cost (Standard Quantity – Actual Quantity)

If standard quantity is revised due to shortage of one material, the formula will be:

= Standard unit cost (Revised Standard Quantity – Actual Quantity).

ii) **When Actual Weight and Standard Weight of Mix are Different**

When quantities of actual material mix and standard material mix are different, the formula will be:

$$\left\{\frac{\text{Total Weight of Actual mix}}{\text{Total Weight of Standard mix}} \times \text{Standard cost of Standard mix}\right\} - \left(\text{Standard costof actual mix}\right)$$

In case the standard is revised due to the shortage of one material then revised standard will be used instead of standard, formula will become:

$$\left\{\frac{\text{Total Weight of Actual mix}}{\text{Total Weight of Revised Standard mix}} \times \text{Standard cost of Revised Standard mix}\right\} - \left(\text{Standard cost of actual mix}\right)$$

5.12.5. Materials Yield Variance

This is a type of material usage variance. It is caused by the difference between actual yield and standard yield.

Materials yield variance is defined as "that portions of the direct materials usage variance which is due to the standard yield specified and the actual yield obtained".

Material Yield Variance = Standard Rate (Actual Yield – Standard Yield)

Standard Rate is calculated as follows:

$$\text{Standard Rate} = \frac{\text{Standard Cost of Standard mix}}{\text{Net standard output i.e., Gross output - Standard loss}}$$

In certain situations, actual mix may be different from standard mix. Following formula is used when the standard is revised:

$$\text{Standard Rate} = \frac{\text{Standard Cost of Revised Standard mix}}{\text{Net standard output}}$$

Material yield variance is considered favourable when actual yield is higher than the standard yield.

For verification purpose, following relationship should match

1) Material Price Variance + Material Usage Variance = Material Cost Variance.
2) Material Mix Variance + Material Sub-usage or Yield Variance = Material Usage Variance.
3) Material Price Variance + Material Mix Variance + Material Sub-usage Variance = Material Cost Variance.
4) Material Usage Variance – Material Mix Variance = Material Sub-usage (or, Yield) Variance.

Example 23: The standard cost card shows the following detail relating to the material requirement to produce 1Kg of groundnut oil:

Quantity of Groundnut Oil	3kg
Price of Groundnut	₹0.76 per kg
Actual Production Data:	
Production during One Month	1,000kg
Quantity Used	3,500kg
Price of Groundnut	₹1.00kg per kg

Calculate and give your opinion about the following variances:

1) Material cost variance
2) Material price variance
3) Material usage variance

Solution:

1) **Material Cost Variance = (Standard Quantity × Standard Price) – (Actual Quantity × Actual Price)**
$= (3,000 \times 0.76) - (3,500 \times 1) = 2,280 - 3,500 = ₹1,220\ (A)$

2) **Material Price Variance = Actual Quantity (Standard Price – Actual Price)**
$= 3,500\ (0.76 - 1.00) = ₹840\ (A)$

3) **Material Usage Variance = Standard Price (Standard Quantity – Actual Quantity)**
$= 0.76\ (3,000 - 3,500) = ₹380\ (A)$

Verification:

Material Cost Variance = Material Price Variance + Material Usage Variance

1,220(A) = 840 (A) + 380(A)

1,220(A) = 1,220(A)

Working Note: Standard quantity for 1,000kg = 1,000 × 3 = 3,000kg

Example 24: From the following data calculate various material variances:

Materials	Standard		Actual	
	Quantity (Units)	Price per Unit	Quantity (Units)	Price per Unit
X	50	7	60	8
Y	100	4	120	3
	150		**180**	

Solution:

1) **Material Cost Variance = (Standard Quantity × Standard Price) – (Actual Quantity × Actual Price)**

Material X	= (50 × 7)– (60 ×8) = 350 – 480	= ₹130 (A)
Material Y	= (100 × 4) – (120 ×3) = 400 – 360	= ₹40 (F)
Material Cost Variance		**= ₹90 (A)**

2) **Material Price Variance = Actual Quantity (Standard Price – Actual Price)**

Material X	= 60 (7 – 8) = 60 (-1)	= ₹60 (A)
Material Y	= 120 (4 – 3) = 120 (1)	= ₹120 (F)
Material Price Variance		**= ₹60 (F)**

3) **Material Usage Variance = Standard Price (Standard Quantity – Actual Quantity)**

Material X	= 7 (50 – 60) = 7 (– 10)	= ₹70 (A)
Material Y	= 4 (100 – 120) = 4 (–20)	= ₹80 (A)
Material Usage Variance		**= ₹150 (A)**

Example 25: From the following particulars calculate:

1) Material Cost Variance
2) Material Price Variance
3) Material Usage Variance

Material	Standard		Actual	
	Units	Price	Units	Price
A	2,020	2	2,160	2.40
B	820	3	760	3.60
C	700	4	760	3.80

Solution:

1) **Material Cost Variance = (Standard Quantity × Standard Price) – (Actual Quantity × Actual Price)**

Material A = (2,020 × 2) – (2,160 × 2.40) = 4,040 – 5,184	=	₹1,144 (A)
Material B = (820 × 3) – (760 × 3.60) = 2,460 – 2,736	=	₹276 (A)
Material C = (700 × 4) – (760 × 3.80) = 2,800 – 2,888	=	₹88 (A)
Material Cost Variance	=	**₹1,508 (A)**

2) **Material Price Variance = Actual Quantity (Standard Price – Actual Price)**

Material A = 2,160× (2 – 2.40)	=	₹864 (A)
Material B = 760×(3 – 3.60)	=	₹456 (A)
Material C = 760× (4 – 3.80)	=	₹152 (F)
Material Price Variance	=	**₹1,168 (A)**

3) **Material Usage Variance = Standard Price (Standard Quantity – Actual Quantity)**

Material A = 2 × (2,020 – 2,160)	=	₹280 (A)
Material B = 3 × (820 – 760)	=	₹180 (F)
Material C = 4 × (700 – 760)	=	₹240 (A)
Material Usage Variance	=	**₹340 (A)**

Example 26: Use following information to calculate (1) Material cost variance, (2) Material usage variance (3) Material price variance, separately for material A and B.

Materials	Standard Quantity (Kg.)	Standard Price (₹)	Actual Quantity (Kg.)	Actual Price (₹)
A	12	4	11	3.50
B	14	6	17	7.50
	26		**28**	

Solution:

1) **Material Cost Variance= (Standard Quantity × Standard Price) – (Actual Quantity × Actual Price)**

Material A	= (12 × 4) – (11 × 3.50) = ₹48 – 38.5	= ₹9.5 (F)
Material B	= (14 × 6) – (17 × 7.50) = ₹84 – 127.5	= ₹43.5 (A)
Material Cost Variance		**= ₹34 (A)**

2) **Material Usage Variance= Standard Price (Standard Quantity – Actual Quantity)**

Material A	= 4 (12 – 11)	= ₹4 (F)
Material B	= 6 (14 – 17)	= ₹18 (A)
Material Usage Variance		**= ₹14 (A)**

3) **Material Price Variance = Actual Quantity (Standard Price – Actual Price)**

Material A	= 11 (4 – 3.50)	= ₹5.5 (F)
Material B	= 17 (6 – 7.50)	= ₹25.5 (A)
Material Price Variance		**= ₹20(A)**

Example 27: From the following particulars calculate:

1) Material Cost Variance
2) Material Usage Variance
3) Material Price Variance

Material	Std. Qty	Std. Price (₹)	Actual Qty.	Actual Price (₹)
A	500	3	450	2.50
B	600	5	650	4.00

Solution:

1) **Material Cost Variance = (Standard Quantity × Standard Price) – (Actual Quantity × Actual Price)**

Material A = (500 × 3) – (450 × 2.50) = 1,500 – 1,125	=	₹375 (F)
Material B = (600 × 5) – (650 × 4.00) = 3,000 – 2,600	=	₹400 (F)
Material Cost Variance	=	**₹775 (F)**

2) **Material Usage Variance = Standard Price (Standard Quantity – Actual Quantity)**

Material A = 3 (500 – 450)	=	₹150 (F)
Material B = 5 (600 – 650)	=	₹250 (A)
Material Usage Variance	=	**₹100 (A)**

3) **Material Price Variance = Actual Quantity (Standard Price – Actual Price)**

Material A = 450 (3 – 2.50)	=	₹225 (F)
Material B = 650 (5 – 4.00)	=	₹650 (F)
Material Price Variance	=	**₹875(F)**

Verification: Material Cost Variance = Material Usage Variance + Material Price Variance
₹775(F) = ₹100(A) + ₹875(F)
₹775(F) = ₹775(F)

Example 28: From the following particulars calculate:
1) Material cost variance
2) Material price variance
3) Material usage variance

Quantity of materials purchased	4000 units
Value of materials purchased	₹14,000
Standard quantity of materials required per tonne of output	40 units
Standard rate of materials	₹3 per unit
Opening stock of materials	200 units
Closing stock of materials	700 units
Output during the period	90 tonnes

Solution:

1) **Material Cost Variance = (Standard Quantity × Standard Price) – (Actual Quantity × Actual Price)**

= (3,600 × 3) – (3,500 × 3.5) = 10,800 – 12,250 = ₹1,450 (A)

2) **Material Price Variance = Actual Quantity (Standard Price – Actual Price)**

= 3,500 × (3 – 3.5) = ₹1,750 (A)

3) **Material Usage Variance = Standard Price (Standard Quantity – Actual Quantity)**

= 3 × (3,600 – 3,500) = ₹300 (F)

Verification:
Material Cost Variance = Material Price Variance + Material Usage Variance
1,450(A) = 1,750(A) – 300(F)

Working Note:

1)	Material Consumed	=	Quantity of Material Purchased + Opening Stock of Material – Closing Stock of Material
		=	4,000 + 200 –700
		=	3,500 units
2)	Actual Rate of Material	=	$\frac{\text{Value of Material Purchased}}{\text{Quantity of Material Purchased}} = \frac{14,000}{4,000}$ = ₹3.50 per unit of material
3)	Standard Quantity for Actual Output	=	Output during the Period × Standard Quantity of Materials Required per Tonne of Output
		=	90 × 40
		=	3,600 units of material

Example 29: Given the cost standards for materials consumption are 40kgs at ₹10 per kg. Compare the variance when actual are:
48kgs at ₹10 per kg.
40kgs at ₹12 per kg.
48kgs at ₹12 per kg.
36kgs for a total cost of ₹360.

Solution:

1) **Material Cost Variance = (Standard Quantity × Standard Price) – (Actual Quantity × Actual Price)**

Case A = (40 × 10) – (48 × 10)	=	₹80(A)
Case B = (40 × 10) – (40 × 12)	=	₹80(A)
Case C = (40 × 10) – (48 × 12)	=	₹176(A)
Case D = (40 × 10) – (36 × 10)	=	₹40(F)
Material Cost Variance	=	**₹296 (A)**

2) **Material Price Variance = Actual Quantity (Standard Price – Actual Price)**

Case A = 48 (10 – 10)	=	0
Case B = 40 (10 – 12)	=	₹80(A)
Case C = 48 (10 – 12)	=	₹96 (A)
Case D= 36 (10 – 10)	=	0
Material Price Variance	=	**₹176 (A)**

3) **Material Usage Variance = Standard Price(Standard Quantity – Actual Quantity)**

Case A =10 (40 – 48)	=	₹80(A)
Case B =10 (40 – 40)	=	0
Case C =10 (40 – 48)	=	₹80(A)
Case D =10 (40 – 36)	=	₹40(F)
Material Usage Variance	=	**₹120(A)**

Verification,
Material Cost Variance = Material Price Variance + Material Usage Variance
₹296 (A) = ₹176 (A) + ₹120(A)
₹296 (A) = ₹296 (A))

Example 30: Use following data to calculate Material Mix Variance:

Materials	Standard		Actual	
	Quantity (units)	Price per unit	Quantity (units)	Price per unit
A	40	1	50	1.50
B	110	2	90	3.00

For revised mix, the use of material A was increased by 5% and that of material B decreased by 10%

Solution:
Material Mix Variance = (Revised Standard Proportion of Actual Input – Actual Proportion) × Standard Price

Material A	= 1 (42 – 50) =1 (–8)	= ₹8 (A)
Material B	= 2 (99 – 90) = 2 (9)	= ₹18 (F)
Material Mix Variance		**= ₹10 (F)**

Working Note:
In this question the standards will be revised. Revised standards will be:
Material A = 40 + (40 × 5/100) = 42 units
Material B = 110 – (110 × 10/100) = 99 units

Example 31: Use the following data to calculate the material price variance, the material usage variance and material mix variance.

Material	Standard	Actual
X	50 units @ ₹40 per unit	60 units @ ₹40 per unit
Y	40 units @ ₹60 per unit	40 units @ ₹65 per unit

Solution:

1) **Material Price Variance = Actual Quantity (Standard Price – Actual Price)**

Material X	= 60 (40 – 40)	= ₹Nil
Material Y	= 40 (60 – 65)	= ₹200 (A)
Material Price Variance		**= ₹200 (A)**

2) **When quantities of actual material mix and standard material mix are different:**
Material Mix Variance = (Revised Standard Proportion of Actual Input – Actual Proportion) × Standard Price

(**See working notes for revised standard proportion of actual input)		
Material A	= (56 – 60) 40	= ₹160 (A)
Material B	= (44 – 40) 60	= ₹240 (F)
Material Mix Variance		**= ₹80 (F)**

3) **Material Usage Variance = Standard Price (Standard Quantity – Actual Quantity)**

Material X	= 40 (50 – 60)	= ₹400 (A)
Material Y	= 60 (40 – 40)	= ₹Nil
Material Usage Variance		**= ₹400(A)**

Working Note:

$$\text{**Revised standard proportion of actual input} = \frac{\text{Standard Mix}}{\text{Total Standard Input}} \times \text{Total Actual Input}$$

$$\text{Material A} = \frac{50}{90} \times 100 = 56 \text{ units} \qquad \text{Material B} = \frac{40}{90} \times 100 = 44 \text{ units}$$

Example 32: Following is the data for standard mix of a product:

Material A	40 units at ₹20 per unit	₹800
Material B	60 units at ₹25 per unit	₹1,500
Material C	80 units at ₹30 per unit	₹2,400
	180	**₹4,700**

The above mentioned mix can produce 10 units of finished product. During the month of December, ten mixes were completed with the following consumption data:

Material A	500 units at ₹25 per unit	₹12,500
Material B	750 units at ₹20 per unit	₹15,000
Material C	600 units at ₹25 per unit	₹15,000
	1,850	**₹42,500**

The actual output was 80 units.
Use the above given information to calculate various material variances.

Solution:

1) **Material Cost Variance = Standard Cost – Actual Cost**
Standard Cost for 100 Units = 4,700 × 10 = ₹47,000
Actual yield is 80 units, so standard cost will be adjusted accordingly.

$$\text{Standard Cost for Actual Yield} = \frac{47,000}{100} \times 80 = ₹37,600.$$

Material Cost Variance = 37,600 – 42,500 = ₹4,900 (A)

2) **Material Price Variance = Actual Quantity (Standard Price – Actual Price)**

Material A = 500 (20 – 25)	= 500 × (–5)	= ₹2,500 (A)
Material B = 750 (25 – 20)	= 750 × 5	= ₹3,750 (F)
Material C = 600 (30 – 25)	= 600 × 5	= ₹3,000 (F)
Material Price Variance		**= ₹4,250 (F)**

3) **Material Mix Variance**

Since standard quantity (180 × 10 = 1,800) and actual quantity (1,850) are different, so the standard will be revised first. Revised standard quantity will be:

$$A = \frac{40}{180} \times 1{,}850 = 411; \quad B = \frac{60}{180} \times 1{,}850 = 617 \text{ (approx.)}; \quad C = \frac{80}{180} \times 1{,}850 = 822 \text{ (approx.)}$$

Material Mix Variance = (Revised Standard Proportion of Actual Input – Actual Proportion) × Standard Price

Material A	= (411 – 500) × ₹20	= ₹1,780 (A)
Material B	= (617 – 750) × ₹25	= ₹3,325 (A)
Material C	= (822 – 600) × ₹30	= ₹6,660 (F)
Material Mix Variance		**= ₹1,555 (F)**

4) **Material Yield Variance = Standard Rate (Actual Yield – Standard Yield)**

$$\text{Standard Rate} = \frac{\text{Standard Cost of Standard Mix}}{\text{Standard Output}} = \frac{4{,}700}{10} = ₹470$$

$$\text{Standard Yield} = \frac{10}{180} \times 1{,}850 = 102.777 \text{ units}$$

Material Yield Variance = 470 (80 – 102.777) = ₹10,705 (A)

5) **Material Usage Variance**

The standard quantity will be revised in proportion to actual output. Revised quantity will be:

$$\text{Material A} = \frac{400}{100} \times 80 = ₹320, \quad \text{Material B} = \frac{600}{100} \times 80 = ₹480, \quad \text{Material C} = \frac{800}{100} \times 80 = ₹640$$

Materials Usage Variance = Standard Price (Standard Quantity – Actual Quantity)

Material A	= ₹20 (320-500)	= ₹3,600 (A)
Material B	= ₹25 (480-750)	= ₹6,750 (A)
Material C	= ₹30 (640-600)	= ₹1,200 (F)
Material Usage Variance		**= ₹9,150 (A)**

Verification:

a) **Material Cost Variance** = Material Price Variance + Material Usage Variance

4,900 (A) = 4,250 (F) + 9,150 (A)

b) **Material Usage Variance** = Material Mix Variance + Material Yield Variance

9,150 (A) = 1,555 (F) + 10,705 (A)

Example 33: Compute material cost, material price, material mix and material usage variance using following given information:

Materials	Standard			Actual		
	Quantity (Kilos)	Unit Price (₹)	Total (₹)	Quantity (Kilos)	Unit Price (₹)	Total (₹)
A	5	1	5	3	2	6
B	10	2	20	6	3	18
C	10	3	30	11	2	22
Total	**25**	**6**	**55**	**20**	**7**	**46**

Solution:

1) **Material Cost Variance = (Standard Quantity for Actual Output × Standard Price) – (Actual Quantity × Actual Price)**

Material A	= (5× 1) – (3 × 2)	=	₹1 (A)
Material B	= (10× 2) – (6 × 3)	=	₹2 (F)
Material C	= (10 × 3) – (11 × 2)	=	₹8 (F)
Material Cost Variance		=	**₹9 (F)**

2) **Material Price Variance = Actual Quantity (Standard Price – Actual Price)**

Material A	= 3(1 – 2)	=	₹3 (A)
Material B	= 6(2 – 3)	=	₹6 (A)
Material C	= 11(3 – 2)	=	₹11 (F)
Material Price Variance		=	**₹2 (F)**

3) **Material Mix Variance = (Revised Standard Proportion of Actual Input – Actual Proportion) × Standard Price**

Revised standard proportion in terms of actual input needs to be calculated for computing material mix variance. **Revised Standard Proportion** will be calculated on the basis of the following formula:

$$\frac{\text{Standard Proportion} \times \text{Actual Input}}{\text{Total Standard Quantity}}$$

$$\text{Material A} = \frac{5}{25} \times 20 = 4 \text{ kg}, \quad \text{Material B} = \frac{10}{25} \times 20 = 8 \text{ kg}, \text{ Material C} = \frac{10}{25} \times 20 = 8 \text{ kg}$$

The following formula is used to calculate mix variance:

Material A	= (4 – 3) × 1	=	₹1 (F)
Material B	= (8 – 6) × 2	=	₹4 (F)
Material C	= (8 – 11) × 3	=	₹9 (A)
Material Mix Variance		=	**₹4 (A)**

4) **Material Sub-usage Variance = (Standard Quantity – Revised Standard Proportion) × Standard Price**

Material A	= (5 – 4) × 1	=	₹1 (F)
Material B	= (10 – 8) × 2	=	₹4 (F)
Material C	= (10 – 8) × 3	=	₹6 (F)
Material Sub-usage Variance		=	**₹11 (F)**

5) **Material Usage (or Quantity) Variance = Standard Price × (Standard Quantity – Actual Quantity)**

Material A	= 1 × (5 – 3)	=	₹2 (F)
Material B	= 2 × (10 – 6)	=	₹8 (F)
Material C	= 3 × (10 – 11)	=	₹3 (A)
Material Usage Variance		=	**₹7 (F)**

Note: Material Usage or Quantity variance is the sum of material mix variance and material sub-usage variance. Therefore,

Material Mix Variance + Material Sub-usage Variance = Material Usage Variance

₹4 (A) + ₹11 (F) = ₹7 (F)

Example 34: The standard cost of a chemical mixture is as under:

8 tonnes of Material A at ₹40 per tonne.

12 tonnes of Material B at ₹60 per tonne.

Standard yield is 90% of input.

Actual cost for a period is as under:

10 tonnes of Material A at ₹30 per tonne.

20 tonnes of Material B at ₹68 per tonne.

Actual yield is 26.5 tonnes.

Compute:

1) Materials Cost variance
2) Materials Usage variance
3) Materials Price variance
4) Materials Mix variance
5) Materials Yield variance

Solution:

Material	Standard			Actual		
	Tonnes	Price (₹)	₹	Tonnes	Price (₹)	₹
A	8	40	320	10	30	300
B	12	60	720	20	68	1,360
Input	20		1,040	30		1,660
Less: Loss 10%	2		–	3.5		–
Output	18		1,040	26.5		1,660

1) **Material Cost Variance = (Standard Quantity for Actual Output × Standard Price) – (Actual Quantity × Actual Rate)**

$$\text{Standard Quantity for Actual Output} = \frac{\text{Standard Quantity}}{\text{Standard Output}} \times \text{Actual Output}$$

$$A = \frac{8}{18} \times 26.5 = ₹11.78$$

$$B = \frac{12}{18} \times 26.5 = ₹17.67$$

Material A	= (11.78 × 40) – (10 × 30) = 471.2 – 300	= ₹171(F)
Material B	= (17.67 × 60) – (20 × 68) = 1,060 – 1,360	= ₹300(A)
Material Cost Variance		**= ₹129(A)**

2) **Material Price Variance = Actual Quantity (Standard Price – Actual Price)**

Material A	= 10 × (40 – 30)	= ₹100 (F)
Material B	= 20 × (60 – 68)	= ₹160 (A)
Material Price Variance		= ₹60 (A)

3) **Material Usage Variance = Standard Price (Standard Quantity for Actual Output – Actual Quantity)**

Material A	= 40 × (11.78 – 10)	= 71(F)
Material B	= 60 × (17.67 – 20)	= 140 (A)
Material Usage Variance		= ₹69 (A)

Verification: Material Cost Variance = Material Price Variance + Material Usage Variance

129(A) = 60(A) + 69(A)

4) **Material Mix Variance** = Standard Price × (Revised Standard Quantity – Actual Quantity)

$$\text{Revised Standard Quantity} = \frac{\text{Total Weight of Actual Mix}}{\text{Total Weight of Standard Mix}} \times \text{Standard Quantity}$$

$$A = \frac{30}{20} \times 8 = ₹12$$

$$B = \frac{30}{20} \times 12 = ₹18$$

Material A	= 40 × (12 – 10)	= ₹80(F)
Material B	= 60 × (18 – 20)	= ₹120(A)
Material Mix Variance		= ₹40(A)

5) **Material Yield Variance** = Standard Cost per Unit of Output (Standard Output for Actual Mix – Actual Output)

$$\text{Standard Cost per Unit of Output} = \frac{\text{Standard Total Cost}}{\text{Standard Output}} = \frac{1{,}040}{18} = ₹57.78$$

$$\text{Standard Output for Actual Mix} = \frac{30}{20} \times 18 = 27 \text{ tonnes}$$

Material Yield Variance = ₹57.78 (27 – 26.50) = ₹29(A)

Verification: Material Usage Variance = Material Mix Variance + Material Yield Variance
= 69(A) = 40(A) + 29(A)

Example 35: From the following information, calculate:
1) Material Price Variance
2) Material Usage Variance
3) Material Mix Variance

Particulars	Standard			Actual		
	Quantity (Kg)	Rate (₹)	(₹)	Quantity (Kg)	Rate (₹)	(₹)
Material X	5	2.00	10.00	4	5.00	20.00
Material Y	3	3.00	9.00	2	3.00	6.00
Material Z	2	4.00	8.00	3	3.00	9.00
Total	**10**		**27.00**	**9.00**		**35.00**

Solution:

1) **Material Price Variance** = Actual Quantity (Standard Price – Actual Price)

Material X	=	4 (2 –5)	=	12 (A)
Material Y	=	2 (3 –3)	=	0
Material Z	=	3 (4 –3)	=	3 (F)
Material Price Variance	=			**9 (A)**

2) **Material Usage Variance = Standard Price (Standard Quantity – Actual Quantity)**

Material X	=	2 (5 – 4)	=	₹2 (F)
Material Y	=	3 (3 – 2)	=	₹3 (F)
Material Z	=	4 (2 – 3)	=	₹4 (A)
Material Usage Variance	=			**₹1 (F)**

3) **Material Mix Variance** = Standard Price × (Revised Standard Quantity – Actual Quantity)

$$\text{Revised Standard Quantity} = \frac{\text{Total Quantity of Actual Mix}}{\text{Total Quantity of Standard Mix}} \times \text{Standard Quantity}$$

$$\text{Material X} = \frac{(4+2+3)}{(5+3+2)} \times 5 = 4.5$$

$$\text{Material Y} = \frac{9}{10} \times 3 = 2.7$$

$$\text{Material Z} = \frac{9}{10} \times 2 = 1.8$$

Material X	=	2 (4.5 – 4)	=	₹1 (F)
Material Y	=	3 (2.7 – 2)	=	₹2.1 (F)
Material Z	=	4 (1.8 – 3)	=	₹4.8 (A)
Material Mix Variance	=			**₹2.6 (F)**

Example 36: From the following data, calculate:

1) Material Cost Variance
2) Material Price Variance
3) Material Usage Variance
4) Material Mix Variance

Material	Standard	Actual
A	100 units @ ₹20 per unit	50 units @ ₹30 per unit
B	200 units @ ₹30 per unit	100 units @ ₹60 per unit
C	200 units @ ₹60 per unit	150 units @ ₹50 per unit

Solution:

1) **Material Cost Variance = (Standard Quantity × Standard Price) – (Actual Quantity × Actual Price)**

Material A	=	(100 ×20) – (50 ×30)	=	2,000 – 1,500	=	₹500(F)
Material B	=	(200 ×30) – (100 ×60)	=	6,000 – 6,000	=	₹ 0
Material C	=	(200 ×60) – (150 ×50)	=	12,000 –7,500	=	₹4,500(F)
Material Cost Variance	=					₹ 5,000(F)

2) **Material Price Variance = Actual Quantity (Standard Price – Actual Price)**

Material A	=	50 (20 – 30)	=	₹500(A)
Material B	=	100 (30 – 60)	=	₹3,000(A)
Material C	=	150 (60 – 50)	=	₹1,500(F)
Material Price Variance	=			**₹2,000(A)**

3) **Material Usage Variance = Standard Price (Standard Quantity – Actual Quantity)**

Material A	=	20 (100 – 50)	=	₹1,000(F)
Material B	=	30 (200 –100)	=	₹3,000(F)
Material C	=	60 (200 – 150)	=	₹3,000(F)
Material Usage Variance	=			**₹7,000 (F)**

Verification,

Material Cost Variance = Material Price Variance + Material Usage Variance

₹5,000 (F) = ₹2,000 (A) + ₹7,000 (F)

₹5,000 (F) = ₹5,000 (F)

4) **Material Mix Variance = Standard Price (Revised Standard Quantity – Actual Quantity)**

$$\text{Revised Standard Quantity} = \frac{\text{Total Quantity of Actual Mix}}{\text{Total Quantity of Standard Mix}} \times \text{Standard Quantity}$$

Material A: $\text{Revised Standard Quantity} = \frac{(50+100+150)}{(100+200+200)} \times 100 = 60$

Material B: $\text{Revised Standard Quantity} = \frac{300}{500} \times 200 = 120$

Material C: Revised Standard Quantity $= \frac{300}{500} \times 200 = 120$

Material A	=	20 (60 – 50)	=	₹200 (F)
Material B	=	30(120 – 100)	=	₹600 (F)
Material C	=	60(120 – 150)	=	₹1,800 (A)
Material Mix Variance	=			₹1,000 (F)

Example 37: From the following data, calculate:
1) Material Cost Variance
2) Material Price Variance
3) Material Usage Variance
4) Material Mix Variance

Material	Standard		Actual	
	Qty. (kgs.)	Rate (Rs.)	Qty. (kgs.)	Rate (Rs.)
X	8,000	1.05	7,500	1.20
Y	3,000	2.15	3,300	2.30
Z	2,000	3.30	2,400	3.50
Total	**13,000**		**13,200**	

Solution:

1) **Material Cost Variance = (Standard Quantity × Standard Price) – (Actual Quantity × Actual Price)**

Material X	=	(8,000 × 1.05) – (7,500 × 1.20) =	8,400 – 9,000	=	₹600(A)
Material Y	=	(3,000 × 2.15) – (3,300 × 2.30)=	6,450 – 7,590	=	₹1,140(A)
Material Z	=	(2,000 × 3.30) – (2,400 × 3.50)=	6,600 – 8,400	=	₹1,800(A)
Material Cost Variance				=	**₹3,540(A)**

2) **Material Price Variance = Actual Quantity (Standard Price – Actual Price)**

Material X	=	7,500 (1.05 – 1.20)	=	₹1,125(A)
Material Y	=	3,300 (2.15 – 2.30)	=	₹495(A)
Material Z	=	2,400 (3.30 – 3.50)	=	₹480(A)
Material Price Variance			=	**₹2,100(A)**

3) **Material Usage Variance = Standard Price (Standard Quantity – Actual Quantity)**

Material X	=	1.05 (8,000 – 7,500)	=	₹525(F)
Material Y	=	2.15 (3,000 – 3,300)	=	₹645(A)
Material Z	=	3.30 (2,000 – 2,400)	=	₹1,320(A)
Material Price Variance			=	**₹1,440(A)**

Verification,

Material Cost Variance = Material Price Variance + Material Usage Variance

3,540(A) = 2,100(A) + 1,440(A)

4) **Material Mix Variance**

Material Mix Variance = Standard Price (Revised Standard Quantity – Actual Quantity)

$$\text{Revised SQ} = \frac{\text{Total Quantity of Actual Mix}}{\text{Total Quantity of Standard Mix}} \times \text{Standard Quantity}$$

Material X: Revised Standard Quantity $= \frac{7,500+3,300+2,400}{8,000+3,000+2,000} \times 8,000 = 8,123$

Material Y: Revised Standard Quantity $= \frac{13,200}{13,000} \times 3,000 = 3,046$

Material Z: Revised Standard Quantity $= \frac{13,200}{13,000} \times 2000 = 2,030$

Material X	=	1.05 (8,123 – 7,500)	=	₹654(F)
Material Y	=	2.15 (3,046 – 3,300)	=	₹546(A)
Material Z	=	3.30 (2,030 – 2,400)	=	₹1,22(A)
Material Mix Variance			=	**₹1,113(A)**

5.13. LABOUR VARIANCES

5.13.1. Labour Cost Variance (LCV)

Labour cost variance is calculated in the same way as Material Cost Variance. It uses same techniques and formulas. Labour cost variance is the difference between actual labour cost and Standard Labour Cost, for a given quantity of production.

According to Terminology of CIMA defines Direct Labour Cost Variance as "The difference between the standard direct labour cost and actual direct labour cost incurred for the production achieved".

In the form of formula:
LCV = Standard Cost – Actual Cost
i.e., (Standard Hours × Standard Rate) – (Actual Hours × Actual Rate)

In case where the actual output is different from standard output, standard labour cost of actual output is calculated and the following formula is used:

LCV = Standard Cost of Actual Production – Actual Cost

Labour cost variances are sub-divided into:
1) Labour rate of pay or wage rate variance, and
2) Labour efficiency or labour time variance.

5.13.2. Labour Rate or Wage Rate Variance (LRV)

Labour Rate Variance is "that portion of the wages variance which is due to the difference between the standard rate specified and the actual rate paid". Personnel department is entrusted with the responsibility of managing labour rate of pay variance. Following is the formula for calculating this variance:
Labour Rate Variance = Actual Hour (Standard Rate – Actual Rate)

In case where actual rate is lower than the standard, the variance is said to be **favourable**. Conversely, when the standard cost is lower than the actual cost, the variance is **adverse**.

5.13.3. Labour Efficiency or Labour Time Variance (LEV)

This variance calculates and explains the difference between standard labour hours and the actual labour hours spent. Following formula is used for the purpose of calculating this variance:

Labour efficiency variance is calculated as follows:
Labour Efficiency Variance = Standard Rate (Standard Hour – Actual Hour)

This variance is considered unfavourable if the actual time taken is more than the standard time calculated while in the opposite condition, the variance is deemed to be positive.

Labour efficiency variance has many variants which are as follows:
1) Labour Mix or Gang Composition Variance,
2) Labour Idle Time Variance, and
3) Labour Yield Variance.

5.13.4. Labour Mix or Gang Composition Variance

This variance explains change in labour cost due to the difference in actual gang composition and standard gang composition. Various reasons such as shortage of a type of labour may lead to the change in gang composition. This variance can be calculated in two different ways:

1) **When Standard and Actual Times of the Labour Mix are Same:** Following formula is used for the purpose of calculating the variance:

   Labour Mix Variance = Standard Cost of Standard Labour Mix – Standard Cost of Actual Labour Mix

   Shortage or non-availability of one grade of labour may necessitate the change in standard labour mix. In such case, following formula is used:

   Labour Mix Variance = Standard Cost of Revised Standard Labour Mix – Standard Cost of Actual Labour Mix

2) **When Standard and Actual Time of Labour Mix are Different:** In such case, the following formula is used:

$$\left[\left(\frac{\text{Total Hour of Actual Labour Mix}}{\text{Total Hour of Standard Labour mix}} \times \text{Standard Cost of Standard Labour Mix}\right) - \left(\text{Standard Cost of Actual Labour Mix}\right)\right]$$

   Revised standard mix is used in the formula, if there is any change in gang composition due to non-availability or shortage of any class of labour:

$$\left[\left(\frac{\text{Total Hour of Actual Labour Mix}}{\text{Total Hour of Revised Standard Labour mix}} \times \text{Standard Cost of Revised Standard Labour Mix}\right) - \left(\text{Standard Cost of Actual Labour Mix}\right)\right]$$

5.13.5. Labour Idle Time Variance

This variance is the standard cost of actual time paid to workers. These are the hours which have not been worked due to abnormal reasons such as shortage of materials, power failure, shutdown for repairing, etc. This variance is calculated as follows:

Labour Idle Time Variance = Standard Rate (Actual Hours Paid – Actual Hours Worked)
= Standard Rate × Actual Hours of Idle Time.

5.13.6. Labour Yield Variance

This variance explains the difference between actual output of workers and standard output of workers specified:

Labour Yield Variance = (Standard Output for Actual Hour – Actual Output) × Standard Cost per unit.

Variance is calculated for the purpose of exercising control. Many variances are controllable while others are not. For controllable variance, proper caution should be exercised to avoid their recurrence, while for non-controllable variance, there should be revision in future standards.

For verifying the answer, formula used is: LCV = LRV + LEV

Example 38: A product requires 10 hours per unit to produce at a standard rate of ₹2 per hour. 1,000 units were produced during March, 2018. Actual time taken to produce was 9000 hours. The actual rate was ₹2.50 per hour.

Calculate:
1) Labour Cost Variance
2) Labour Rate Variance
3) Labour Efficiency Variance

Solution:
Standard Hours = 10 hours per unit
Standard Rate = ₹2 per hour
Actual Production = 1000 units
Actual Hours = 9000 hours
Actual Rate = ₹2.50 per hours

1) **Labour Cost Variance = (Standard Hour × Standard Rate) – (Actual Hour × Actual Rate)**
= (10,000 × 2) – (9,000 × 2.50)
= 20,000 – 22,500 = ₹2,500 (A)

2) **Labour Rate Variance = (Standard Rate – Actual Rate) × Actual Hours**
= (2 – 2.50) × 9,000
= ₹4,500 (A)

3) **Labour Efficiency Variance = Standard Rate (Standard Hour – Actual Hour)**
= 2 × (10,000 – 9,000)
= 2 × 1,000 = ₹2,000 (F)

Verifications
Labour Cost Variance = Labour Rate Variance + Labour Efficiency Variance
₹2,500 (A) = ₹4,500 (A) + ₹2,000 (A)

Working Note:
Standard hours for actual production:
For 1 unit standard hours = 10
1000 units standard hours = 10 × 1000 = 10000 hours

Example 39: Ash Ltd., Pune produces a product; the Standard Cost Card of the same discloses the following information:

Standard Rate per hour – ₹26
Standard Time for a unit of output hours – 5

The Actual Cost Data for a particular period are as follows:

Total Number of Units Produced	200 units
Actual Hours Worked	900 hours
Total Labour Cost	₹27,000

You are required to calculate:
1) Labour Cost Variance,
2) Labour Rate Variance, and
3) Labour Efficiency Variance.

Also verify your results.

Solution:

1) **Labour Cost Variance = (Standard Hours for Actual Output × Standard Rate) – (Actual Hours × Actual Rate)**

Standard Hours for Actual Output = Standard Hours Per Unit × Actual Quantity.

Standard Hours for Actual Output = 5 hours × 200 units = 1,000 hours

Actual Rate = Total Labour Cost ÷ Actual Hours Worked.

$$\text{Actual Rate} = \frac{27,000}{900} = ₹30$$

Labour Cost Variance = (1,000 × 26) – (900 × 30) = 26,000 – 27,000 = ₹1,000 (A)

1) **Labour Rate Variance**	=	**Actual Hours (Standard Rate – Actual Rate)**
	=	900 (26 – 30) = ₹3,600 (A)
2) **Labour Efficiency Variance**	=	Standard Rate (Standard Hours for Actual Output – Actual Hours)
	=	26(1,000 – 900) = ₹2,600 (F)

Verification,

Labour Cost Variance = Labour Rate Variance + Labour Efficiency Variance

1,000 (A) = 3,600 (A) + 2,600 (F)

1,000 (A) = 1,000 (A)

Example 40: Following data pertains to two group of workmen and their standard and actual working hours and rate:

	Standard Hours	Rate per hour	Total
Worker A	25 hrs.	2	50
Worker B	10 hrs.	3	30
			₹80

	Actual Hours	Rate per hour	Total
Worker A	20 hrs.	2.50	50
Worker B	15 hrs.	4	60
			₹110

Calculate

1) Labour cost variance,
2) Labour efficiency variance,
3) Labour rate variance.

Solution:

1) **Labour Cost Variance = (Standard Hours × Standard Rate) – (Actual Hours × Actual Rate)**

Worker A	= (25 × 2) – (20 × 2.50)	=	Nil
Worker B	= (10× 3) – (15 × 4)	=	₹30(A)
Labour Cost Variance		=	**₹30 (A)**

2) **Labour Efficiency Variance = Standard Rate (Standard Hours – Actual Hours)**

Worker A	= 2(25 – 20)	=	₹10 (F)
Worker B	**= 3 (10** – 15)	=	₹15 (A)
Labour Efficiency Variance		=	**₹5 (A)**

3) **Labour Rate Variance = Actual Hours (Standard Rate – Actual Rate)**

Worker A	= 20 (2.00 – 2.50)	= ₹10 (A)
Worker B	= 15 (3 – 4)	= ₹15 (A)
Labour Rate Variance		= **₹25 (A)**

Verification:
Labour Cost Variance = Labour Efficiency Variance + Labour Rate Variance
₹30 (A) = ₹5 (A) + ₹25 (A)

Example 41: Following is the data given for a product:

Standard hours	20 hours
Standard rate	₹5 per hour
Actual output	1,000 units
Actual hours	20,500 hours
Actual rate	₹4 per hour

Calculate the following:
1) Labour cost variance,
2) Labour efficiency variance
3) Labour rate variance

Solution:
1) **Labour Cost Variance = (Standard Hour for Actual Output × Standard Rate) – (Actual Hour × Actual Rate)**
= (20,000 × 5) – (20,500 × 4)
= 1,00,000 – 82,000 = ₹18,000(F)

2) **Labour Efficiency Variance = Standard Rate (Standard Hour – Actual Hour)**
= 5 (20,000 – 20,500)
= ₹2,500(A)

3) **Labour Rate Variance = Actual Hours (Standard Rate – Actual Rate)**
= 20,500 (5 – 4)
= ₹20,500 (F)

Verifications:
Labour Cost Variance = Labour Efficiency Variance + Labour Rate Variance
₹18,000 (F) = ₹2,500 (A) + ₹20,500 (F)

Working Note:
Standard hours for actual output
For 1 unit standard hours = 20 hours
1,000 units standard hours = 20 × 1,000 = 20,000 hours

Example 42: Using the following information calculate each of three labour variances for each department:

Particulars	Dept. A	Dept. B
Gross wages direct (₹)	25,000	20,000
Standard hours produced	10,000	7,500
Standard rate per hour (₹)	2	4
Actual hours worked	8,500	8,000

Solution:

1) **Labour Cost Variance = (Standard Hours × Standard Rate) – (Actual Hours × Actual Rate)**

Department A	(10,000 × 2) – (8,500 × 2.94)	=	₹4,990 (A)
Department B	(7,500 × 4) – (8,000 × 2.50)	=	₹10,000 (F)
Labour Cost Variance		=	**₹5,010 (F)**

2) **Labour Rate Variance = Actual Hours (Standard Rate – Actual Rate)**

Department A	= 8,500 (2 – 2.94)	=	₹7,990 (A)
Department B	= 8,000 (4 – 2.50)	=	₹12,000 (F)
Labour Rate Variance		=	**₹4,010 (F)**

3) **Labour Efficiency Variance = (Standard Hours – Actual Hours) × Standard Rate**

Department A	= (10,000 – 8,500) × 2	=	₹3,000 (F)
Department B	= (7,500 – 8,000) × 4	=	₹2,000 (A)
Labour Efficiency Variance		=	**₹1,000 (F)**

Working Note:

Calculation of Actual Rate = $\frac{\text{Gross Wages}}{\text{Actual Hour Worked}}$

$$A = \frac{25,000}{8,500} = 2.94$$

$$B = \frac{20,000}{8,000} = 2.50$$

Example 43: Following is the data pertaining to a job schedule to be completed in 30 weeks:

Category of Worker	Standard		Actual	
	No. of Labourers	Weekly Wage Rate per Labour (₹)	No. of Labourers	Weekly Wage Rate per Labour (₹)
Skilled	80	50	75	40
Semi-Skilled	50	40	40	45
Unskilled	55	60	60	50

Actual time taken is 45 weeks. Calculate various labour variances.

Solution:

Calculation of Standard Costs

Category of Workers	Weeks	Standard Rate	₹
Skilled	80 × 30 = ₹2,400	50	1,20,000
Semi-Skilled	50 × 30 = ₹1,500	40	60,000
Unskilled	55 × 30 = ₹1,650	60	99,000
	₹5,550		**2,79,000**

Calculation of Actual Costs

Category of Workers	Weeks	Actual Rate	₹
Skilled	75 × 45 = ₹3,375	40	1,35,000
Semi-Skilled	40 × 45 = ₹1,800	45	81,000
Unskilled	60 × 45 = ₹2,700	50	135,000
	₹7,875		**3,51,000**

1) **Labour Cost Variance = Standard Cost – Actual Cost**
= ₹2,79,000 – 3,51,000 = ₹72,000 (A)

2) **Labour Rate Variance = Actual Hours (Standard Rate – Actual Rate)**

Skilled	= 3,375 (50 – 40)	= ₹33,750 (F)
Semi-skilled	= 1,800 (40 – 45)	= ₹9,000 (A)
Unskilled	= 2,700 (60 – 50)	= ₹27,000 (F)
Labour Rate Variance		**= ₹51,750 (F)**

3) **Labour Efficiency Variance = Standard Rate (Standard Hours – Actual Hours)**

Skilled	= 50 (2,400 – 3,375)	= ₹48,750 (A)
Semi-skilled	= 40 (1,500 – 1,800)	= ₹12,000 (A)
Unskilled	= 60 (1,650 – 2,700)	= ₹63,000 (A)
Labour Efficiency Variance		**= ₹1,23,750 (A)**

Verifications:
Labour Cost Variance = Labour Efficiency Variance + Labour Rate Variance
₹72,000 (A) = ₹1,23,750 (A) + ₹51,750 (F)

Example 44: In a factory, 100 workers are engaged and the average rate of wages is 50 paise per hour Standard working hours per week are 40 and the standard performance is 10 units per gang hour.

During a week in March, wages paid for 50 workers were at the rate of 50 paise per hour 10 workers at 70 paise per hour and 40 workers at 40 paise per hour. Actual output was 380 units.

The factory did not work for five hours due to break down of machinery. Calculate Labour cost variance, Labour rate variance, Labour efficiency variance, Idle time variance.

Solution:

1) **Labour Cost Variance = Standard Labour Cost – Actual Labour Cost**
Now,
Standard Labour Cost = 100 men are working in the factory and so one gang hour means 100 hours. As one gang hour's production is 10 units, so one unit will require 10 hours time. Thus, for the production of 380 units will take (380 ×10) = 3800 hours at standard rate of 50 paise per hour will come to 3,800 × 0.50 = ₹1,900.

Actual Labour Cost =

50 workers × 40hrs. = 2,000hrs. × 0.50	= ₹1,000
10 workers × 40hrs. = 400 hrs. × 0.70	= ₹280
40 workers × 40hrs. = 1,600hrs. × 0.40	= ₹640
4,000hrs.	= ₹1,920

∴ Labour Cost Variance = 1,900 – 1,920 = ₹20(A)

2) **Labour Rate Variance = Actual Hours (Standard Rate – Actual Rate)**

2,000 (0.50 – 0.50)	=	₹0
400 (0.50 – 0.70)	=	₹80(A)
1,600 (0.50 – 0.40)	=	₹160(F)
Labour Rate Variance		₹80(F)

3) **Labour Efficiency Variance = Standard Rate (Standard Hours – Actual Hours)**
Labour Efficiency Variance = 0.50 (3,800 – 3,500) =150(F)

100 men × 40 hrs.	=	4000 hrs.
Less: 100 men's Idle Time (100 × 5)	=	500 hrs.
		3,500 hrs.

4) **Idle Time Variance = Standard Rate (Actual Hours Paid – Actual Hours Worked)**
= 0.50 (3,500 – 4,000) = 250(A)

Verification:
Labour Cost Variance = Labour Rate Variance + Labour Efficiency Variance + Idle Time Variance
20(A) = 80(F) + 150(F) + 250(A)
20(A) = 20(A)

Example 45: Following is the labour budget for producing 1,000 units of product 'A'

		Total Standard Hours	Total Standard Cost
50 Men	₹6 per hour for 70 hrs.	3,500	21,000
40 Women	₹5 per hour for 60 hrs.	2,400	12,000
20 Boys	₹3 per hour for 30 hrs.	600	1,800
		6,500	**34,800**

Following is the actual data:
Articles produced, 1,000

		Total Actual Hours	Total Actual Cost
50 Men	₹6 per hour for 70 hrs.	3,500	21,000
35 Women	₹5 per hour for 60 hrs.	2,100	10,500
30 Boys	₹4 per hour for 30 hrs.	900	3,600
		6,500	**35,100**

You are required to calculate following:
1) Labour cost variance,
2) Labour rate variance
3) Labour efficiency variance,
4) Labour mix variance.

Solution:
1) **Labour Cost Variance = (Standard Hours for Actual Output × Standard Rate) – (Actual Hours × Actual Rate)**

Men	= (3,500 × 6.00) – (3,500 × 6.00)	=	Nil
Women	= (2,400 × 5.00) – (2,100 × 5.00)	=	₹1,500 (F)
Boys	= (600 × 3.00) – (900 × 4.00)	=	₹1,800 (A)
Labour Cost Variance		=	**₹300 (A)**

2) **Labour Rate Variance = Actual Hours (Standard Rate – Actual Rate)**

Men	= 3,500 (6 – 6)	=	Nil
Women	= 2,100 (5 – 5)	=	Nil
Boys	= 900 (3 – 4)	=	₹900 (A)
Labour Rate Variance		=	**₹900 (A)**

3) **Labour Efficiency Variance = Standard Rate × (Standard Hours for Actual Production – Revised Standard Hours)**

Men	= 6 × (3,500 – 3,500)	=	Nil
Women	= 5 × (2,400 – 2,400)	=	Nil
Boys	= 3 × (600 – 600)	=	Nil
Labour Efficiency Variance		=	**Nil**

4) **Labour Mix Variance = (Revised Standard Hours – Actual Hours) × Standard Rate of Actual Total Hours**

Men	= (3,500 – 3,500) × 6	=	Nil
Women	= (2,400 – 2,100) × 5	=	₹1,500 (F)
Boys	= (600 – 900) × 3	=	₹900 (A)
Labour Mix Variance		=	**₹600 (F)**

Note: Revised standard production is same as standard hours because the actual total hours are equal to total standard hours.

Example 46: Following is the Standard labour hours and rate for production data for Product X:

Particulars	Hours	Rate (₹)	Total (₹)
Skilled Worker	6	3.00 per hour	18
Semi – skilled Worker	5	2.00 per hour	10
Unskilled Worker	8	1.50 per hour	12
			40

The actual produced article is 1,000:

Particulars	Hours	Rate (₹)	Total (₹)
Skilled Worker	5,500	3.00 per hour	16,500
Semi – skilled Worker	4,500	2.25 per hour	10,125
Unskilled Worker	8,000	1.75 per hour	14,000
			40,625

Calculate the following:

1) Labour cost variance,
2) Labour rate variance,
3) Labour mix variance,
4) Labour efficiency variance.

Solution:

1) **Labour Cost Variance = (Standard Hours for Actual Output × Standard Rate) – (Actual Hours × Actual Rate)**

Standard Hours for Actual Production

Skilled Worker – For 1 unit standard hours = 6

For 1,000 units standard hours = 6 × 1,000 = 6,000 hours

Semi-skilled Worker – For 1 unit standard hours = 5

For 1,000 units standard hours = 5 × 1,000 = 5,000 hours

Unskilled Worker – For 1 unit standard hours = 8

For 1,000 units standard hours = 8 × 1,000 = 8,000 hours

Skilled Worker	= (6,000 × 3.00) – (5,500 × 3)	= 18,000 – 16,500 =	₹1,500 (F)
Semi-skilled Worker	= (5,000 × 2.00) – (4,500 × 2.25)	= 10,000 – 10,125 =	₹125 (A)
Unskilled Worker	= (8,000 × 1.50) – (8,000 × 1.75)	= 12,000 – 14,000 =	₹2,000 (A)
Labour Cost Variance		=	**₹625 (A)**

2) **Labour Rate Variance = Actual Hours (Standard Rate – Actual Rate)**

Skilled Worker	= 5,500 (3.00 – 3.00)	=	Nil
Semi-skilled Worker	= 4,500 (2.00 – 2.25)	=	₹1,125 (A)
Unskilled Worker	= 8,000 (1.50 – 1.75)	=	₹2,000 (A)
Labour Rate Variance		=	**₹3,125 (A)**

3) **Labour Mix Variance = (Revised Standard Mix of Actual Hours Worked – Actual Mix) × Standard Rate**
For calculation of labour mix variance we will have to calculate revised standard hours in terms of total actual hours worked.

$$\frac{\text{Standard Mix}}{\text{Total Standard Hours}} \times \text{Total Actual Hours}$$

i) Skilled Worker $= \frac{6,000}{19,000} \times 18,000 = 5,684$

ii) Semi-skilled Worker $= \frac{5,000}{19,000} \times 18,000 = 4,737$

iii) Unskilled Worker $= \frac{8,000}{19,000} \times 18,000 = 7,579$

Skilled Workers	= (5,684 – 5,500) × 3.00	=	₹552 (F)
Semi-skilled Workers	= (4,737 – 4,500) × 2.00	=	₹474 (F)
Unskilled Workers	= (7,579 – 8,000) × 1.50	=	₹631.50 (A)
Labour Mix Variance		=	**₹394.50 (F)**

4) **Labour Efficiency Variance = Standard Rate (Standard Hours for Actual Output – Revised Standard Hours)**

Skilled Workers	= 3 (6,000 – 5,684)	=	₹948 (F)
Semi-skilled Workers	= 2 (5,000 – 4,737)	=	₹526 (F)
Unskilled Workers	= 1.50 (8,000 – 7,579)	=	₹631.50 (F)
Labour Efficiency Variance		=	**₹2,105.5 (F)**

Example 47: The Standard labour employment and actual labour employment and the actual labour engaged in a week for a job are as under:

Particulars	Skilled Worker	Semi-Skilled Workers	Unskilled Workers
Standard no. of workers in the gang	32	12	6
Actual no. of workers employed	28	18	4
Standard wage rate per hour (₹)	3.00	2.00	1.00
Actual wage rate per hour (₹)	4.00	3.00	2.00

During 40 hours working week, the gang produced 1800 standard labour hours of work.

Calculate:
1) Labour Cost Variance
2) Labour Rate Variance
3) Labour Efficiency Variance
4) Labour Mix Variance

Solution:

Particulars	Standard					Actual				
	No. of Workers	Hour Per Week	Total Hours	Rate (₹)	Amount (₹)	No. of Workers	Hour Per Week	Total Hours	Rate (₹)	Amount (₹)
Skilled Worker	32	40	1,280	3	3,840	28	40	1,120	4	4,480
Semi-skilled Worker	12	40	480	2	960	18	40	720	3	2,160
Unskilled Worker	6	40	240	1	240	4	40	160	2	320
			2,000		**5,040**			**2,000**		**6,960**
Less: Loss (Bal. Fig.)			–		–			200		–
			2,000		**5,040**			**1,800**		**6,960**

1) **Labour Cost Variance = Standard Labour Cost For Actual Hours Worked – Actual Labour Cost for Actual Hours Worked**

$$\textbf{Standard Labour Cost for Actual Hours Worked} = \frac{\textbf{Standard Total Cost}}{\textbf{Standard Total Hours}} \times \textbf{Actual Total Hours}$$

$$= \frac{5040}{2000} \times 1800 = 4536$$

Labour Cost Variance	4,536 – 6,960	= ₹2,424(A)

2) **Labour Rate Variance = (Standard Rate – Actual Rate) Actual Hours**

Skilled Worker	= (3 – 4) × 1,120	=	₹1,120 (A)
Semi-skilled Worker	= (2 – 3) × 720	=	₹720 (A)
Unskilled Worker	= (1 – 2) × 160	=	₹160 (A)
Labour Rate Variance			**₹2,000 (A)**

3) **Labour Efficiency Variance = Standard Rate [Standard Hours for Actual Output – Actual Hours]**

$$\textbf{Standard Hours for Actual Output} = \frac{\textbf{Standard Total Cost}}{\textbf{Standard Total Hours}} \times \textbf{Actual Total Hours}$$

$$\text{Skilled Workers} = \frac{1,280}{2000} \times 1800 = 1152$$

$$\text{Semi-skilled Worker} = \frac{480}{2000} \times 1800 = 432$$

$$\text{Unskilled Workers} = \frac{240}{2000} \times 1800 = 216$$

Skilled Worker	= 3 (1,152 – 1,120)	=	₹96 (F)
Semi-skilled Worker	= 2 (432 – 720)	=	₹576 (A)
Unskilled Worker	= 1 (216 – 160)	=	₹56 (F)
Labour Efficiency Variance			**₹424 (A)**

4) **Labour Mix Variance = Standard Rate [Standard Labour Hours For Actual Input Hours – Actual Hours]**

$$\textbf{Standard Labour Hours For Actual Input Hours} = \frac{\textbf{Standard Total hours of skilled workers}}{\textbf{Standard Total Hours}} \times \textbf{Standard Total Hours}$$

$$\text{Skilled Workers} = \frac{1,280}{2000} \times 2000 = 1280$$

$$\text{Semi-skilled Worker} = \frac{480}{2000} \times 2000 = 480$$

$$\text{Unskilled Workers} = \frac{240}{2000} \times 2000 = 240$$

Skilled Worker	= 3 (1,280 – 1,120)	=	₹480 (F)
Semi-skilled Worker	= 2 (480 – 720)	=	₹480 (A)
Unskilled Worker	= 1 (240 – 160)	=	₹80 (F)
Labour Mix Variance			**₹80 (F)**

5.14. EXERCISE

5.14.1. Very Short Answer Type Questions

1) Define the term 'budget'.
2) What do you mean by budgetary control?
3) Define cash budget.
4) Name the sources of cash receipts.
5) Name the sources of cash payments.
6) What is flexible budget?
7) What is sales budget?
8) What do you understand by standard costs?
9) Define the term 'variance'.
10) Give the forms of variances.
11) How calculate material price variance?

5.14.2. Short Answer Type Questions

1) What are the objectives of budgetary control?
2) Discuss the advantages of material budget.
3) Discuss the features of flexible budget.
4) Write a note on concept of standard costs?
5) What are the objectives of standard costing?
6) What are the uses of variance analysis?
7) What is material price variance? How it is calculated?

5.14.3. Long Answer Type Questions

1) Discuss the concept of budget and its characteristics.
2) Explain the advantages and limitations of budgetary control.
3) Discuss the classification of functional budget in detail.
4) Define production budget. What are the factors considered in preparing a production budget?
5) What are the characteristics of the cash budget? Also explain the format of the cash budget.
6) What are the advantages and disadvantages of the flexible budget?
7) What do you mean by standard costing? Discuss the difference between standard costing and budgetary control.
8) Discuss the importance and limitations of standard costing.
9) What do you understand by Variance Analysis? Explain the classification of variance.
10) Explain the term 'material cost variance'. Discuss the types of material cost variance.
11) What are the different types of the labour cost variances? Also explain their causes?

5.14.4. Practical Questions

Cash Budget

1) From the following forecast of income and expenditure, prepare a cash budget for the months January to April, 2019.

Months	Sales (Credit) (₹)	Purchases (Credit) (₹)	Wages (₹)	Manufacturing Expenses (₹)	Administrative Expenses (₹)	Selling Expenses (₹)
2018 Nov.	30,000	15,000	3,000	1,150	1,060	500
" Dec.	35,000	20,000	3,200	1,225	1,040	550
2019 Jan.	25,000	15,000	2,500	990	1,100	600
" Feb.	30,000	20,000	3,000	1,050	1,150	620
" March	35,000	22,500	2,400	1,100	1,220	570
" April	40,000	25,000	2,600	1,200	1,180	710

Additional Information:

i) The customers are allowed a credit period of 2 months.
ii) A dividend of ₹10,000 is payable in April.

iii) Capital expenditure to be incurred: Plant purchased on 15th January for ₹5,000; a Building has been purchased on 1st March and the payments are to be made in monthly installments of ₹2,000 each.

iv) The creditors are allowing a credit of 2 months.

v) Wages are paid on the 1st of the next month.

vi) Lag in payment of other expenses is one month.

vii) Balance of cash in hand on 1st January 2019 is ₹15,000.

[Ans: Closing Balance: January = ₹18,985, February = ₹28,795, March = ₹30,975, April = ₹23,685]

2) A newly formed Company desires to prepare a Cash Budget for 4 months from March to June 2018. The following information is available.

Months	Sales (₹)	Purchases (₹)	Wages (₹)	Overheads (₹)
Jan	20,000	20,000	4,000	4,000
Feb	22,000	14,000	4,400	4,200
March	23,000	14,000	4,600	4,300
April	36,000	22,000	4,600	4,500
May	30,000	20,000	4,000	4,100
June	40,000	25,000	5,000	4,800

Additional Information:

i) Cash balance on 1st Jan. 2018 ₹10,000.

ii) New machinery is to be installed at a cost at ₹20,000 in the month of February which is to be paid in two equal instalments, March and April.

iii) ₹12,000 is to be collected in March as 2nd call money.

iv) Period of credit allowed by creditors is 2 months and allowed to customers 1 month.

v) 50% of sales and purchases are for cash.

vi) Delay in payment of wages ½ month overheads are paid in the same month.

[Ans: Closing Balance: March = ₹10,800, April = ₹5,700, May = ₹13,300, June = ₹15,500]

Flexible Budget

3) A factory which expects to operate 7,000 hours, i.e., at 70% level of activity, furnishes details of expenses as under:

Particulars	₹
Variable expenses	1,260
Semi-variable expenses	1,200
Fixed Expenses	1,800

The semi-variable expenses go up by 10% between 85% and 95% activity and by 20% above 95% activity. Construct a flexible budget for 80, 90 and 100 per cent activities.

[Ans: Recovery rate per hour: On 70% Capacity = 0.61; On 80% Capacity = 0.55; On 90% capacity = 0.53 and on 100% capacity = 0.50]

4) Draft a flexible budget for overhead expenses on the basis of following information and determine the overhead rates at 70%, 80% and 90% plant capacity.

Particulars	70% Capacity (₹)	80% Capacity (₹)	90% Capacity (₹)
Variable Overheads:			
Indirect Labour	–	24,000	–
Stores including spares	–	8,000	–
Semi-Variable Overheads:			
Power (30% fixed and 70% variable)	–	40,000	–
Repairs and maintenance (80% fixed and 20% variable)	–	4,000	–
Fixed Overheads :			
Depreciation	–	22,000	–
Insurance	–	6,000	–
Salaries	–	20,000	–
Total Overheads	–	**1,24,000**	–

[Ans: Total overheads: At 70% capacity = ₹1,15,900; At 80% capacity = ₹1,24,000; At 90% capacity = ₹1,32,100]

5) Good luck Ltd. is currently operating at 75% of its capacity. In the past two years, the levels of operations were 55% and 65% respectively. Presently, the production is 75,000 units. The company is planning for 85% capacity level during 2018-2019. The cost details are as follows:

Particulars	55% (₹)	65% (₹)	75% (₹)
Direct Materials	11,00,000	13,00,000	15,00,000
Direct Labour	5,50,000	6,50,000	7,50,000
Factory Overheads	3,10,000	3,30,000	3,50,000
Selling Overheads	3,20,000	3,60,000	4,00,000
Administrative Overheads	1,60,000	1,60,000	1,60,000
	24,40,000	**28,00,000**	**31,60,000**

Profit is estimated @ 20% on sales.
The following increases in costs are expected during the year:

Particulars	In Percentage
Direct Materials	8
Direct Labour	5
Variable Factory Overheads	5
Variable Selling Overheads	8
Fixed Factory Overheads	10
Fixed Selling Overheads	15
Administrative Overheads	10

Prepare flexible budget for the period 2013-2014 at 85% level of capacity. Also ascertain profit and contribution.

[Ans: 85% level of capacity: Profit = ₹9,46,300; Contribution = ₹14,57,300]

Material Budget

6) Draw a Material Procurement Budget (quantitative) from the following information:

Estimated sales of product 40,000 units.

Each unit of the product requires 3 units of material A and 5 units of material B. Estimated opening balance at the commencement of the next year:

Particulars	Units
Finished product	5,000
Material A	12,000
Material B	20,000
Materials on order:	
Material A	7,000
Material B	11,000

The desirable closing balances at the end of the next year:

Particulars	Units
Finished Product	7,000
Material A	15,000
Material B	25,000
Materials on order:	
Material A	8,000
Material B	10,000

[Ans: Production = 42,000 finished units; Purchases: A = 1,30,000 units, B = 2,14,000 units]

Sales Budget

7) A company sells two products A and B, which are manufactured in one plant. During the year 2018, it plans to sell the following quantities of each product:

Sales Budget Units

	I Quarter	II Quarter	III Quarter	IV Quarter
A	25,000	75,000	1,25,000	45,000
B	20,000	40,000	45,000	42,000

The company plans to sell product A throughout the year at a price of ₹10 per unit and product B at a price of ₹16 per unit. A study of the past experience reveals that the company has lost 3% of its billed revenue each year because of return (constituting 2% of loss of revenue) allowances and bad debts (1% loss).

You are required to prepare a sales budget incorporating the above information.

[Ans:

	I Quarter (₹)	II Quarter (₹)	III Quarter (₹)	IV Quarter (₹)
A	2,42,500	7,27,500	12,12,500	4,36,500
B	3,10,400	6,20,800	6,98,400	6,59,600
Total Sales	**5,52,900**	**13,48,300**	**19,10,900**	**10,96,100**

Production Budget

8) X Company Ltd. manufactures two products A and B. Forecast of the number of units to be sold in the first seven months of the year is given below:

Month	Product A	Product B
January	1,000	2,800
February	1,200	2,800
March	1,600	2,400
April	2,000	2,000
May	2,400	1,600
June	2,400	1,600
July	2,000	1,800

It is anticipated that:

i) There will be no work-in-progress at the end of every month.

ii) Finished units equal to half of the sales for the next month will be in stock at the end of each month (including previous December).

Budgeted production and costs for the whole year are as follows:

Particulars	Product A	Product B
Products (units)	22,000	24,000
Direct Materials Cost per Unit	12.50	19
Direct Labour Cost per Unit	4.50	7
Total Factory Overhead Apportioned	66,000	96,000

Prepare for the six months period ending 30th June, 2018:

i) Production budget for each month; and

ii) Summarised production cost budget.

[Ans: Production of Product A = 11,100 units and that of B = 12,700 units, Total Cost of A = ₹2,20,000 and B = 3,81,000]

Material Variance

9) The standard quantity and standard price of raw material required for one unit of product S are given below:

Material	Quantity (Unit)	Price (₹)
A	3	6
B	5	3

The actual production and relevant data are as under:

Output 1,600 units of product S.

Material	Quantity (Total for 800 Units of Output)	Total Cost (₹)
A	10,000	1,30,000
B	14,000	84,000

Calculate material cost, price and usage, and mix variance.

[Ans: Material Cost Variance = ₹1,61,200 (A); Material Price Variance = ₹1,12,000 (A); Material Usage Variance = ₹37,200 (A); Material Mix Variance = ₹78,0750 (F)]

10) The standard mix of a product P is shown below:

Raw Material	Units @ ₹
X	15 @ 10 each
Y	35 @ 15 each

Standard Loss is 10% of input.

Actual data:

Raw Material	Units @ ₹
X	17 @ 9 each
Y	33 @18 each

Actual Loss is 15% of input. Calculate material variances.

[Ans: Material Cost Variance = ₹156.5 (A); Material Price Variance = ₹65 (A), Material Quantity Variance = ₹74.5 (A), Material Mix Variance = ₹685 (F), Material Yield Variance = ₹759.5 (A)]

11) Kuber Ltd. produces an article by blending two basic raw materials. It operates a standard-costing system and the following standards have been set for the raw materials:

Material	Standard Mix	Price per kg
X	40%	₹4.00
Y	60%	₹3.00

The standard loss in processing is 15%.

During January 2019, the company produced 1,700kgs of finished product (output). The position of stock and purchases for the month of January 2019 is as follows:

Material	Stock on 1.1.19 (Kg)	Stock on 31.01.19 (Kg)	Purchased during January 2019	
			Kg	₹
X	35	5	800	3,400
Y	40	50	1,200	6,000

You are require to calculate the following:

i) Material-price variance,
ii) Material-usage variance,
iii) Material-yield variance,
iv) Material-mix variance, and
v) Total material-cost variance.

[Ans: Material Price Variance = ₹376 (F), Material Usage Variance = ₹90 (A), Material Yield Variance = ₹68 (A), Material Mix Variance = ₹22 (A); Total Material Cost Variance = ₹286 (F)]

12) From the following data for May, 2019, calculate:

i) Material Cost Variance
ii) Material Price Variance
iii) Material Usage Variance
iv) Material Mix Variance

Material	Standard		Actual	
	Qty. (Kgs.)	Rate (₹)	Qty. (Kgs.)	Rate (₹)
X	4,000	1.05	3,750	1.20
Y	1,500	2.15	1,650	2.30
Z	1,000	3.30	1,200	3.50
	6,500		**6,600**	

[Ans: Material Cost Variance: X = ₹300 (A), Y = ₹570 (A), Z = ₹900 (A); Material Price Variance: X = ₹525 (A), Y = ₹247.5(A), Z = ₹240 (A); Material Usage Variance: X = ₹262.5 (F), Y = ₹322.5 (A), Z = ₹660 (A); Material Mix Variance X = ₹247.5 (A), Y = ₹273 (A), Z = ₹396 (F)]

Labour Variance

13) Find out the labour rate variance, efficiency variance and labour cost variance using the following data:
Standard – 72 hours @ ₹5 per hour
Actual – 70 hours @ ₹5.50 per hour

[Ans: Labour Rate Variance = ₹35(A); Labour Efficiency Variance = ₹10(F); Labour Cost Variance = ₹25(A)]

14) The standard and actual figures of X metal works are as under:

Standard time for the job	500 hours
Standard rate per hour	0.25 p.
Actual time taken	450 hours
Actual wages paid	₹180

Calculate:
i) Labour rate variance
ii) Labour efficiency variance
iii) Labour cost variance

[Ans: Labour Rate Variance = ₹67.5(A); Labour Efficiency Variance = ₹12.5(F); Labour Cost Variance = ₹55(A)]

15) A product requires 20 hours per unit to produce at a standard rate of ₹4 per hour. 2,000 units were produced during March, 2019. Actual time taken to produce was 18,000 hours. The actual rate was ₹5 per hour.

Calculate:
i) Labour Cost Variance
ii) Labour Rate Variance
iii) Labour Efficiency Variance

[Ans: Labour Cost Variance = ₹70,000(F); Labour Rate Variance = ₹18,000(A); Labour Efficiency Variance = ₹88,000(F)]

16) Standard labour hours required to produce one unit of product Z are 10 and standard wage rate is ₹10 per labour hour. During the month ended 30th April, 2016, 200 workers were engaged for 50 days for 16 hours daily. Due to power failure all the workers remained idle for 6 hours on 17th April, 2016. Actual output for the month of April, 2016 was 8,000 units. The actual wage rate was ₹11 per labour hours. Compute labour variances.

[Ans: Labour Cost Variance = ₹9,60,000 (A); Labour Rate Variance = ₹1,60,000 (A); Labour Efficiency Variance = ₹7,80,000 (A), Labour Idle Time Variance = ₹12,000 (A)]

SAVITRIBAI PHULE PUNE UNIVERSITY

MBA - FIRST SEMESTER EXAMINATION, 2014

ACCOUNTING FOR BUSINESS DECISIONS

Time: 2½ Hours **Max. Marks: 50**

Note: 1) All questions are compulsory.
2) Each question has an internal option.
3) Each question carries 10 marks.
4) Figures to the right indicate marks for question/sub-question.
5) Answers to the questions should be specific and to the point.
6) Use of calculator is permitted (as applicable).

Ques 1 a) "Management Accounting provides immense help in management decision-making." Elaborate. (10)

Ans: Meaning of Management Accounting
Refer Unit-1, Page No. 23

Importance of Management accounting in management decision making
Refer Unit-1, Page No. 23

Or

Ques 1 b) Explain the basic accounting concepts and conventions in financial accounting. (10)

Ans: Basic Accounting Concepts in Financial Accounting
Refer Unit-1, Page No. 25

Basic Accounting Conventions in Financial Accounting
Refer Unit-1, Page No. 27

Ques 2 a) How would you analyse the financial position of a company from the point of view of: (10)
i) Investor,
ii) Creditor; and
iii) Financial executive of the company.

Ans: Out of Syllabus

Or

Ques 2 b) Briefly explain the guidelines of Schedule VI of the Companies Act, 1956 for the following items with regard to the general instructions for preparation of Balance Sheet. (10)
i) Delivery Vehicle,
ii) Accounts Payable,
iii) Prepaid Insurance,
iv) Income Tax Payable and
v) Trademark.

Ans: Out of Syllabus

Ques 3 a) "Cost Accounting System is neither unnecessary nor expensive, rather it is profitable investment." Comment. (10)

Ans: "Cost Accounting System is neither unnecessary nor expensive, rather it is profitable investment." This statement is more clear with the help of following important advantages of a good cost accounting system.
For Detail Refer Unit-3, Page No. 94

Or

Ques 3 b) i) "A good system of costing serve as a means of control over expenditure and helps to secure economy in manufacture." Discuss. (05)

Ans: "A good system of costing serve as a means of control over expenditure and helps to secure economy in manufacture." It is an integral part of a good management. While introducing a costing system different steps are involved. For Detail Refer Unit-3, Page No. 94

Ques 3 b) ii) Classify the following items into Factory Overheads, Administration Overheads and Selling and Distribution Overheads. (05)

Loose tools	**Income Tax**	**Postage and Telegram**
Rent, Rate and Taxes	**Land and Building**	**Delivery Charges**
Indirect Material	**Packing Expenses**	**Director's Fees**
Warehouse Rent	**Welfare Expenses**	**Overtime Wages**
Advertising	**General Expenses**	**Collection Charges**
Property Tax on Building	**Carriage Outward**	**Dividend Paid**
Labour Expenses	**Bad Debts**	**Unproductive Wages**

Ans: Refer Unit-3, Page No. 100

Ques 4 a) B Co. Ltd. uses a periodic inventory system. It has maintained the following records of purchases and sales of an item: (10)

Jan. 1	**Opening stock of inventory 120 units @ ₹18.00**
Mar. 3	**Purchases 30 units @ ₹18.50**
May 22	**Sold 80 units**
June 1	**Purchases 60 units @ ₹19.20**
Aug. 8	**Sold 80 units**
Oct. 18	**Purchases 40 units @ ₹21.00**
Oct. 24	**Spoilage 20 units**

You are required to compute the cost of ending inventory and cost of goods sold using:

i) **Last-in first-out method, and**

ii) **Weighted Average method.**

Ans: Out of Syllabus

Or

Ques 4 b) The Modern Company is having four departments A, B and C are producing departments and D is servicing department. (10)

The actual costs for a period are as follows:

	₹
Rent	2,000
Repairs	1,200
Depreciation	900
Supervision charges	3,000
Lighting	200
Insurance of stock	1,000
Employees liability insurance	300
Power	1,800

The following data is also available in respect of four departments:

Particulars	A	B	C	D
Area Sq. Ft.	150	110	90	50
No. of Light Points	15	11	9	3
No. of Workers	24	16	12	8
Total Wages (₹)	8,000	6,000	4,000	2,000
Value of Plant (₹)	24,000	18,000	12,000	6,000
Value of Stock (₹)	15,000	9,000	6,000	–
H.P. of Machine	12	9	6	3

Apportion the costs of the various departments on the most equitable basis.

Ans: Out of Syllabus

Ques 5 a) Himmatlal Co. Ltd., Pune provides the following information for the year 2014-15: (10)
Sales – 15,000 units
Fixed Cost – ₹34,000
Sales Value – ₹1,50,000
Variable Cost – ₹6 per unit

You are required to calculate:

i) **P/V Ratio, BEP in ₹ and Margin of Safety.**

ii) **Also calculate and comment on the effect of the following:**
   1) **Decrease of 10% in selling price.**
   2) **Increase of 10% in variable cost.**

Ans: Refer Unit-4, Page No. 136

Or

Ques 5 b) A newly started company wishes to prepare cash budget from January. Prepare a cash budget for the first six months from the following estimated revenue and expenses: (10)

Month	Total Sales (₹)	Purchases (₹)	Wages (₹)	Production Overheads (₹)	Selling and Distribution Overheads (₹)
January	20,000	20,000	4,000	3,200	800
February	22,000	14,000	4,400	3,300	900
March	28,000	14,000	4,600	3,400	900
April	36,000	22,000	4,600	3,500	1,000
May	30,000	20,000	4,000	3,200	900
June	40,000	25,000	5,000	3,600	1,200

Cash balance on 1st January was ₹10,000. A new machinery is to be installed at ₹20,000 on credit, to be repaid by two equal instalments in March and April.

Sales commission @ 5% on total sales is to be paid within a month following actual sales.

₹10,000 being the amount of 2nd call may be received in March. Share premium amounting to ₹2,000 is also obtainable with the 2nd call.

Period of credit allowed by suppliers – 2 months.

Period of credit allowed by customers – 1 months.
Delay in payment of wages ½ month and overheads 1 month.
Assume cash sales to 50% of total sales.

Ans: Refer Unit-5, Page No. 176

SAVITRIBAI PHULE PUNE UNIVERSITY

MBA - FIRST SEMESTER EXAMINATION, 2015

ACCOUNTING FOR BUSINESS DECISIONS

Time: 2½ Hours **Max. Marks: 50**

Note: 1) All questions are compulsory.
2) Each question carries 10 marks.

Ques 1 a) "Management accounting is an important tool for planning, control and decision-making." Explain this statement. (10)

Ans: Importance of Management accounting in planning, control and decision-making
Refer Unit-1, Page No. 23

Or

Ques 1 b) Write short notes on (any Two): (10)
i) Going Concern Concept

Ans: Refer Unit-1, Page No. 25

ii) Business Entity Concept

Ans: Refer Unit-1, Page No. 25

iii) Convention of Consistency

Ans: Refer Unit-1, Page No. 27

Ques 2 a) Explain the importance and objectives of financial statements. (10)
Ans: Importance of Financial Statements
Refer Unit-2, Page No. 53

Objectives of Financial Statement
Refer Unit-2, Page No. 53

Or

Ques 2 b) Give the disclosure requirements of following items of Balance Sheet as per schedule VI of companies Act: (10)
i) Fixed Assets
ii) Reserves and Surplus

Ans: Out of Syllabus

Ques 3 a) Define Cost Accounting? Also explain the various types of costs? (10)

Ans: Meaning and Definition of Cost Accounting
Refer Unit-3, Page No. 91

Various Types/Classification of Costs
Refer Unit-3, Page No. 101

Or

Ques 3 b) Prepare Cost Sheet from the following information:

Particulars		₹
Raw materials consumed		40,000
Wages paid to labourers		10,000
Directly chargeable expenses		2,000
Oil and Waste		100
Wages of Foreman		1,000
Storekeepers Wages		500
Electric Power		200
Lighting:		
Factory	500	
Office	200	700
Rent:		
Factory	2,000	
Office	1,000	3,000
Repairs and Renewals:		
Factory Plant	500	
Machinery	1,000	
Office Premises	200	1,700
Depreciation:		
Office Premises	500	
Plant and Machinery	200	700
Consumable stores		1,000
Manager's salary		2,000
Director's fees		500
Office printing and Stationery		200
Telephone charges		50
Postage and Telegrams		100
Salesmen's Commission and Salary		500
Travelling Expenses		200
Advertising		500
Warehouse charges		200
Carriage outward		150

Ans: Refer Unit-3, Page No. 110

Ques 4 a) The following transactions took place in respect of a material item:

Data	Receipt Quantity	Rate (₹)	Issue Quantity
March 2	200	20	
March 3	300	24	
March 15			250
March 18	250	26	
March 20			200

Prepare a stores Ledger A/C, using:
i) Simple average method
ii) Weighted average method

Ans: Out of Syllabus

Or

Ques 4 b) X Ltd. has two production departments and two service departments and provides you the following data: (10)

Particulars	Production Dept.		Service Dept.	
	P_1	P_2	S_1	S_2
Direct Materials	40,000	30,000	20,000	10,000
Direct Wages	15,000	20,000	5,000	10,000
Floor Area (sq. Feet)	5,000	4,000	3,000	2,000
Value of Plant and Machinery	50,000	60,000	20,000	10,000
Value of Stock	35,000	25,000	5,000	5,000
No. of Workers	10	50	25	25
No. of Light Points	200	50	25	25
Horse Power of Machines	50	25	15	10

The indirect expenses for the period were:

Items	₹
Factory Rent, Rates, Taxes and Repairs	14,000
Depreciation, Insurance and Repairs of Machinery	28,000
Insurance of stock	7,00
Supervision and staff welfare expenses	2,000
Stores Overheads	1,000
Lighting and Heating	3,000
Power	1,000

Prepare the statement showing the apportionment of overheads.

Ans: Out of Syllabus

Ques 5 a) The sales turnover and profit during two years were as follows:

Year	Sales (₹)	Profit (₹)
2011	1,40,000	15,000
2012	1,60,000	20,000

You are required to calculate:
i) P/V Ratio
ii) Sales required to earn a profit of ₹40,000
iii) Profit when sales are ₹1,20,000.

Ans: Refer Unit-4, Page No. 132

Or

Ques 5 b) The expenses for the production of 5,000 units in a factory are given as follows:

	Per Unit (₹)
Materials	50
Labour	20
Variable Overheads	15
Fixed Overheads (₹50,000)	10
Administrative Expenses (5% Variable)	10
Selling Expenses (20% Fixed)	6
Distribution Expenses (10% Fixed)	5
Total Cost of Sales per unit	116

You are required to prepare a budget for the production of 7,000 units.

Ans: Refer Unit-5, Page No. 181

SAVITRIBAI PHULE PUNE UNIVERSITY

MBA - FIRST SEMESTER EXAMINATION - 2016

ACCOUNTING FOR BUSINESS DECISIONS

Time: $2^1/_4$ Hours **Max. Marks: 50**

Note: 1) All questions are compulsory.
2) Each question has an internal option.
3) Each question carries 10 marks.
4) Figures to the right indicates marks for question/sub-question.
5) Use of simple calculator is permitted (as applicable).

Ques 1) "Financial Accounting and Management Accounting are not same." Explain. (10)

Ans: Difference between Financial Accounting and Management Accounting
Refer Unit 1, Page No. 24

Or

Ques 1) Write in detail accounting conventions with suitable example.

Ans: Accounting Conventions with Suitable Example
Refer Unit 1, Page No. 27

Ques 2) Following is the Trial Balance of ABC as on 31/03/2015. Prepare Trading Account, Profit and Loss Account for the year ended 31/03/2015 and prepare Balance Sheet as on that date. (10)

Debit Balance	₹	Credit Balance	₹
Opening stock	25,000	Creditors	25,000
Plant	1,64,000	Bills payable	25,000
Rent	20,000	Capital	3,25,000
Salaries	30,000	Sales	4,00,000
Wages	20,000		
Carnage Inward	2,000		
Carriage Outward	3,000		
Factory Rent	5,000		
Purchases	2,25,000		
Insurance	1,000		
Furniture	80,000		
Debtors	2,00,000		
	7,75,000		7,75,000

Additional Information:
a) **Closing stock valued ₹25,000 as per market value and as per book value ₹20,000.**
b) **Depreciation on furniture ₹8,000.**

Ans: Refer Unit 2, Page No. 76

Or

Ques 2) Following is the Trial Balance and DEF as on 31/03/2010. Prepare Trading Account, Profit and Loss Account for the year ended 31/03/2010 and Balance Sheet as on that date.

Debit Balance	₹	Credit Balance	₹
Opening Stock	15,000	Bank Loan	50,000
Machinery	2,00,000	Capital	2,75,000
Insurance	1,000	Creditors	35,000
Printing and Stationery	5,000	Sales	1,86,500
Telephone Charges	500	Returns Outward	1,500
Wages	25,000		
Advertising	15,000		
Purchases	1,00,000		
Returns Inward	1,000		
Motor Van	30,000		
Debtors	1,50,000		
Motive and Power	5,500		
	5,48,000		5,48,000

Additional Information:
a) **Closing stock valued at ₹5,000.**
b) **Wages outstanding ₹5,000 and prepaid insurance ₹500.**

Ans: Refer Unit 2, Page No. 77

Ques 3 a) Explain sunk costs with suitable example. (05)

Ans: Sunk Costs with Suitable Example
Refer Unit 3, Page No. 104

Ques 3 b) From the following information you are required to calculate prime cost, works cost, total cost of production, cost of sales and profit. (05)

Particulars	₹	Particulars	₹
Direct Labour	1,00,000	Direct Expenses	1,50,000
Factory Overheads	2,00,000	Office Overheads	1,80,000
Direct Material	5,00,000	Sales	15,00,000
Selling and Distribution Overheads	50,000		

Ans: Ans: Refer Unit 3, Page No. 107

Or

Ques 3 a) State in brief objectives of cost accounting.

Ans: Objectives of Cost Accounting
Refer Unit 3, Page No. 91

Ques 3 b) Prepare a cost sheet showing prime cost, works cost, total cost of production, cost of sales and profit from the following information related to Chennai Pvt. Ltd., Chennai.

Particulars	₹	Particulars	₹
Sales	5,00,000	Direct Wages	50,000
Direct Expenses	50,000	Factory Rent	20,000
Storekeeper Salary	10,000	Salary	1,00,000
Direct Material	2,00,000	Director's Fees	30,000
Showroom Rent	7,000		

Ans: Refer Unit 3, Page No. 108

Ques 4) ABC Ltd. furnishes following data for the month of January, 2013. Prepare stores ledger account, pricing the material issue on the basis of LIFO method and ascertain the value of closing stock. (10)

Date	Particulars
1st	Opening stock – 200 units @ ₹350 per unit
9th	Purchases – 350 units @ ₹ 300 per unit
12th	Issues – 400 units
19th	Purchases – 600 units @ ₹320 per unit
20th	Issues – 250 units
22th	Purchases – 400 units @ ₹340 per unit
24th	Issues – 400 units
25th	Purchases – 100 units @ ₹300 per unit
28th	Spoilage – 20 units

Ans: Out of Syllabus

Or

Ques 4) Aim Ltd. is divided into four departments P1, P2, P3 are producing departments and S1 is a service department.

The actual costs for a period are as follows:

Particulars	₹	Particulars	₹
Rent	2,000	Supervision	3,000
Repairs to Plant	1,200	Fire Insurance Stock	1,000
Depreciation of Plant	900	Power	1,800
Insurance of Employees	300	Light	240
Wages	15,000	Materials	10,000

Following information is available in respect of the four departments:

Particulars	P1	P2	P3	SI
Area (Sq. Meters)	1,500	1,100	900	500
No. of Employees	20	15	10	5
Total Wages (₹)	6,000	4,000	3,000	2,000
Value of Plant (₹)	24,000	18,000	12,000	6,000
Value of Stock (₹)	15,000	9,000	6,000	–
H.P. Plant	24	18	12	6
Total Materials (₹)	3,000	3,000	4,000	–

Apportion the costs to the various departments on the most equitable basis.

Ans: Out of Syllabus

Ques 5) KLM Industries furnishes following information for the level of output of 1,000 units for the year 2010. (10)

Selling Price per Unit	=	₹100
Variable Cost per Unit	=	₹40
Total Fixed Cost	=	₹30,000

Find:

a) **P/V Ratio**
b) **BEP in units**
c) **BEP in sales**
d) **The amount of sales required to earn profit of ₹42,000 (assuming fixed cost in total remains same).**

Ans: Refer Unit 4, Page No. 133

Or

Ques 5) From the following prepare cash budget for 3 months for January. February and March, 2012.

Month	Total Sales (₹)	Materials (₹)	Wages (₹)	Overheads	
				Production (₹)	Selling and Distt. (₹)
January	40,000	42,000	8,000	6,400	1,600
February	44,000	28,000	8,800	6,600	1,800
March	48,000	28,000	9,200	6,600	1,600
April	52,000	24,000	9,200	6,800	1,800
May	56,000	24,000	9,600	7,000	1,800
June	60,000	32,000	9,600	7,200	2,000

1) **Cash balance on January 1, 2012 was ₹20,000.**
2) **A new machine is to be installed at ₹60,000 on credit, to be repaid by two equal instalments in March and April, 2012.**
3) **Period of credit allowed by suppliers 1 month.**
4) **Period of credit allowed to customers 1 month.**
5) **Delay in payment of overheads 1 month.**
6) **Delay in payment of wages 1 month.**
7) **Assume cash sales to be 50% of total sales.**

Ans: Refer Unit 5, Page No. 175

SAVITRIBAI PHULE PUNE UNIVERSITY

MBA - FIRST SEMESTER EXAMINATION - 2017

ACCOUNTING FOR BUSINESS DECISIONS

Time: 2¼ Hours **Max. Mark: 50**

Note: 1) All questions are compulsory.
2) Each question has an internal option.
3) Each question carries 10 marks.
4) Figures to the right indicate marks for questions/sub questions.
5) Use of simple calculator is permitted.

Ques 1) Is there any similarity between Management Accounting and Financial Accounting? (10)

Ans: Refer Unit 1, Page No. 23

Or

Explain any five accounting concepts with suitable example. (10)

Ans: Accounting Concepts with Suitable Example
Refer Unit-1, Page No. 25

Ques 2) The following is the trial balance of Mr. Shah. Prepare the final account as on 31st March 2017. (10)

Particulars	Dr. (₹)	Cr. (₹)
Capital		2,00,000
Land and Building	87,000	
Plant and Machinery	17,500	
Goodwill	20,000	
Drawings	22,600	
Cash in Hand	1,795	
Opening Stock	27,000	
Wages	10,000	
Purchases	69,000	
Carriage Inward	600	
Travellers Commission	6,000	
Insurance Premium	2,000	
Motor Car	3,000	
Carriage Outward	1,400	
Sales		94,000
Salaries	15,000	
Bank Charges	105	
RDD		1,500
Debtors	20,000	
Creditors		7,500
	3,03,000	3,03,000

Adjustments:
a) **On 31st March 2017 the stock was valued at ₹46,000.**
b) **Depreciation on Plant and Machinery 10% and Motor car 20%.**
c) **Create a Reserve for Doubtful Debts at 10% on Debtors.**
d) **Insurance premium amounting to ₹800 is prepaid.**
e) **Outstanding salaries amount to ₹1,000.**

Ans: Refer Unit 2, Page No. 82

Or

The following balances were extracted from the books of Mr. Ashok as on 31st March 2017. Prepare Trading and Profit and Loss Account for the year ended 31st March 2017 and the Balance Sheets as on that date after taking into account the following adjustments: (10)

Particulars	Dr.(₹)	Cr. (₹)
Capital		3,30,000
Drawings	24,450	
Opening Stock	2,00,000	
Bills Receivable	25,000	
Purchases	2,75,000	
Sales		4,20,000
Bills Payable		60,000
Return Outwards		4,500
Return Inward	5,000	
Plant and Machinery	1,00,000	
Loose Tools	25,000	
Land and Building	25,000	
Sundry Debtors	1,25,000	
Sundry Creditors		1,40,000
Cash at Bank	77,550	
Salaries and Wages	50,000	
Repairs and Renewals	7,500	
Insurance	3,000	
Power and Fuel	3,500	
Printing and Stationery	2,000	
General Expenses	6,500	
	9,54,500	9,54,500

Adjustments:
a) **Closing stock was worth ₹1,30,000.**
b) **Depreciation Land and Building 15% and Plant and Machinery by 5%.**
c) **Provide for Bad and Doubtful Debts at 5% on Debtors.**
d) **Insurance paid in advance to the extent of ₹750.**
e) **Revalue total at ₹20,000.**
f) **Provide for outstanding expenses as follows:**
g) **Salaries ₹2,500 and Wages ₹1,000.**

Ans: Refer Unit 2, Page No. 83

Ques 3 a) Explain in brief classification of cost as per elements. (05)

Ans: Classification of Cost as per Elements
Refer Unit-3, Page No. 101

Ques 3 b) From the following prepare cost sheet. **(05)**

Particulars	₹
Sales	5,00,000
Direct Wages	1,10,000
Factory Rent	30,000
Direct Material	2,90,000
Showroom Rent	15,000
Direct Expenses	10,000
Printing and Stationery	5,000
Audit Fees	15,000

Ans: Refer Unit-3, Page No. 108

Or

Ques 3 a) Write a short note on Relevant and Irrelevant Cost. **(05)**

Ans: Relevant and Irrelevant Cost
Refer Unit-3, Page No. 104

Ques 3 b) From the following information you are required to calculate prime cost, works cost, total cost of production, cost of sales and profit. **(05)**

Particulars	₹
Sales	1,00,000
Direct Material	40,000
Direct Expenses	2,000
Factory Overheads	7,000
Direct Wages	10,000
Selling and Distribution Overheads	2,000
Administrative Overheads	5,000

Ans: Refer Unit-3, Page No. 109

Ques 4) Modern Company Ltd. furnishes following data for the month of July 2017.
Prepare store ledger account, pricing the material issue on the basis of LIFO Method and ascertain the value of closing stock. **(10)**

Date	Particulars
1	Opening stock – 100 units at ₹1.80
5	Purchases – 150 units at ₹1.50
8	Issues – 200 units
12	Purchases – 300 units at ₹1.60
18	Issues – 250 units
22	Purchases – 400 units at ₹1.70
29	Issues – 400 units
30	Spoilage – 50 units

Ans: Out of Syllabus

Or

The Ultra-Modern Company Ltd. is divided into four departments A, B and C are production departments and D is service department. You are required to allocate the following common overheads expenses on the basis of information given below. **(10)**

Particulars	₹
Power	10,700
Rent and Rates	32,000
Lighting	5,000
Insurance of Factory Building	3,600
Insurance of Machinery	7,200
Depreciation of Machinery	36,000
Supervision charges	25,000

Other Information:

Particulars	Production Dept.			Service Dept.
	A	B	C	D
Horse Power Hours Ratio	25	35	30	10
Space Occupied sq. ft	800	600	400	200
Value of Machinery (₹)	40,000	60,000	30,000	70,000
Lighting Points	50	30	20	25
Time Devoted for Supervision	20%	30%	25%	25%

Ans: Out of Syllabus

Ques 5) The sales turnover and profits during two years were as follows: **(10)**

Year	Sales	Profit
2016	45,00,000	5,00,000
2017	50,00,000	7,00,000

You are required to calculate:
a) P/V Ratio.
b) Fixed cost.
c) Break Even Point (₹)
d) Sales to Earn Profit of ₹10,00,000.
e) Margin of Safety.

Ans: Refer Unit-4, Page No. 136

Or

From the following information, prepare a flexible budget at 60% and 80% capacity. **(10)**

Particulars	70% Capacity (₹)
Variable Overheads:	
Indirect Material	5,000
Indirect Labour	15,000
Semi-Variable Overheads:	
Electricity (60% Variable)	50,000
Repairs and Maintenance (35% Fixed)	5,000
Fixed Overheads:	
Salaries	10,000
Depreciation on Machine	14,000
Insurance	6,000
	1,05,000

Ans: Refer Unit-5, Page No. 183

SAVITRIBAI PHULE PUNE UNIVERSITY

MBA - FIRST SEMESTER EXAMINATION - 2018

ACCOUNTING FOR BUSINESS DECISIONS

Time: 2¼ Hours **Max. Mark: 50**

Note: 1) All questions are compulsory.
2) Each question has an internal option.
3) Each question carries 10 marks.
4) Figures to the right indicate lull marks for question/sub-question.
5) Use of simple calculator is permitted.

Ques 1 a) Explain different forms of business organisation? (05)

Ans: Different Forms of Business Organisation
Refer Unit-1, Page No. 11

Ques 1 b) Explain the accounting convention. (05)

Ans: Accounting Convention
Refer Unit-1, Page No. 27

Or

Ques 1 a) Define accounting. Who are the users of accounting information? (05)

Ans: Meaning and Definition of Accounting
Refer Unit-1, Page No. 14

Users of Accounting Information
Refer Unit-1, Page No. 21

Ques 1 b) Explain the use of Tally in Accounting. (05)

Ans: Out of Syllabus

Ques 2) From the following trial balance of Rajan, prepare trading and profit and loss account for the year ended 31.3.2016 and the Balance Sheet on at that date after taking into account the adjustments given below: (10)

Trial Balance as on 31.3.2016

Dr. Particulars	₹	Particulars	₹ Cr.
Drawings	7,600	Capital	2,90,000
Purchases	89,000	Sales	1,50,000
Sales Return	2,800	Purchases Return	4,500
Stock (1.4.2015)	12,000	Interest	3,500
Wages	8,000	Sundry creditors	12,000
Building	2,20,000	Bills payable	7,000
Freight	20,000		
Trade Expenses	2,000		
Advertisement	2,400		
Taxes and Insurance	1,300		
Sundry Debtors	65,000		
Bills Receivables	15,000		
Cash at Bank	12,000		
Cash in Hand	1,900		
Salaries	8,000		
	4,67,000		4,67,000

Adjustments:
1) **Stock on 31.3.16 was valued at ₹15,000.**
2) **Insurance was prepaid of ₹400.**
3) **Salary outstanding ₹2000 and taxes outstanding ₹1300.**
4) **Depreciate Building at 2% P.A.**

Ans: Refer Unit-2, Page No. 80

Or

Ques 2) Following Trial Balance is prepared on 31.3.16 from Asha's book. Prepare Trading and Profit and Loss Account for the year ended 31.3.16 and the Balance Sheet as on that date after taking into account the adjustment given below. (10)

Trial Balance as on 31.3.2016

Dr. Particulars	₹	Particulars	₹ Cr.
Cash in hand	8,000	Sales	5,00,000
Cash at Bank	12,000	Creditors	40,000
Wages	20,000	Capital	3,20,000
Machinery	2,12,000	Commission	2,800
Bills Receivables	40,000		
Opening stock	84,000		
Purchases	3,30,000		
Sales Return	6,000		
Salaries	32,000		
Debtors	86,000		
Taxes and Insurance	22,000		
Bad debts	6,000		
Freight	4,800		
	8,62,800		8,62,800

Adjustments:
1) **Closing stock ₹90,000.**
2) **Outstanding salaries ₹3,200 and wages ₹2000.**
3) **Prepaid insurance ₹900 and accrued commission ₹1200.**
4) **Charge 10% depreciation on machinery.**

Ans: Refer Unit-2, Page No. 81

Ques 3 a) What do you mean by cost accounting? Explain objectives of cost accounting. (05)

Ans: Meaning and Definition of Cost Accounting
Refer Unit-3, Page No. 91

Objectives of Cost Accounting
Refer Unit-3, Page No. 91

Ques 3 b) Calculate material consumed, prime cost and works cost – Opening stock of raw material ₹40,000, closing stock of raw materials ₹50,000, purchase of raw material ₹4,00,000, Direct labour ₹3,00,000, Indirect labour ₹10,000, carriage outward ₹20,000, lubricant ₹10,000, Factory rent ₹60,000. (05)

Ans: Refer Unit-3, Page No. 106

Or

Ques 3 a) Write short note on classification of costs. (05)

Ans: Classification of Costs
Refer Unit-3, Page No. 101

Ques 3 b) Identify whether following expenses are factory/office and administration/selling and distribution overheads. (05)

i) Show room rent.
ii) Cleaning charges.
iii) Storekeeper's wages.
iv) Labour welfare expenses.
v) Legal fees.
vi) Travelling expenses.
vii) General manager's salary.
viii) Consumable stores.
ix) Sundry expenses
x) Counting house salary.

Ans: Refer Unit-3, Page No. 100

Ques 4) The following transactions took place in respect of material A for March 2013:

Date	Receipts Rate	Receipts Units	Issues Units
1/3/13	1	400	–
10/3/13	1.20	600	–
16/3/13	–	–	500
19/3/13	1.30	500	–
25/3/13	–	–	400

Prepare stores ledger accounts, pricing the issues at simple average method. (10)

Ans: Out of Syllabus

Or

Ques 4) M&Co. has 3 production departments and 2 service departments. The expenses are as given below: (10)

Consumable stores ₹15,400, Supervision ₹22,800, Rent ₹10,000, Insurance ₹2,000, Depreciation ₹30,000, Power ₹9,000, light and heat ₹4000.

The following information is available.

Bases	Production Departments			Service Departments	
	Machine	Assembly	Finishing	Stores	Repairs
Direct material	34%	39%	13%	4%	10%
Direct wages	35%	22%	27%	10%	7%
Area (sq.ft)	5,250	3,500	4,375	1,750	2,625
Asset value (₹)	2,00,000	2,25,000	50,000	12,500	12,500
H. P × hours × LF	10,800	7,200	–	–	–

Ques 4 a) Prepare the primary distribution statement using the most appropriate basis for apportionment.

Ques 4 b) The Machine shop, Assembly shop and Finishing Department have issued stores requisitions in the ratio of 9:6:5 and repairs requests in the ratio of 2:3:1 prepare the Secondary Distribution Statement on direct distribution basis.

Ans: Out of Syllabus

Ques 5) The turnover and profits during the two periods were as follows:

	Sales (₹)	Profits (₹)
Period-I	40 lac	4 lac
Period-II	60 lac	8 lac

Assuming that the cost structure and selling prices remain the same in the two periods.

Calculate:
Profit Volume Ratio, BEP sales, the sales required to earn profit of ₹10 lac, margin of safety in period II. (10)

Ans: Refer Unit-4, Page No. 139

Or

Ques 5) In a factory, 100 workers are engaged and the average rate of wages is 50 paise per hour Standard working hours per week are 40 and the standard performance is 10 units per gang hour.

During a week in March, wages paid for 50 workers were at the rate of 50 paise per hour 10 workers at 10 paise per hour and 40 workers at 40 paise per hour. Actual output was 380 units.

The factory did not work for five hours due to break down of machinery. Calculate Labour cost variance, Labour rate variance, Labour efficiency variance, Idle time variance, Labour yield variance. (10)

Ans: Refer Unit-5, Page No. 208

UNIT-WISE CLASSIFICATION OF PREVIOUS YEAR QUESTION PAPERS

UNIT I

First Semester Examination, 2014

1 a) "Management Accounting provides immense help in management decision-making." Elaborate.

1 b) Explain the basic accounting concepts and conventions in financial accounting.

First Semester Examination, 2015

1 a) "Management accounting is an important tool for planning, control and decision-making." Explain this statement.

1 b) Write short notes on (any Two):
i) Going Concern Concept
ii) Business Entity Concept
iii) Convention of Consistency

First Semester Examination, 2016

1) "Financial Accounting and Management Accounting are not same." Explain.

Or

1) Write in detail accounting conventions with suitable example.

First Semester Examination, 2017

1) Is there any similarity between Management Accounting and Financial Accounting?

Or

Explain any five accounting concepts with suitable example.

First Semester Examination, 2018

1 a) Explain different forms of business organisation?

1 b) Explain the accounting convention.

1 a) Define accounting. Who are the users of accounting information?

UNIT II

First Semester Examination, 2014

First Semester Examination, 2015

2 a) Explain the importance and objectives of financial statements.

First Semester Examination, 2016

2) Following is the Trial Balance of ABC as on 31/03/2015. Prepare Trading Account, Profit and Loss Account for the year ended 31/03/2015 and prepare Balance Sheet as on that date.

Debit Balance	₹	Credit Balance	₹
Opening stock	25,000	Creditors	25,000
Plant	1,64,000	Bills payable	25,000
Rent	20,000	Capital	3,25,000
Salaries	30,000	Sales	4,00,000
Wages	20,000		
Carnage Inward	2,000		
Carriage Outward	3,000		
Factory Rent	5,000		
Purchases	2,25,000		
Insurance	1,000		
Furniture	80,000		
Debtors	2,00,000		
	7,75,000		**7,75,000**

Additional Information:

a) Closing stock valued ₹25,000 as per market value and as per book value ₹20,000.

b) Depreciation on furniture ₹8,000.

Or

2) Following is the Trial Balance and DEF as on 31/03/2010. Prepare Trading Account, Profit and Loss Account for the year ended 31/03/2010 and Balance Sheet as on that date.

Debit Balance	₹	Credit Balance	₹
Opening Stock	15,000	Bank Loan	50,000
Machinery	2,00,000	Capital	2,75,000
Insurance	1,000	Creditors	35,000
Printing and Stationery	5,000	Sales	1,86,500
Telephone Charges	500	Returns Outward	1,500
Wages	25,000		
Advertising	15,000		
Purchases	1,00,000		
Returns Inward	1,000		
Motor Van	30,000		
Debtors	1,50,000		
Motive and Power	5,500		
	5,48,000		**5,48,000**

Additional Information:

a) Closing stock valued at ₹5,000.

b) Wages outstanding ₹5,000 and prepaid insurance ₹500.

First Semester Examination, 2017

2) The following is the trial balance of Mr. Shah. Prepare the final account as on 31st March 2017.

Particulars	Dr. (₹)	Cr. (₹)
Capital		2,00,000
Land and Building	87,000	
Plant and Machinery	17,500	
Goodwill	20,000	
Drawings	22,600	
Cash in Hand	1,795	
Opening Stock	27,000	
Wages	10,000	
Purchases	69,000	
Carriage Inward	600	
Travellers Commission	6,000	
Insurance Premium	2,000	

Motor Car	3,000	
Carriage Outward	1,400	
Sales		94,000
Salaries	15,000	
Bank Charges	105	
RDD		1,500
Debtors	20,000	
Creditors		7,500
	3,03,000	**3,03,000**

Adjustments:

a) On 31st March 2017 the stock was valued at ₹46,000.
b) Depreciation on Plant and Machinery 10% and Motor car 20%.
c) Create a Reserve for Doubtful Debts at 10% on Debtors.
d) Insurance premium amounting to ₹800 is prepaid.
e) Outstanding salaries amount to ₹1,000.

Or

The following balances were extracted from the books of Mr. Ashok as on 31st March 2017. Prepare Trading and Profit and Loss Account for the year ended 31st March 2017 and the Balance Sheets as on that date after taking into account the following adjustments:

Particulars	Dr.(₹)	Cr. (₹)
Capital		3,30,000
Drawings	24,450	
Opening Stock	2,00,000	
Bills Receivable	25,000	
Purchases	2,75,000	
Sales		4,20,000
Bills Payable		60,000
Return Outwards		4,500
Return Inward	5,000	
Plant and Machinery	1,00,000	
Loose Tools	25,000	
Land and Building	25,000	
Sundry Debtors	1,25,000	
Sundry Creditors		1,40,000
Cash at Bank	77,550	
Salaries and Wages	50,000	
Repairs and Renewals	7,500	
Insurance	3,000	
Power and Fuel	3,500	
Printing and Stationery	2,000	
General Expenses	6,500	
	9,54,500	**9,54,500**

Adjustments:

a) Closing stock was worth ₹1,30,000.
b) Depreciation Land and Building 15% and Plant and Machinery by 5%.
c) Provide for Bad and Doubtful Debts at 5% on Debtors.
d) Insurance paid in advance to the extent of ₹750.
e) Revalue total at ₹20,000.
f) Provide for outstanding expenses as follows: Salaries ₹2,500 and Wages ₹1,000.

First Semester Examination, 2018

2) From the following trial balance of Rajan, prepare trading and profit and loss account for the year ended 31.3.2016 and the Balance Sheet on at that date after taking into account the adjustments given below:

Dr. Trial Balance as on 31.3.2016 Cr.

Particulars	₹	Particulars	₹
Drawings	7,600	Capital	2,90,000
Purchases	89,000	Sales	1,50,000
Sales Return	2,800	Purchases Return	4,500
Stock (1.4.2015)	12,000	Interest	3,500
Wages	8,000	Sundry creditors	12,000
Building	2,20,000	Bills payable	7,000
Freight	20,000		
Trade Expenses	2,000		
Advertisement	2,400		
Taxes and Insurance	1,300		
Sundry Debtors	65,000		
Bills Receivables	15,000		
Cash at Bank	12,000		
Cash in Hand	1,900		
Salaries	8,000		
	4,67,000		**4,67,000**

Adjustments:

1) Stock on 31.3.16 was valued at ₹15,000.
2) Insurance was prepaid of ₹400.
3) Salary outstanding ₹2000 and taxes outstanding ₹1300.
4) Depreciate Building at 2% P.A.

Or

2) Following Trial Balance is prepared on 31.3.16 from Asha's book. Prepare Trading and Profit and Loss Account for the year ended 31.3.16 and the Balance Sheet as on that date after taking into account the adjustment given below.

Dr. Trial Balance as on 31.3.2016 Cr.

Particulars	₹	Particulars	₹
Cash in hand	8,000	Sales	5,00,000
Cash at Bank	12,000	Creditors	40,000
Wages	20,000	Capital	3,20,000
Machinery	2,12,000	Commission	2,800
Bills Receivables	40,000		
Opening stock	84,000		
Purchases	3,30,000		
Sales Return	6,000		
Salaries	32,000		
Debtors	86,000		
Taxes	22,000		
Bad debts	6,000		
Freight	4,800		
	8,62,800		**8,62,800**

Adjustments:

1) Closing stock ₹90,000.
2) Outstanding salaries ₹3,200 and wages ₹2000.
3) Prepaid insurance ₹900 and accrued commission ₹1200.
4) Charge 10% depreciation on machinery.

UNIT III

First Semester Examination, 2014

3 a) "Cost Accounting System is neither unnecessary nor expensive, rather it is profitable investment." Comment.

3 b) i) "A good system of costing serve as a means of control over expenditure and helps to secure economy in manufacture." Discuss.

3 b) ii) Classify the following items into Factory Overheads, Administration Overheads and Selling and Distribution Overheads.

Loose tools	Income Tax	Postage and Telegram
Rent, Rate and Taxes	Land and Building	Delivery Charges
Indirect Material	Packing Expenses	Director's Fees
Warehouse Rent	Welfare Expenses	Overtime Wages
Advertising	General Expenses	Collection Charges
Property Tax on Building	Carriage Outward	Dividend Paid
Labour Expenses	Bad Debts	Unproductive Wages

First Semester Examination, 2015

3 a) Define Cost Accounting? Also explain the various types of costs?

3 b) Prepare Cost Sheet from the following information:

Particulars		₹
Raw materials consumed		40,000
Wages paid to labourers		10,000
Directly chargeable expenses		2,000
Oil and Waste		100
Wages of Foreman		1,000
Storekeepers Wages		500
Electric Power		200
Lighting:		
Factory	500	
Office	200	700
Rent:		
Factory	2,000	
Office	1,000	3,000
Repairs and Renewals:		
Factory Plant	500	
Machinery	1,000	
Office premises	200	1,700
Depreciation:		
Office Premises	500	
Plant and Machinery	200	700
Consumable stores		1,000
Manager's salary		2,000
Director's fees		500
Office printing and Stationery		200
Telephone charges		50
Postage and Telegrams		100
Salesmen's Commission and Salary		500
Travelling Expenses		200
Advertising		500
Warehouse charges		200
Carriage outward		150

First Semester Examination, 2016

3 a) Explain sunk costs with suitable example.

3 b) From the following information you are required to calculate prime cost, works cost, total cost of production, cost of sales and profit.

Particulars	₹	Particulars	₹
Direct Labour	1,00,000	Direct Expenses	1,50,000
Factory Overheads	2,00,000	Office Overheads	1,80,000
Direct Material	5,00,000	Sales	15,00,000
Selling and Distribution Overheads	50,000		

Or

3 a) State in brief objectives of cost accounting.

3 b) Prepare a cost sheet showing prime cost, works cost, total cost of production, cost of sales and profit from the following information related to Chennai Pvt. Ltd., Chennai.

Particulars	₹	Particulars	₹
Sales	5,00,000	Direct Wages	50,000
Direct Expenses	50,000	Factory Rent	20,000
Storekeeper Salary	10,000	Salary	1,00,000
Direct Material	2,00,000	Director's Fees	30,000
Showroom Rent	7,000		

First Semester Examination, 2017

3 a) Explain in brief classification of cost as per elements.

3 b) From the following prepare cost sheet.

Particulars	₹
Sales	5,00,000
Direct Wages	1,10,000
Factory Rent	30,000
Direct Material	2,90,000
Showroom Rent	15,000
Direct Expenses	10,000
Printing and Stationery	5,000
Audit Fees	15,000

Or

3 a) Write a short note on Relevant and Irrelevant Cost.

3 b) From the following information you are required to calculate prime cost, works cost, total cost of production, cost of sales and profit.

Particulars	₹
Sales	1,00,000
Direct Material	40,000
Direct Expenses	2,000
Factory Overheads	7,000
Direct Wages	10,000
Selling and Distribution Overheads	2,000
Administrative Overheads	5,000

First Semester Examination, 2018

3 a) What do you mean by cost accounting? Explain objectives of cost accounting.

3 b) Calculate material consumed, prime cost and works cost – Opening stock of raw material ₹40,000, closing stock of raw materials ₹50,000, purchase of raw material ₹4,00,000, Direct labour ₹3,00,000, Indirect labour ₹10,000, carriage outward ₹20,000, lubricant ₹10,000, Factory rent ₹60,000.

3 a) Write short note on classification of costs.

3 b) Identify whether following expenses are factory/office and administration/selling and distribution overheads.

i) Show room rent.
ii) Cleaning charges.
iii) Storekeeper's wages.
iv) Labour welfare expenses.
v) Legal fees.
vi) Travelling expenses.
vii) General manager's salary.
viii) Consumable stores.
ix) Sundry expenses
x) Counting house salary.

UNIT IV

First Semester Examination, 2014

5 a) Himmatlal Co. Ltd., Pune provides the following information for the year 2014-15:

Sales – 15,000 units
Fixed Cost – ₹34,000
Sales Value – ₹1,50,000
Variable Cost – ₹6 per unit

You are required to calculate:

1) P/V Ratio, BEP in ₹ and Margin of Safety.
2) Also calculate and comment on the effect of the following:
   i) Decrease of 10% in selling price.
   ii) Increase of 10% in variable cost.

First Semester Examination, 2015

5 a) The sales turnover and profit during two years were as follows:

Year	Sales (₹)	Profit (₹)
2011	1,40,000	15,000
2012	1,60,000	20,000

You are required to calculate:

i) P/V Ratio
ii) Sales required to earn a profit of ₹40,000
iii) Profit when sales are ₹1,20,000.

First Semester Examination, 2016

Ques 5) KLM Industries furnishes following information for the level of output of 1,000 units for the year 2010. (10)

Selling Price per Unit = ₹100
Variable Cost per Unit = ₹40
Total Fixed Cost = ₹30,000

Find:

a) P/V Ratio
b) BEP in units
c) BEP in sales
d) The amount of sales required to earn profit of ₹42,000 (assuming fixed cost in total remains same).

First Semester Examination, 2017

5) The sales turnover and profits during two years were as follows:

Year	Sales	Profit
2016	45,00,000	5,00,000
2017	50,00,000	7,00,000

You are required to calculate:

a) P/V Ratio.
b) Fixed cost.
c) Break Even Point (₹)
d) Sales to Earn Profit of ₹10,00,000.
e) Margin of Safety.

First Semester Examination, 2018

5) The turnover and profits during the two periods were as follows:

	Sales (₹)	Profits (₹)
Period-I	40 lac	4 lac
Period-II	60 lac	8 lac

Assuming that the cost structure and selling prices remain the same in the two periods.

Calculate:
Profit Volume Ratio, BEP sales, the sales required to earn profit of ₹10 lac, margin of safety in period II.

UNIT V

First Semester Examination, 2014

5 b) A newly started company wishes to prepare cash budget from January. Prepare a cash budget for the first six months from the following estimated revenue and expenses:

Month	Total Sales (₹)	Purchases (₹)	Wages (₹)	Production Overheads (₹)	Selling and Distribution Overheads (₹)
January	20,000	20,000	4,000	3,200	800
February	22,000	14,000	4,400	3,300	900
March	28,000	14,000	4,600	3,400	900
April	36,000	22,000	4,600	3,500	1,000
May	30,000	20,000	4,000	3,200	900
June	40,000	25,000	5,000	3,600	1,200

Cash balance on 1st January was ₹10,000. A new machinery is to be installed at ₹20,000 on credit, to be repaid by two equal instalments in March and April.

Sales commission @ 5% on total sales is to be paid within a month following actual sales.

₹10,000 being the amount of 2nd call may be received in March. Share premium amounting to ₹2,000 is also obtainable with the 2nd call.

Period of credit allowed by suppliers – 2 months.

Period of credit allowed by customers – 1 months.

Delay in payment of wages $^1/_2$ month and overheads 1 month.

Assume cash sales to 50% of total sales.

First Semester Examination, 2015

5 b) The expenses for the production of 5,000 units in a factory are given as follows:

Particulars	Per Unit (₹)
Materials	50
Labour	20
Variable Overheads	15
Fixed Overheads (₹50,000)	10
Administrative Expenses (5% Variable)	10
Selling Expenses (20% Fixed)	6
Distribution Expenses (10% Fixed)	5
Total Cost of Sales per unit	116

You are required to prepare a budget for the production of 7,000 units.

First Semester Examination, 2016

5) From the following prepare cash budget for 3 months for January. February and March, 2012.

Month	Total Sales (₹)	Materials (₹)	Wages (₹)	Overheads	
				Production (₹)	Selling and Distt. (₹)
January	40,000	42,000	8,000	6,400	1,600
February	44,000	28,000	8,800	6,600	1,800
March	48,000	28,000	9,200	6,600	1,600
April	52,000	24,000	9,200	6,800	1,800
May	56,000	24,000	9,600	7,000	1,800
June	60,000	32,000	9,600	7,200	2,000

1) Cash balance on January 1, 2012 was ₹20,000.
2) A new machine is to be installed at ₹60,000 on credit, to be repaid by two equal instalments in March and April, 2012.
3) Period of credit allowed by suppliers 1 month.
4) Period of credit allowed to customers 1 month.
5) Delay in payment of overheads 1 month.
6) Delay in payment of wages 1 month.
7) Assume cash sales to be 50% of total sales.

First Semester Examination, 2017

5) From the following information, prepare a flexible budget at 60% and 80% capacity.

Particulars	70% Capacity (₹)
Variable Overheads:	
Indirect Material	5,000
Indirect Labour	15,000
Semi-Variable Overheads:	
Electricity (60% Variable)	50,000
Repairs and Maintenance (35% Fixed)	5,000
Fixed Overheads:	
Salaries	10,000
Depreciation on Machine	14,000
Insurance	6,000
	1,05,000

First Semester Examination, 2018

5) In a factory, 100 workers are engaged and the average rate of wages is 50 paise per hour Standard working hours per week are 40 and the standard performance is 10 units per gang hour.

During a week in March, wages paid for 50 workers were at the rate of 50 paise per hour 10 workers at 10 paise per hour and 40 workers at 40 paise per hour. Actual output was 380 units.

The factory did not work for five hours due to break down of machinery. Calculate Labour cost variance, Labour rate variance, Labour efficiency variance, Idle time variance, Labour yield variance.

MODEL QUESTION PAPER-I

MANAGERIAL ACCOUNTING

MBA - FIRST SEMESTER EXAMINATION

Time: 2½ Hours **Max. Marks: 50**

Note: 1) All questions shall be compulsory with internal choice within the questions.
2) FIVE questions each carries 10 marks.
3) Figures to the right indicate marks for that question/sub-question.

1) **Answer any 5 out of 8:** (2 marks each)
   a) Write a short note on business entity concept.
   b) What do you mean by accounting equation?
   c) Write a short note on Relevant and Irrelevant Cost.
   d) What is cost unit?
   e) Define marginal costing.
   f) Define P/V ratio.
   g) What is procurement budget?
   h) Define the term 'variance'.

2) **Answer any 2 out of 3:** (5 marks each)
   a) Explain different forms of business organisation?
   b) Explain any five accounting concepts with suitable example.
   c) Write short note on classification of costs.

3) a) What do you mean by cost accounting? Explain objectives of cost accounting. (10)

Or

b) The following are the balances extracted from the books of ABC as on 31st December, 2018. From these balances, prepare his Trading and Profit and Loss Account and Balance Sheet as on that date:

Particulars	Dr. (₹)	Cr. (₹)
Opening Stock	12,000	
Purchases	40,000	
Sales		86,000
Discount		400
Sales Return	6,000	
Buildings	50,000	
Debtors	16,000	
Salaries	2,400	
Office Expenses	1,200	
Wages	10,000	
Purchase Return		4,000
Interest		800
Travelling Expenses	400	
Fire Insurance Premium	800	
Machinery	20,000	
Carriage on Purchases	700	
Commission	400	
Cash in hand	2,300	
Rent and Taxes	1,800	
Capital		62,000
Creditors		10,800
Total	**1,64,000**	**1,64,000**

Adjustments:

i) Closing Stock was valued at ₹16,000.
ii) Wages ₹2,000 and salaries ₹1,200 are outstanding.
iii) Rent for two months at the rate of ₹500 per month is outstanding.
iv) Depreciate Buildings by 5% and machinery by 10%.
v) Prepaid Insurance ₹200. (10)

4) a) A factory produces a standard product. The following information is given to you from which you are required to prepare a cost sheet for January 2019:

Particulars	₹
Raw materials consumed	2,91,000
Direct wages	1,29,000
Other direct expenses	81,000
Factory overheads	80% of direct wages
Office overheads	10% of works cost
Selling & Distribution expenses	₹20 per unit sold
Units produced and sold during the month	10,000

Find the selling price per unit when the profit earned is at 20% on selling price. (10)

Or

b) KLM Industries furnishes following information for the level of output of 1,000 units for the year 2018.

Selling Price per Unit	=	₹100
Variable Cost per Unit	=	₹40
Total Fixed Cost	=	₹30,000

Find:

a) P/V Ratio
b) BEP in units
c) BEP in sales
d) The amount of sales required to earn profit of ₹42,000 (assuming fixed cost in total remains same). (10)

5) a) The expenses for the production of 5,000 units in a factory are given as follows:

Particulars	Per Unit (₹)
Materials	50
Labour	20
Variable Overheads	15
Fixed Overheads (₹50,000)	10
Administrative Expenses (5% Variable)	10
Selling Expenses (20% Fixed)	6
Distribution Expenses (10% Fixed)	5
Total Cost of Sales per unit	**116**

You are required to prepare a budget for the production of 7,000 units. (10)

Or

b) Compute material cost, material price, material mix and material usage variance using following given information: (10)

Materials	Standard			Actual		
	Quantity (Kilos)	Unit Price (₹)	Total (₹)	Quantity (Kilos)	Unit Price (₹)	Total (₹)
A	5	1	5	3	2	6
B	10	2	20	6	3	18
C	10	3	30	11	2	22
Total	**25**	**6**	**55**	**20**	**7**	**46**

MODEL QUESTION PAPER-II

MANAGERIAL ACCOUNTING

MBA - FIRST SEMESTER EXAMINATION

Time: 2½ Hours **Max. Marks: 50**

Note: 1) All questions shall be compulsory with internal choice within the questions.
2) FIVE questions each carries 10 marks.
3) Figures to the right indicate marks for that question/sub-question.

1) **Answer any 5 out of 8:** (2 marks each)
   a) What do you understand by business decisions?
   b) What is trial balance?
   c) What do you mean by capital and revenue receipts?
   d) Explain sunk costs with suitable example.
   e) What is cost centre?
   f) What is contribution?
   g) Define the term 'budget'.
   h) What do you understand by standard costing?

2) **Answer any 2 out of 3:** (5 marks each)
   a) Define accounting. Who are the users of accounting information?
   b) Explain the importance and objectives of financial statements.
   c) Give the difference between capital expenditure and revenue expenditure.

3) a) Explain the basic accounting concepts and conventions in financial accounting. (10)

Or

b) The following Trial Balance was extracted from the books of M/s ABC on 31st December, 2018. You are required to prepare Trading and Profit and Loss Account for the year ending on 31st December, 2018 and a Balance Sheet as on that date:

Particulars	Dr. (₹)	Cr. (₹)
Capital		30,000
Drawings	5,000	
Debtors and Creditors	20,000	10,000
Loan from Bank		9,500
Interest on loan	300	
Cash in hand	2,000	
Provision for bad debts		700
Stock (1.1.2018)	6,800	
Motor vehicle	10,000	
Cash at bank	3,500	
Land and building	12,000	
Purchases and Sales	66,000	1,10,000
Returns	8,000	1,500
Carriage Outwards	2,500	
Carriage Inwards	3,000	
Salaries	9,000	
Rent and Insurance	3,000	
General expenses	6,900	
Bad Debts	500	
Discount		500
Bills Receivable and Payable	6,000	2,000
Rent Received		300
Total	**1,64,500**	**1,64,500**

Adjustments:
i) Stock on 31-12-2018 ₹7,000.
ii) Depreciation on Land and Building @ 2.5% and on Motor Vehicle @ 20% per annum.

iii) Salaries outstanding ₹200.
iv) Prepaid insurance ₹200.
v) Provision for bad debts is to be maintained at 5% on debtors. (10)

4) a) Prepare a cost sheet showing prime cost, works cost, total cost of production, cost of sales and profit from the following information related to Chennai Pvt. Ltd., Chennai. (10)

Particulars	₹	Particulars	₹
Sales	5,00,000	Direct Wages	50,000
Direct Expenses	50,000	Factory Rent	20,000
Storekeeper Salary	10,000	Salary	1,00,000
Direct Material	2,00,000	Director's Fees	30,000
Showroom Rent	7,000		

Or

b) The sales turnover and profits during two years were as follows: (10)

Year	Sales	Profit
2017	45,00,000	5,00,000
2018	50,00,000	7,00,000

You are required to calculate:
a) P/V Ratio.
b) Fixed cost.
c) Break Even Point (₹)
d) Sales to Earn Profit of ₹10,00,000.
e) Margin of Safety.

5) a) From the following prepare cash budget for 3 months for January. February and March, 2019.

Month	Total Sales (₹)	Materials (₹)	Wages (₹)	Overheads	
				Production (₹)	Selling and Distt. (₹)
January	40,000	42,000	8,000	6,400	1,600
February	44,000	28,000	8,800	6,600	1,800
March	48,000	28,000	9,200	6,600	1,600
April	52,000	24,000	9,200	6,800	1,800
May	56,000	24,000	9,600	7,000	1,800
June	60,000	32,000	9,600	7,200	2,000

1) Cash balance on January 1, 2019 was ₹20,000.
2) A new machine is to be installed at ₹60,000 on credit, to be repaid by two equal instalments in March and April, 2019.
3) Period of credit allowed by suppliers 1 month.
4) Period of credit allowed to customers 1 month.
5) Delay in payment of overheads 1 month.
6) Delay in payment of wages 1 month.
7) Assume cash sales to be 50% of total sales. (10)

Or

b) Following data pertains to two group of workmen and their standard and actual working hours and rate:

	Standard Hours	Rate per Hour	Total
Worker A	25 hrs.	2	50
Worker B	10 hrs.	3	30
			₹80

	Actual Hours	Rate per Hour	Total
Worker A	20 hrs.	2.50	50
Worker B	15 hrs.	4	60
			₹110

Calculate: (10)
1) Labour cost variance, 2) Labour efficiency variance, 3) Labour rate variance.

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